Spectrum Team

اللجنة الأكاديمية

حلول الكتاب لمادة

مكونات الآلة و أليتها





تواصل معنا



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Instructor's Solutions Manual
to accompany

Machines and Mechanisms: Applied Kinematic Analysis
Fourth Edition



MECHATRONICS SPECTRUM

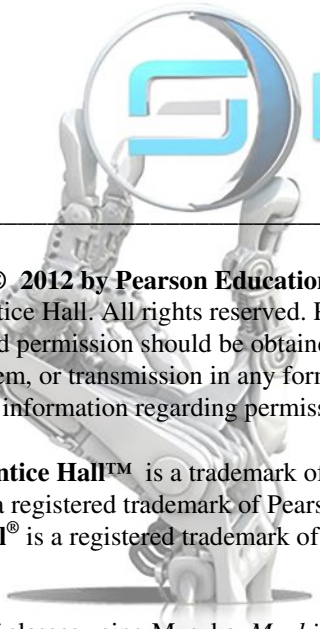
David Myszka

TEAM

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Upper Saddle River, New Jersey
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MECHATRONICS SPECTRUM TEAM

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Solution Manual to Accompany:

Machines and Mechanisms

Applied Kinematic Analysis

4th Edition

David H. Myszka, Ph.D., P.E.

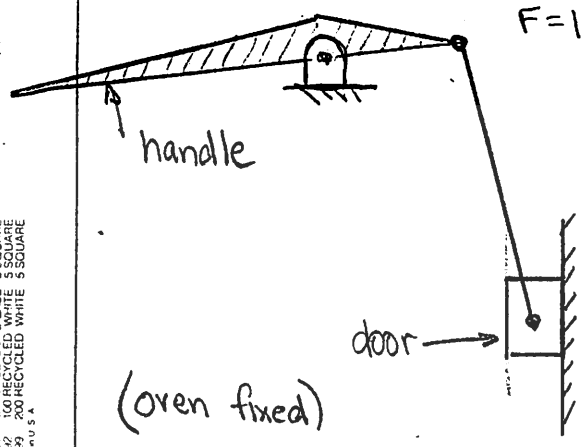
University of Dayton
300 College Park
Dayton OH, 45469
dmyszka@udayton.edu

SPECTRUM TEAM

CHAPTER 1

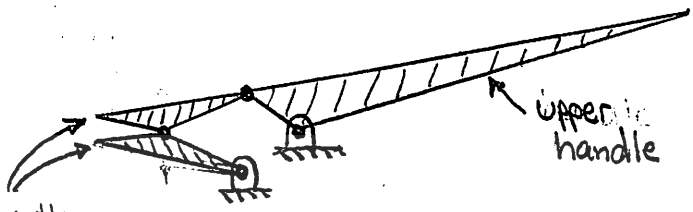
1-1

n=4
J=4
F=1



(oven fixed)

1-2



(lower handle fixed)

n=4, J=4, F=1

1-3

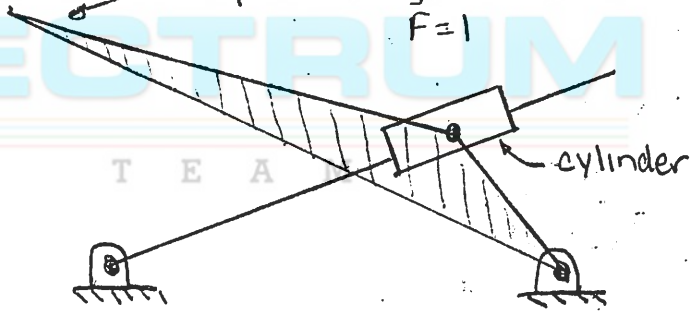
n=4, J=4, F=1

(seat back fixed)



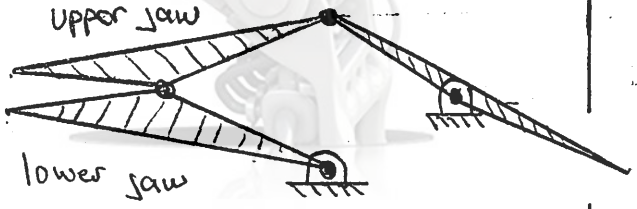
1-4

n=4
J=4
F=1



1-5

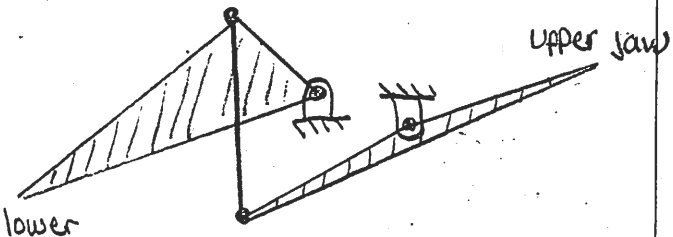
n=4, J=4, F=1



(upper handle fixed)

1-6

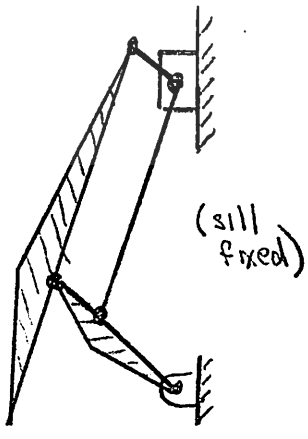
n=4, J=4, F=1



(upper handle/lower jaw fixed)

1-7

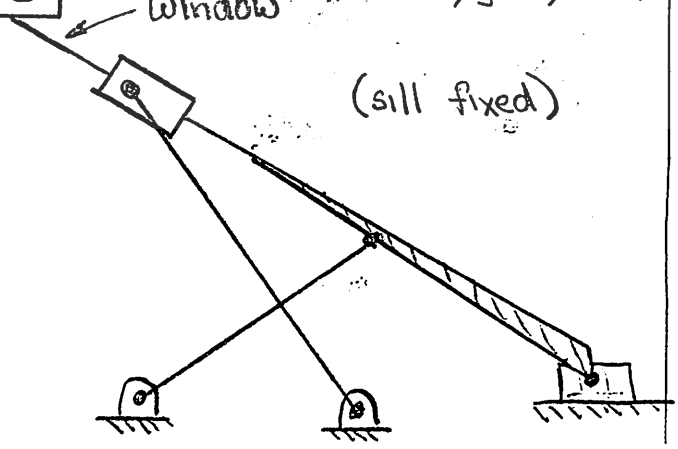
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J=7
F=1



(sill fixed)

1-8

n=6, J=7, F=1

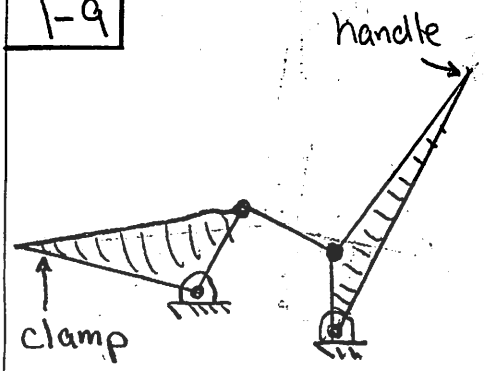


(sill fixed)

MECHATRONICS SPECTRUM TEAM

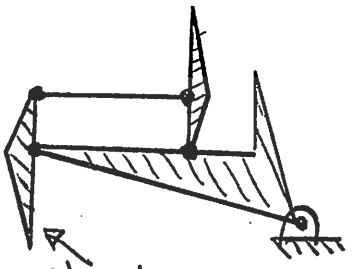
42-381 50 SHEETS EYE-EASE® 5 SQUARE
42-382 100 SHEETS EYE-EASE® 5 SQUARE
42-389 200 SHEETS EYE-EASE® 5 SQUARE
42-392 100 RECYCLED WHITE 5 SQUARE
42 399 200 RECYCLED WHITE 5 SQUARE
Made in U.S.A.
National® Brand

1-9



n=4
j=4
F=1

1-10



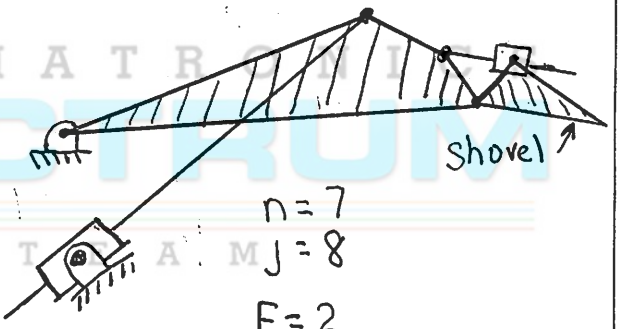
n=5
j=5
F=2

1-11



n=4
j=4
F=1

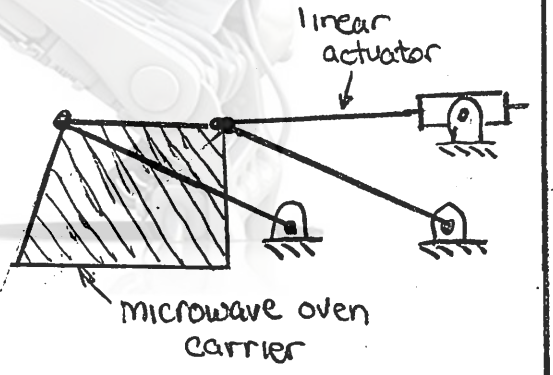
1-12



n=7
j=8
F=2

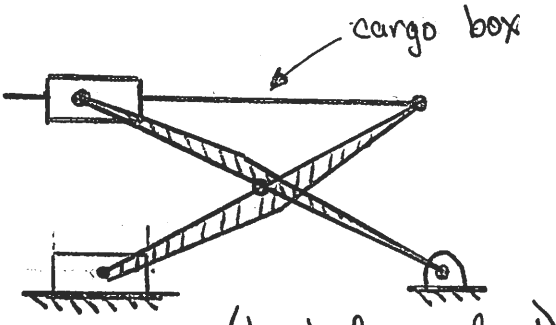
1-13

n=6, j=7, F=1

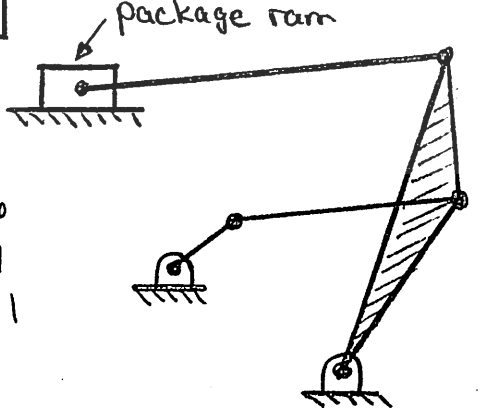


1-14

n=6, j=7, F=1



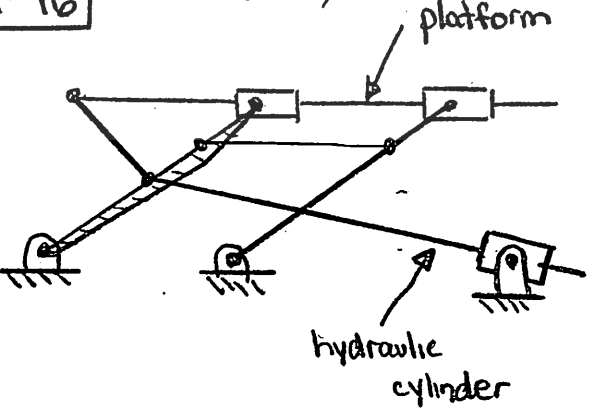
1-15



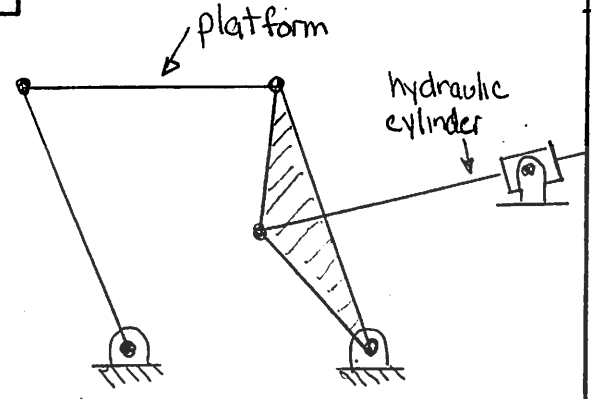
n=6
j=7
F=1

1-16

n=10, j=13, F=1

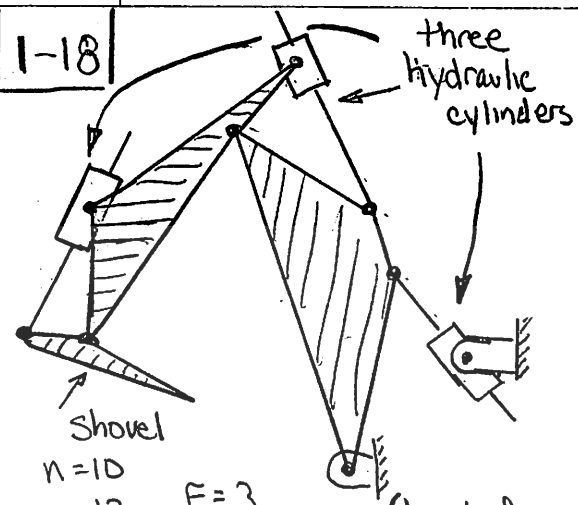


1-17



$n = 6$, $J_p = 7$, $F = 1$

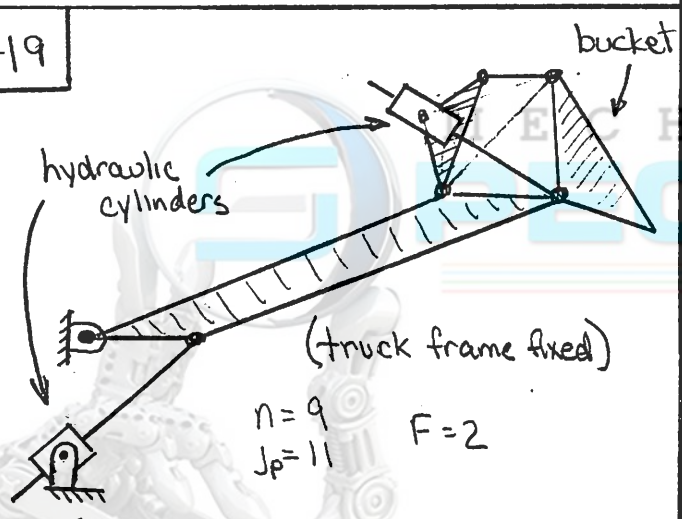
1-18



$n = 10$
$J = 12$ $F = 3$

(truck frame fixed)

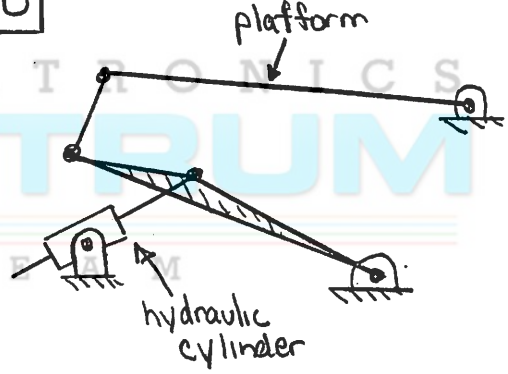
1-19



(truck frame fixed)

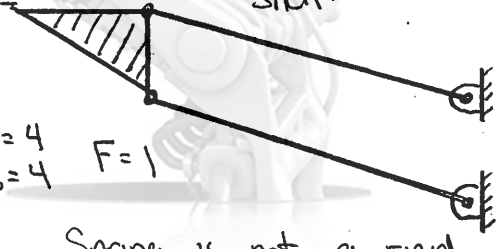
$n = 9$
$J_p = 11$ $F = 2$

1-20



$n = 6$, $J_p = 7$, $F = 1$

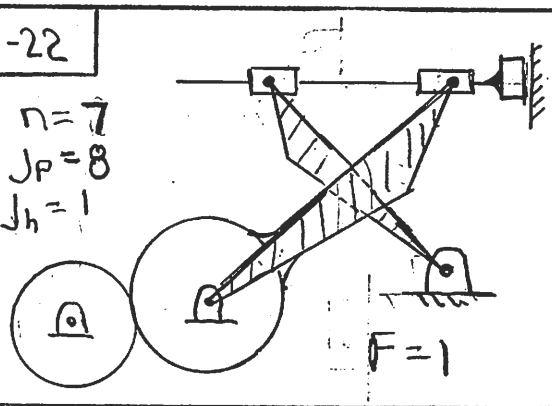
1-21



$n = 4$
$J_p = 4$ $F = 1$

Spring is <u>not</u> a rigid body

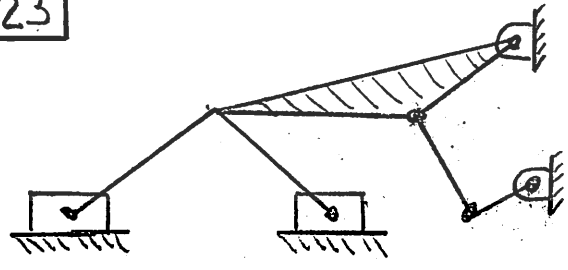
1-22



$n = 7$
$J_p = 8$
$J_h = 1$

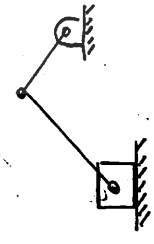
$F = 1$

1-23



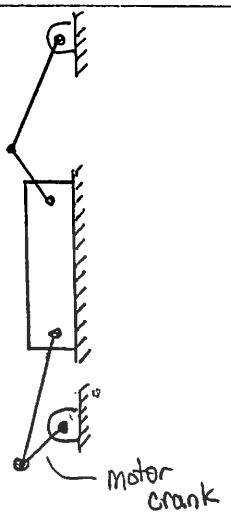
$n = 8$
$J_p = 10$
$F = 1$

1-24



$n = 4$
$J_p = 4$
$F = 1$

1-25



$n = 6$
$J = 7$
$F = 1$

1-26 Links = 4 Joints = 4 (3 pins, 1 sliding)

$DOF = 3(4-1) - 2(4) = 1$ //

1-27 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-28 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-29 Links = 4 Joints = 4 (3 pins, 1 sliding)

$DOF = 3(4-1) - 2(4) = 1$ //

1-30 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-31 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-32 Links = 6 Joints = 7 (6 pins, 1 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-33 Links = 6 Joints = 7 (5 pins, 2 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-34 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-35 Links = 5 Joints = 5 (5 pins)

$DOF = 3(5-1) - 2(5) = 2$ //

1-36 Links = 4 Joints = 4 (3 pins, 1 sliding)

$DOF = 3(4-1) - 2(4) = 1$ //

1-37 Links = 7 Joints = 8 (6 pins, 2 sliding)

$DOF = 3(7-1) - 2(8) = 2$ //

1-38 Links = 6 Joints = 7 (5 pins, 1 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-39 Links = 6 Joints = 7 (5 pins, 2 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-40 Links = 6 Joints = 7 (6 pins, 1 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-41 Links = 10 Joints = 13 (10 pins, 3 sliding)

$DOF = 3(10-1) - 2(13) = 1$ //

1-42 Links = 6 Joints = 7 (6 pins, 1 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

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1-43 Links = 10 Joints = 12 (9 pins, 3 sliding)

$DOF = 3(10-1) - 2(12) = 3$ //

1-44 Links = 9 Joints = 11 (9 pins, 2 sliding)

$DOF = 3(9-1) - 2(11) = 2$ //

1-45 Links = 6 Joints = 7 (6 pins, 2 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-46 Links = 4 Joints = 4 (4 pins)

$DOF = 3(4-1) - 2(4) = 1$ //

1-47 Links = 8 Primary Joints = 10 (6 pins, 4 sliding)
Higher Joints = 1 (gear)

$DOF = 3(8-1) - 2(10) - 1 = 1$ //

1-48 Links = 8 Joints = 10

$DOF = 3(8-1) - 2(10) = 1$ //

1-49 Links = 4 Simple joints = 4 (3 pin + 1 sliding)

$DOF = 3(4-1) - 2(4) = 1$ //

1-50 Links = 6 joints = 7 (6 pins, 1 sliding)

$DOF = 3(6-1) - 2(7) = 1$ //

1-51 $s = 1.5, l = 14, p = 12, q = 4$

s = side

$1.5 + 14 < 14 + 4$

$15.5 < 18$ (yes)

⇒ crank-rocker

1-52 $s = 4, l = 12, p = 12, q = 5$

s = side

$4 + 12 < 12 + 5$

$16 < 17$ (yes)

⇒ crank-rocker

1-53 $s = 3, l = 12, p = 8, q = 4$

s = side

$3 + 12 < 8 + 4$

$15 < 12$ (no)

⇒ triple-rocker

1-54 $s = 3, l = 12, p = 12, q = 5$

s = side

$3 + 12 < 12 + 5$

$15 < 17$ (yes)

⇒ crank-rocker

MECHATRONICS SPECTRUM TEAM

Answers to the Chapter 1 Case Study Questions:

Case 1-1.

1. As link A rotates clockwise, 90^0, slide C will move to the left.
2. As link A rotates clockwise, 90^0, the ball trapped in slide C will drop down the lower, left chute.
3. As link A continues another 90^0, clockwise, link A will be oriented straight down and slide C will return to the original position shown in the figure.
4. This device allows one feeder bowl to distribute balls to two separate stations.
5. The chamfers on slide C allow relief as a ball drops into the empty slot as slide C moves under chute D.
6. This device would be useful in that only one feeder bowl will need to be filled and monitored.

Case 1-2.

1. As handle A is rotated counterclockwise, flapper C rotates clockwise.
2. As flapper C is raised, the water in the tank is allowed to flow through the opening.
3. A buoyancy force will offset the water pressure, keeping the flapper in the upper position.
4. As the water level lowers to the level of the flapper, the flapper will lower with the water line.
5. Item D floats on top of the water. As the water level lowers, float D also lowers.
6. As item D rotates counterclockwise, item F is also rotates counterclockwise.
7. Item F is a valve that controls water flow, In the upper position, it fills the tank. In the lower position, water flow is cut off.
8. These mechanisms allow a rapid flow of water from the tank, slowly refill the tank, then shut-off the water flow.
9. The water pressure in residential areas does produce the required flow rate for a water closet.

Case 1-3.

1. In the shown configuration, the water level in bucket B_1 is raising, as the flow into the bucket exceeds the amount that is leaking from the holes at the bottom of the bucket.
2. In the shown configuration, the water level in bucket B_2 is lowering, as the water is leaking from the holes at the bottom of the bucket.
3. If bucket B_2 were forced upward, rocker arm C would rotate clockwise.
4. If bucket B_2 were forced upward, rocker arm R would rotate counterclockwise.
5. Rocker arm R controls a directional valve, channeling the water flow to either the upper pipe or the lower pipe.
6. As water drains from one bucket, making it lighter, and fills the other, making it heavier, the weight shift cases the rocker C to rotate and reverse the direction of the water flow. The process repeats itself and rotates rocker C back to the original position. The continual motion is oscillation of rocker C.
7. As rocker C rotates, channel S moves between left and right positions. This allows the steel rod, which is constantly moving, to be coiled onto a reel placed on the left side or a reel on placed on the right side.
8. Since water is abundant and a common cooling medium in most foundries, water flow can be used to drive some machinery.

Answers to the Chapter 2 Case Study Questions:

Case 2-1.

1. As handle A is rotated, moving threaded rod B to the left, grip C also moves to the left and slightly upward. Notice that links E and F are pivoting in the middle, thus grip C is constrained to a swinging motion.
2. As handle A is rotated, moving threaded rod B to the left, grip D moves to the right and slightly downward. Since links E and F are pivoting in the middle, grip D will have motion opposing grip C.
3. The purpose of this mechanism is to serve as a machining clamp for the workpiece.
4. The spring, G, pulling on link D would cause is to return to an upward and rightward position.
5. The purpose of spring G is, ultimately, to keep a positive contact between the threaded rod and link C.
6. Links E and F have a peculiar configuration to avoid interference with the workpiece, throughout the range of motion of the clamp.
7. Such a device could be called a machining clamp.
8. Since link C is moving in a swinging motion, the rounded end on the threaded rod, assures a consistent point contact with link C.



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Chapter 3.

3-1 $\cos\theta = 5/18 \Rightarrow \theta = 73.8°$ //

3-2 $A = \sqrt{(18)^2 - (5)^2} = 17.3$ in //

3-3 $\tan 60° = \frac{x}{6} \Rightarrow x = 10.4$ in //

3-4 $\theta = 180 - (60 + 90) = 30°$ //

$\cos 60° = \frac{6}{R} \Rightarrow R = 12$ in //

3-5 a) $\tan\theta = \frac{6}{4} \Rightarrow \theta = 56.3°$ //

$R = \sqrt{(6)^2 + (4)^2} = 7.2$ in //

b) $\theta = 180 - (90 + 60) = 30°$ //

$\cos 30° = \frac{5}{R} \Rightarrow R = 5.8$ in //

c) $\tan\theta = \frac{5}{8} \Rightarrow \theta =$

$R = \sqrt{(5)^2 + (8)^2} = 9.4$ in //

3-6 $\tan\beta = \frac{x}{y} = \frac{150}{275} \Rightarrow \beta = 28.6°$ //

hypotenuse = $2s$

$2s = \sqrt{(150)^2 + (275)^2} \Rightarrow s = 156.6$ mm //

3-7 $\tan 35° \approx \frac{x}{16} \Rightarrow x = 11.2$ in //

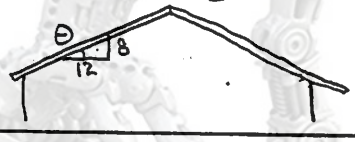
$2s = \sqrt{(16)^2 + (11.2)^2} \Rightarrow s = 9.8$ in //

3-8 $s = 10 \Rightarrow$ hypotenuse $= 20$ in

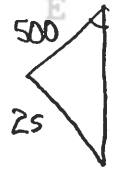
$\sin 35° = \frac{x}{20} \Rightarrow x = 11.5$ in //

$\cos 35° = \frac{y}{20} \Rightarrow y = 16.4$ in //

3-9 $\tan\theta = \frac{8}{12} \Rightarrow \theta = 33.7°$ //



3-10

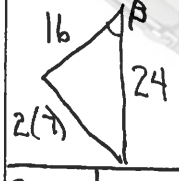


law of cosines

$2s = \sqrt{(500)^2 + (850)^2 - 2(500)(850)\cos 35}$

$s = 175$ mm //

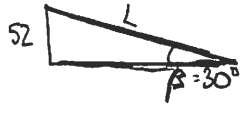
3-11



$\cos\beta = \frac{(16)^2 + (14)^2 - (24)^2}{2(16)(14)}$

$\beta = 42.9°$ //

3-12



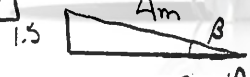
$L = 52/\sin 30$

$= 104$ in //

$104\text{ in}\left(\frac{1\text{ ft}}{12\text{ in}}\right)$

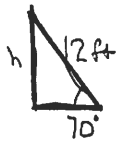
$= 8$ ft 8 in //

3-13



$\sin\beta = 1.5/4 \Rightarrow \beta = 22.0°$ //

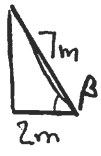
3-14



$\sin 70° = \frac{h}{12}$

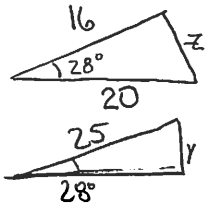
$h = 11.3$ ft //

3-15



$\cos\beta = \frac{2}{7} \Rightarrow \beta = 73.4°$ //

3-16



law of cosines

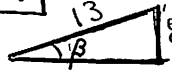
$z = \sqrt{(16)^2 + (20)^2 - 2(16)(20)\cos 28}$

$= 9.5$ ft //

$\sin 28° = y/25$

$\Rightarrow y = 11.7$ ft //

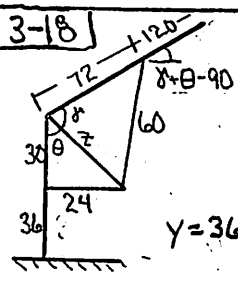
3-17



$\sin\beta = 8/13$

$\Rightarrow \beta = 38.0°$ //

3-18



$\tan\theta = 24/30 \Rightarrow \theta = 38.7°$

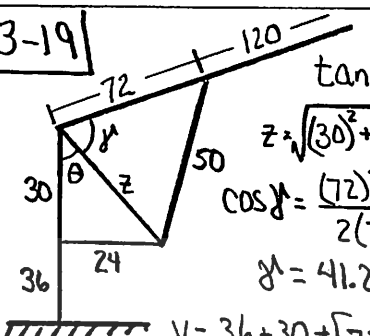
$z = \sqrt{(24)^2 + (30)^2} = 38.4$ in

$\cos\gamma = \frac{(72)^2 + (38.4)^2 - (60)^2}{2(72)(38.4)} \Rightarrow \gamma = 56.$

$y = 36 + 30 + [72 + 120]\sin(38.7 + 56.4 - 90) = 83.1$ //

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3-19



$\tan\theta = \frac{24}{30} \Rightarrow \theta = 38.7°$

$z = \sqrt{(30)^2 + (24)^2} = 38.4$ in

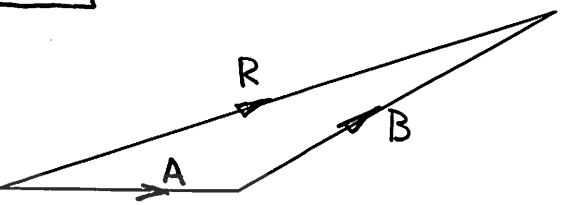
$\cos\gamma = \frac{(72)^2 + (38.4)^2 - (50)^2}{2(72)(38.4)}$

$\gamma = 41.2°$

$y = 36 + 30 + [72 + 120]\sin(\theta + \gamma - 90)$

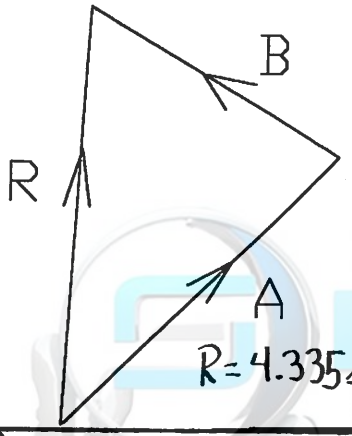
$= 66 + (192)\sin(-10.1°) = 32.3$ in

3-20



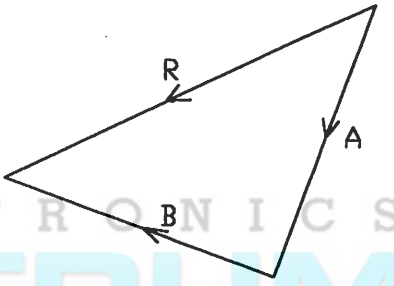
measuring: $R = 24.18 \angle 18.1°$

3-21



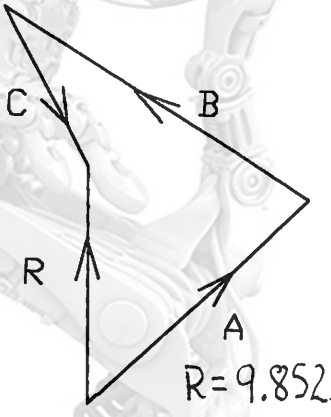
$R = 4.335 \angle 86.9°$

3-22



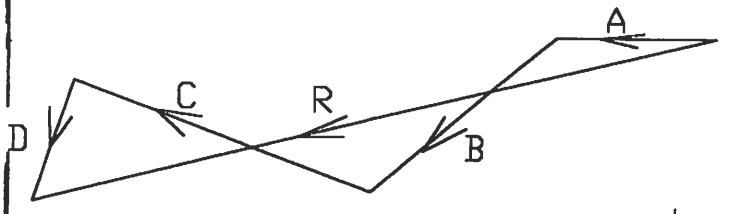
$R = 212.13 \angle 25°$

3-23



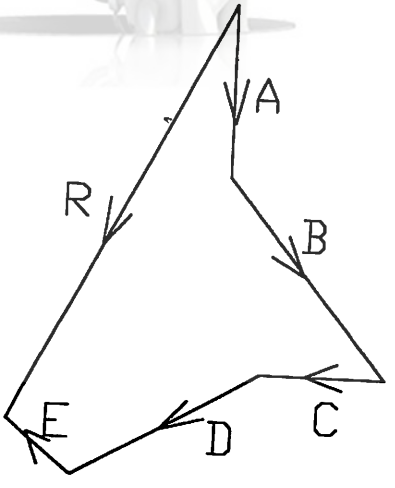
$R = 9.852 \angle 87.66$

3-24



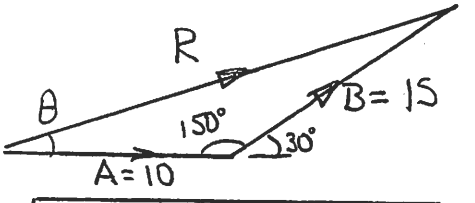
$R = 221.20 \angle 13.49°$

3-25



$R = 110.14 \angle 62.97°$

3-26



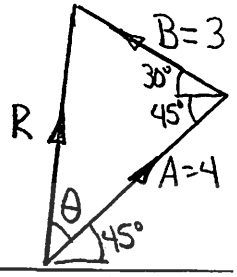
$R = \sqrt{(10)^2 + (15)^2 - 2(10)(15)\cos 150} = 24.18$

$\theta = \sin^{-1}\left(\frac{15}{24.18}\sin 150\right) = 18.07°$

$R = 24.18 \angle 18.07°$

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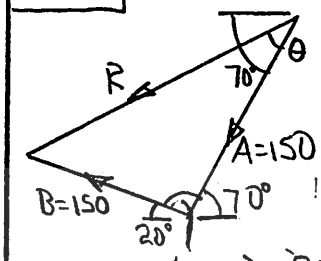
$R=\sqrt{(4)^2+(3)^2-2(4)(3)\cos 75^\circ} = 4.335$

$\theta = \sin^{-1}\left[\frac{3}{4.335}\sin 75\right] = 41.9^\circ$

$(45+41.9) = 86.9^\circ$

$R = 4.335 \angle 86.9^\circ$

3-28



$R=\sqrt{150^2+150^2-2(150)(150)\cos 90}$

$= 212.13$

$\theta = \sin\left[\frac{150}{212.13}\sin 90\right]$

$= 45^\circ$

$180-(20+70) = 90$

$(70-45) = 25$

$R = 212.13 \quad 25^\circ$

3-29

$R_x = 12.5\cos 45 - 15\cos 30 + 7.5\cos 60$

$= -0.40$

$R_y = 12.5\sin 45 + 15\sin 30 - 7.5\sin 60 = 9.84$



$R=\sqrt{(-0.4)^2+(9.84)^2}$ 9.85

$\theta = \tan^{-1}\left(\frac{9.84}{0.40}\right) = 87.66^\circ$

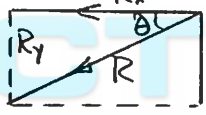
$R = 9.85 \quad 87.67^\circ$

3-30

$R_x = -50 - 75\cos 40 - 100\cos 20$

$-40\cos 70 = -215.1$

$R_y = 0 - 75\sin 40 + 100\sin 20 - 40\sin 70 = -51.6$



$R=\sqrt{(-215.1)^2+(-51.6)^2}$

$= 221.2$

$\theta = \tan^{-1} 51.6/215.1 = 13.4$

$R = 221.2 \quad 13.4^\circ$

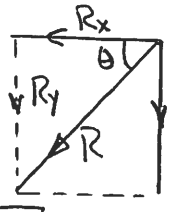
3-31

$R_x = 0 + 60\sin 40 - 30 - 50\cos 30$

$-20\sin 50 = -50.05$

$R_y = -40 - 60\cos 40 - 0 - 50\sin 30 + 20\cos 50$

$= -98.11$



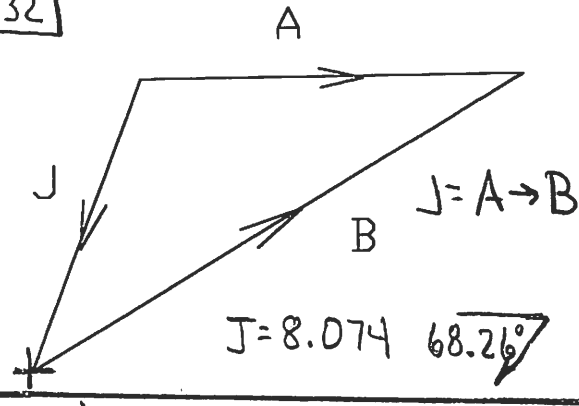
$R=\sqrt{(-50.05)^2+(98.11)^2}$

$= 110.14$

$\theta = \tan^{-1} 98.11/50.05 = 62.97^\circ$

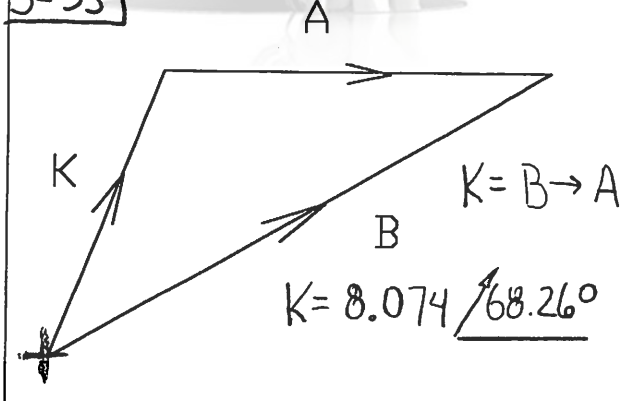
$R = 110.14 \quad 62.97^\circ$

3-32



$J = 8.074 \quad 68.26^\circ$

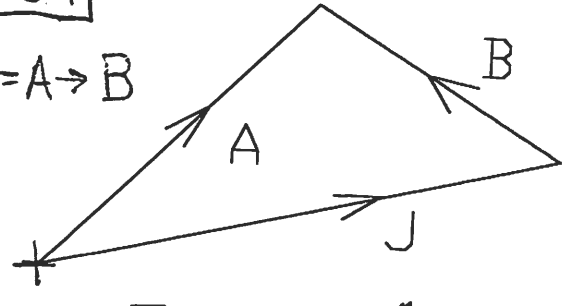
3-33



$K = 8.074 \angle 68.26^\circ$

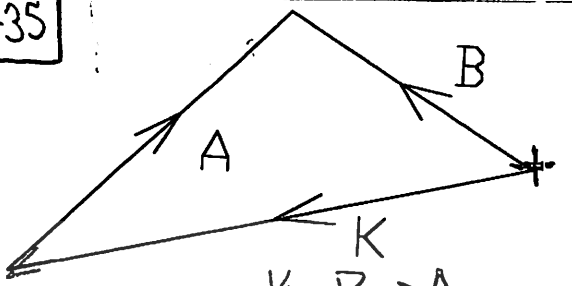
3-34

$J = A \rightarrow B$



$J = 5.587 \angle 13.76^\circ$

3-35



$K = B \rightarrow A$

Measuring:

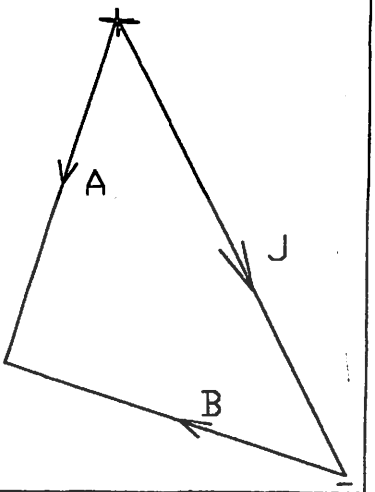
$K = 5.587 \quad 13.76°$

3-36

$J = A \rightarrow B$

Measuring:

$J = 212.13 \quad 65°$

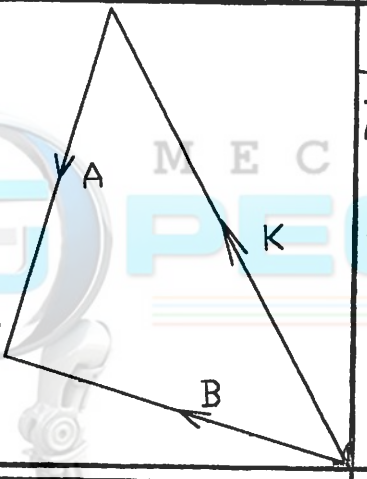


3-37

$K = B \rightarrow A$

Measuring:

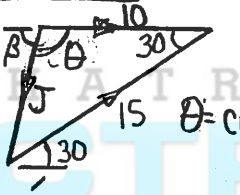
$K = 212.13 \quad 65°$



3-38

$J = \sqrt{10^2 + 15^2 - 2(10)(15)\cos 30}$

$= 8.074$



$\theta = \cos^{-1}\left(\frac{8.074^2 + 10^2 - 15^2}{2(8.074)(10)}\right) = 111.74°$

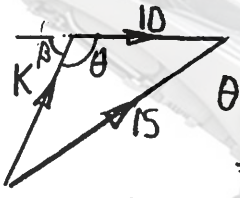
$\beta = 180 - 111.74 = 68.26°$

$J = 8.074 \quad 68.26°$

3-39

$K = \sqrt{10^2 + 15^2 - 2(10)(15)\cos 30}$

$= 8.074$



$\theta = \cos^{-1}\left(\frac{8.074^2 + 10^2 - 15^2}{2(8.074)(10)}\right)$

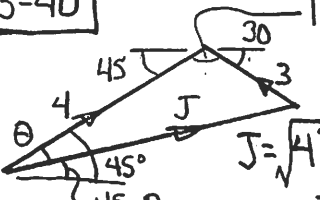
$= 111.74°$

$\beta = 180 - \theta = 68.26$

$K = 8.074 \quad 68.26°$

3-40

$180 - (45 + 30) = 105$



$J = \sqrt{4^2 + 3^2 - 2(4)(3)\cos 105}$

$= 5.587$

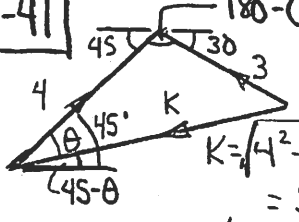
$\theta = \sin^{-1}\left(\frac{3}{5.587}\sin 105\right) = 31.24°$

$45 - \theta = 13.76°$

$J = 5.587 \quad 13.76°$

3-41

$180 - (45 + 30) = 105$



$K = \sqrt{4^2 + 3^2 - 2(4)(3)\cos 105}$

$= 5.587$

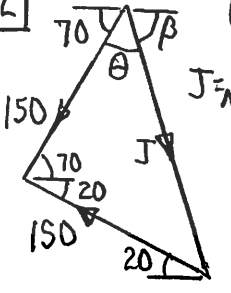
$\theta = \sin^{-1}\left(\frac{3}{5.587}\sin 105\right) = 31.24°$

$45 - \theta = 13.76°$

$K = 5.587 \quad 13.76°$

3-42

$(70 + 20) = 90°$



$J = \sqrt{150^2 + 150^2 - 2(150)(150)\cos 90}$

$= 212.13$

$\theta = \sin^{-1}\left(\frac{150}{212.13}\sin 90\right)$

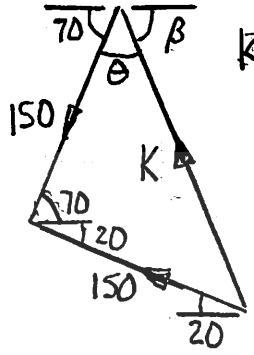
$= 45°$

$\beta = 180 - (\theta + 70) = 65°$

$J = 212.13 \quad 65°$

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3-43



$70+20=90$

$K=\sqrt{150^2+150^2-2(150)(150)\cos 90}$

$=212.13$

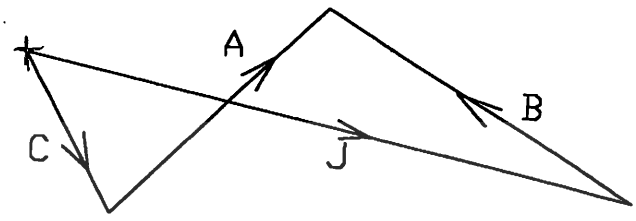
$\theta=\sin^{-1}\left(\frac{150}{212.13}\sin 90\right)$

$=45°$

$\beta=180-(\theta+70)=65°$

$K=212.13 \quad \underline{65°}\uparrow$

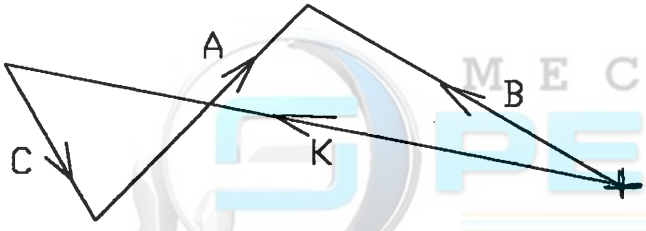
3-44 $J=C\nrightarrow A\rightarrow B$



Measuring:

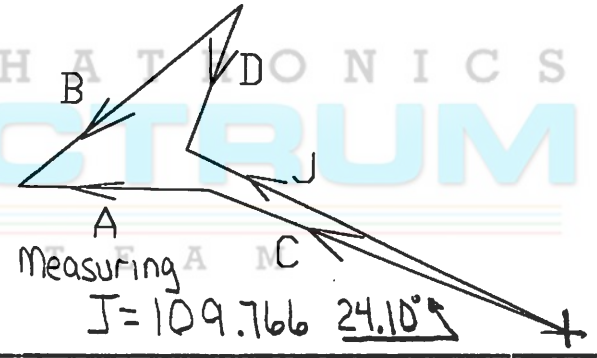
$J=26.094 \quad \underline{11.40°}\downarrow$

3-45 $K=B\rightarrow A\rightarrow C$



Measuring: $K=26.094 \quad \underline{11.40°}\uparrow$

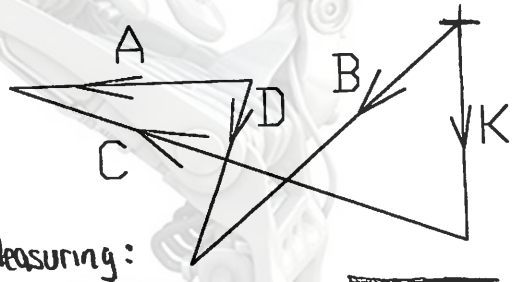
3-46 $J=C\nrightarrow A\rightarrow B\nrightarrow D$



Measuring

$J=109.766 \quad \underline{24.10°}\uparrow$

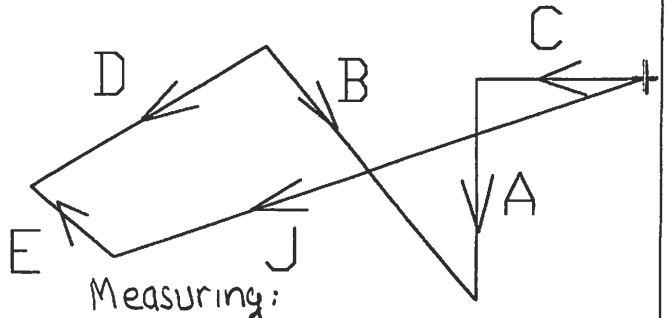
3-47 $K=B\rightarrow D\nrightarrow A\rightarrow C$



Measuring:

$K=44.824 \quad \underline{89.75°}\downarrow$

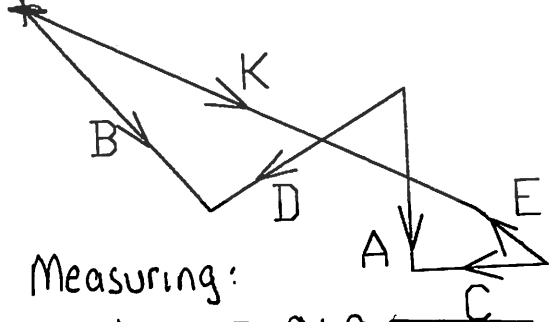
3-48 $J=C\nrightarrow A\rightarrow B\nrightarrow D\rightarrow E$



Measuring:

$J=101.679 \quad \underline{18.28°}\downarrow$

3-49 $K=B\rightarrow D\nrightarrow A\rightarrow C\nrightarrow E$

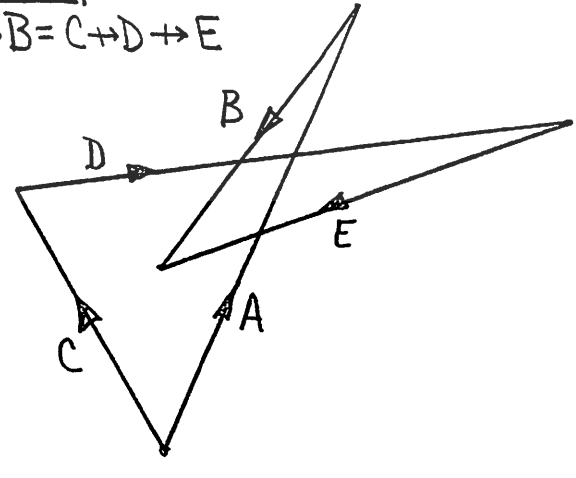


Measuring:

$K=107.869 \quad \underline{26.49°}\downarrow$

3-50 $A\rightarrow D\nrightarrow B=C\nrightarrow E$

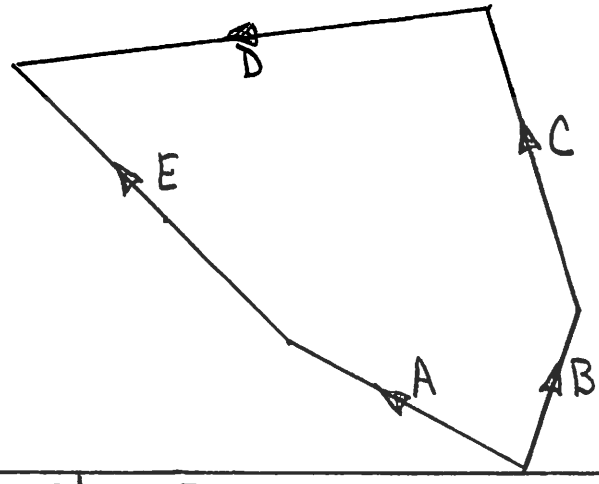
$A\nrightarrow B=C\nrightarrow D\nrightarrow E$



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3-51 $A \to C \leftrightarrow E = D \leftrightarrow B$

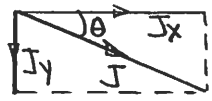
$A \leftrightarrow E = B \leftrightarrow C \leftrightarrow D$



3-52 $J = C \leftrightarrow A \to B$

$J_x = (7.5\cos 60) + (12.5\cos 45) - (-15\cos 30)$
$= 25.58$

$J_y = (-7.5\sin 60) + (12.5\sin 45) - (15\sin 30)$
$= -5.16$



$J = \sqrt{(25.58)^2 + (-5.16)^2}$
$= 26.10$

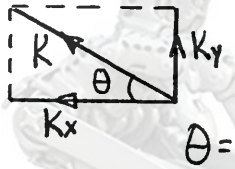
$\theta = \tan^{-1}(5.16/25.58) = 11.40°$

$J = 26.10 \angle 11.40°$

3-53 $K = B \to A \to C$

$K_x = (-15\cos 30) - (12.5\cos 45) - (7.5\cos 60)$
$= -25.58$

$K_y = (15\sin 30) - (12.5\sin 45) - (-7.5\sin 60)$
$= 5.16$



$K = \sqrt{(-25.58)^2 + (5.16)^2}$
$= 26.10$

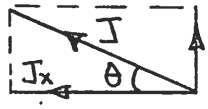
$\theta = \tan^{-1}(5.16/25.58) = 11.40°$

$K = 26.10 \angle 11.40°$

3-54 $J = C \leftrightarrow A \to B \leftrightarrow D$

$J_x = (-100\cos 20) + (-50) - (-75\cos 40) + (-40\cos 70) = -100.2$

$J_y = (100\sin 20) + (0) - (-75\sin 40) + (-40\sin 70) = 44.8$



$J = \sqrt{(-100.2)^2 + (44.8)^2}$
$= 109.8$

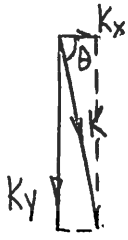
$\theta = \tan^{-1}(44.8/109.8) = 24.1$

$J = 109.8 \angle 24.1°$

3-55 $K = B \to D \leftrightarrow A \to C$

$K_x = (-75\cos 40) - (-40\cos 70) + (-50) - (-100\cos 20) = 0.20$

$K_y = (-75\sin 40) - (-40\sin 70) + (0) - (100\sin 20) = -44.82$



$K = \sqrt{(0.20)^2 + (-44.82)^2} = 44.82$

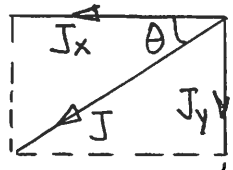
$\theta = \tan^{-1}(44.82/0.2) = 89.7°$

$K = 44.82 \angle 89.7°$

3-56 $J = C \leftrightarrow A \to B \leftrightarrow D \to E$

$J_x = (-30) + (0) - (60\sin 40) + (-50\cos 30) - (-20\sin 50) = -96.55$

$J_y = 0 + (-40) - (-60\cos 40) + (-50\sin 30) - (20\cos 50) = -31.89$



$J = \sqrt{(-96.55)^2 + (-31.89)^2}$
$= 101.68$

$\theta = \tan^{-1}(31.89/96.55) = 18.3°$

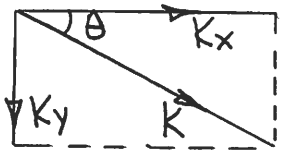
$J = 101.68 \angle 18.3°$

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3-57 $K = B \to D \nrightarrow A \to C \nrightarrow E$

$K_x = (60\sin 40) - (-50\cos 30) + (0) - (-30) + (-20\sin 50) = 96.54$

$K_y = (-60\cos 40) - (-50\sin 30) + (-40) - 0 + (20\cos 50) = -48.11$



$K = \sqrt{(96.54)^2 + (-48.11)^2} = 107.86$

$\theta = \tan^{-1}\left(\frac{48.11}{96.54}\right) = 26.5°$

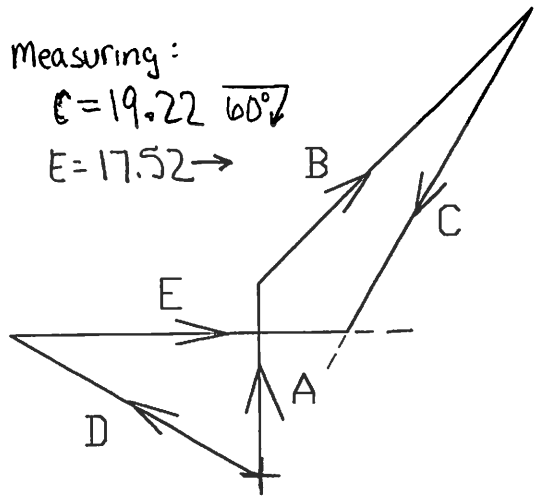
$K = 107.86 \angle 26.5°$

3-58 $A \nrightarrow B \nrightarrow C = D \nrightarrow E$

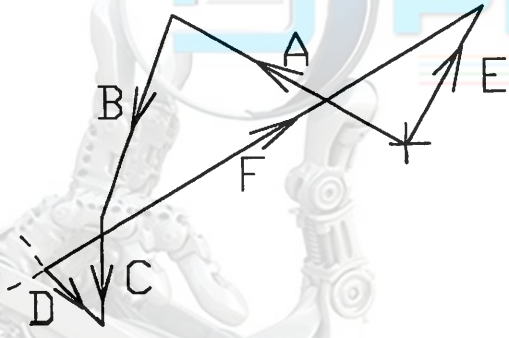
Measuring:

$C = 19.22 \angle 60°$

$E = 17.52 \to$



3-59 $A \nrightarrow B \nrightarrow C \to D = E \to F$



Measuring:

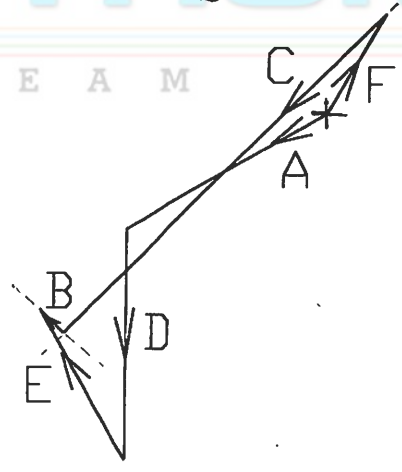
$D = 38.12 \angle 45°$

$F = 238.9 \angle 30°$

3-60 $A \to B \to C \nrightarrow D = \to E \nrightarrow F$

rewriting:

$A \nrightarrow D \nrightarrow E \to B = F \nrightarrow C$



measuring

$B = 8.81 \angle 45°$ $C = 117.7 \angle 45°$

3-61 $A \nrightarrow B \nrightarrow C = D \nrightarrow E$

assume directions $C \nearrow$ $E \to$

horiz:

$0 + (20\cos 45) + C\cos 60 = -15\cos 30 + E$

Vert:

$10 + (20\sin 45) + C\sin 60 = 15\sin 30 + 0$

$C = -19.2$

substituting $\Rightarrow E = 17.5$

$C = 19.22 \angle 60°$ & $E = 17.52 \to$

3-62 $A \nrightarrow B \nrightarrow C \to D = E \to F$

horiz assume directions $D \nwarrow$, $F \nearrow$

$(-125\cos 30) + (-100\cos 70) + (0) - (-D\cos 45) = (75\cos 60) - (F\cos 30)$

$D = (179.95 - F\cos 30)/\cos 45$ ①

Vert:

$(125\sin 30) + (-100\sin 70) + (-50) - (D\sin 45) = (75\sin 60) - (F\sin 30)$

$D = (F\sin 30 - 146.42)/\sin 45$ ②

using ① & ② simultaneous

$F = 23.89$, $D = -38.12$

$F = 238.9 \angle 30°$ $D = 38.12 \angle 45°$

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3-63 $A \rightarrow B \rightarrow C \nrightarrow D = \rightarrow E \nrightarrow F$

assume directions $B \nwarrow$, $C \nearrow$

horiz:

$(-60\cos 30) - (-B\cos 45) - (C\cos 45) + 0$

$= -(-45\cos 60) + (30\sin 30)$

$B = (C\cos 45 + 89.46)/\cos 45$ — ①

vert:

$(-60\sin 30) - (B\sin 45) - (C\sin 45) + (-60)$

$= -(45\sin 60) + (30\cos 30)$

$B = (-77.0 - C\sin 45)/\sin 45$ — ②

Solving ① & ② simultaneously

$B = 8.81$, $C = -117.7$

$B = 8.81$ 45° $C = 117.7$ 45°

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MECHATRONICS
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Answers to the Chapter 3 Case Study Questions:

Case 3-1.

1. As key 2 is pressed, rocker plate A rotates back, or clockwise in the left end-view.
2. Spring C provides resistance to rotating the rocker plate, counterclockwise in the left end view.
3. Spring B provides resistance to pressing the keys, 1 and 2.
4. As key 2 is pressed, the rocker plate A rotates back, and releases key 1. the Spring under key 1 forces key 1 into an upward position.
5. The purpose of this mechanism is to hamper two keys from being in the downward position at the same time.
6. Spring B, acting on key 1, is in compression and forcing button 1 upward.
7. Spring C is in tension, forcing rocker plate A to rotate counterclockwise in the right end view. Stop D prevents any further rotation than the position shown.
8. A cassette tape player uses similar mechanical means to prevent the play, rewind or fast forward buttons to be pressed at the same time.
9. As mentioned, pin D serves as a stop for rocker plate A. It prevents further clockwise rotation as seen from the right end view.

Case 3-2.

1. As driveshaft A turns, collar B also turns because of a keyed connection.
2. The motion of the shaft and collar is transmitted to gear C because the protrusion of link D is seated into the notch in collar B.
3. If link D were forced upward, the protrusion would come out of the notch in collar B. Nothing would be left driving gear C, so it would stop rotating.
4. Link D would dislodge from the notch if gear C would be exposed to significant resistance to rotation.
5. The tendency of link D to have upward motion must overcome the spring tension.
6. This device is intended to stop the rotation of the gear, if significant resistance is encountered.
7. Such a device is called a slip clutch.
8. This device would stop the winding mechanism if wire became jammed, thus supplying resistance on gear C.
9. To reset this device, link D must be place under link F and aligned with the notch in collar B.
10. The spring must be in tension. Therefore, it pulls link F to the right, and link E to the left.

Case 3-3.

1. As handle A is rotated, moving threaded rod B to the left, grip C also moves to the left and slightly upward. Notice that links E and F are pivoting in the middle, thus grip C is constrained to a swinging motion.
2. As handle A is rotated, moving threaded rod B to the left, grip D moves to the right and slightly downward. Since links E and F are pivoting in the middle, grip D will have motion opposing grip C.
3. The purpose of this mechanism is to serve as a machining clamp for the workpiece.
4. The spring, G, pulling on link D would cause is to return to an upward and rightward position.
5. The purpose of spring G is, ultimately, to keep a positive contact between the threaded rod and link C.
6. Links E and F have a peculiar configuration to avoid interference with the worikpiece, throughout the range of motion of the clamp.
7. Such a device could be called a machining clamp.
8. Since link C is moving in a swinging motion, the rounded end on the threaded rod, assures a consistent point contact with link C.

MECHATRONICS
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4-1 $t = .1s$

$x = 3\cos[50(.1)+40] = 2.121"$

$t = 1.5s$

$x = 3\cos[50(1.5)+40] = -1.268"$

$\Delta x = (-1.268) - (2.121) = -3.389 = 3.389" \leftarrow$

4-2 $t = 3.8s$

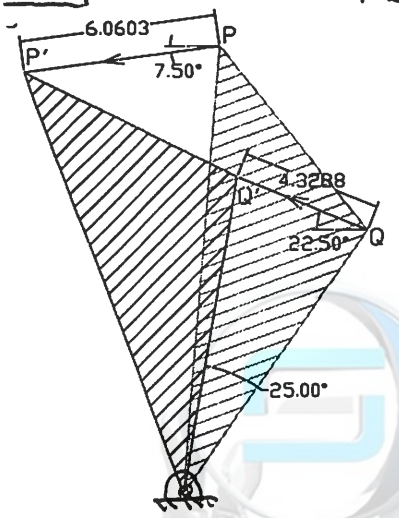
$x = 3\cos[50(3.8)+40] = -1.928$ in

$t = 4.7s$

$x = 3\cos[50(4.7)+40] = .261$ in

$\Delta x = .261 - (-1.928) = +2.189 = 2.189$ in $\rightarrow$

4-3

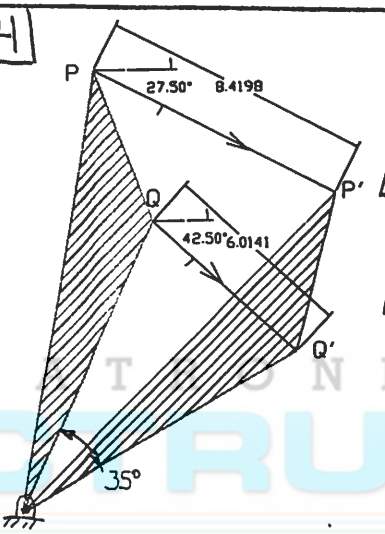


Measuring:

$\Delta R_P = 6.060$ 7.50°

$\Delta R_Q = 4.329$ 22.50°

4-4



Measuring:

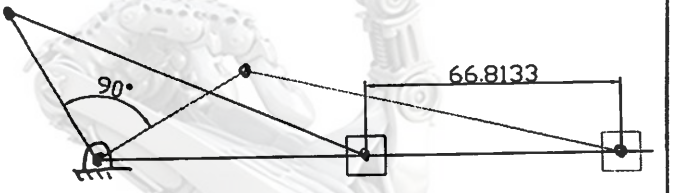
$\Delta R_P = 8.420$

27.50°

$\Delta R_Q = 6.014$

42.50°

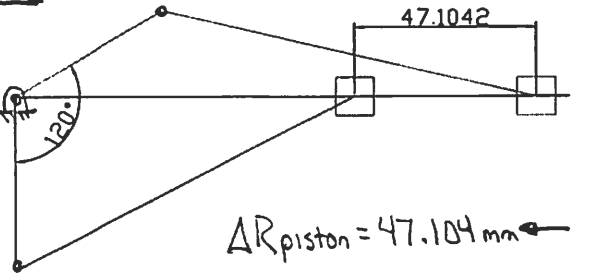
4-5



measuring:

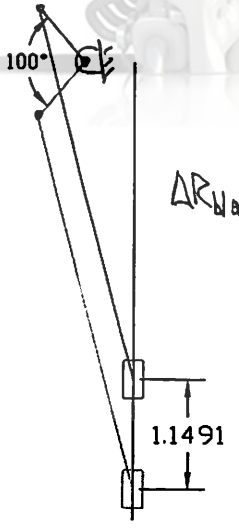
$\Delta R_{piston} = 66.813$ mm $\leftarrow$

4-6



$\Delta R_{piston} = 47.104$ mm $\leftarrow$

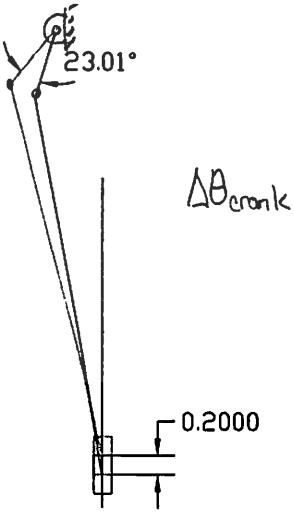
4-7



measuring:

$\Delta R_{blade} = 1.149"$ $\uparrow$

4-8

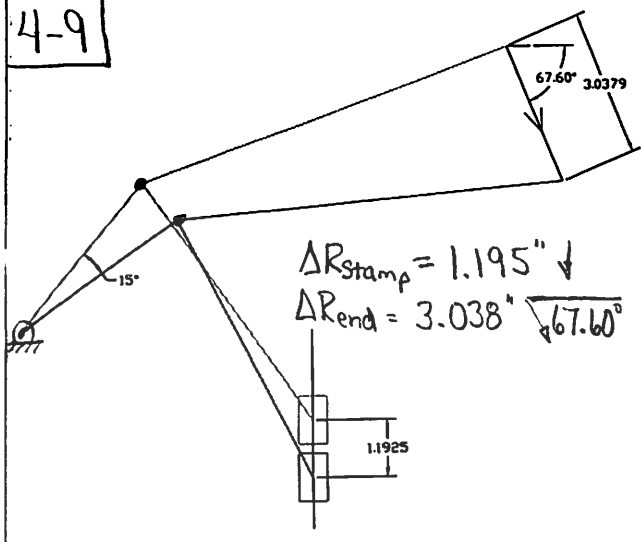


$\Delta\theta_{crank} = 23.01°$, ccw

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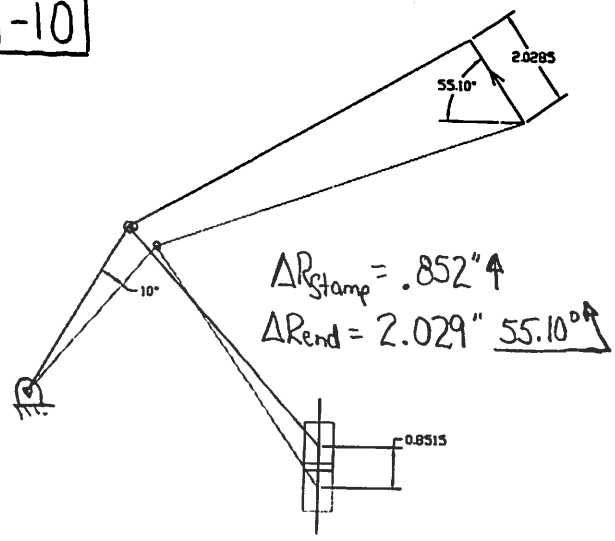
4-9



$\Delta R_{stamp} = 1.195"$ ↓

$\Delta R_{end} = 3.038"$ ↘ 67.60°

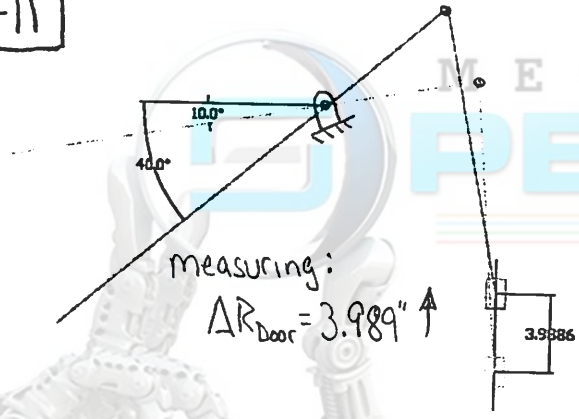
4-10



$\Delta R_{stamp} = .852"$ ↑

$\Delta R_{end} = 2.029"$ 55.10° ↗

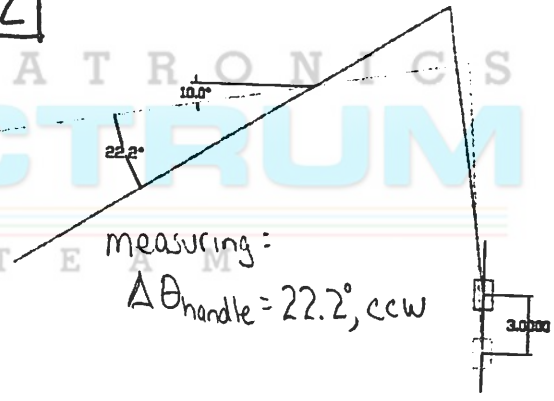
4-11



measuring:

$\Delta R_{Door} = 3.989"$ ↑

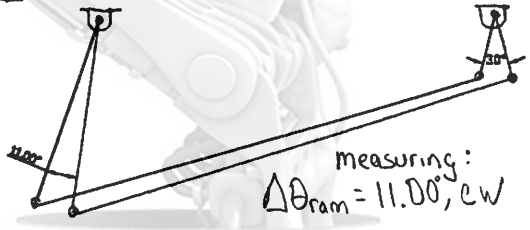
4-12



measuring:

$\Delta\theta_{handle} = 22.2°$, ccw

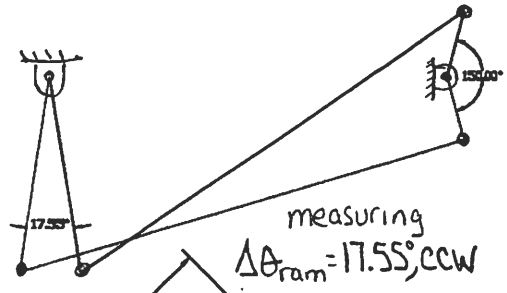
4-13



measuring:

$\Delta\theta_{ram} = 11.00°$, cw

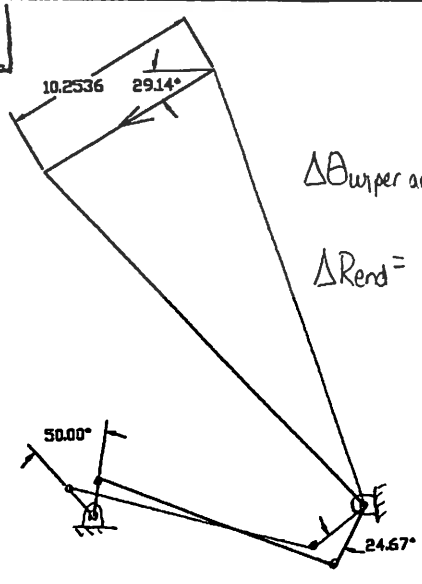
4-14



measuring

$\Delta\theta_{ram} = 17.55°$, ccw

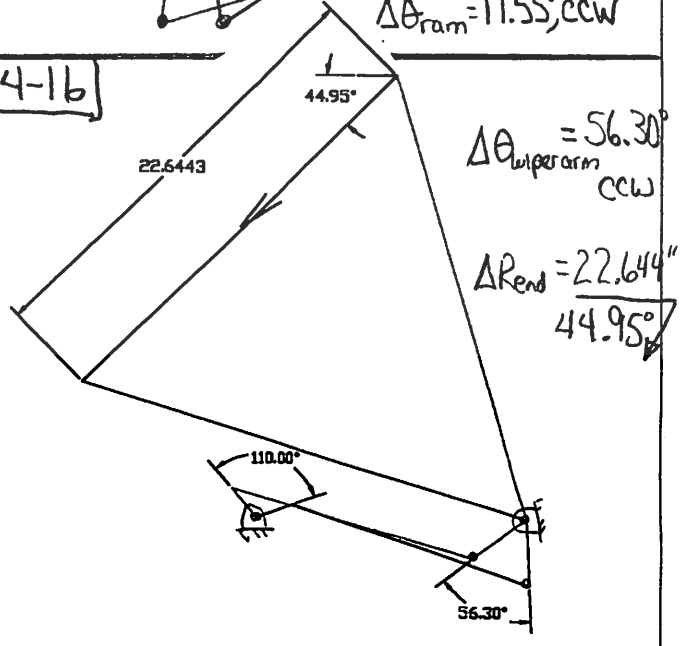
4-15



$\Delta\theta_{wiper\ arm} = 24.67°$ ccw

$\Delta R_{end} = 10.254"$ ↙ 29.14°

4-16

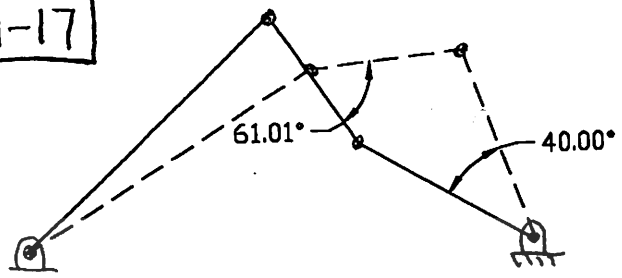


$\Delta\theta_{wiper\ arm} = 56.30°$ ccw

$\Delta R_{end} = 22.644"$ ↙ 44.95°

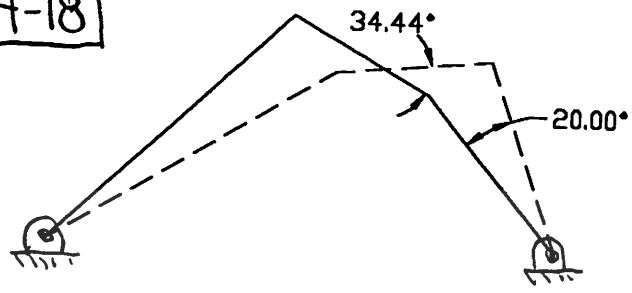
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4-17



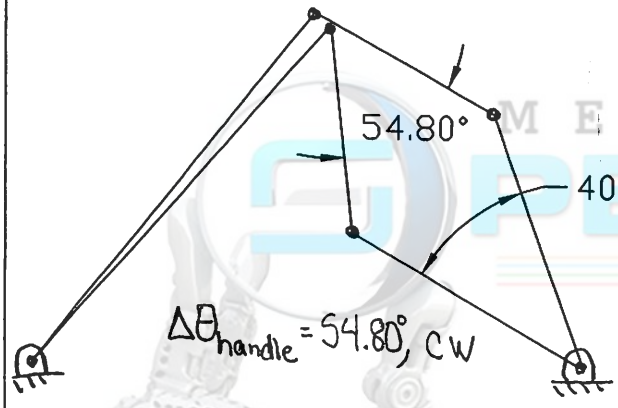
$\Delta\theta_{handle} = 61.01°, CCW$

4-18



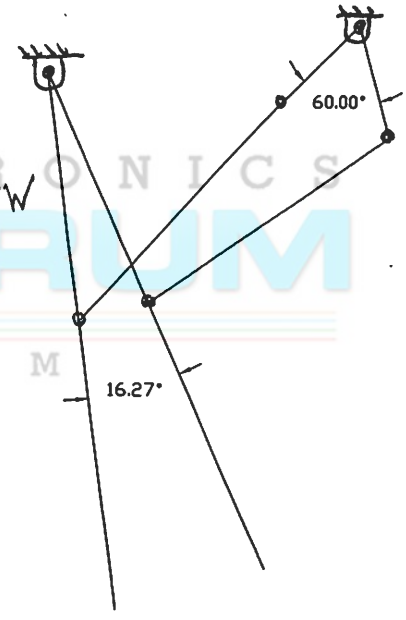
$\Delta\theta_{handle} = 34.44°, CCW$

4-19



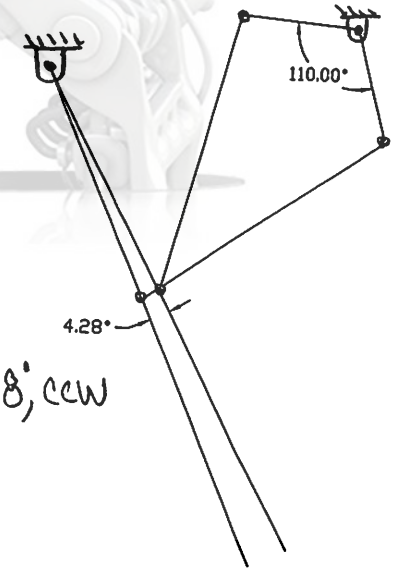
$\Delta\theta_{handle} = 54.80°, CW$

4-20



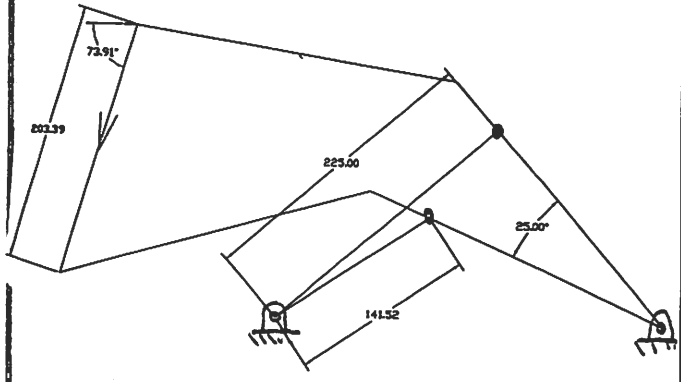
$\Delta\theta_{wheel} = 16.27°, CW$

4-21



$\Delta\theta_{wheel} = 4.28°, CCW$

4-22



$\Delta L_{cyl} = 225 - 141.52 = 83.48^{mm}$

$\Delta Vol = \frac{\pi}{4}(25)^2(83.48) = 40978 mm^3$

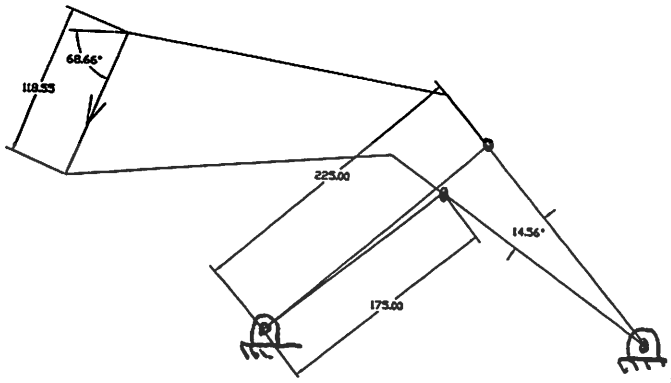
$= 40.97 cc\ (cm^3)$

$\Delta Rend = 203.39^{mm}$ $\boxed{73.91°}$

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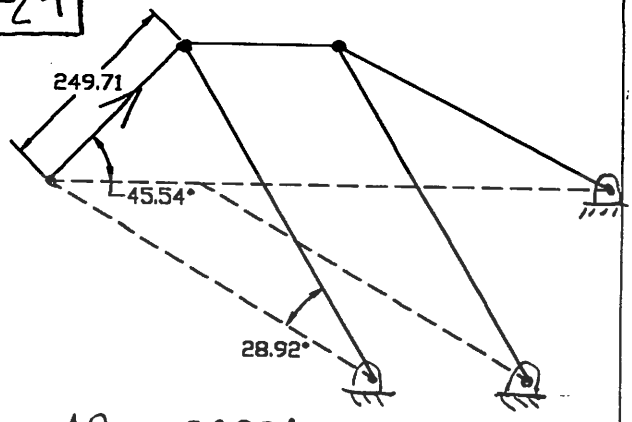
4-23



$\Delta\theta_{pedal} = 14.56°$, ccw

$\Delta R_{end} = 118.55^{mm}$ $\nearrow 68.66°$

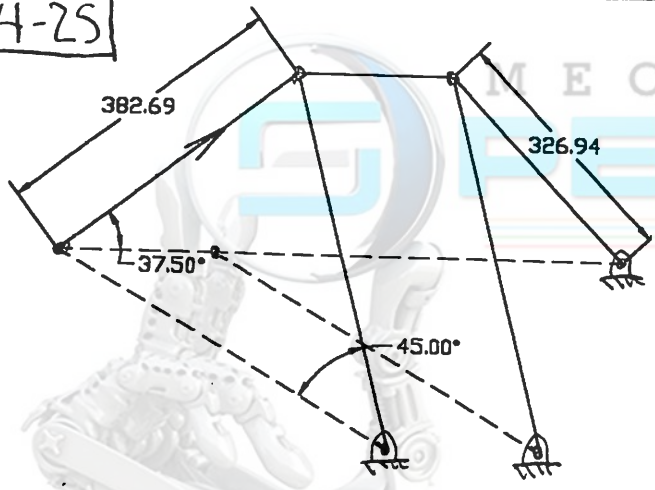
4-24



$\Delta\theta_{leg} = 28.92°$, cw

$\Delta R_{carrier} = 249.71^{mm}$ $\nearrow 45.54°$

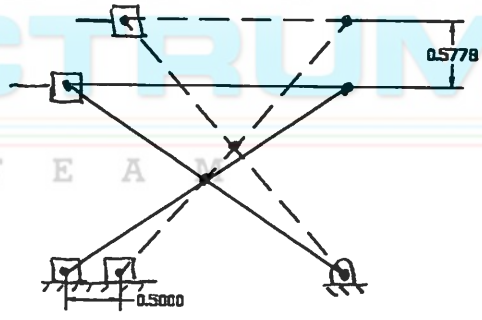
4-25



$\Delta L_{actuator} = 533 - 326.94 = 206.06^{mm}$

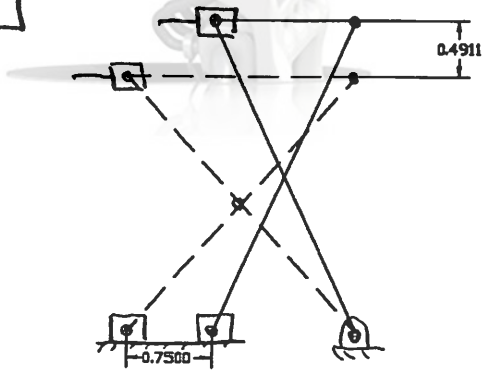
$\Delta R_{end} = 382.69$ $\nearrow 37.50°$

4-26



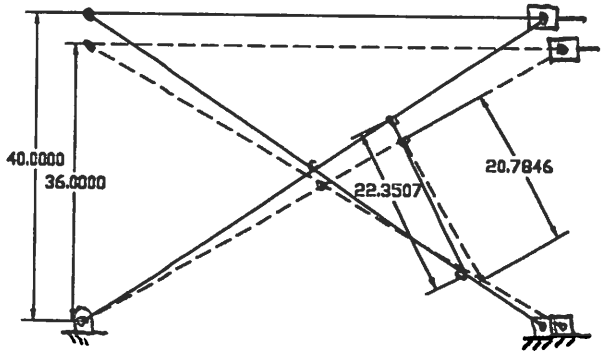
$\Delta R_{box} = .5778^{m}$ ↓

4-27



$\Delta R_{box} = .4911^{m}$ ↑

4-28

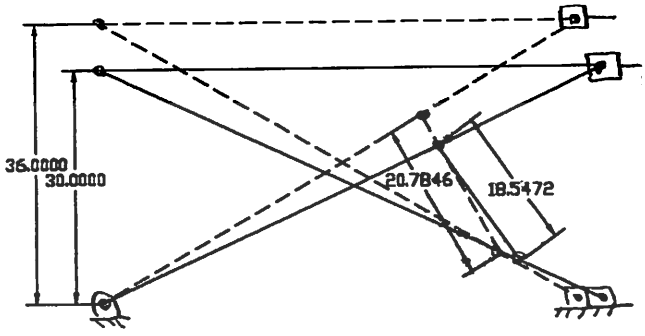


$\Delta L_{cyl} = 22.3507 - 20.7846$

$= 1.5661$ in longer

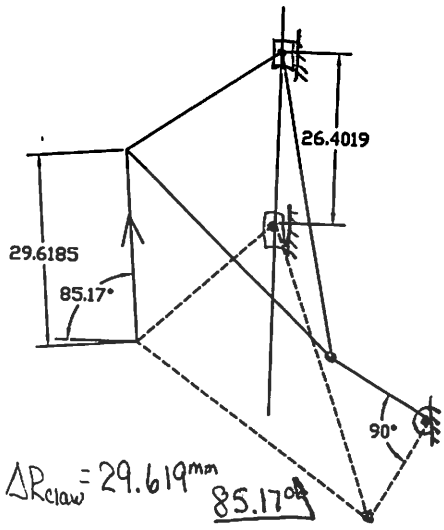
MECHATRONICS SPECTRUM TEAM

4-29



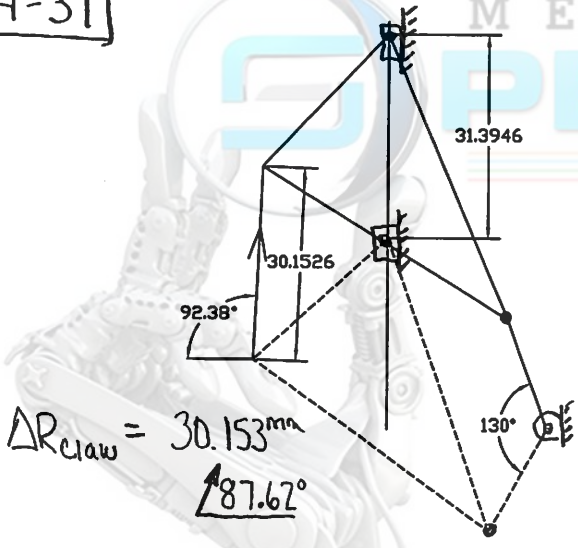
$\Delta L_{cyl} = 18.5472 - 20.7846$
$= -2.2374$
$= 2.2374$ shorter

4-30



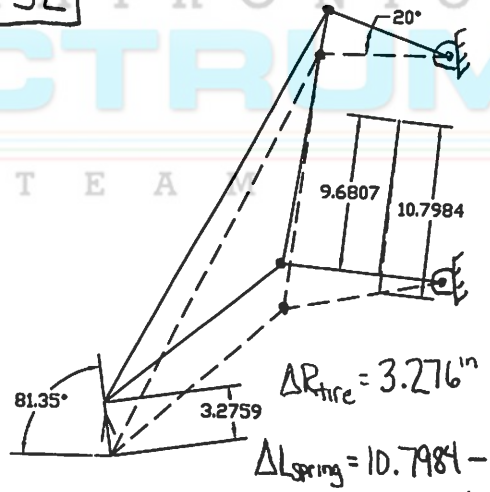
$\Delta R_{claw} = 29.619^{mm}$ 85.17°

4-31



$\Delta R_{claw} = 30.153^{mm}$ 87.62°

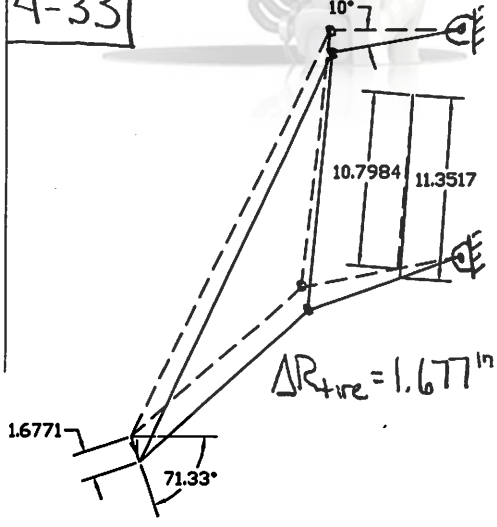
4-32



$\Delta R_{tire} = 3.276^{in}$ 81.35°

$\Delta L_{spring} = 10.7984 - 9.6807$
$= 1.1177^{in}$ (shorter)

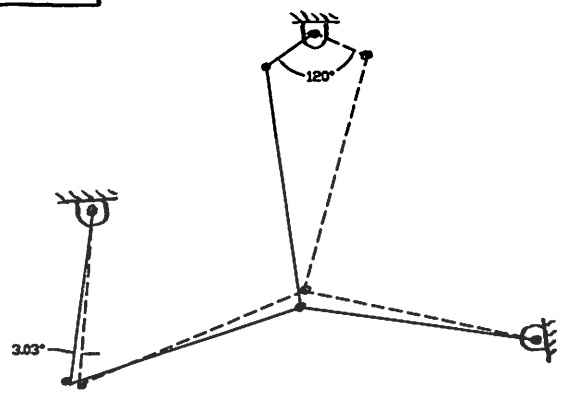
4-33



$\Delta R_{tire} = 1.677^{in}$ 71.33°

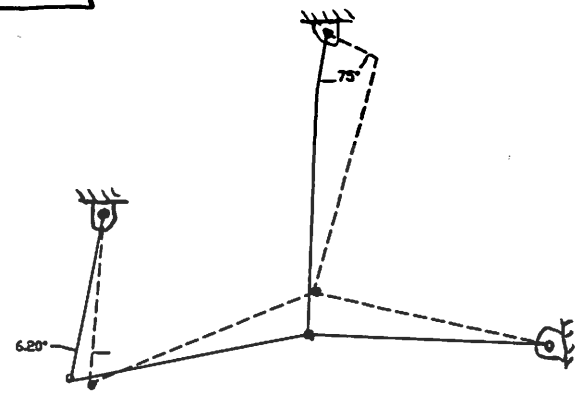
MECHATRONICS SPECTRUM TEAM

4-34



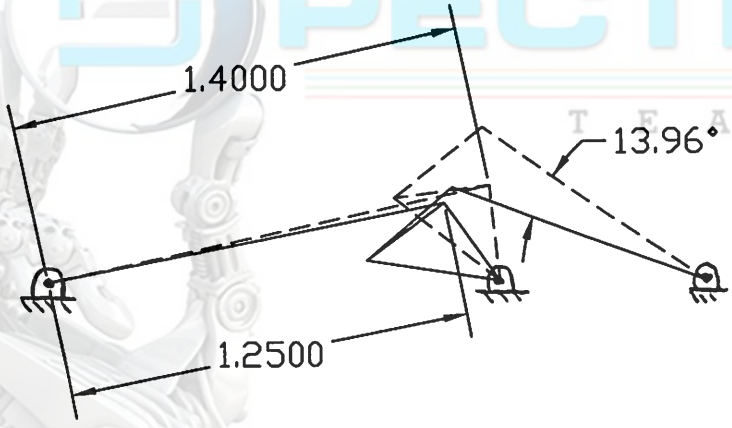
$\Delta\theta_{ram} = 3.03°$, cw

4-35



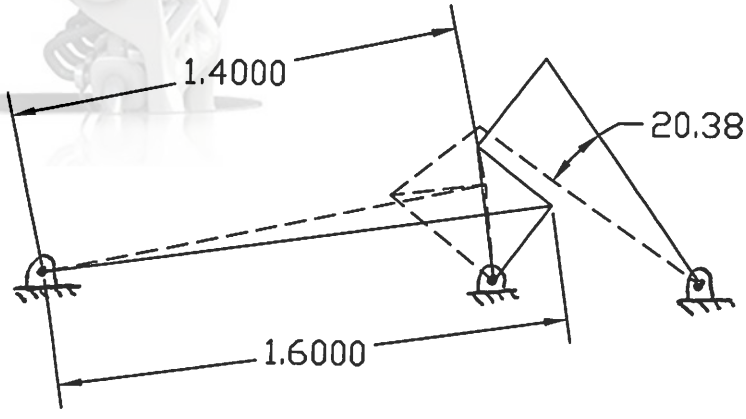
$\Delta\theta_{ram} = 6.20°$, cw

4-36



$\Delta\theta_{bed} = 13.96°$, ccw

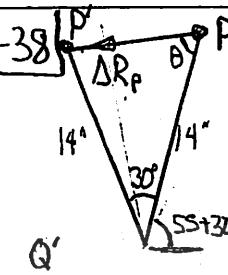
4-37



$\Delta\theta_{bed} = 20.38°$, cw

MECHATRONICS SPECTRUM TEAM

4-38

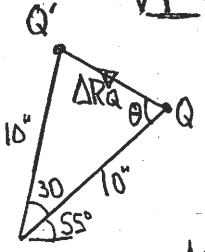


$\Delta R_P = \sqrt{14^2+14^2-2(14)(14)\cos 30} = 7.247"$

$\theta = \sin^{-1}\left(\frac{14}{7.247}\sin 30\right) = 75.0°$

$(55+30) - 75 = 10°$

$\Delta R_P = 7.247"$ 10° ↙



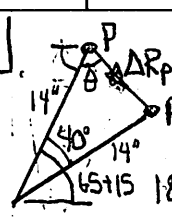
$\Delta R_Q = \sqrt{10^2+10^2-2(10)(10)\cos 30} = 5.176"$

$\theta = \sin^{-1}\left(\frac{10}{5.176}\sin 30\right) = 75°$

$75 - 55 = 20°$

$\Delta R_Q = 5.176"$ 20° ↖

4-39

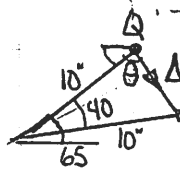


$\Delta R_P = \sqrt{14^2+14^2-2(14)(14)\cos 40} = 9.577"$

$\theta = \sin^{-1}\left(\frac{14}{9.577}\sin 40\right) = 70°$

$180 - (65+15) - 70 = 30°$

$\Delta R_P = 9.577"$ 30° ↖



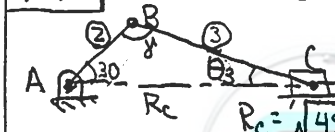
$\Delta R_Q = \sqrt{10^2+10^2-2(10)(10)\cos 40} = 6.840"$

$\theta = \sin^{-1}\left(\frac{10}{6.840}\sin 40\right) = 70°$

$180 - (65) - 70 = 45°$

$\Delta R_Q = 6.840"$ 45° ↖

4-40 Current configuration:

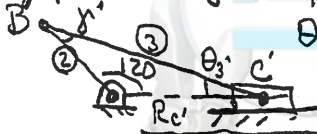


$\theta_3 = \sin^{-1}\left(\frac{45}{100}\sin 30\right) = 13°$

$\gamma = 180 - (30+13) = 137°$

$R_c = \sqrt{45^2+100^2-2(45)(100)\cos 137} = 136.41\ mm$

displaced config:



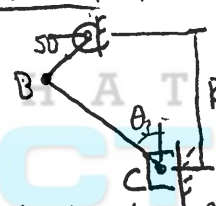
$\theta_3' = \sin^{-1}\left(\frac{45}{100}\sin 120\right) = 22.94°$

$\gamma' = 180 - (120+22.94) = 37.06°$

$R_c' = \sqrt{45^2+100^2-2(45)(100)\cos(37.06)} = 69.59mm$

$\Delta R_c = 69.59 - 136.41 = 66.82mm$ ←

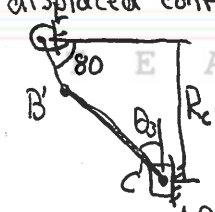
4-41 Current configuration:



$\theta_3 = \sin^{-1}\left[\frac{.5+.75\cos 50}{4}\right] = 14.21°$

$R_c = .75\sin 50 + 4\cos 14.21 = 4.452"$

displaced config:

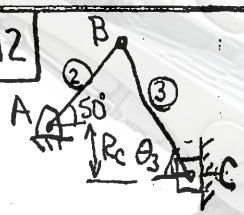


$\theta_3' = \sin^{-1}\left[\frac{.5-.75\cos 80}{4}\right] = 5.30°$

$R_c' = .75\sin 80 + 4\cos 5.30° = 4.722"$

$\Delta R_c = 4.722 - 4.452 = .270"$ ↓

4-42

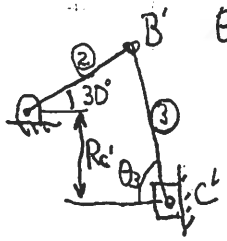


Current configuration:

$\theta_3 = \cos^{-1}\left[\frac{6-4\cos 50}{6}\right] = 55.15°$

$R_c = 6\sin 55.15 - 4\sin 50 = 1.860"$

displaced config:

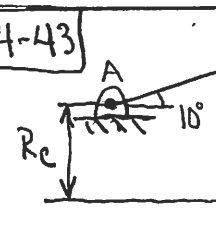


$\theta_3 = \cos^{-1}\left[\frac{6-4\cos 30}{6}\right] = 65.0°$

$R_c' = 6\sin 65 - 4\sin 30 = 3.438"$

$\Delta R_c = 3.438 - 1.860 = 1.578"$ ↓

4-43

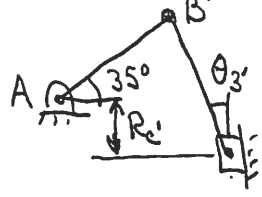


Current configuration:

$\theta_3 = \sin^{-1}\left[\frac{9-8\cos 10}{15}\right] = 4.29°$

$R_c = 15\cos\theta_3 - 8\sin 10 = 13.569in$

displaced config:



$\theta_3' = \sin^{-1}\left[\frac{9-8\cos 35}{15}\right] = 9.39°$

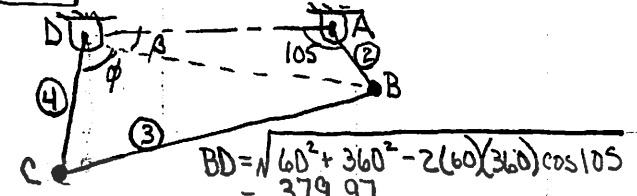
$R_c' = 15\cos\theta_3' - 8\sin 35 = 8.246in$

$\Delta R_c = 13.569 - 8.246 = 5.323in$ ↑

13-782 42-381 42-387 42-389 42-392 42-399 National® Brand
500 SHEETS, FILLER, 5 SQUARE
50 SHEETS EYE-EASE® 5 SQUARE
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200 SHEETS EYE-EASE® 5 SQUARE
100 RECYCLED WHITE 5 SQUARE
200 RECYCLED WHITE 5 SQUARE
Made in U.S.A.

4-44

Original configuration:

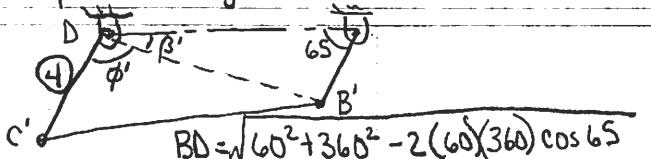


$BD = \sqrt{60^2 + 360^2 - 2(60)(360)\cos 105} = 379.97$

$\beta = \sin^{-1}\left[\frac{60}{379.97}\sin 105\right] = 8.77°$

$\phi = \cos^{-1}\left[\frac{379.97^2 + 180^2 - 420^2}{2(379.97)(180)}\right] = 89.84°$

displaced configuration



$BD = \sqrt{60^2 + 360^2 - 2(60)(360)\cos 65} = 339.03$

$\beta' = \sin^{-1}\left[\frac{60}{339.03}\sin 65\right] = 9.23°$

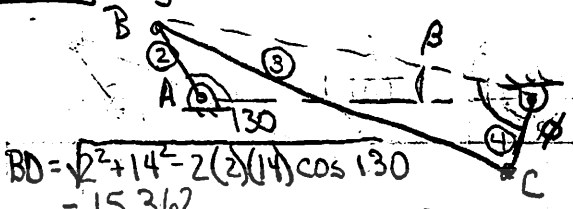
$\phi' = \cos^{-1}\left[\frac{339.03^2 + 180^2 - 420^2}{2(339.03)(180)}\right] = 103.77°$

$\Delta\theta_4 = (\beta' + \phi') - (\beta + \phi)$

$= (103.77 + 9.23) - (8.77 + 89.84) = 14.39°$, CW

4-45

Original configuration:

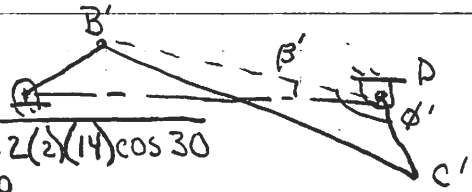


$BD = \sqrt{2^2 + 14^2 - 2(2)(14)\cos 130} = 15.362$

$\beta = \sin^{-1}\left[(2/15.362)\sin 130\right] = 5.72°$

$\phi = \cos^{-1}\left[\frac{15.362^2 + 3.5^2 - 13^2}{2(15.362)(3.5)}\right] = 42.53$

displaced configuration:



$BD' = \sqrt{2^2 + 14^2 - 2(2)(14)\cos 30} = 12.309$

$\beta' = \sin^{-1}\left[(2/12.309)\sin 30\right] = 4.66°$

$\phi' = \cos^{-1}\left[\frac{12.309^2 + 3.5^2 - 13^2}{2(12.309)(3.5)}\right] = 93.49°$

$\Delta\theta_4 = (\phi' - \beta') - (\phi - \beta)$

$= (93.49 - 4.66) - (42.53 - 5.72) = 52.02°$, CCW

4-46

Original configuration



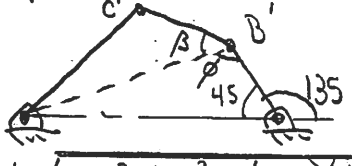
$BD = \sqrt{10.6^2 + 4.2^2 - 2(10.6)(4.2)\cos 70} = 9.977\text{ cm}$

$\phi = \sin^{-1}\left[(10.6/9.977)\sin 70\right] = 86.73°$

$\beta = \cos^{-1}\left[\frac{9.977^2 + 3.2^2 - 7^2}{2(9.977)(3.2)}\right] = 17.85°$

$\theta_3 = 110 - [86.73 + 17.85] = 5.42°$

displaced configuration:



$BD' = \sqrt{10.6^2 + 4.2^2 - 2(10.6)(4.2)\cos 45} = 8.188\text{ cm}$

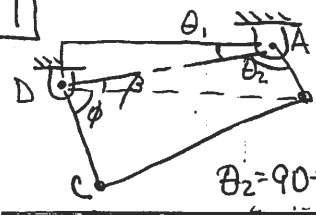
$\phi' = \sin^{-1}\left[(10.6/8.188)\sin 45\right] = 66.26°$

$\beta' = \cos^{-1}\left[\frac{8.188^2 + 3.2^2 - 7^2}{2(8.188)(3.2)}\right] = 57.34°$

$\theta_3 = 135 - [66.26 + 57.34] = -11.40$

$\Delta\theta_3 = 11.40 + 5.42 = 16.82°$, CW

4-47



$AD = \sqrt{32^2 + 5^2} = 32.388'$

$\theta_1 = \tan^{-1}(5/32) = 8.88°$

$\theta_2 = 90 + 15 - 8.88 = 96.12°$

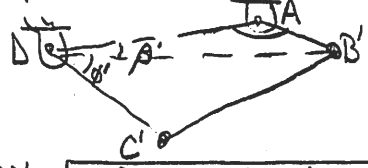
$BD = \sqrt{32.388^2 + 12^2 - 2(32.388)(12)\cos 96.12°} = 35.719''$

$\beta = \sin^{-1}\left[(12/35.719)\sin 96.12\right] = 19.51°$

$\phi = \cos^{-1}\left[\frac{26^2 + 35.719^2 - 30^2}{2(26)(35.719)}\right] = 55.51°$

$\theta_4 = (19.51 + 55.51) - 8.88 = 66.14°$

displaced configuration:



$\theta_2' = \theta_2 + 60 = 156.12°$

$BD' = \sqrt{32.388^2 + 12^2 - 2(32.388)(12)\cos 156.12} = 43.632''$

$\beta' = \sin^{-1}\left[(12/43.632)\sin 156.12\right] = 6.39°$

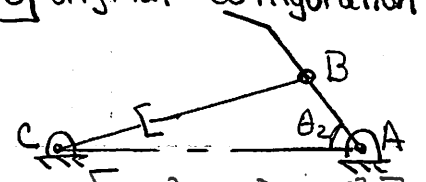
$\phi' = \cos^{-1}\left[\frac{26^2 + 43.632^2 - 30^2}{2(26)(43.632)}\right] = 42.32°$

$\theta_4' = (6.39 + 92.32) - 8.88 = 39.83°$

$\Delta\theta_4 = 66.14 - 39.83 = 26.31°$, CCW

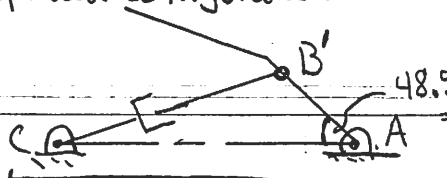
4-48

original configuration



$$\theta_2 = \cos^{-1}\left[\frac{200^2+300^2-225^2}{2(200)(300)}\right] = 48.59°$$

displaced configuration



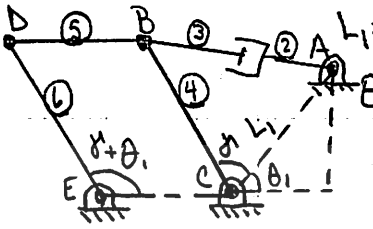
$$B'C = \sqrt{200^2+300^2-2(200)(300)\cos 28.59} = 156.95$$

$$\Delta L = 225 - 156.95 = 68.05\ mm$$

$$V = \frac{\pi}{4}(25)^2(68.05) = 33{,}406\ mm^3$$

4-49

original configuration:

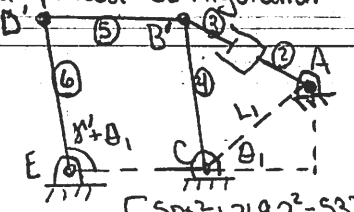


$$L_1 = \sqrt{100^2+250^2} = 269.2\ mm$$

$$\theta_1 = \tan^{-1}\left(\frac{250}{100}\right) = 68.2°$$

$$\gamma = \cos^{-1}\left[\frac{500^2+269.2^2-533^2}{2(500)(269.2)}\right] = 81.8°$$

displaced configuration:

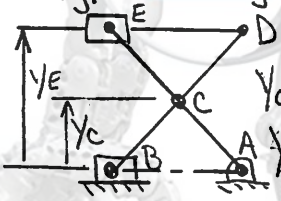


$$\gamma' = \cos^{-1}\left[\frac{500^2+269.2^2-533^2}{2(500)(269.2)}\right] = 58.2°$$

$$\Delta\theta_6 = (\gamma'+\theta_1) - (\gamma+\theta_1) = 23.6°\ CW$$

4-50

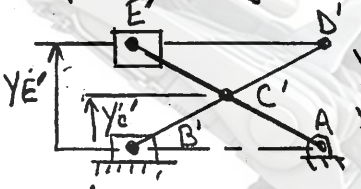
original configuration:



$$Y_C = \sqrt{1.5^2-1^2} = 1.118^m$$

$$Y_E = 2.236^m$$

displaced configuration:



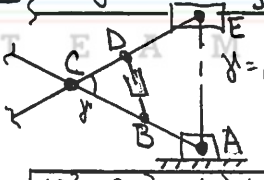
$$Y_C' = \sqrt{1.5^2-.75^2} = 1.299^m$$

$$Y_E' = 2.598^m$$

$$\Delta Y_E = 2.598 - 2.236 = .362^m$$

4-51

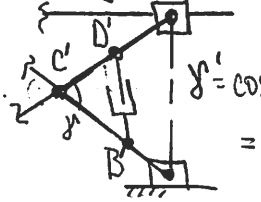
original config:



$$\gamma = \cos^{-1}\left[\frac{36^2+36^2-36^2}{2(36)(36)}\right] = 60°$$

$$DB = \sqrt{12^2+24^2-2(12)(24)\cos 60} = 20.784''$$

displaced configuration:



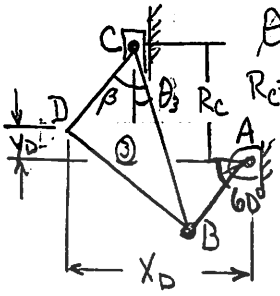
$$\gamma' = \cos^{-1}\left[\frac{36^2+36^2-45^2}{2(36)(36)}\right] = 77.36°$$

$$D'B' = \sqrt{12^2+24^2-2(12)(24)\cos 77.36} = 24.371''$$

$$\Delta DB = 24.371 - 20.784 = 3.587''$$

500 SHEETS FILLER 5 SQUARE
500 SHEETS EYE/EASE 5 SQUARE
100 SHEETS EYE/EASE 5 SQUARE
200 SHEETS EYE/EASE 5 SQUARE
200 SHEETS EYE/EASE 5 SQUARE
100 RECYCLED WHITE 5 SQUARE
200 RECYCLED WHITE 5 SQUARE
13-782 42-381 42-383 42-382 42-399
Made in U.S.A.
National Brand
MECHANICS
PEARSON

4-52 original configuration:



$\theta_3 = \sin^{-1}\left[\frac{25-18\cos 60}{48}\right] = 19.47°$

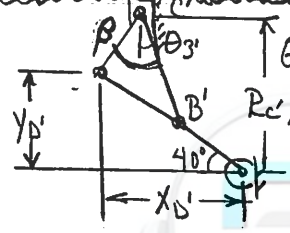
$R_C = 48\cos\theta_3 - 18\sin 60 = 29.67\ mm$

$\beta = \cos^{-1}\left[\frac{48^2+28^2-45^2}{2(48)(28)}\right] = 66.70°$

$X_D = 18\cos 60 + 48\sin\theta_3 + 28\sin(\beta-\theta_3) = 45.55\ mm$

$Y_D = -18\sin 60 + 48\cos\theta_3 - 28\cos(\beta-\theta_3) = 10.65\ mm$

displaced configuration:



$\theta_{3'} = \sin^{-1}\left[\frac{25-18\cos 40}{48}\right] = 13.51°$

$R_{C'} = 48\cos\theta_3 + 18\sin 40 = 22.78\ mm$

$X_{D'} = 18\cos 40 + 48\sin\theta_{3'} + 28\sin(\beta-\theta_{3'}) = 47.42$

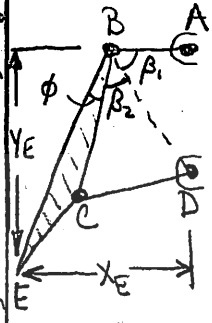
$Y_{D'} = 18\sin 40 + 48\cos\theta_{3'} - 28\cos(\beta-\theta_{3'}) = 41.47$

$\Delta R_D = \sqrt{(47.42-45.55)^2+(41.47-10.65)^2} = 30.87\ mm$

$\theta_D = \tan^{-1}\frac{(41.47-10.65)}{(47.42-45.55)} = 86.52°$

$\Delta R_D = 30.87\ mm \angle 86.52°$

4-53 original configuration:



$BD = \sqrt{8^2+14^2-2(8)(14)\cos 90} = 16.124"$

$\beta_1 = \sin^{-1}\left(\frac{14}{16.12}\sin 90\right) = 60.26°$

$\beta_2 = \cos^{-1}\left(\frac{16.12^2+16^2-10^2}{2(16.12)(16)}\right) = 36.28°$

$\phi = \cos^{-1}\left(\frac{16^2+28^2-14^2}{2(16)(28)}\right) = 19.62°$

$X_E = 8 + 28\sin(\phi+\beta_1+\beta_2-90) = 20.345"$

$Y_E = 28\cos(\phi+\beta_1+\beta_2-90) = 25.132"$

displaced configuration:

$B'D = \sqrt{8^2+14^2-2(8)(14)\cos 105} = 17.832"$

$\beta_1' = \sin^{-1}\left(\frac{14}{17.832}\sin 105\right) = 49.32°$

$\beta_2' = \cos^{-1}\left(\frac{17.832^2+16^2-10^2}{2(17.832)(16)}\right) = 33.84°$

$X_{E'} = 8 + 28\sin(\phi+\beta_1'+\beta_2'-90) = 14.194"$

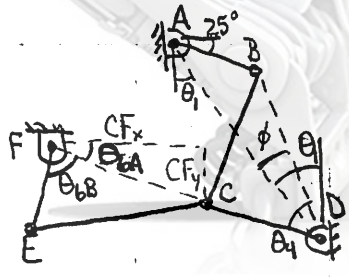
$Y_{E'} = 28\cos(\phi+\beta_1'+\beta_2'-90) = 27.306"$

$\Delta R_E = \sqrt{(14.194-20.345)^2+(27.306-25.132)^2} = 6.524"$

$\theta_E = \tan^{-1}\frac{(27.306-25.132)}{(14.194-20.345)} = 19.46°$

$\Delta R_E = 6.524" \angle 19.46°$

4-54 original configuration:



$AD = \sqrt{230^2+320^2} = 394.08 \quad \theta_1 = \tan^{-1}\left(\frac{230}{320}\right) = 35.71°$

$BD = \sqrt{60^2+394.08^2-2(60)(394.08)\cos(90-25-35.71)} = 343.00$

$\theta_4 = \cos^{-1}\left(\frac{250^2+343^2-250^2}{2(250)(343)}\right) = 46.69°$

$\phi = \cos^{-1}\left(\frac{343^2+394.08^2-60^2}{2(343)(394.08)}\right) = 4.91°$

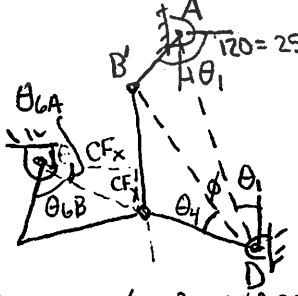
$CF_x = 230+230-250\sin(\theta_1-\phi+\theta_4) = 215.93$

$CF_y = 320-180-250\cos(\theta_1-\phi+\theta_4) = 85.84$

$CF = \sqrt{215.93^2+85.84^2} = 232.37 \quad \theta_{CA} = \tan^{-1}(85.84/215.93) = 21.68°$

$\theta_{CB} = \cos^{-1}\left(\frac{180^2+232.37^2-250^2}{2(180)(232.37)}\right) = 73.40°$

displaced configuration:



$B'D = \sqrt{60^2+394.08^2-2(60)(394.08)\cos(120-90+\theta_1)} = 373.4$

$\theta_4' = \cos^{-1}\left(\frac{250^2+373.4^2-250^2}{2(250)(373.4)}\right) = 41.69°$

$\phi' = \cos^{-1}\left(\frac{373.4^2+394.08^2-60^2}{2(373.4)(394.08)}\right) = 8.42°$

$CF_x = 230+230-250\sin(\theta_1+\phi+\theta_4) = 210.67$

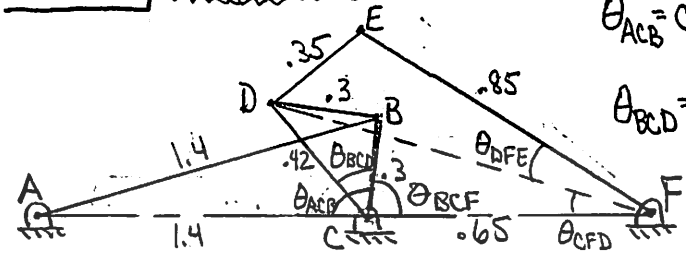
$CF_y = 320-180-250\cos(\theta_1+\phi+\theta_4) = 121.74$

$CF = \sqrt{210.67^2+121.74^2} = 243.32 \quad \theta_{CA} = \tan^{-1}(121.74/210.67) = 30.02°$

$\theta_{CB} = \cos^{-1}\left(\frac{180^2+243.32^2-250^2}{2(180)(243.32)}\right) = 70.59° \quad \Delta\theta_C = (70.59+30.02)-(73.40+21.68) = 5.53°, CW$

4-55

Original configuration



$$\theta_{ACB} = \cos^{-1}\left[\frac{(1.4)^2+(.3)^2-(1.4)}{2(1.4)(.3)}\right] = 83.85^\circ$$

$$\theta_{BCD} = \cos^{-1}\left[\frac{(.3)^2+(.42)^2-(.3)^2}{2(.3)(.42)}\right] = 45.57^\circ$$

$$\theta_{BCF} = 180 - \theta_{ACB} = 96.15^\circ$$

$$\theta_{DCF} = \theta_{BCF} + \theta_{ACB} = 141.72^\circ$$

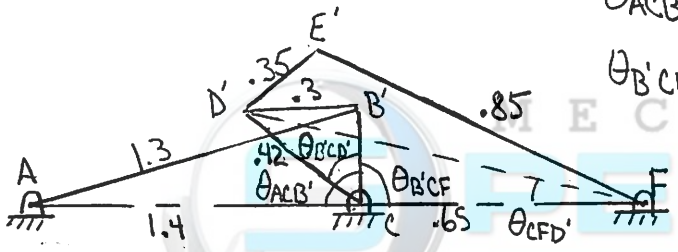
$$DF = \sqrt{(.65)^2+(.42)^2-2(.65)(.42)\cos 141.72} = 1.0136$$

$$\theta_{CFD} = \sin^{-1}\left[\frac{.42}{1.0136}\sin 141.72\right] = 14.87$$

$$\theta_{DFE} = \cos^{-1}\left[\frac{(.85)^2+(1.0136)^2-(.35)^2}{2(.85)(1.0136)}\right] = 19.19^\circ$$

$$\theta_{CFE} = \theta_{CFD} + \theta_{DFE} = 34.06^\circ$$

displaced configuration



$$\theta_{ACB'} = \cos^{-1}\left[\frac{(1.4)^2+(.3)^2-(1.3)^2}{2(1.4)(.3)}\right] = 64.62^\circ$$

$$\theta_{B'CD'} = \theta_{BCD} = 45.57$$

$$\theta_{B'CF} = 180 - \theta_{ACB'} = 115.38^\circ$$

$$\theta_{D'CF} = \theta_{B'CF} + \theta_{ACB'} = 160.95^\circ$$

$$D'F = \sqrt{(.65)^2+(.42)^2-2(.65)(.42)\cos 160.95} = 1.0559$$

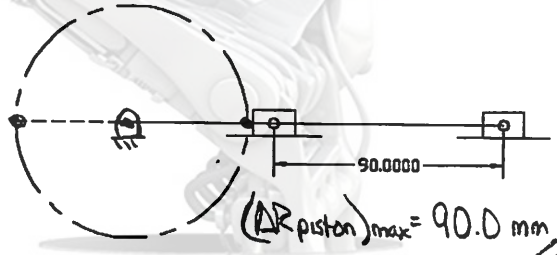
$$\theta_{CFD'} = \sin^{-1}\left[\frac{.42}{1.0559}\sin 160.95\right] = 7.46^\circ$$

$$\theta_{D'FE'} = \cos^{-1}\left[\frac{(.85)^2+(1.0559)^2-(.35)^2}{2(.85)(1.0559)}\right] = 17.18^\circ$$

$$\theta_{CFE'} = \theta_{CFD'} + \theta_{D'FE'} = 24.64$$

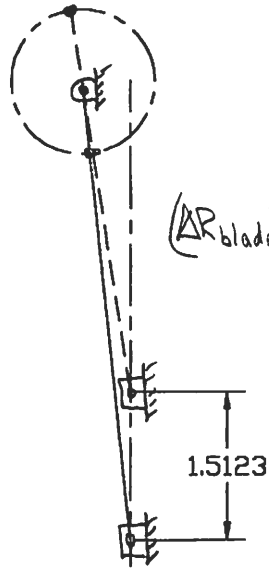
$$\Delta\theta_{CFE} = 24.64 - 34.06 = -9.41^\circ = 9.41^\circ, \text{ CW}$$

4-56



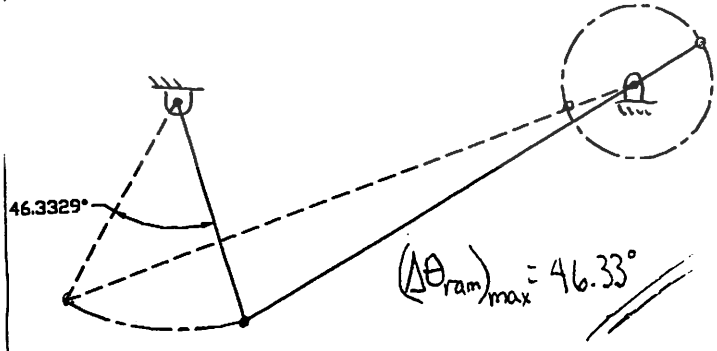
$(\Delta R_{piston})_{max} = 90.0$ mm

4-57



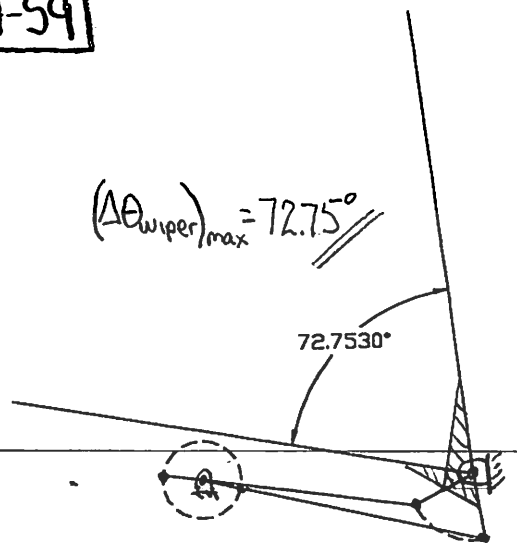
$(\Delta R_{blade})_{max} = 1.5123$ in

4-58



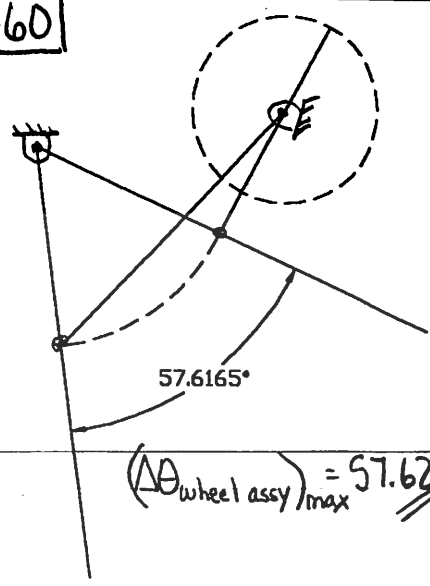
$(\Delta\theta_{ram})_{max} = 46.33^\circ$

4-59



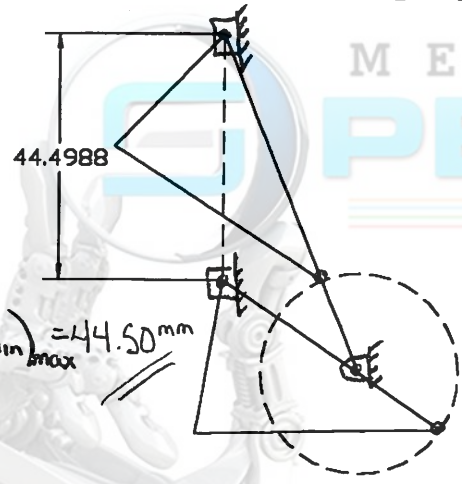
$(\Delta\theta_{wiper})_{max} = 72.75°$

4-60



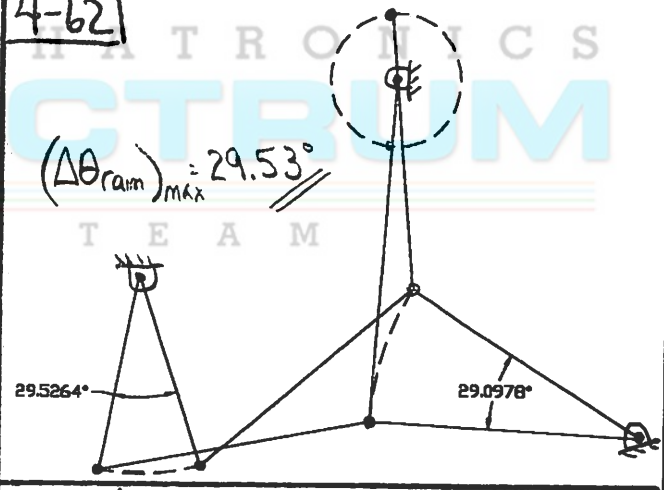
$(\Delta\theta_{wheel\ assy})_{max} = 57.62°$

4-61



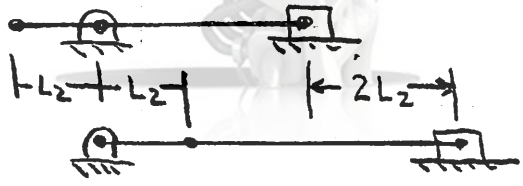
$(\Delta R_{slide\ pin})_{max} = 44.50^{mm}$

4-62



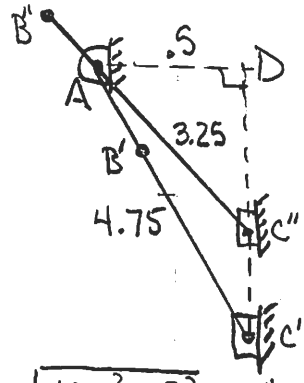
$(\Delta\theta_{ram})_{max} = 29.53°$

4-63



$(\Delta R_{piston})_{max} = 2(45) = 90mm$

4-64



$DC' = \sqrt{4.75^2 - .5^2} = 4.724"$

$DC'' = \sqrt{3.25^2 - .5^2} = 3.211"$

$(\Delta R_{blade})_{max} = 4.724 - 3.211 = 1.513"$

MECHATRONICS SPECTRUM TEAM

Answers to the Chapter 4 Case Study Questions:

Case 4-1.

1. As wheel C is rotated clockwise, the motion of pin D is circular, clockwise, motion about the center of the wheel.
2. As wheel C is rotated clockwise, the motion of pin P is oscillating, along an arc, centered about the joint on the tapped sliding link.
3. As wheel C is rotated clockwise, the motion of pin K is oscillating translation, since item U is constrained to sliding on the frame.
4. Turning the hand wheel will either raise or lower the tapped slide. As the tapped slide is raised, for example, the path of pin P will be at a higher level. The path of pin K will be pulled to the left. Pin P and K will also tend to oscillate through slightly greater displacements.
5. Turning the hand-wheel will move the oscillating motion of slide U, either leftward or rightward. The range of motion will also alter, slightly.
6. This device can be used as a running adjustment to machines that require an oscillating sliding motion, such as a surface grinder.

Case 4-2.

1. The rotary motion of the bowl, coupled with friction, brings the screw blanks to the fingers.
2. The blanks have a head, which can be efficiently oriented with a parallel finger. As the blanks approach the fingers, the blanks that happen to randomly be aligned with the fingers will tend to ride up the track.
3. The track intermittently rotates to a raised position, then back down into the bowl.
4. A connecting arm E, couples the motion of link D to link B.
5. The motion of link D is restricted to pure translation, and link B is constrined to pure rotation. Therefore, a connecting link, or a slotted joint, is required to join the two components.
6. As track B is in an elevated potision, the screw blanks are able to slide down towards C.
7. The tip of the fingers contacting the bowl is the lowesst position. A stop on F would serve as an effective method to prevent the ffinger tips to drag on the bottom of the bowl.
8. As screw blanks are congested in the outlet track, and fingers B, item F contacts the heads of the screw blanks. This contact prohibits the full lowering of the fingers.
9. As screw blanks are congested in the outlet track, and fingers B, item F contacts the heads of the screw blanks. This contact prohibits the fingers B from dipping into the bowl. Therefore, no other blanks attempt to enter the fingers.
10. This device is a feeder track, that automatically stops feeding screw blanks when the outlet track is jammed.
11. This device must be driven by a mechanism that can compensate for a shorter displacement required when the track is conjected. A slider-crank mechanism, coupled with a spring is one option.

Case 4-3.

1. Bar B, and table A, is restricted to horizontal translation.
2. The top of link C can only move horizontally and the bottom can only move vertically. So as cylinder L is shortened, the top of link C moves toward the right, and the bottom of link C moves downward. Therefore, link C moves to the right, and rotates clockwise.
3. Point K moves to the right, and downward.
4. Since the top of link C is constrained from moving vertically, and the bottom is constrained from moving horizontally, the bottom must be able to move vertically if any motion is to occur. Therefore, point F must ride in slot G.
5. The purpose of this mechanism is to move table A left and right. The motion of the cylinder is amplified and the bottom linkage assists to support a load placed on table A.
6. Turning the threaded rod end moves the "zone" of motion of table A, pushing it either leftward or to the right.

5-1

$$Q = \frac{t_1}{t_2} = \frac{1.1s}{.8s} = 1.375$$

$$\beta = 180^\circ\left(\frac{1.375-1}{1.375+1}\right) = 28.4^\circ$$

$$\omega = \frac{1\,rev}{(1.1+.8)s} = 0.526\,\frac{rev}{sec} = 31.6\,rpm$$

5-2

$$Q = \frac{t_1}{t_2} = \frac{0.35s}{0.2s} = 1.75$$

$$\beta = 180^\circ\left(\frac{1.75-1}{1.75+1}\right) = 49.1^\circ$$

$$\omega = \frac{1\,rev}{(0.35+0.2)s} = 1.81\,\frac{rev}{sec} = 109\,rpm$$

5-3

$$Q = \frac{t_1}{t_2} = \frac{0.041s}{0.027s} = 1.519$$

$$\beta = 180^\circ\left(\frac{1.519-1}{1.519+1}\right) = 36.1^\circ$$

$$\omega = \frac{1\,rev}{(.041+.027)s} = 14.7\,\frac{rev}{sec} = 882.4\,rpm$$

5-4

$\omega = 180\,rpm = 3\,rev/sec \Rightarrow .33\,sec/rev$

$$Q = \frac{180+25}{180-25} = 1.322$$

$$Q = \frac{t_1}{.33-t_1} \Rightarrow t_1 = \frac{.33Q}{Q+1} = .188s$$

$$t_2 = \frac{t_1}{Q} = .142\,s$$

5-5

$\omega = 75\,rpm = 1.25\,\frac{rev}{sec} \Rightarrow .8\,sec/rev$

$$Q = \frac{180+37}{180-37} = 1.517$$

$$Q = \frac{t_1}{.8-t_1} \Rightarrow t_1 = \frac{.8Q}{Q+1} = .482s$$

$$t_2 = \frac{t_1}{Q} = .318s$$

5-6

$\omega = 500\,rpm = 8.33\,\frac{rev}{sec} \Rightarrow .12\,\frac{sec}{rev}$

$$Q = \frac{180+20}{180-20} = 1.25$$

$$Q = \frac{t_1}{.12-t_1} \Rightarrow t_1 = \frac{.12Q}{Q+1} = .067\,sec$$

$$t_2 = \frac{t_1}{Q} = .053sec$$

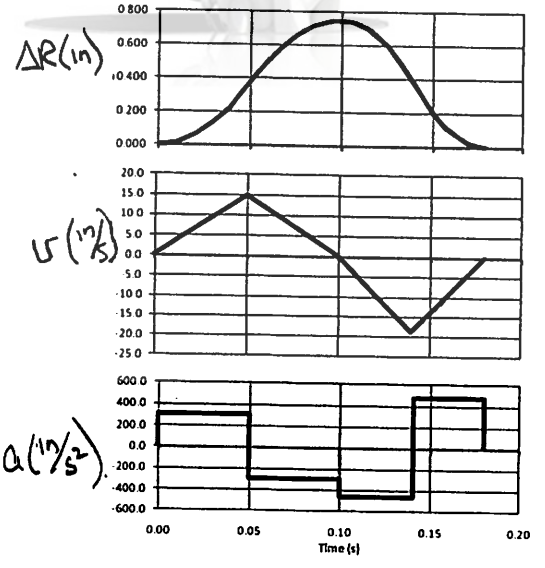
5-7

$$Q = \frac{0.10\,s}{0.08\,s} = 1.25$$

$$\omega = \frac{1\,rev}{(0.10+0.08)s}\left(\frac{60s}{1min}\right) = 333\,rpm$$

$$v_{peak} = 2\frac{0.75\,in}{(0.08s)} = 18.75\,in/s$$

$$a_{peak} = 4\frac{0.75\,in}{(0.08s)^2} = 468.8\,in/s^2$$



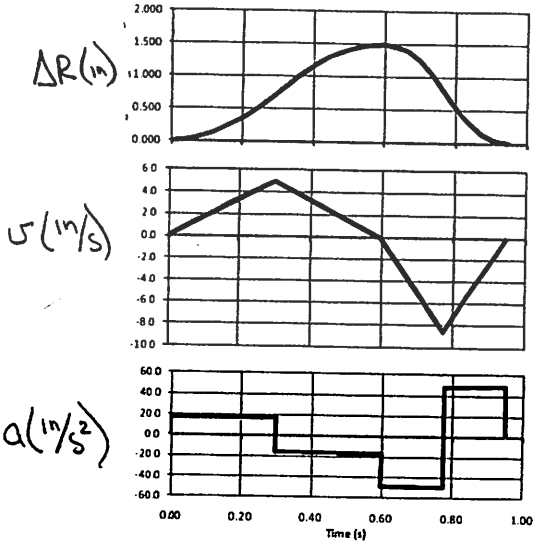
5-8

$$Q = \frac{0.60s}{0.35s} = 1.714$$

$$\omega = \frac{1\,rev}{(.60+.35)s}\left(\frac{60s}{1min}\right) = 63.2\,rpm$$

$$v_{peak} = 2\frac{1.5\,in}{(0.35s)} = 8.57\,in/s$$

$$a_{peak} = 4\frac{1.5\,in}{(0.35s)^2} = 49.0\,in/s^2$$



5-9 $Q = \frac{.8s}{.6s} = 1.33$

$\omega = \frac{1\,rev}{.8+.6\,s}\left(\frac{60s}{1\,min}\right) = 42.86\,rpm$

Conveyor:

$v_{peak} = 2(6in)/(.6s) = 20\,in/s$

$a_{peak} = 4(6in)/(.6s)^2 = 66.7\,in/s^2$

Stamp: outward

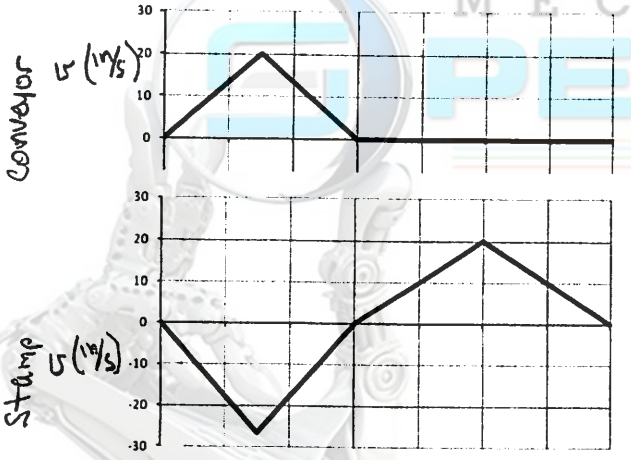
$v_{peak} = 2(8in)/(.8s) = 20\,in/s$

$a_{peak} = 4(8in)/(.8s)^2 = 50\,in/s^2$

Stamp: return

$v_{peak} = 2(8in)/(.6s) = 26.7\,in/s$

$a_{peak} = 4(8in)/(.6s)^2 = 88.9\,in/s^2$



5-10 $Q = \frac{.25s}{.12s} = 2.083$

$\omega = \frac{1\,rev}{.25+.12\,s}\left(\frac{60s}{1\,min}\right) = 162.2\,rpm$

Conveyor

$v_{peak} = 2(2in)/(.12s) = 33.3\,in/s$

$a_{peak} = 4(2in)/(.12s)^2 = 555.5\,in/s^2$

Press: outward

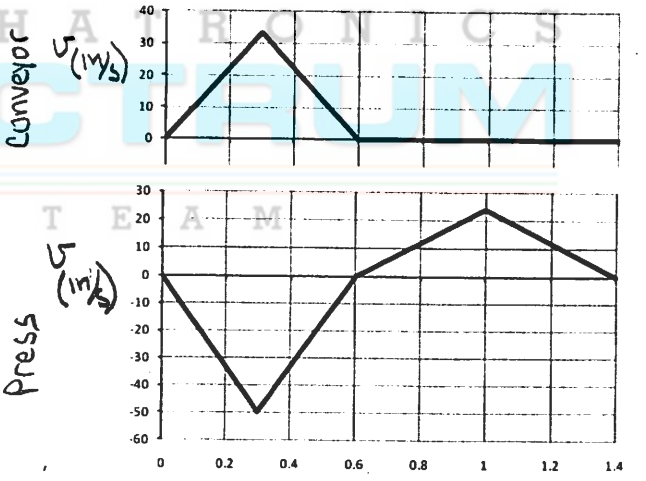
$v_{peak} = 2(3in)/(.25s) = 24\,in/s$

$a_{peak} = 4(3in)/(.25s)^2 = 192\,in/s^2$

Press: return

$v_{peak} = 2(3in)/(.12s) = 50\,in/s$

$a_{peak} = 4(3in)/(.12s)^2 = 833.3\,in/s^2$



5-11 Q=1 (inline, slider-crank)

$l_2 = \frac{s}{2} = 1\,in$

any l_3 will work, use $l_3 = 5(l_2) = 5in$

$\omega = \frac{1\,rev}{1.2\,s}\left(\frac{60\,sec}{1\,min}\right) = 50\,rpm$

5-12 Q=1 (inline, slider-crank)

$l_2 = \frac{s}{2} = 4\,mm$

any l_3 will work, use $l_3 = 5(l_2) = 20\,mm$

$\omega = \frac{1\,rev}{.08\,sec}\left(\frac{60\,sec}{1\,min}\right) = 750\,rpm$

5-13 Q=1 (inline, slider-crank)

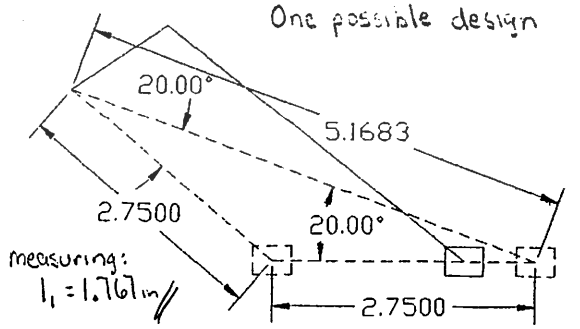
$l_2 = \frac{.9mm}{5} = .45\,mm$ (micro-machine)

any l_3 will work, use $l_3 = 5(.45) = 2.25\,mm$

$\omega = \frac{1\,rev}{.4\,sec}\left(\frac{60\,sec}{1\,min}\right) = 150\,rpm$

5-14 $Q = 1.25 \Rightarrow \beta = 180\frac{(1.25-1)}{(1.25+1)} = 20°$

One possible design



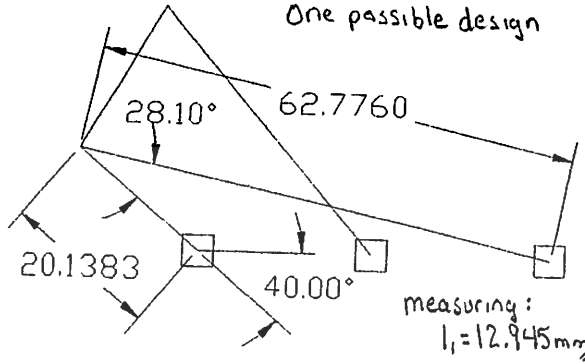
$2l_2 = 5.1683 - 2.75 \Rightarrow l_2 = 1.209\,in$

$l_3 = 5.1683 - 1.209 = 3.959\,in$

$\omega = \frac{1\,rev}{.6\,s}\left(\frac{60\,sec}{1\,min}\right) = 100\,rpm$

5-15 $Q = 1.37 \Rightarrow \beta = 180\left(\frac{1.37-1}{1.37+1}\right) = 28.1°$

One possible design



measuring: $l_1 = 12.945\,mm$

$2l_2 = 62.776 - 20.1383 \Rightarrow l_2 = 21.319\,mm$

$l_3 = 62.776 - l_2 = 41.457\,mm$

$\omega = \frac{1\,rev}{3.4\,s}\left(\frac{60\,s}{1\,min}\right) = 17.65\,rpm$

5-16 $Q = 1.15 \Rightarrow \beta = 180\frac{(1.15-1)}{(1.15+1)} = 12.6°$

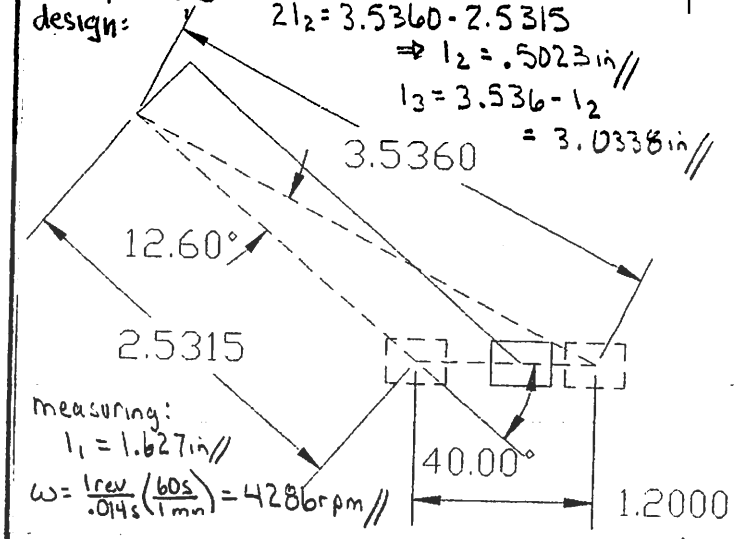
One possible design:

$2l_2 = 3.5360 - 2.5315$

$\Rightarrow l_2 = .5023\,in$

$l_3 = 3.536 - l_2$

$= 3.0338\,in$



measuring: $l_1 = 1.627\,in$

$\omega = \frac{1\,rev}{.014\,s}\left(\frac{60\,s}{1\,mn}\right) = 4286\,rpm$

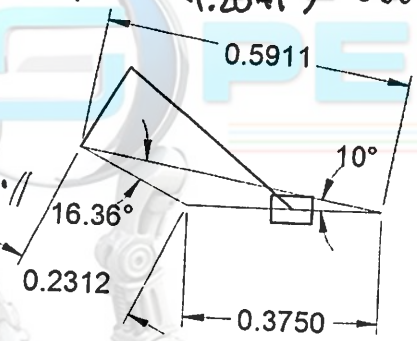
5-17 $Q = 1.20 \Rightarrow \beta = 180\left(\frac{1.20-1}{1.20+1}\right) = 16.36°$

One possible design:

measuring $l_1 = .1026\,in$

$\omega = \frac{1\,rev}{.025\,s}\left(\frac{60\,sec}{1\,min}\right)$

$= 2400\,rpm$



$2l_2 = .5911 - .2312 \Rightarrow l_2 = .1799\,in$

$l_3 = .5911 - l_2 = .4112\,in$

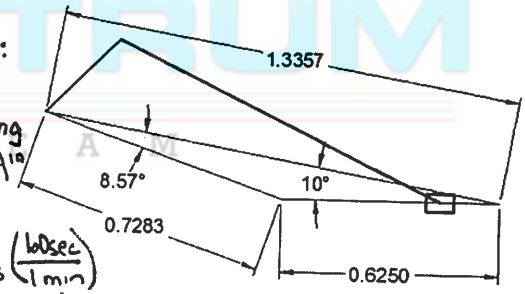
5-18 $Q = 1.10 \Rightarrow \beta = 180\left(\frac{1.10-1}{1.10+1}\right) = 8.57°$

one design:

measuring $l_1 = .2319\,in$

$\omega = \frac{1\,rev}{.033\,s}\left(\frac{60\,sec}{1\,min}\right)$

$= 1818\,rpm$

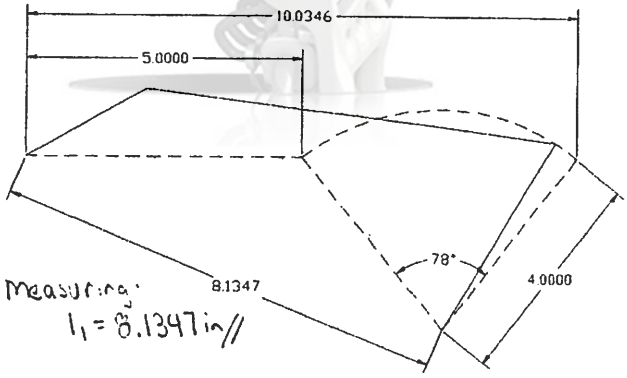


$2l_2 = 1.3357 - .7283 \Rightarrow l_2 = .3037\,in$

$l_3 = 1.3357 - l_2 = 1.032\,in$

5-19 $Q = 1$ (inline, crank-rocker)

One possible design: use $l_4 = 4.00\,in$



measuring: $l_1 = 8.1347\,in$

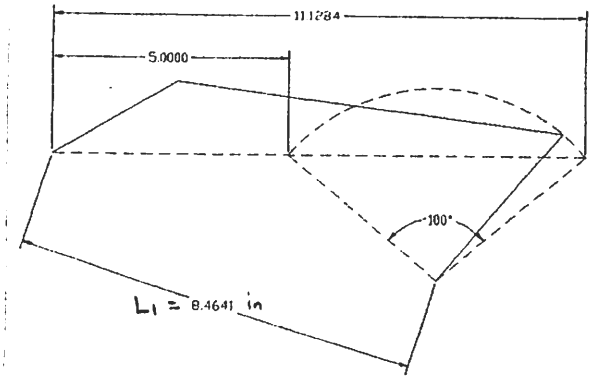
$2l_2 = 10.0346 - 5.000 \Rightarrow l_2 = 2.5173\,in$

$l_3 = 10.0346 - l_2 \Rightarrow l_3 = 7.5173\,in$

$\omega = \frac{1\,rev}{1.2\,s}\left(\frac{60\,sec}{1\,min}\right) = 50\,rpm$

5-20 $Q = 1$ (in-line, crank-rocker)

One possible design: use $l_4 = 4.00\,in$



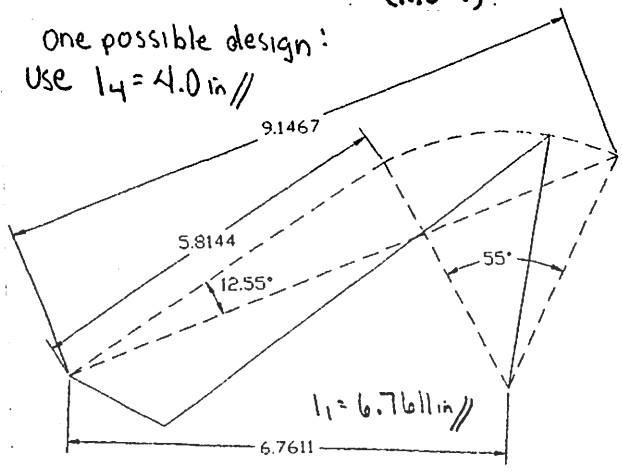
$2l_2 = 11.1284 - 5.000 \Rightarrow l_2 = 3.064\,in$

$l_3 = 11.1284 - l_2 \Rightarrow l_3 = 8.064\,in$

$\omega = \frac{1\,rev}{3.5\,s}\left(\frac{60\,sec}{1\,min}\right) = 17.14\,rpm$

5-21 $Q = 1.15 \Rightarrow \beta = 180\frac{(1.15-1)}{(1.15+1)} = 12.55°$

One possible design:
Use $l_4 = 4.0$ in //

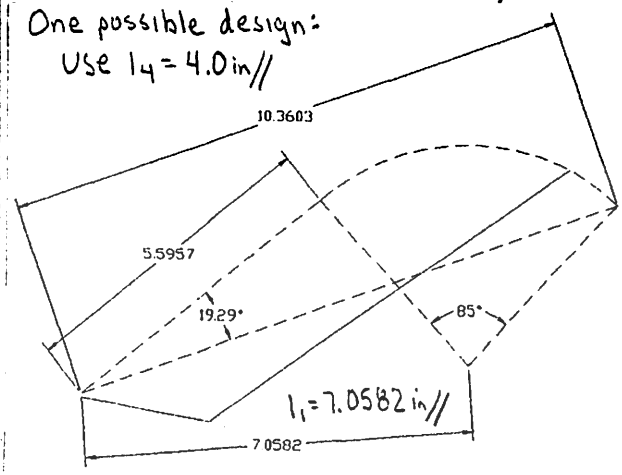


$l_1 = 6.7611$ in //

$2l_2 = 9.1467 - 5.8144 \Rightarrow l_2 = 1.666$ in //
$l_3 = 9.1467 - l_2 \Rightarrow l_3 = 7.4806$ in //
$\omega = \frac{(1\text{ rev})}{.45\text{s}}\left(\frac{60\text{s}}{1\text{ min}}\right) = 133.3$ rpm //

5-22 $Q = 1.24 \Rightarrow \beta = 180\frac{(1.24-1)}{(1.24+1)} = 19.29°$

One possible design:
Use $l_4 = 4.0$ in //

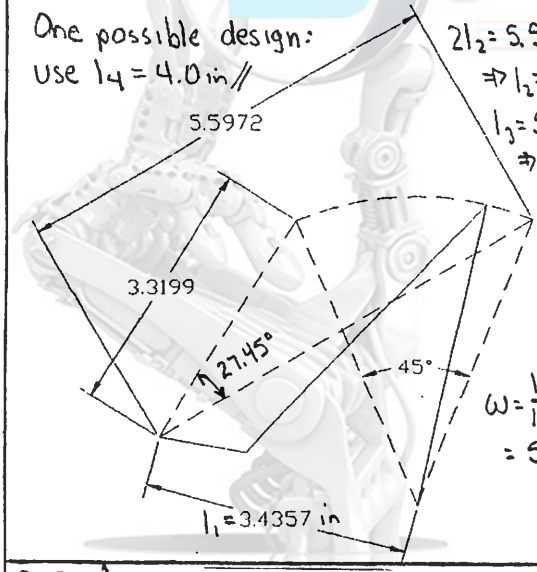


$l_1 = 7.0582$ in //

$2l_2 = 10.3603 - 5.5957 \Rightarrow l_2 = 2.3823$ in //
$l_3 = 10.3603 - l_2 \Rightarrow l_3 = 7.978$ in //
$\omega = \frac{(1\text{ rev})}{1.8\text{s}}\left(\frac{60\text{ s}}{1\text{ min}}\right) = 33.3$ rpm //

5-23 $Q = 1.36 \Rightarrow \beta = 180\frac{(1.36-1)}{(1.36+1)} = 27.45°$

One possible design:
Use $l_4 = 4.0$ in //

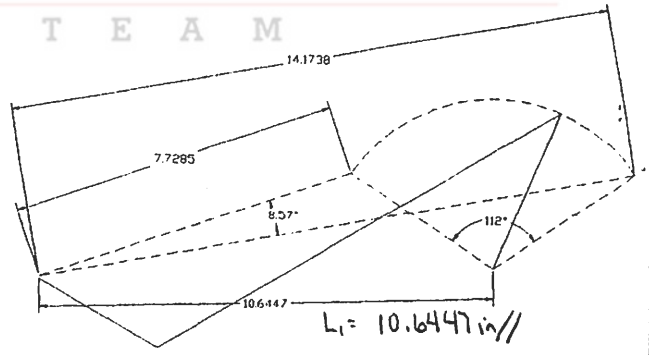


$2l_2 = 5.5972 - 3.3119 \Rightarrow l_2 = 1.1387$ in //
$l_3 = 5.5972 - l_2 \Rightarrow l_3 = 4.459$ in //

$\omega = \frac{1\text{rev}}{1.2\text{s}}\left(\frac{60\text{sec}}{1\text{ min}}\right) = 50$ rpm //

5-24 $Q = 1.1 \Rightarrow \beta = 180°\frac{(1.1-1)}{(1.1+1)} = 8.57°$

One possible design:

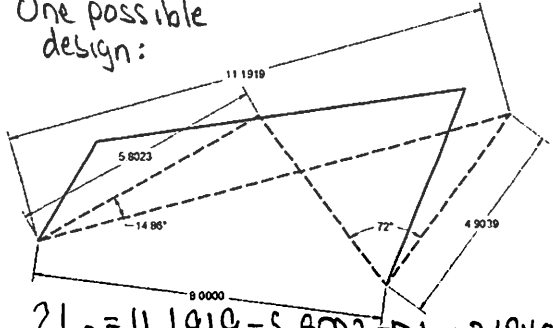


$L_1 = 10.6447$ in //

$2l_2 = 14.1738 - 7.7285 \Rightarrow l_2 = 3.223$ in //
$l_3 = 14.1738 - l_2 \Rightarrow l_3 = 10.9511$ in //
$\omega = \frac{(1\text{ rev})}{(.3\text{ s})}\left(\frac{60\text{ s}}{1\text{min}}\right) = 200$ rpm //

5-25 $Q = 1.18 \Rightarrow \beta = 180\frac{(1.18-1)}{(1.18+1)} = 14.86°$

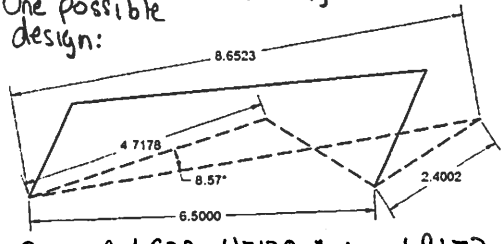
One possible design:



$2L_2 = 11.1919 - 5.8023 \Rightarrow L_2 = 2.6948$
$L_3 = 11.1919 - L_2 = 8.4971$
$\omega_2 = \left(\frac{1\text{rev}}{0.08\text{sec}}\right)\left(\frac{60\text{sec}}{1\text{ min}}\right) = 750$ rpm

5-26 $Q = 1.10 \Rightarrow \beta = 180\frac{(1.10-1)}{(1.10+1)} = 8.57°$

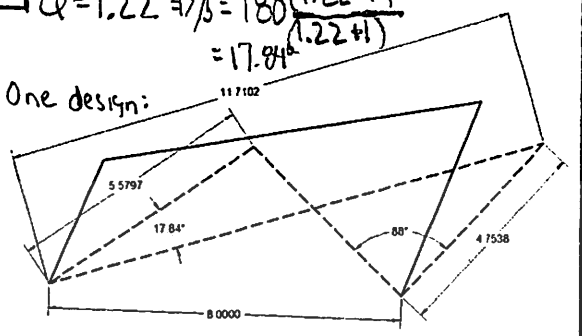
One possible design:



$2L_2 = 8.6523 - 4.7178 \Rightarrow L_2 = 1.9673$
$L_3 = 8.6523 - L_2 = 6.6851$
$\omega_2 = \left(\frac{1\text{rev}}{0.2\text{sec}}\right)\left(\frac{60\text{sec}}{1\text{ min}}\right) = 300$ rpm

5-27 $Q = 1.22 \Rightarrow \beta = 180\left(\frac{1.22-1}{1.22+1}\right) = 17.84°$

One design:



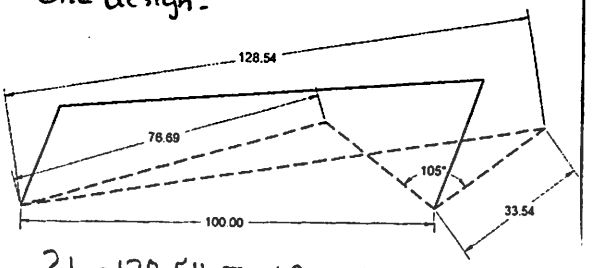
$2L_2 = 11.7102 - 5.5797 \Rightarrow L_2 = 3.0653$

$L_3 = 11.7102 - L_2 = 8.6450$

$\omega_2 = \left(\frac{1\,rev}{0.75\,s}\right)\left(\frac{60\,sec}{1\,min}\right) = 80\,rpm$

5-28 $Q = 1.08 \Rightarrow \beta = 180\,\frac{(1.08-1)}{(1.08+1)} = 6.92°$

One design:



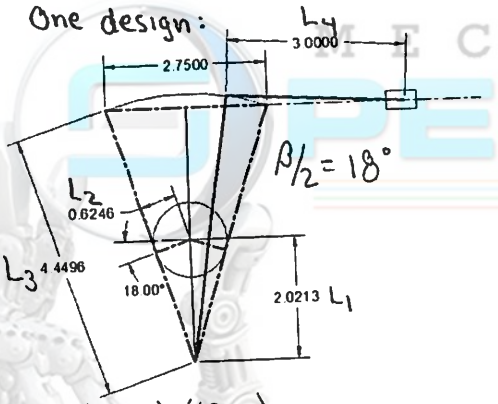
$2L_2 = 128.54 - 76.69 \Rightarrow L_2 = 25.93$

$L_3 = 128.54 - L_2 = 102.61$

$\omega_2 = \left(\frac{1\,rev}{1.5\,sec}\right)\left(\frac{60\,sec}{1\,min}\right) = 40\,rpm$

5-29 $Q = 1.50 \Rightarrow \beta = 180°\left(\frac{1.50-1}{1.50+1}\right) = 36°$

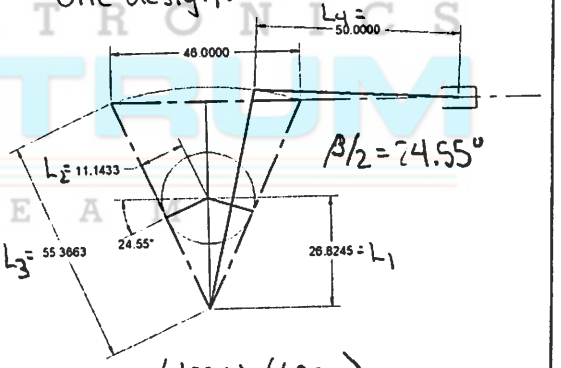
One design:



$\omega_2 = \left(\frac{1\,rev}{0.6\,sec}\right)\left(\frac{60\,sec}{1\,min}\right) = 100\,rpm$

5-30 $Q = 1.75 \Rightarrow \beta = 180°\left(\frac{1.75-1}{1.75+1}\right) = 49.09°$

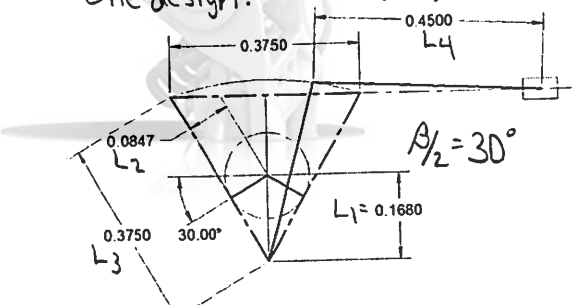
One design:



$\omega_2 = \left(\frac{1\,rev}{3.4\,sec}\right)\left(\frac{60\,sec}{1\,min}\right) = 17.6\,rpm$

5-31 $Q = 2.0 \Rightarrow \beta = 180°\frac{(2-1)}{(2+1)} = 60°$

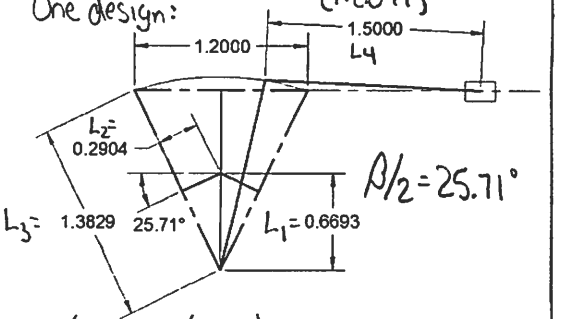
One design:



$\omega_2 = \frac{(1\,rev)}{(.014\,sec)}\frac{(60\,sec)}{(1\,min)} = 4286\,rpm$

5-32 $Q = 1.80 \Rightarrow \beta = 180°\frac{(1.80-1)}{(1.80+1)} = 51.43°$

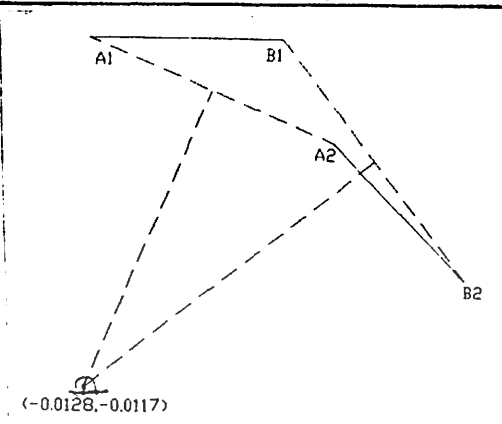
One design:



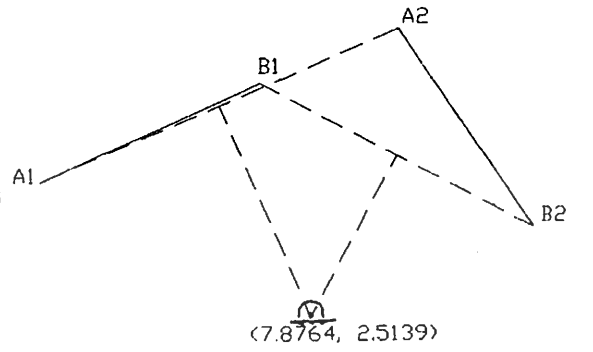
$\omega_2 = \left(\frac{1\,rev}{.25\,sec}\right)\left(\frac{60\,sec}{1\,min}\right) = 240\,rpm$

MECHATRONICS SPECTRUM TEAM

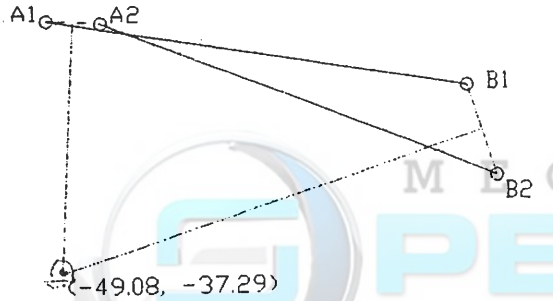
5-33



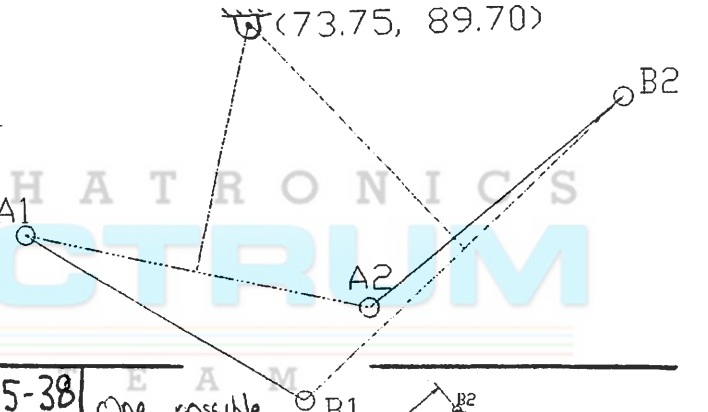
5-34



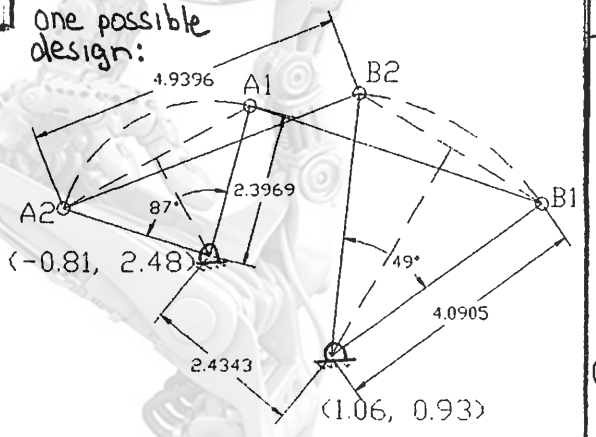
5-35



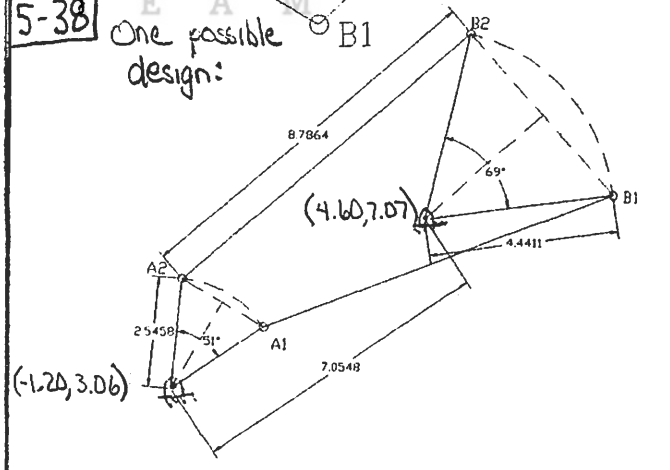
5-36.



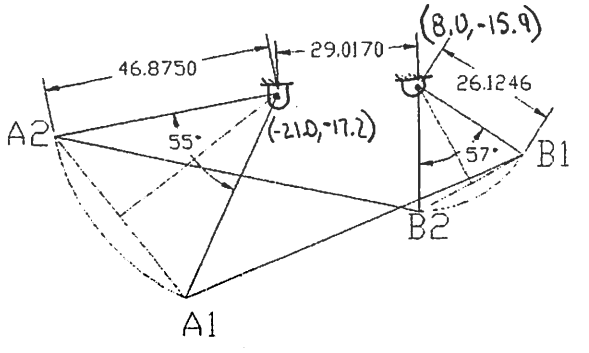
5-37 One possible design:



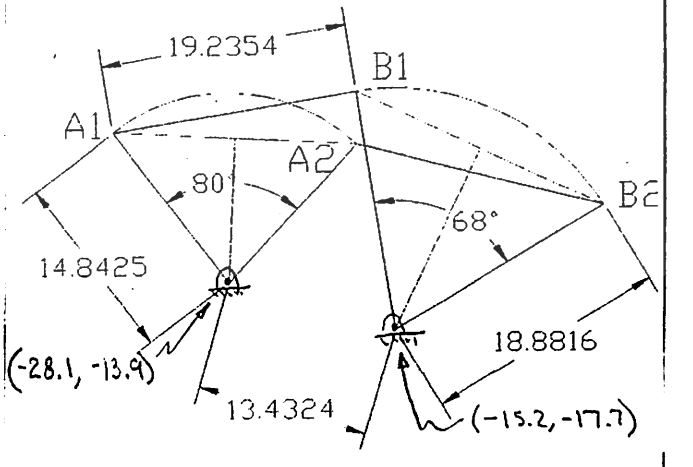
5-38 One possible design:



5-39 One possible design:

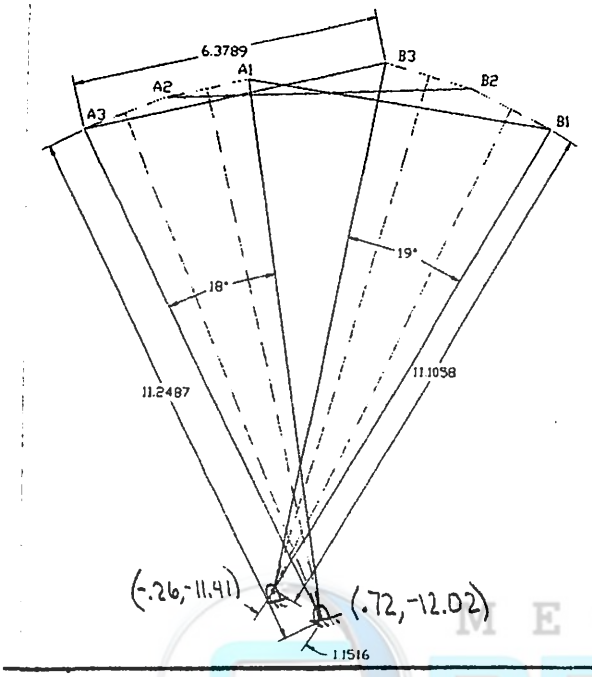


5-40 One possible design:

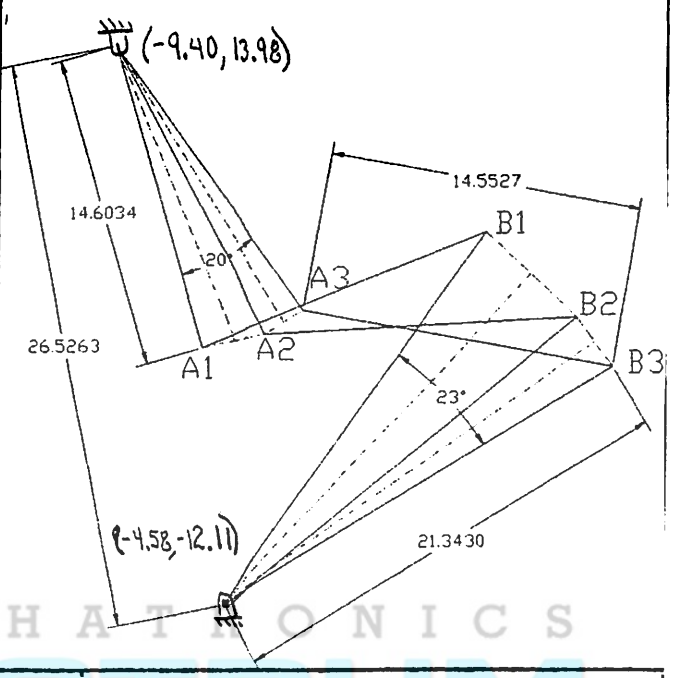


MECHATRONICS SPECTRUM TEAM

5-41



5-42



MECHATRONICS
SPECTRUM
TEAM

Answers to the Chapter 5 Case Study Questions:

Case 5-1.

1. As shaft A rotates 90^0, clockwise, the eccentric of lobe B moves down to the 6 o'clock position.
2. As shaft A rotates 90^0, clockwise, link C will move into a vertical position.
3. A slot is necessary at point E, to accommodate for the vertical motion of link C.
4. As shaft A rotates 90^0, clockwise, pin H will move toward the left, and drop downward.
5. As shaft A rotates 90^0, clockwise, pin I will move toward the left.
6. The mechanism has 6 links, 5 pin joints, 2 sliding joints, and one degree of freedom.
7. As thread G pulls roller E downward, the distance from the drive shaft to the effective pivot will lengthen. As a result, the angle at which link C will oscillate is reduced.
8. As thread G pulls roller E downward, the distance from the drive shaft to the effective pivot will lengthen. As a result, the horizontal distance at which point H will oscillate is reduced.
9. The purpose of the mechanism is to move block I back and forth. However, the speed of travel is faster in one direction, yielding quick return action.

Case 5-2.

1. As rod A moves to the right, sliding block B is also driven to the right.
2. As rod A moves to the right, and the roller C reaches groove D, the roller falls into the groove. As a result, the bellcrank link rotates clockwise. The sliding block B will still move to the right, but at a slower speed.
3. As rod A reverses and moves to the left, the bellcrank link rotates counterclockwise, pulling roller C from groove D. The sliding block B also moves to the left, but at a relatively slow speed.
4. As rod A oscillates horizontally, sliding block B also oscillates horizontally. However at the right extreme end of the stroke, sliding block B moves at a slower speed.
5. The purpose of the mechanism is to take a single-speed linear motion and create two-speed linear motion.
6. Rod A can be driven with a slider-crank mechanism, a pneumatic or hydraulic cylinder, a cam follower, etc.
7. The adjustment slots at E allow the user to modify the location where the speed of sliding block B will change.

MECHATRONICS SPECTRUM TEAM

6-1 $V = \frac{\Delta R}{\Delta t} = \frac{25\text{ ft}}{15\text{ s}} = 1.67\text{ ft/s}$

6-2 $\Delta t = \frac{\Delta R}{V} = \frac{15\text{ in}}{2\text{ ft/min}}\left(\frac{1\text{ ft}}{12\text{ in}}\right)\left(\frac{60\text{ s}}{1\text{ min}}\right)$
$= 37.5\text{ sec}$

6-3 $V = \frac{1\text{ mi}}{4\text{ min}}\left(\frac{60\text{ min}}{1\text{ hr}}\right) = 15\text{ mph}$

6-4 $V = \frac{100\text{ m}}{10\text{ s}}\left(\frac{3600\text{ s}}{1\text{ hr}}\right)\left(\frac{1\text{ mi}}{1609.3\text{ m}}\right) = 22.37\text{ mph}$

6-5 $\omega = \frac{\Delta\theta}{\Delta t} = \frac{270^\circ}{2\text{ s}} = 135\text{ deg/sec}$
$135\text{ deg/sec}\left(\frac{2\pi\text{ rad}}{360\text{ deg}}\right) = 2.36\text{ rad/s}$
$2.36\text{ rad/s}\left(\frac{1\text{ rev}}{2\pi\text{ rad}}\right)\left(\frac{60\text{ sec}}{1\text{ min}}\right) = 22.5\text{ rpm}$

6-6 hour: $\omega = \frac{1\text{ rev}}{12\text{ hr}}\left(\frac{1\text{ hr}}{60\text{ min}}\right) = .00139\text{ rpm}$
min: $\omega = \frac{1\text{ rev}}{60\text{ min}} = .0167\text{ rpm}$
sec: $\omega = \frac{1\text{ rev}}{1\text{ min}} = 1\text{ rpm}$

6-7 $\Delta R = \int v\,dt$
$= \frac{1}{2}(8)(2) + 8(9-2) + \frac{1}{2}(8)(11-9)$
$= 72\text{ in}$

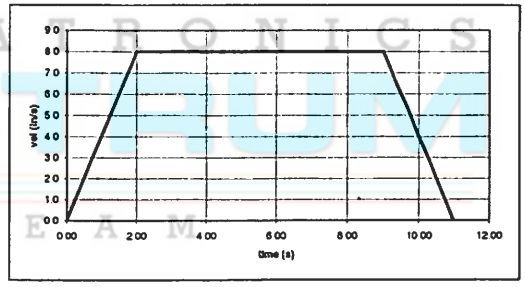
6-8

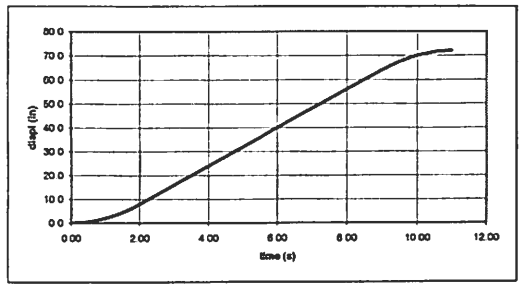
Stage 1: $0 \le t < 2$
$V = 4t \quad \Delta R = \int v\,dt = 2t^2$

Stage 2: $2 \le t < 9$
$V = 8 \quad \Delta R = 8t + c$ (at $t=2$, $s=8$) $\Rightarrow c = -8$
$= 8t + 8$

Stage 3: $9 \le t < 11$
$V = -4t + (11)(4) = -4t + 44$
$\Delta R = -2t^2 + 44t + k$ (at $t=9$, $s=64$) $\Rightarrow k = -170$
$= -2t^2 + 44t - 170$

t (s)	v (in/s)	s (in)
0.00	0.0	0.0
0.25	1.0	0.1
0.50	2.0	0.5
0.75	3.0	1.1
1.00	4.0	2.0
1.25	5.0	3.1
1.50	6.0	4.5
1.75	7.0	6.1
2.00	8.0	8.0
9.00	8.0	64.0
9.25	7.0	65.9
9.50	6.0	67.5
9.75	5.0	68.9
10.00	4.0	70.0
10.25	3.0	70.9
10.50	2.0	71.5
10.75	1.0	71.9
11.00	0.0	72.0





6-9 $\Delta R = \int v\,dt$
$= \frac{1}{2}(2)(2) + 2(4-2) + \frac{1}{2}(2)(5.5-4)$
$= 7.5\text{ in}$

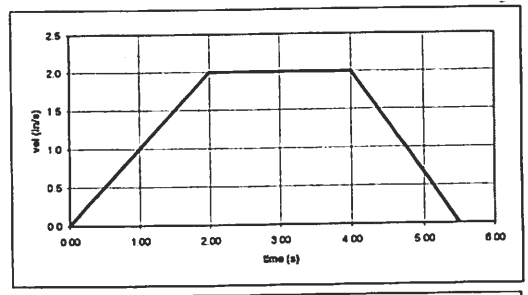
Stage 1: $0 \le t < 2$
$V = t \quad s = \int v\,dt = \frac{1}{2}t^2$

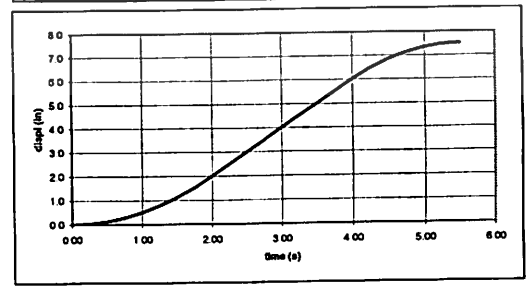
Stage 2: $2 \le t \le 4$
$V = 2 \quad s = 2t + c$ (at $t=2$, $s=2$) $\Rightarrow c = -2$
$= 2t - 2$

Stage 3: $4 \le t \le 5.5$
$V = -1.33t + (5.5)(1.33)$
$= -1.33t + 7.33$
$s = -.666t^2 + 7.33t + k$ (at $t=4$, $s=6$) $\Rightarrow k = -12.66$
$= -.666t^2 + 7.33t - 12.66$

t (s)	v (in/s)	s (in)
0.00	0.0	0.0
0.25	0.3	0.0
0.50	0.5	0.1
0.75	0.8	0.3
1.00	1.0	0.5
1.25	1.3	0.8
1.50	1.5	1.1
1.75	1.8	1.5
2.00	2.0	2.0
4.00	2.0	6.0
4.25	1.7	6.5
4.50	1.3	6.8
4.75	1.0	7.1
5.00	0.7	7.3
5.25	0.3	7.5
5.50	0.0	7.5

6-10





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6-11

$$\omega = \frac{V}{r} = \frac{10\ ft/min}{8\ in}\left(\frac{12\ in}{1\ ft}\right) = 15\ rad/min$$

$$\left(15\ rad/min\right)\left(\frac{1\ rev}{2\pi\ rad}\right) = 2.38\ rpm, ccw$$

6-12

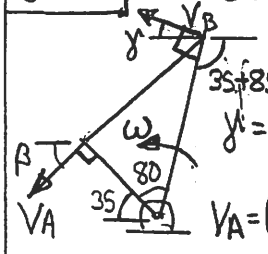
$$\omega = 30\ rpm\left(\frac{2\pi\ rad}{1\ rev}\right) = 188.5\ rad/min$$

$$V = r\omega = (8\ in)(188.5\ rad/min)\left(\frac{1\ ft}{12\ in}\right)$$

$$= 125.66\ ft/min$$

6-13

$$\omega = 300\ rpm\left(\frac{\pi}{30}\right) = 31.42\ rad/s, ccw$$



$\beta = 90 - 35 = 55°$

$\gamma = 35 + 85 + 90 - 180 = 25°$

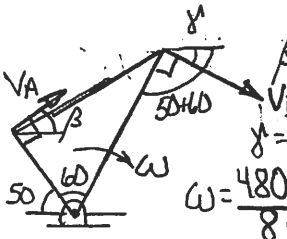
$V_A = (8")(31.42\ rad/s)$

$= 251.36\ in/s$ $55°$

$V_B = (18")(31.42\ r/s) = 565.56\ in/s$ $25°$

6-14

$V_A = 40\ ft/s\left(\frac{12\ in}{1\ ft}\right) = 480\ in/s$ $\angle 40°$



$\beta = 90 - 50 = 40$

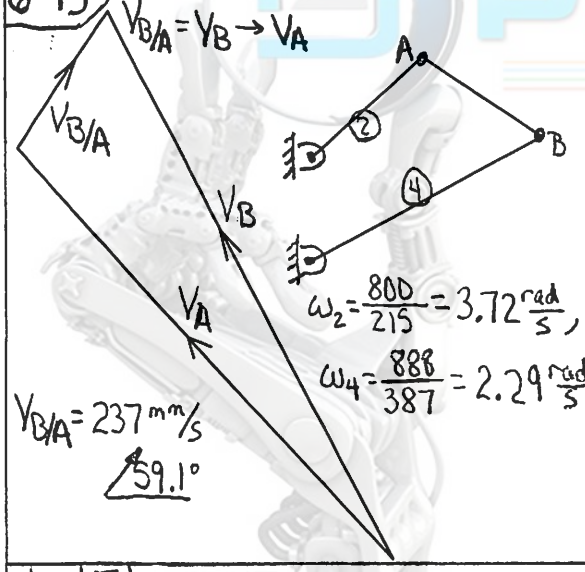
$\gamma = 50 + 60 - 90 = 20°$

$\omega = \frac{480\ in/s}{8\ in} = 60\ rad/s$, CW

$V_B = (18")(60\ rad/s)\left(\frac{1\ ft}{12\ in}\right) = 90\ ft/s$ $\angle 20°$

6-15

$V_{B/A} = V_B \rightarrow V_A$



$\omega_2 = \frac{800}{215} = 3.72\ rad/s$, CCW

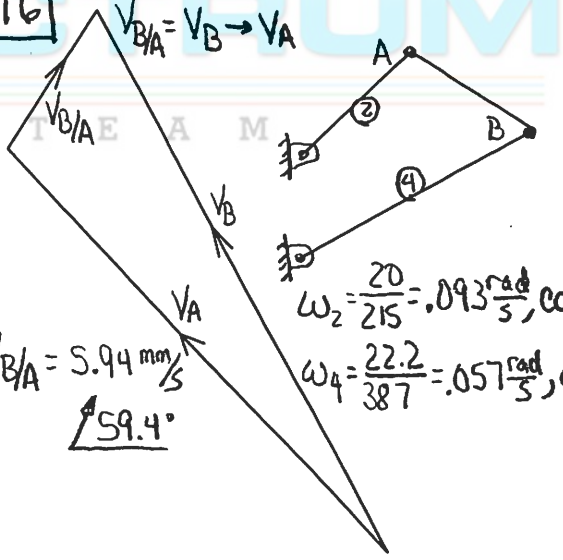
$\omega_4 = \frac{888}{387} = 2.29\ rad/s$, CCW

$V_{B/A} = 237\ mm/s$

$\angle 59.1°$

6-16

$V_{B/A} = V_B \rightarrow V_A$



$\omega_2 = \frac{20}{215} = .093\ rad/s$, CCW

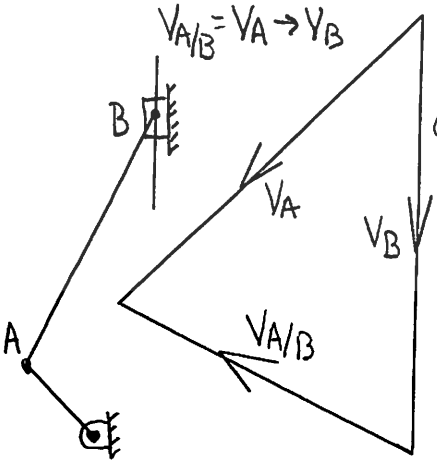
$\omega_4 = \frac{22.2}{387} = .057\ rad/s$, CCW

$V_{B/A} = 5.94\ mm/s$

$\angle 59.4°$

6-17

$V_{A/B} = V_A \rightarrow V_B$



$\omega_2 = \frac{380\ ft/s}{.9\ in}\left(\frac{12\ in}{1\ ft}\right)$

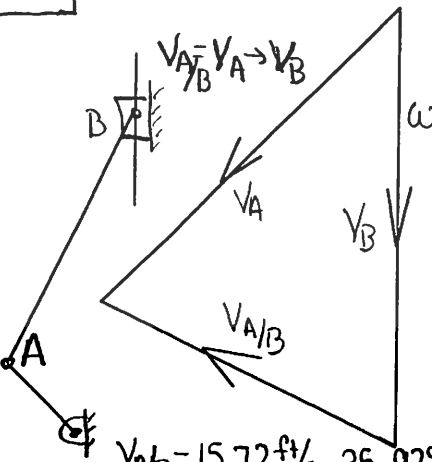
$= 5067\ rad/s$, CCW

$= 530.6\ rpm$, CCW

$V_{A/B} = 298.8\ ft/s$ 25.92

6-18

$V_{A/B} = V_A \rightarrow V_B$



$\omega_2 = \frac{20\ ft/s}{.9\ in}\left(\frac{12\ in}{1\ ft}\right)$

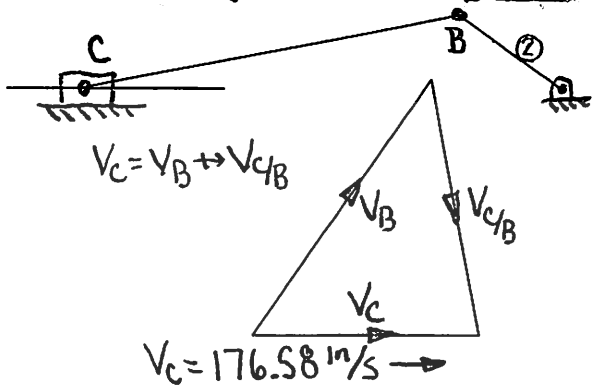
$= 160\ rad/s$, CCW

$= 1528\ rpm$, CCW

$V_{A/B} = 15.72\ ft/s$ $25.92°$

6-19 $\omega_2 = 1150\left(\frac{\pi}{30}\right) = 120.43 \frac{rad}{s}$, CW

$V_B = (120.43)(2) = 240.86 \frac{in}{s}$ ∠55°

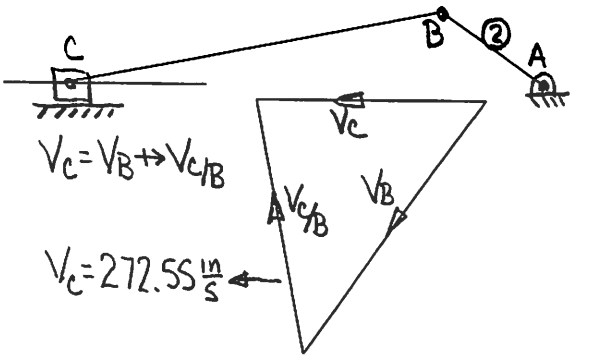


$V_C = V_B \leftrightarrow V_{C/B}$

$V_C = 176.58 \frac{in}{s} \rightarrow$

6-20 $\omega_2 = 1775\left(\frac{\pi}{30}\right) = 185.88 \frac{rad}{sec}$, CCW

$V_B = (185.88)(2) = 371.76 \frac{in}{s}$ ∠55°

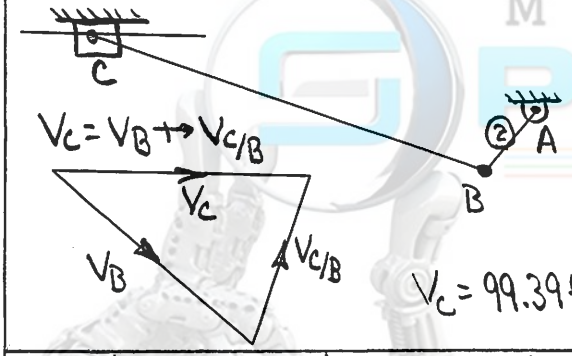


$V_C = V_B \leftrightarrow V_{C/B}$

$V_C = 272.55 \frac{in}{s} \leftarrow$

6-21 $\omega_2 = 1500\left(\frac{\pi}{30}\right) = 157.08 \frac{rad}{s}$, CCW

$V_B = (157.08)(.65) = 102.1 \frac{in}{s}$ ∠40°

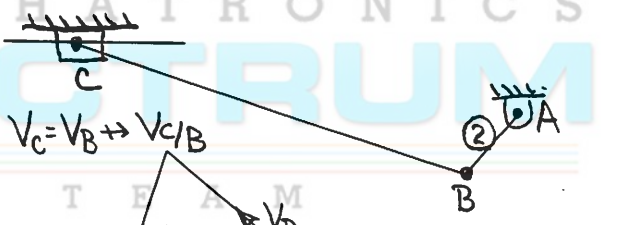


$V_C = V_B \leftrightarrow V_{C/B}$

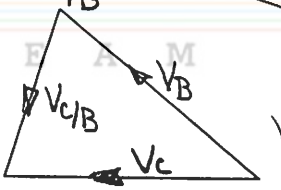
$V_C = 99.39 \frac{in}{s} \rightarrow$

6-22 $\omega_2 = 900\left(\frac{\pi}{30}\right) = 94.25 \frac{rad}{s}$, CW

$V_B = (94.25)(.65) = 61.26 \frac{in}{s}$ ∠40°



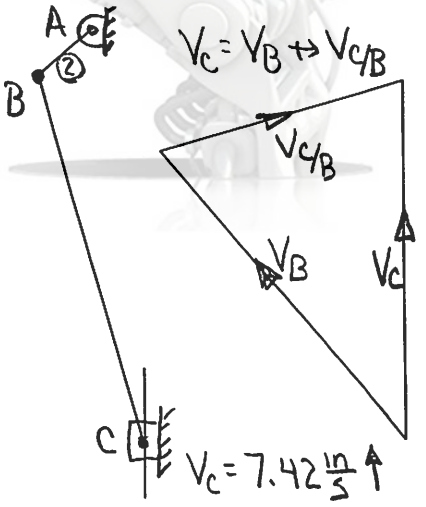
$V_C = V_B \leftrightarrow V_{C/B}$



$V_C = 59.63 \frac{in}{s} \leftarrow$

6-23 $\omega_2 = 100\left(\frac{\pi}{30}\right) = 10.47 \frac{rad}{s}$, CW

$V_B = (10.47)(.75) = 7.85 \frac{in}{s}$ ∠50°

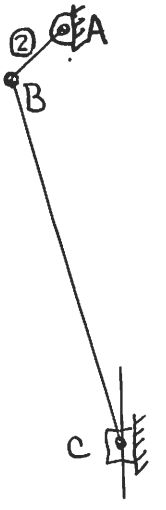


$V_C = V_B \leftrightarrow V_{C/B}$

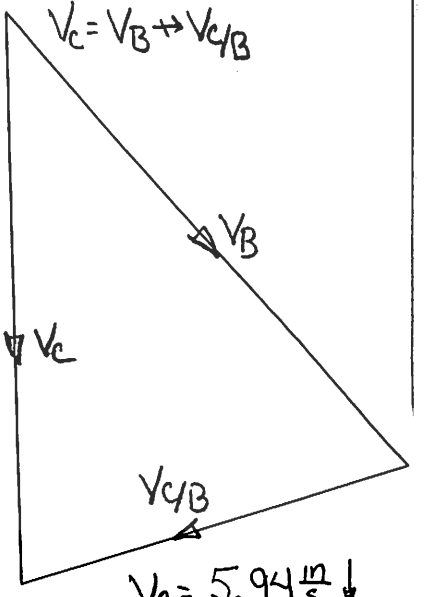
$V_C = 7.42 \frac{in}{s} \uparrow$

6-24 $\omega_2 = (80)\left(\frac{\pi}{30}\right) = 8.38 \frac{rad}{s}$, CCW

$V_B = (8.38)(.75) = 6.28 \frac{in}{s}$ ∠50



$V_C = V_B \leftrightarrow V_{C/B}$



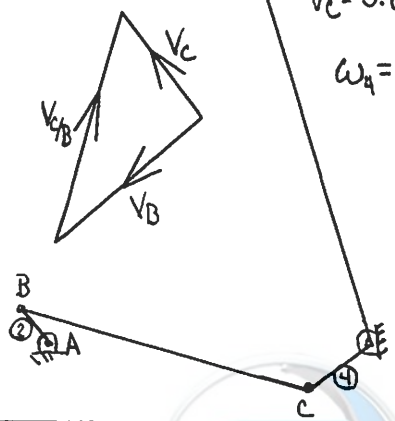
$V_C = 5.94 \frac{in}{s} \downarrow$

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6-25 $\omega_2 = 40\text{rpm}\left(\frac{\pi}{30}\right) = 4.189\frac{\text{rad}}{\text{s}}$, ccw

$V_B = 2''(4.189) = 8.378\frac{\text{in}}{\text{s}}$ $40°$ ↙

$V_C = V_B + V_{C/B}$



$V_C = 5.833\frac{\text{in}}{\text{s}}$ $53.19°$ ↑

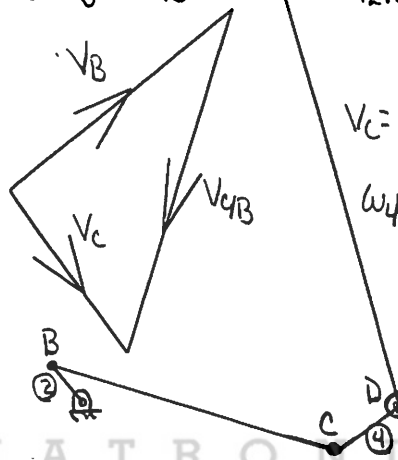
$\omega_4 = \frac{5.833}{3.5} = 1.667\frac{\text{rad}}{\text{s}}$, cw

6-26 $\omega_2 = 60\text{rpm}\left(\frac{\pi}{30}\right) = 6.283\frac{\text{rad}}{\text{s}}$, cw

$V_B = 2''(6.283)$

$= 12.566\frac{\text{in}}{\text{s}}$ ↗ $40°$

$V_C = V_B + V_{C/B}$



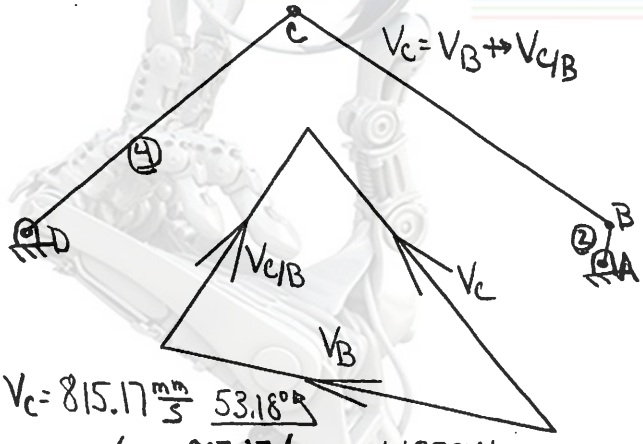
$V_C = 8.750\frac{\text{in}}{\text{s}}$ ↘ $53.19°$

$\omega_4 = \frac{8.750}{3.50} = 2.50\frac{\text{rad}}{\text{s}}$, ccw



6-27 $\omega_2 = 100\text{rpm}\left(\frac{\pi}{30}\right) = 10.472\frac{\text{rad}}{\text{s}}$, ccw

$V_B = 80^{\text{mm}}(10.472) = 837.76\frac{\text{mm}}{\text{s}}$ $15°$ ↖

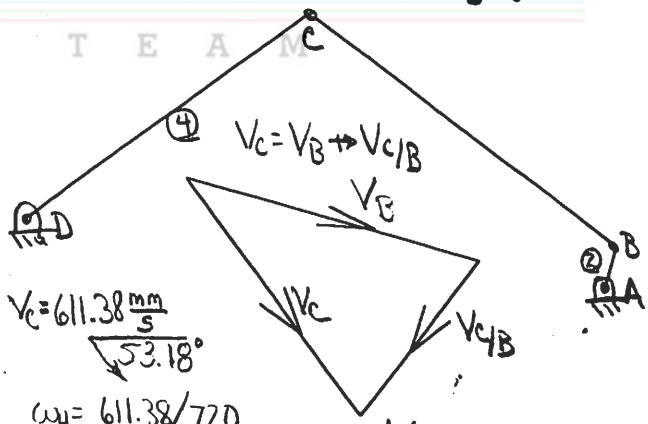


$V_C = 815.17\frac{\text{mm}}{\text{s}}$ $53.18°$ ↖

$\omega_4 = 817.17/720 = 1.132\text{rad/s}$, ccw

6-28 $\omega_2 = 75\text{rpm}\left(\frac{\pi}{30}\right) = 7.854\frac{\text{rad}}{\text{s}}$, cw

$V_B = 80^{\text{mm}}(7.854) = 628.32\frac{\text{mm}}{\text{s}}$ ↙ $75°$



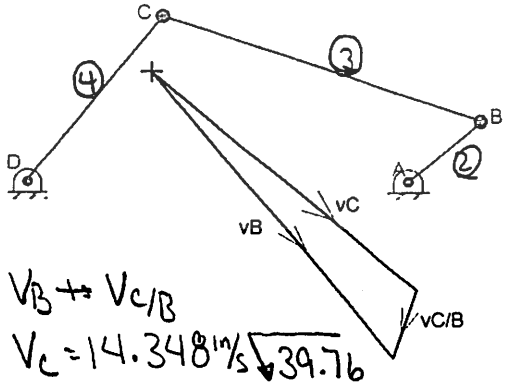
$V_C = 611.38\frac{\text{mm}}{\text{s}}$ ↘ $53.18°$

$\omega_4 = 611.38/720 = .850\text{ rad/s}$, cw

6-29

$\omega_2 = 50\text{rpm}(\pi/30) = 5.236\text{rad/s}$, cw

$V_B = 2''(5.236) = 15.708\text{ in/s}$ ↘ $50°$



$V_C = V_B + V_{C/B}$

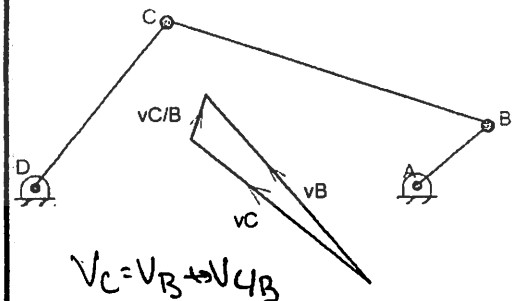
$V_C = 14.348\text{in/s}$ ↘ 39.76

$\omega_4 = 14.348/4.5 = 3.189\text{rad/s}$, cw

6-30

$\omega_2 = 35\text{rpm}(\pi/30) = 3.665\text{rad/s}$, ccw

$V_B = 2''(3.665) = 7.330\text{in/s}$ $50°$ ↖



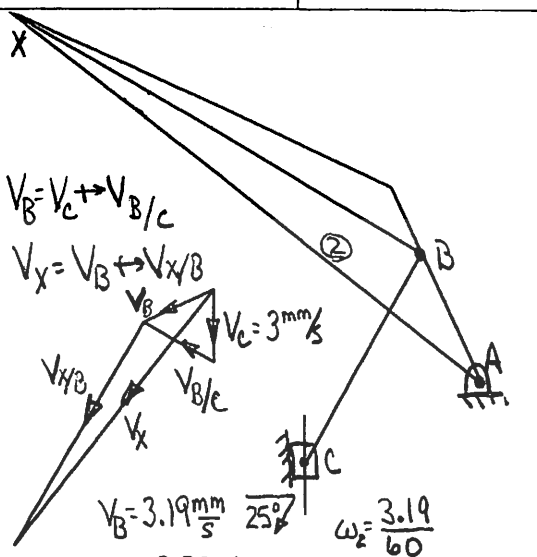
$V_C = V_B + V_{C/B}$

$V_C = 10.044\text{in/s}$ $39.76°$ ↖

$\omega_4 = 10.044/4.5 = 2.232\text{rad/s}$, ccw

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6-31



$V_B = 3.19\,\frac{mm}{s}$ $25°$

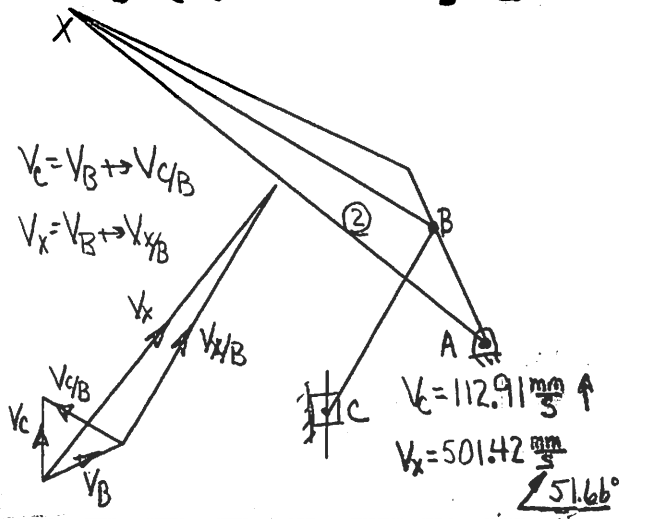
$V_X = 13.32\,\frac{mm}{s}$ $51.66°$

$\omega_2 = \frac{3.19}{60} = .053\,r/s, CCW$

6-32

$\omega_2 = 2\,rad/s, CW$

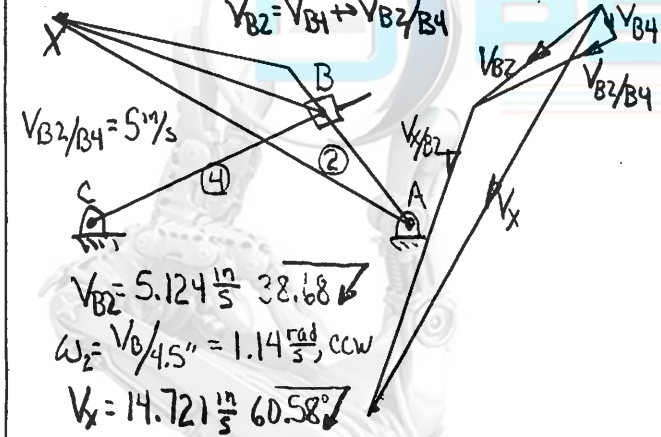
$V_B = (2\,r/s)(60\,mm) = 120\,\frac{mm}{s}$ $25°$



$V_C = 112.91\,\frac{mm}{s}$ ↑

$V_X = 501.42\,\frac{mm}{s}$ $51.66°$

6-33



$V_{B2} = 5.124\,\frac{in}{s}$ $38.68°$

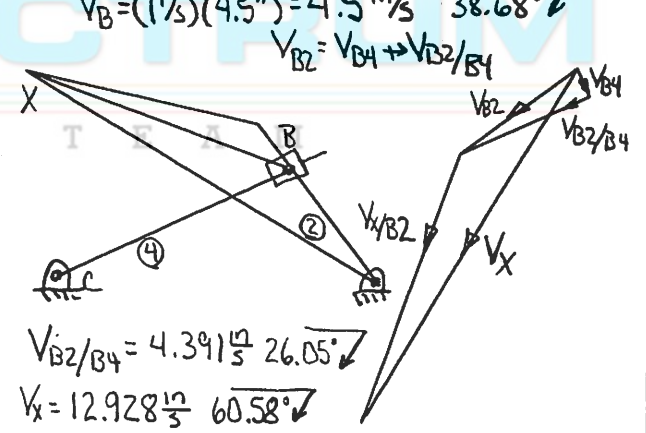
$\omega_2 = V_B/4.5'' = 1.14\,\frac{rad}{s}, CCW$

$V_X = 14.721\,\frac{in}{s}$ $60.58°$

6-34

$\omega_2 = 1\,r/s, CCW$

$V_B = (1\,r/s)(4.5'') = 4.5\,in/s$ $38.68°$

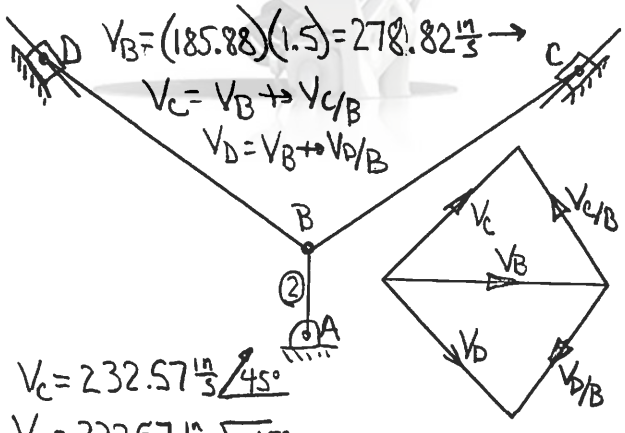


$V_{B2/B4} = 4.391\,\frac{in}{s}$ $26.05°$

$V_X = 12.928\,\frac{in}{s}$ $60.58°$

6-35

$\omega_2 = 1775\,rpm\left(\frac{\pi}{30}\right) = 185.88\,rad/s, CW$



$V_C = 232.57\,\frac{in}{s}$ $45°$

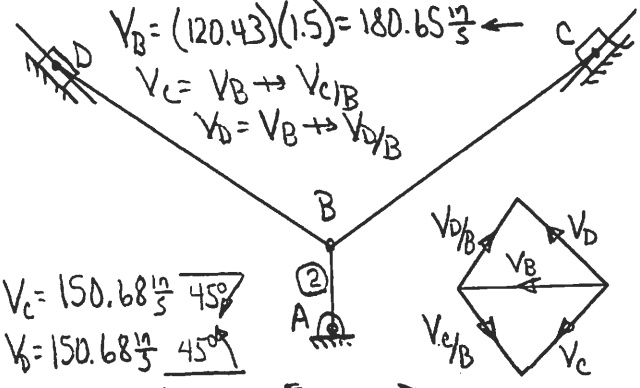
$V_D = 232.57\,\frac{in}{s}$ $45°$

Vol Flow $= (232.57\,\frac{in}{s})\left[\frac{\pi}{4}(1.0'')^2\right] = 182.66\,in^3/s$

6-36

$\omega_2 = 1150\,rpm\left(\frac{\pi}{30}\right) = 120.43\,\frac{rad}{s}, CCW$

$V_B = (120.43)(1.5) = 180.65\,\frac{in}{s} \leftarrow$



$V_C = 150.68\,\frac{in}{s}$ $45°$

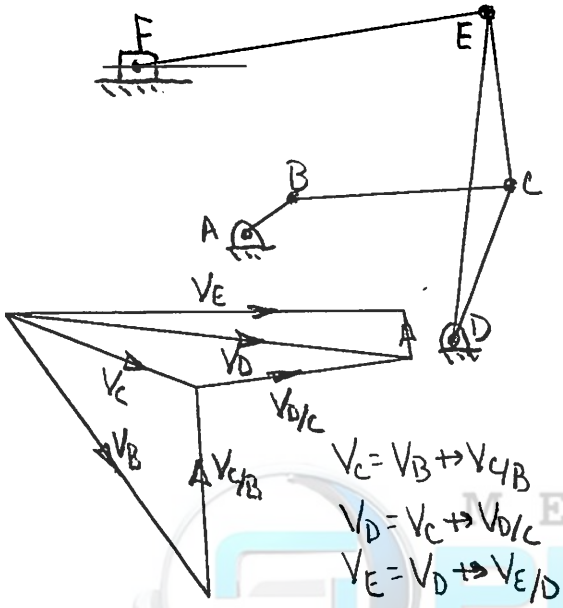
$V_D = 150.68\,\frac{in}{s}$ $45°$

Vol Flow $= (150.68\,\frac{in}{s})\left[\frac{\pi}{4}(1.0'')^2\right]$

$= 118.34\,in^3/s$

6-37 $\omega_2 = 40 \text{ rpm}\left(\frac{\pi}{30}\right) = 4.189 \frac{\text{rad}}{\text{s}}$, cw

$V_B = (4.189)(.1\text{m}) = .4189 \text{ m/s} \nwarrow 55°$



$V_C = V_B + V_{C/B}$

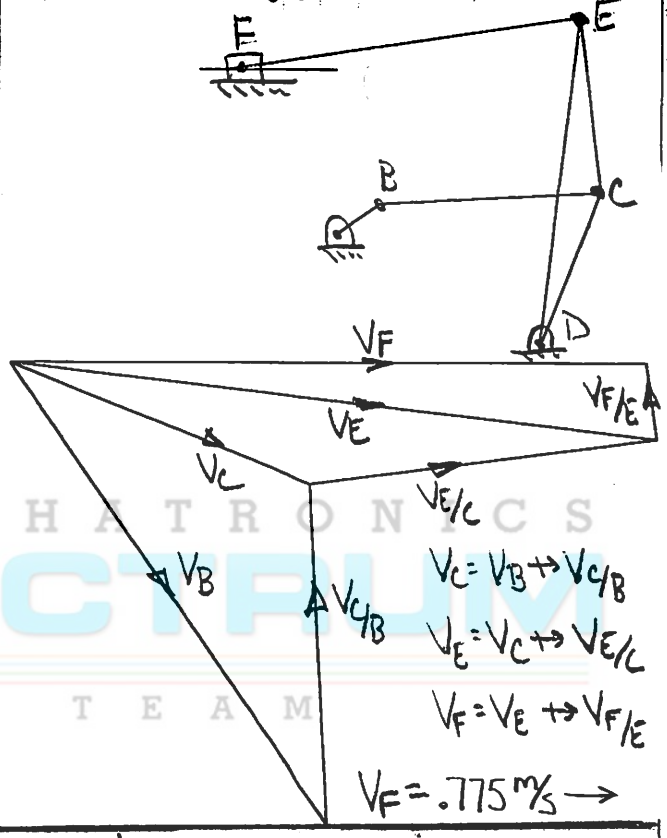
$V_D = V_C + V_{D/C}$

$V_E = V_D + V_{E/D}$

$V_E = .477 \frac{\text{m}}{\text{s}} \rightarrow$

6-38 $\omega_2 = 65 \text{ rpm}\left(\frac{\pi}{30}\right) = 6.807 \text{ r/s}$, cw

$V_B = (6.807 \frac{\text{rad}}{\text{s}})(.1\text{m}) = .6807 \frac{\text{m}}{\text{s}} \searrow 55°$



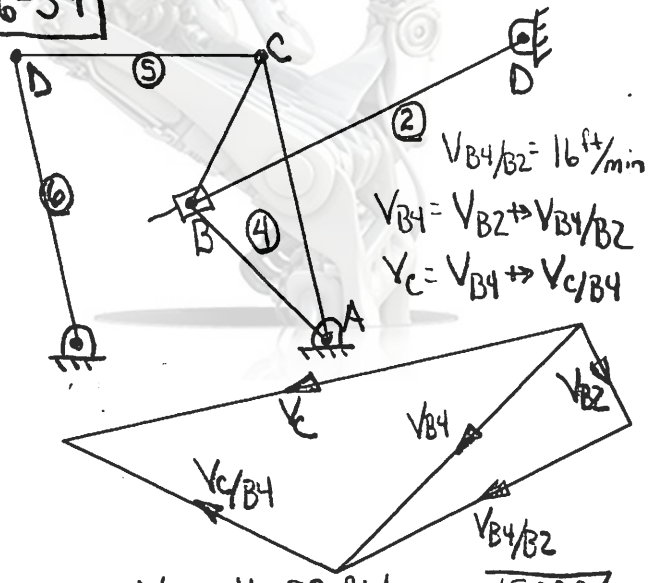
$V_C = V_B + V_{C/B}$

$V_E = V_C + V_{E/C}$

$V_F = V_E + V_{F/E}$

$V_F = .775 \text{ m/s} \rightarrow$

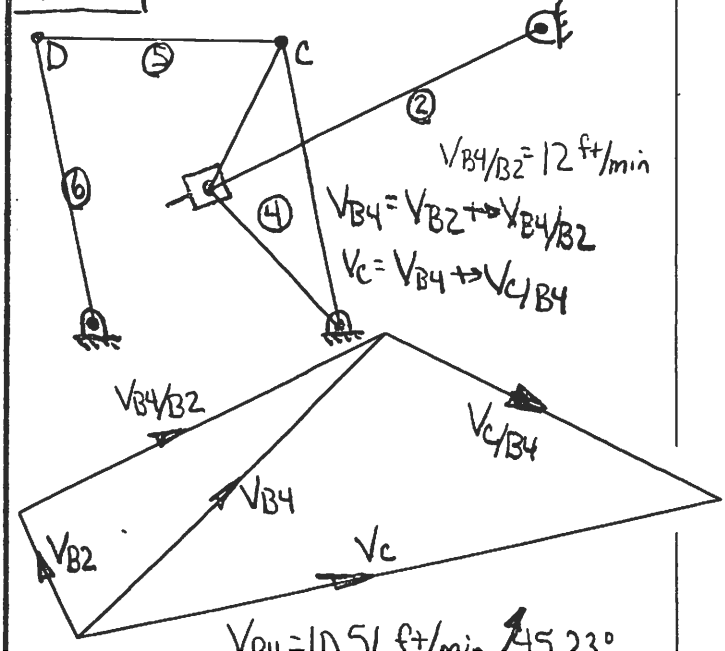
6-39



$V_{B4} = 16.89 \text{ ft/min}$ $45.23°$

$V_C = 25.58 \text{ ft/min}$ $12.88°$

6-40

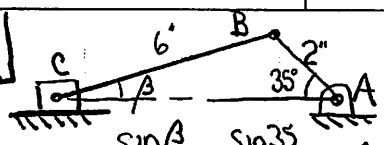


$V_{B4} = 10.56 \text{ ft/min}$ $45.23°$

$V_C = 15.99 \text{ ft/min}$ $12.88°$

MECHATRONICS SPECTRUM TEAM

6-41

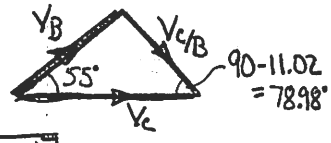


$$\frac{\sin\beta}{2} = \frac{\sin 35}{6} \Rightarrow \beta = 11.02°$$

$$\omega_2 = 950 \text{ rpm}\left(\frac{\pi}{30}\right) = 99.48 \text{ rad/s, CW}$$

$$V_B = (99.48)(2) = 198.97 \text{ in/s} \angle 55°$$

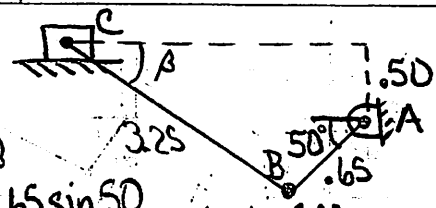
$$V_C = V_B + V_{C/B}$$



$$\frac{V_B}{\sin 78.98} = \frac{V_C}{\sin[180-(55+78.98)]}$$

$$\Rightarrow V_C = 145.86 \text{ in/s} \rightarrow$$

6-42

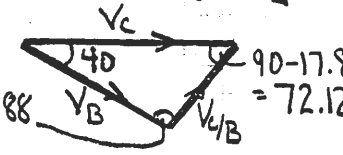


$$3.25\sin\beta = .50 + .65\sin 50 \Rightarrow \beta = 17.88°$$

$$\omega_2 = 1700 \text{ rpm}\left(\frac{\pi}{30}\right) = 178.02 \text{ rad/s, CCW}$$

$$V_B = (178.02)(.65) = 115.72 \frac{\text{in}}{\text{s}} \angle 40$$

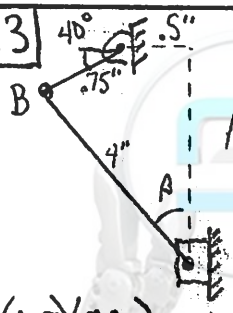
$$V_C = V_B + V_{C/B}$$



$$180-[40+72.12] = 67.88$$

$$\frac{V_B}{\sin 72.12} = \frac{V_C}{\sin 67.88} \Rightarrow V_C = 112.64 \frac{\text{in}}{\text{s}} \rightarrow$$

6-43

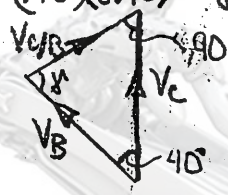


$$.5 + .75\cos 40 = 4\sin\beta$$

$$\beta = \sin^{-1}\left[\frac{.5+.75\cos 40}{4}\right] = 15.58°$$

$$\omega_2 = 85 \text{ rpm}\left(\frac{\pi}{30}\right) = 8.90 \text{ r/s, CW}$$

$$V_B = (.75)(8.90) = 6.68 \text{ in/s} \angle 50°$$



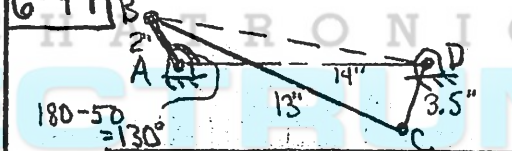
$$90-\beta = 74.42$$

$$\gamma = 180-[40+74.42] = 65.58°$$

$$V_C/\sin 65.58 = V_B/\sin 74.42$$

$$\Rightarrow V_C = 6.31 \text{ in/s} \uparrow$$

6-44



$$BD = \sqrt{(2)^2+(14)^2-2(2)(14)\cos 130} = 15.36"$$

$$\frac{\sin\frac{ADB}{2}}{2} = \frac{\sin 130}{15.36} \Rightarrow ADB = 5.72°$$

$$ABD = 180-[130+5.72] = 44.28°$$

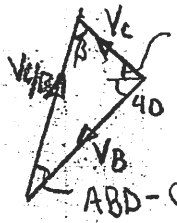
$$BDC = \cos^{-1}\left[\frac{15.36^2+3.5^2-13^2}{2(15.36)(3.5)}\right] = 42.57°$$

$$BCD = \cos^{-1}\left[\frac{13^2+3.5^2-15.36^2}{2(13)(3.5)}\right] = 126.93°$$

$$CBD = 180-[42.57+126.93] = 10.5°$$

$$\omega_2 = 45 \text{ rpm}\left(\frac{\pi}{30}\right) = 4.71 \text{ rad/s, CCW}$$

$$V_B = (4.71)(2") = 9.42 \text{ in/s} \angle 40°$$



$$90-[BDC-ADB] = 53.15°$$

$$\beta = 180-[53.15+40+33.78] = 53.07°$$

$$ABD - CBD = 33.78$$

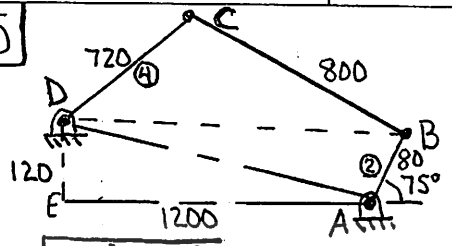
$$\frac{V_C}{\sin(33.78)} = \frac{V_B}{\sin(53.07)}$$

$$\Rightarrow V_C = 6.55 \text{ in/s} \angle 53.15°$$

$$\omega_4 = 6.55/3.5 = 1.88 \text{ rad/s, CW}$$

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6-45



$AD = \sqrt{1200^2 + 120^2} = 1206.00\text{ mm}$

$\angle DAE = \tan^{-1}(120/1200) = 5.71°$

$\angle ADE = 90 - \angle DAE = 84.29°$

$\angle BAD = 180 - (75 + \angle DAE) = 99.29°$

$BD = \sqrt{1206^2 + 80^2 - 2(1206)(80)\cos 99.29}$

$= 1221.47\text{ mm}$

$\frac{\sin\angle ABD}{1206.00} = \frac{\sin 99.29}{1221.47} \Rightarrow \angle ABD = 77.00°$

$\angle ADB = 180 - [\angle BAD + \angle ABD] = 3.71°$

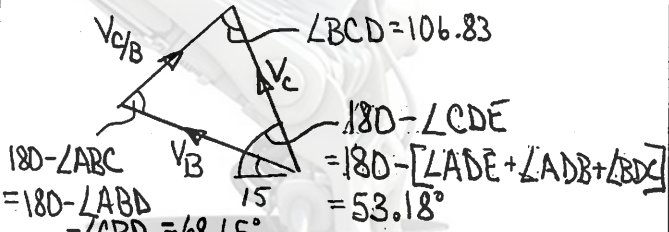
$\angle CBD = \cos^{-1}\left[\frac{800^2 + 1221.47^2 - 720^2}{2(800)(1221.47)}\right] = 34.35°$

$\angle BDC = \cos^{-1}\left[\frac{720^2 + 1221.47^2 - 800^2}{2(720)(1221.47)}\right] = 38.82°$

$\angle BCD = 180 - [\angle CBD + \angle BDC] = 106.83$

$\omega_2 = 90\text{ rpm}\left(\frac{\pi}{30}\right) = 9.42\text{ rad/s, ccw}$

$V_B = (9.42)(80) = 754.0\text{ mm/s}\ \angle 15°$

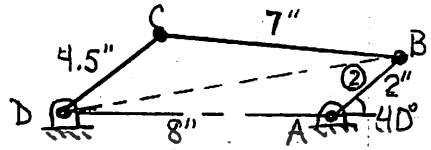


$\frac{V_C}{\sin 68.65} = \frac{754.0}{\sin 106.83} \Rightarrow V_C = 733.68\text{ mm/s}\ \angle 53.18°$

$\omega_4 = 733.68/720$

$= 1.02\text{ rad/s, ccw}$

6-46



$BD = \sqrt{8^2 + 2^2 - 2(8)(2)\cos 140} = 9.618"$

$\frac{\sin\angle ADB}{2} = \frac{\sin 140}{9.618} \Rightarrow \angle ADB = 7.68°$

$\angle ABD = 180 - [140 + \angle ADB] = 32.32°$

$\angle CBD = \cos^{-1}\left[\frac{6^2 + 9.658^2 - 5^2}{2(6)(9.658)}\right] = 25.77°$

$\angle BCD = \cos^{-1}\left[\frac{7^2 + 4.5^2 - 9.618^2}{2(7)(4.5)}\right] = 111.66°$

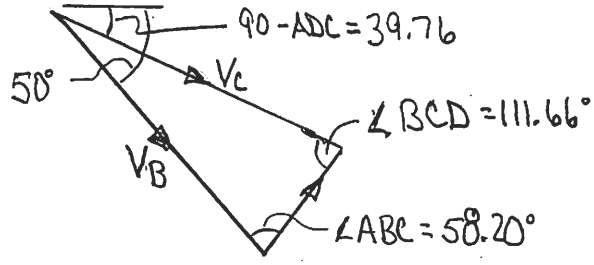
$\angle BDC = 180 - [\angle CBD + \angle BCD] = 42.56°$

$\angle ABC = \angle ABD + \angle CBD = 58.20°$

$\angle ADC = \angle ADB + \angle BDC = 50.24$

$\omega_2 = 60\text{ rpm}\left(\frac{\pi}{30}\right) = 6.283\text{ rad/s, cw}$

$V_B = (6.283)(2) = 12.56\text{ in/s}\ \angle 50°$



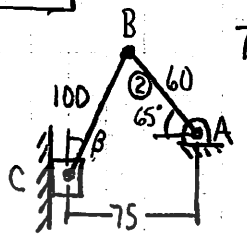
$\frac{V_C}{\sin 58.20} = \frac{12.56}{\sin 111.66}$

$\Rightarrow V_C = 11.49\text{ in/s}\ \angle 39.76°$

$\omega_4 = \frac{11.49}{4.5} = 2.55\text{ rad/s, cw}$

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6-47

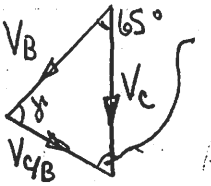


$75 = 60\cos 65 + 100\sin\beta$

$\beta = \sin^{-1}\left[\frac{75 - 60\cos 65}{100}\right]$

$= 29.76°$

$V_C = V_B + V_{C/B}$ $\quad V_C = 2\ mm/s \downarrow$



$90 - \beta = 60.24°$

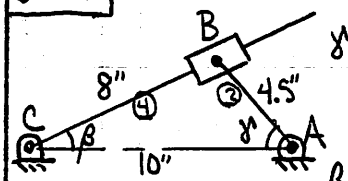
$\gamma = 180 - [65 + 60.24]$

$= 54.76$

$\frac{V_B}{\sin 60.24} = \frac{V_C}{\sin 54.76} \Rightarrow V_B = 2.13\ \frac{mm}{s}$ $25°$

$\omega_2 = \frac{2.13}{60} = .035\ rad/s$, CCW

6-48



$\gamma = \cos^{-1}\left[\frac{4.5^2 + 10^2 - 8^2}{2(4.5)(10)}\right]$

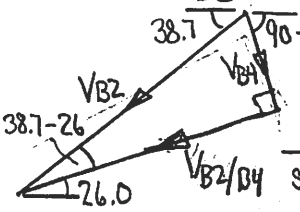
$= 51.3°$

$\beta = \cos^{-1}\left[\frac{10^2 + 8^2 - 4.5^2}{2(10)(8)}\right]$

$= 26.0°$

$V_{B2} = V_{B4} + V_{B2/B4}$

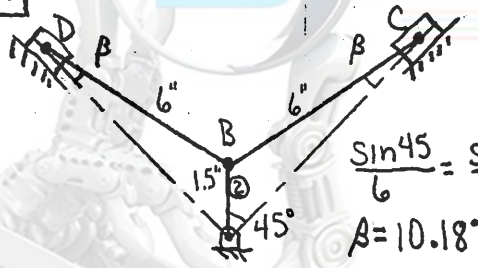
$V_{B2} = 1\ rad/s\,(4.5) = 4.5\ in/s$ $\quad 90 - 51.3 = 38.7°$



$\frac{V_{B2/B4}}{\sin[180 - (38.7 + 64)]} = \frac{V_{B2}}{\sin 90}$

$V_{B4/B2} = 4.39\ in/s$ $\quad 26.0°$

6-49



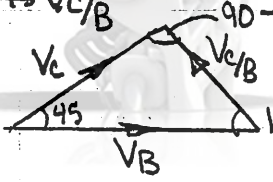
$\frac{\sin 45}{6} = \frac{\sin\beta}{1.5}$

$\beta = 10.18°$

$\omega_2 = 2000\ rpm\left(\frac{\pi}{30}\right) = 209.44\ rad/s$, CW

$V_B = (209.44)(1.5) = 314.16\ in/s \rightarrow$

$V_C = V_B + V_{C/B}$



$90 - \beta = 79.82°$

$180 - [45 + 79.82] = 55.18°$

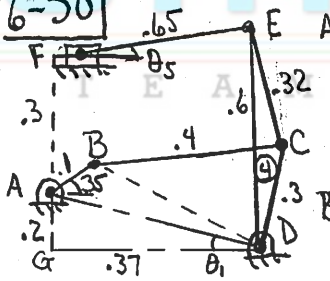
$\frac{V_B}{\sin 79.82} = \frac{V_C}{\sin 55.18}$

$V_C = 262.0\ in/s$ $\quad 45°$

$V_D = 262.0\ in/s$ $\quad 45°$

Vol Flow $= (262\ in/s)\left[\frac{\pi}{4}(1.0")^2\right] = 205.81\ in^3/s$

6-50



$AD = \sqrt{.2^2 + .37^2} = .421\ m$

$\theta_1 = \tan^{-1}\left(\frac{.2}{.37}\right) = 28.39°$

$\angle BAD = 35 + \theta_1 = 63.39°$

$BD = \sqrt{.421^2 + .1^2 - 2(.421)(.1)\cos 63.39}$

$= .387\ m$

$\frac{\sin\angle ADB}{.1} = \frac{\sin 63.39}{.387} \Rightarrow \angle ADB = 13.36°$

$\angle ABD = 180 - [\angle BAD + \angle ADB] = 103.25°$

$\angle CBD = \cos^{-1}\left[\frac{.4^2 + .387^2 - .3^2}{2(.4)(.387)}\right] = 44.77°$

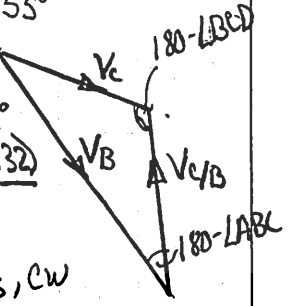
$\angle BDC = \cos^{-1}\left[\frac{.3^2 + .387^2 - .4^2}{2(.3)(.387)}\right] = 69.91°$

$\angle BCD = 180 - [\angle CBD + \angle BDC] = 65.32°$

$\omega_2 = 80\ rpm\,(\pi/30) = 8.377\ rad/s$, CW

$V_B = (8.377)(.1) = .8377\ m/s$ $\quad 55°$

$V_C = V_B + V_{C/B}$



$\angle ABC = \angle ABD + \angle CBD = 148.02°$

$\frac{\sin(180 - 148.02)}{V_C} = \frac{\sin(180 - 65.32)}{.8377}$

$V_C = .4883\ m/s$

$\omega_4 = (.4883)/.3 = 1.627\ rad/s$, CW

$V_E = (1.627)(.6) = .9765\ m/s$

$\angle CDE = \cos^{-1}\left[\frac{.3^2 + .6^2 - .32^2}{2(.3)(.6)}\right] = 15.08°$

$\angle EDG = \theta_1 + \angle ADB + \angle BDC - \angle CDE = 96.58°$

$V_E = .9765\ m/s$ $\quad 6.58°$

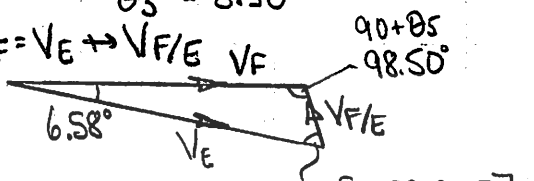
MECHATRONICS SPECTRUM TEAM

6-50 (con't)

$.5 + .65\sin\theta_5 = .6\sin 96.58$

$\theta_5 = 8.50°$

$V_F = V_E + V_{F/E}$

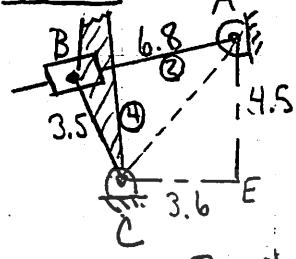


$180 - [6.58 + 98.50] = 74.92°$

$$\frac{V_F}{\sin 74.92} = \frac{.9765}{\sin 98.5}$$

$\Rightarrow V_F = .9533\ m/s \rightarrow$

6-51



$AC = \sqrt{4.5^2 + 3.6^2} = 5.76$

$\angle ACE = \tan^{-1}(4.5/3.6) = 51.34°$

$\angle CAE = 90 - \angle ACE = 38.66°$

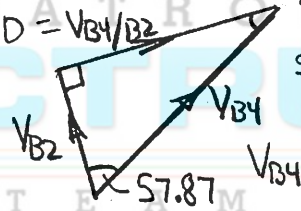
$\angle ACB = \cos^{-1}\left[\frac{3.5^2 + 5.76^2 - 6.8^2}{2(3.5)(5.76)}\right] = 91.15°$

$\angle ABC = \cos^{-1}\left[\frac{3.5^2 + 6.8^2 - 5.76^2}{2(3.5)(6.8)}\right] = 57.87°$

$\angle BAC = 180 - [\angle ACB + \angle ABC] = 30.98°$

$\angle BAE = \angle CAE + \angle BAC = 69.64$

$V_{B4} = V_{B2} + V_{B4/B2}$

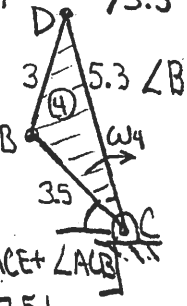


$\sin 57.87 = \frac{10}{V_{B4}}$

$V_{B4} = 11.81\ \frac{ft}{min}$ $\angle 52.49$

$\angle ACE + \angle ACB - 90$

$\omega_4 = 11.81/3.5 = 3.37\ rad/min$, CW



$\angle BCD = \cos^{-1}\left[\frac{3.5^2 + 5.3^2 - 3^2}{2(3.5)(5.3)}\right] = 32.36°$

$180 - [\angle ACE + \angle ACB] = 37.51$

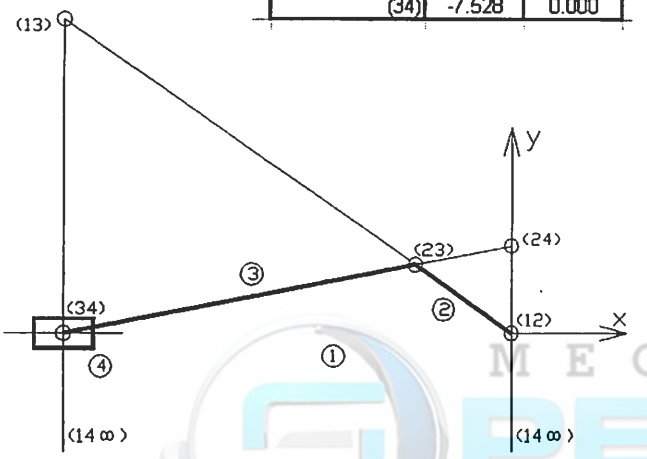
$V_D = (3.37)(5.3) = 17.86\ ft/min$ $\angle 20.13°$

$90 - [37.51 + 32.36]$

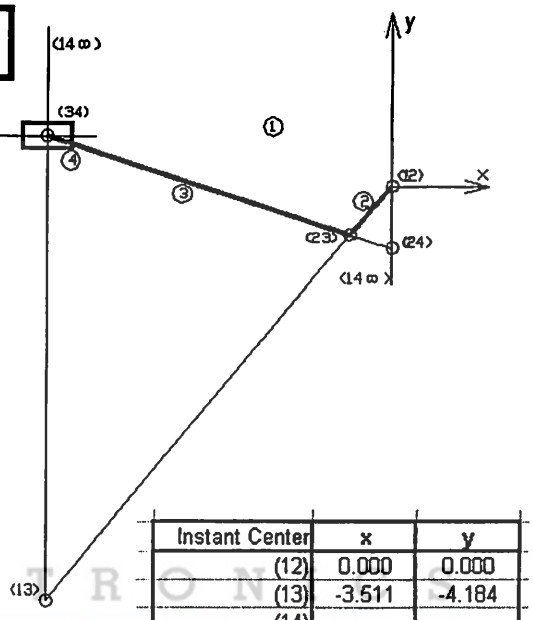
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6-52.

Instant Center	x	y
(12)	0.000	0.000
(13)	-7.528	5.271
(14)		
(23)	-1.638	1.147
(24)	0.000	1.466
(34)	-7.528	0.000



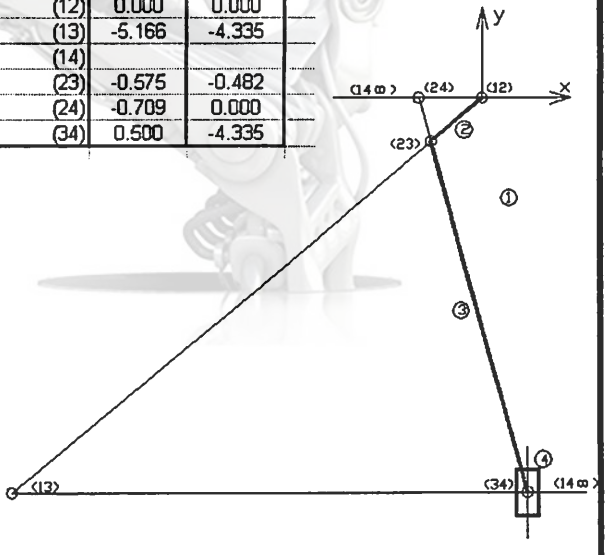
6-53.



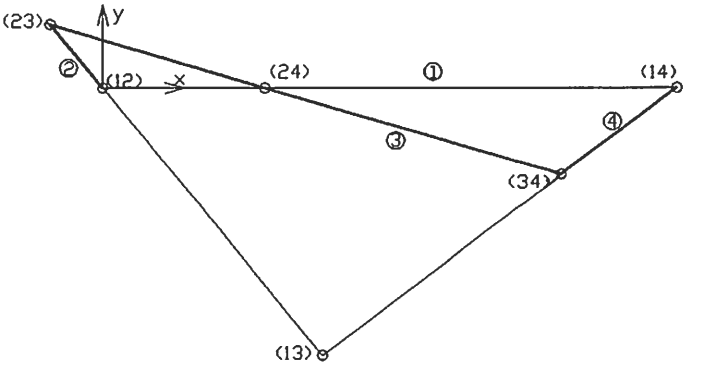
Instant Center	x	y
(12)	0.000	0.000
(13)	-3.511	-4.184
(14)		
(23)	-0.418	-0.498
(24)	0.000	-0.633
(34)	-3.511	0.500

6-54.

Instant Center	x	y
(12)	0.000	0.000
(13)	-5.166	-4.335
(14)		
(23)	-0.575	-0.482
(24)	-0.709	0.000
(34)	0.500	-4.335



6-55.

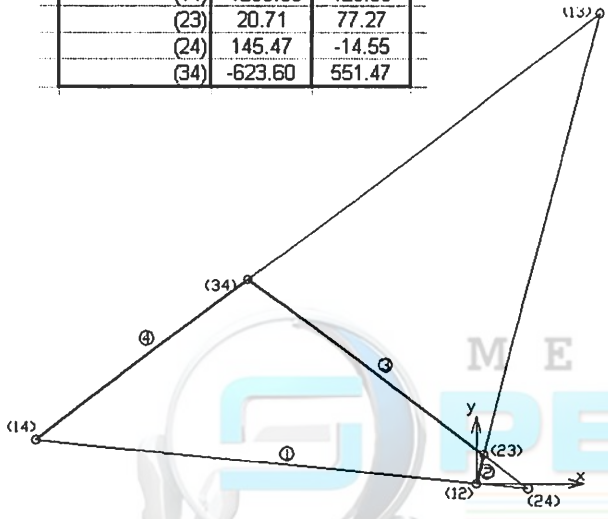


Instant Center	x	y
(12)	0.000	0.000
(13)	5.400	-6.435
(14)	14.000	0.000
(23)	-1.286	1.532
(24)	3.985	0.000
(34)	11.198	-2.097

MECHATRONICS SPECTRUM TEAM

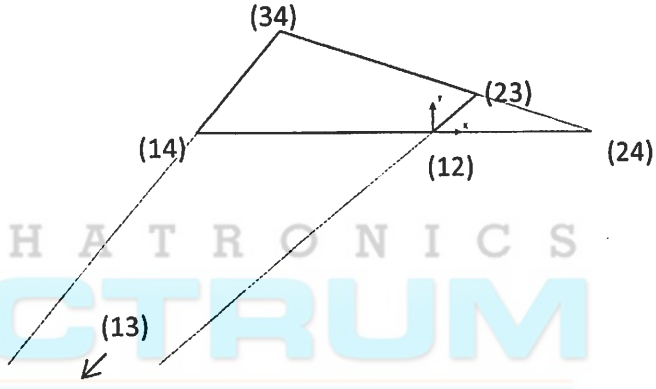
6-56.

Instant Center	x	y
(12)	0.00	0.00
(13)	341.31	1273.78
(14)	-1200.00	120.00
(23)	20.71	77.27
(24)	145.47	-14.55
(34)	-623.60	551.47



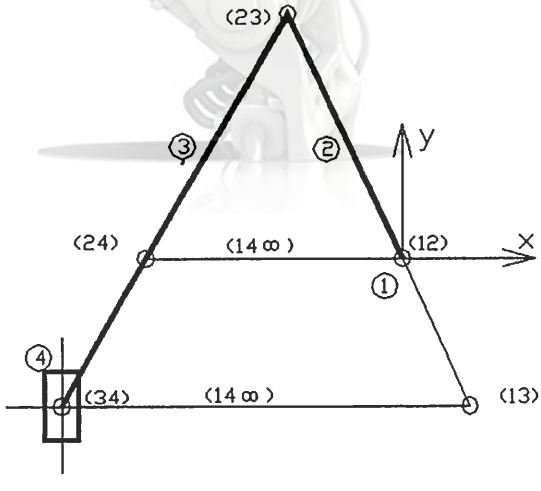
6-57.

Instant Center	X	Y
(12)	0.000	0.000
(13)	-26.5024	-22.2381
(14)	-8.000	0.000
(23)	1.5321	1.2856
(24)	5.4674	0.0000
(34)	-5.1219	3.4592



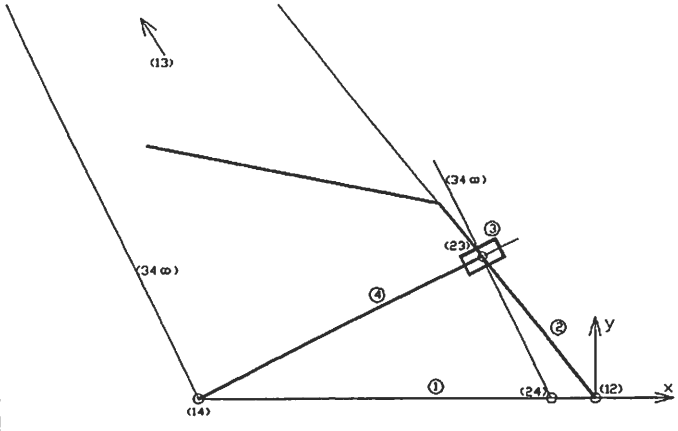
6-58.

Instant Center	x	y
(12)	0.00	0.00
(13)	15.12	-32.43
(14)		
(23)	-25.36	54.38
(24)	-56.45	0.00
(34)	-75.00	-32.43



6-59.

Instant Center	x	y
(12)	0.000	0.000
(13)	-25.670	32.062
(14)	-10.000	0.000
(23)	-2.813	3.513
(24)	-1.096	0.000
(34)		



MECHATRONICS SPECTRUM TEAM

Answers to the Chapter 6 Case Study Questions:

Case 6-1.

1. As gear A is forced to rotate counterclockwise, mating gear B will rotate clockwise. Mating, external, gears will always rotate in opposite directions.
2. As gear A rotates counterclockwise, stud pin C rotates clockwise.
3. As gear A rotates counterclockwise, lever D, which pivots about its right end, oscillates rotationally.
4. Link D drives link E, in a rocker-rocker configuration. Since link E is shorter than link D, link E will have a larger throw angle and link D.
5. Link D will be placed in its lowest extreme position when the center of the slot is tangent to the bottom of the path of pin C.
6. Link D will be placed in its upper extreme position when the center of the slot is tangent to the top of the path of pin C.
7. It takes approximately 210 degrees of gear B to rotate link D (and E) upward and about 150 degrees to return downward.
8. With the imbalance observed in 7, it takes longer to rotate link E up than it takes to lower link E.
9. The motion of link E is "quick-return."

Case 6-2.

1. As wheel C is forced to rotate counterclockwise, pin D, which is attached to wheel C, will rotate counterclockwise.
2. As wheel C is forced to rotate counterclockwise, link G, which pivots at its middle, oscillates rotationally.
3. The left portion of link G will be placed in its upper extreme position when the center of the slot is tangent to the bottom of the path of pin D.
4. The left portion of link D will be placed in its lowest extreme position when the center of the slot is tangent to the top of the path of pin D.
5. It takes approximately 200 degrees of wheel C to rotate link G counterclockwise and about 160 degrees to return.
6. It takes approximately 25% more time to rotate link G counterclockwise than it takes to rotate link G clockwise.
7. The cyclical motion of link N is rotational oscillation. It exhibits relatively slow counterclockwise rotation, and a relatively quick clockwise rotation.
8. The cyclical motion of table R is translational oscillation. It exhibits relatively slow movement to the left, and relatively quick movement to the right.
9. The mechanism creates quick return motion, moving the workpiece quickly through a grinding pass.
10. The screw threads can be used to adjust the position of the translational oscillation of the table.
11. The mechanism has eight links, eight pin joints and two sliding joints, giving one degree of freedom.

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Case 6-3.

1. Since link E is riding in slot J, disk F is driven counterclockwise.
2. At the instant shown, link E and disk F are rotating counterclockwise. This motion is pulling strap G towards the left.
3. At the instant shown, link E and disk F are rotating counterclockwise. This motion is pulling strap G and slide A towards the left.
4. As link E contacts the ramped pad M, it disengages from disk F. The compressed spring N is allowed to expand.
5. As link E contacts the ramped pad M, it continues to rotate counterclockwise, but is raised off of disk F.
6. As link E contacts the ramped pad M, it disengages from disk F. The energy stored in the compressed spring N drives disk F quickly clockwise.
7. As link E contacts the ramped pad M, it disengages from disk F. The energy stored in the compressed spring N drives slide A quickly to the right.
8. As link E continues to rotate, it falls off of ramped pad M but slides on top of disk F. Disk F remains motionless once the energy in spring N is released.
9. As link E catches a second slot at K, it again engages disk F. Disk F is again driven counterclockwise.
10. The continual motion of slide A is translational oscillation. It exhibits relatively slow motion to the left, then rapid motion to the right, followed by a short pause before the cycle is repeated.

7-1

$$a = \frac{\Delta v}{\Delta t} = \frac{45\ ft/min}{.5\ sec}\left(\frac{1\ min}{60\ sec}\right) = 1.5\ \frac{ft}{s^2} \rightarrow$$

7-2 $v_1 = 0 \quad v_2 = 60\ \frac{mi}{hr}\left(\frac{1\ hr}{3600\ s}\right)\left(\frac{5280\ ft}{1\ mi}\right)$

$= 88\ ft/s$

$$a = \frac{\Delta v}{\Delta t} = \frac{88\ ft/s}{8.3\ s} = 10.60\ ft/s^2$$

7-3 $v_f^2 = v_i^2 + 2a\Delta R$

$(0)^2 = (12\ ft/s)^2 + 2(-10\ ft/s^2)\Delta R$

$\Delta R = 7.2\ ft$

7-4

$$a = \frac{\Delta v}{\Delta t} = \frac{(300-100)\ mm/s}{.2\ s} = 1000\ mm/s \uparrow$$

$\Delta R = \frac{1}{2}at^2 + v_i t = \frac{1}{2}(1000)(.2)^2 + 100(.2)$

$= 40\ mm \uparrow$

7-5

$$a = \frac{\Delta v}{\Delta t} = \frac{.75\ m/s}{1.5\ s} = .50\ m/s^2 \leftarrow$$

$\Delta R = \frac{1}{2}(.50\ m/s^2)(1.5s)^2 + (0)(1.5) = .563\ m \leftarrow$

7-6 $\omega_1 = 0 \quad \omega_2 = 750\ rpm\left(\frac{\pi}{30}\right)$

$= 78.54\ rad/s$

$$\alpha = \frac{\Delta \omega}{\Delta t} = \frac{78.54\ r/s}{8\ s} = 9.82\ rad/s^2,\ CW$$

7-7 $\omega_1 = 10{,}000\ rpm\left(\frac{\pi}{30}\right) = 1047.2\ r/s$

$$\alpha = \frac{0 - 1047.2\ rad/s}{2\ min\left(\frac{60s}{1\ min}\right)} = -8.73\ rad/s^2$$

$= 8.73\ rad/s^2, CCW$

$\Delta\theta = \frac{1}{2}(-8.73)(120s)^2 + 1047.2(120s)$

$= 62784\ rad\left(\frac{1\ rev}{2\pi\ rad}\right) = 9992.4\ rev, CW$

7-8 $\omega_1 = 1000\ rpm\left(\frac{\pi}{30}\right) = 104.72\ r/s$

$\omega_2 = 2500\ rpm\left(\frac{\pi}{30}\right) = 261.80\ r/s$

$$\alpha = \frac{261.8 - 104.72}{20} = 7.85\ rad/s^2,\ CW$$

7-9 $\omega_1 = 1100\ rpm\left(\frac{\pi}{30}\right) = 115.19\ rad/s$

$\omega_2 = 800\ rpm\left(\frac{\pi}{30}\right) = 83.78\ rad/s$

$\Delta\theta = 400\ rev\left(\frac{2\pi\ rad}{1\ rev}\right) = 2513.27\ rad$

$\omega_2^2 = \omega_1^2 + 2\alpha\Delta\theta$

$(83.78)^2 = (115.19)^2 + 2\alpha(2513.27)$

$\alpha = -1.243\ rad/s^2 = 1.243\ \frac{rad}{s^2},\ CW$

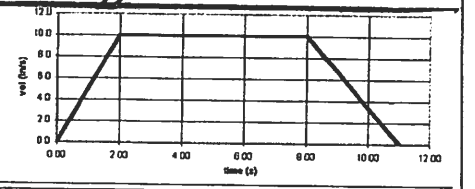
7-10

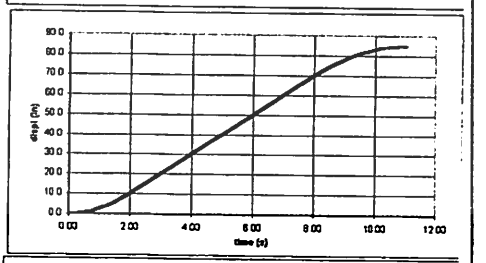
$$a_1 = \frac{\Delta v}{\Delta t} = \frac{10\ in/s}{2s} = 5\ in/s^2$$

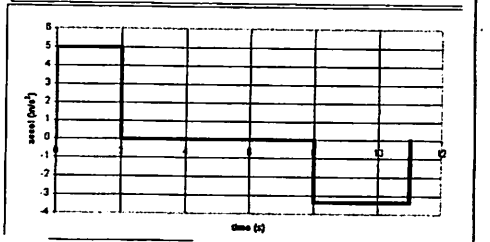
$$a_2 = \frac{0-10}{3s} = -3.33\ in/s^2$$

$\Delta R = \int v\,dt = \frac{1}{2}(10)(2) + 10(8-2) + \frac{1}{2}(10)(11-8)$

$= 85\ in$







7-11

Stage 1: $0 \le t \le 2$

$a = 5 \quad v = 5t \quad s = 2.5t^2$

Stage 2: $2 \le t \le 8$ at $t = 2$

$a = 0 \quad v = 10 \quad s = 10t + c$ $\quad s = 10,\ c = -10$

Stage 3: $8 \le t \le 11$

$a = -3.33, \quad v = -3.33t + k_1$ { at $t = 8$, $v = 10$, $k_1 = 36.67$

$s = -1.666t^2 + 36.67t + k_2$

at $t = 8,\ s = 70 \Rightarrow k_2 = -116.70$

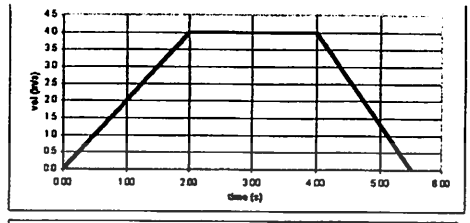
t (s)	a (in/s2)	v (in/s)	s (in)
0	0	0	0
0.00	5.0	0.0	0.0
0.25	5.0	1.3	0.2
0.50	5.0	2.5	0.6
0.75	5.0	3.8	1.4
1.00	5.0	5.0	2.5
1.25	5.0	6.3	3.9
1.50	5.0	7.5	5.6
1.75	5.0	8.8	7.7
2.00	5.0	10.0	10.0
2.00	0.0	10.0	10.0
8.00	0.0	10.0	70.0
8.00	-3.33	10.0	70.0
8.25	-3.33	9.2	72.4
8.50	-3.33	8.4	74.6
8.75	-3.33	7.5	76.6
9.00	-3.33	6.7	78.3
9.25	-3.33	5.9	79.9
9.50	-3.33	5.0	81.3
9.75	-3.33	4.2	82.4
10.00	-3.33	3.4	83.3
10.25	-3.33	2.5	84.1
10.50	-3.33	1.7	84.6
10.75	-3.33	0.9	84.9
11.00	-3.33	0.0	85.0
11.00	0.00	0.0	85.0

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7-12 $a_1 = \frac{\Delta v}{\Delta t} = \frac{4\ in/s}{2\ s} = 2\ in/s^2$

$a_2 = \frac{0 - 4\ in/s}{1.5\ s} = -2.667\ in/s^2$

$\Delta R = \int v\,dt = \frac{1}{2}(2)(2) + (4)(4-2) + \frac{1}{2}(4)(1.5) = 15\ in$



7-13 Stage 1: $0 \le t < 2$

$a = 2\ in/s^2$ $v = 2t$ $\Delta R = t^2$

Stage 2: $2 \le t < 4$

$a = 0$ $v = 4$ $\Delta R = 4t + C$

at $t = 2$, $\Delta R = 4 \Rightarrow C = -4$

Stage 3: $4 \le t \le 5.5$

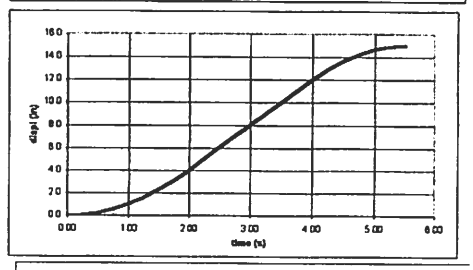
$a = -2.667$ $v = -2.667t + k_1$

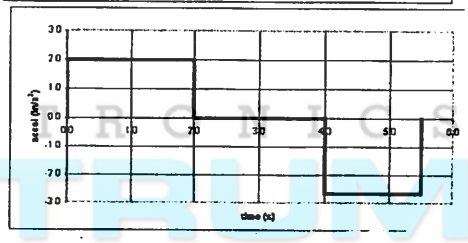
at $t = 4$, $v = 4 \Rightarrow k_1 = 14.667$

$\Delta R = -1.333t^2 + 14.667t + k_2$

at $t = 4$, $\Delta R = 12 \Rightarrow k_2 = -25.33$

t (s)	a (in/s2)	v (in/s)	s (in)
0.0	0.0	0.0	0.0
0.00	2.0	0.0	0.0
0.25	2.0	0.5	0.1
0.50	2.0	1.0	0.3
0.75	2.0	1.5	0.6
1.00	2.0	2.0	1.0
1.25	2.0	2.5	1.6
1.50	2.0	3.0	2.3
1.75	2.0	3.5	3.1
2.00	2.0	4.0	4.0
2.00	0.0	4.0	4.0
4.00	0.0	4.0	12.0
4.00	-2.67	4.0	12.0
4.25	-2.67	3.3	12.9
4.50	-2.67	2.7	13.7
4.75	-2.67	2.0	14.3
5.00	-2.67	1.3	14.7
5.25	-2.67	0.7	14.9
5.50	-2.67	0.0	15.0
5.50	0.00	0.0	15.0





7-14 Stage 1:

$a_1 = 6\ in/s^2$, $v_1 = 4\ in/s$ $a_1 = \frac{\Delta v_1}{\Delta t_1} \Rightarrow \Delta t_1 = \frac{\Delta v_1}{a_1} = \frac{4}{6} = .667s$

$\Delta R_1 = \frac{1}{2}(4\ in/s)(.667s) = 1.333\ in$

Stage 3:

$a_3 = -6\ in/s$, $\Delta v_3 = -4\ in/s$, $\Delta t_3 = \frac{-4}{-6} = .667s$

$\Delta R_3 = 1.333\ in$

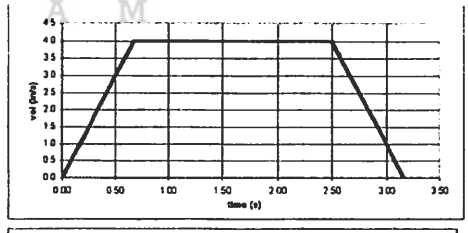
Stage 2: $10\ in = \Delta R_1 + \Delta R_2 + \Delta R_3$

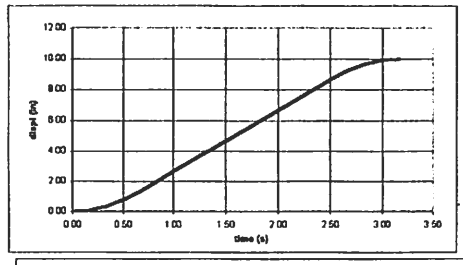
$\Rightarrow \Delta R_2 = 7.333\ in$

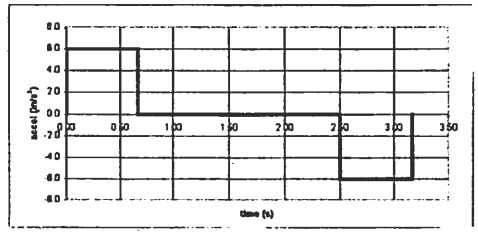
$V_2 = 4\ in/s = \frac{\Delta R_2}{\Delta t_2} \Rightarrow \Delta t_2 = \frac{\Delta R_2}{V_2} = \frac{7.333\ in}{4\ in}$

$= 1.833s$

t (s)	a (in/s2)	v (in/s)	ΔR (in)
0.00	0.0	0.0	0.00
0.00	6.0	0.0	0.00
0.17	6.0	1.0	0.08
0.33	6.0	2.0	0.33
0.50	6.0	3.0	0.75
0.67	6.0	4.0	1.33
0.67	0.0	4.0	1.33
2.50	0.0	4.0	8.66
2.50	-6.0	4.0	8.66
2.67	-6.0	3.0	9.25
2.83	-6.0	2.0	9.66
3.00	-6.0	1.0	9.91
3.17	-6.0	0.0	10.00
3.17	0.00	0.0	10.00







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7-15

Stage 1:

$a_1 = 100\ mm/s^2$, $V_1 = 50\ mm/s$ $\quad a_1 = \frac{\Delta V_1}{\Delta t_1} \Rightarrow \Delta t_1 = \frac{\Delta V_1}{a_1} = \frac{50}{100} = .5s$

$\Delta R_1 = \frac{1}{2}(100)(.5)^2 = 12.5\ mm$

Stage 3:

$a_3 = -100\ mm/s^2$, $V_3 = -50\ mm/s$, $\Delta t_3 = \frac{-50}{-100} = .5s$

$\Delta R_3 = 12.5\ mm$

stage 2:

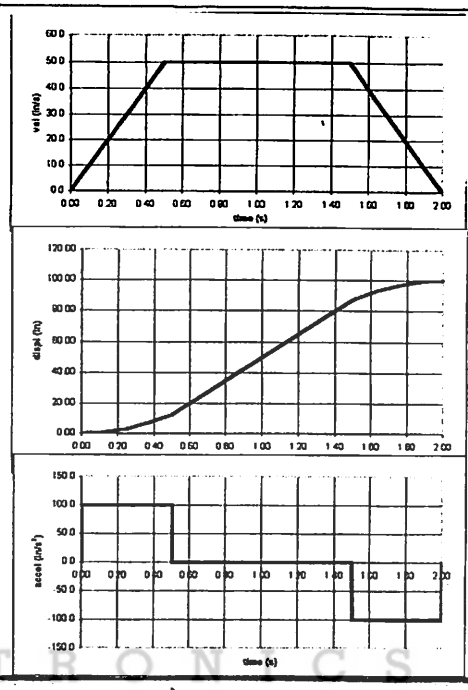
$75\ mm = \Delta R_1 + \Delta R_2 + \Delta R_3$

$\Delta R_2 = 50\ mm$

$V_2 = 50\ mm/s = \frac{\Delta R_2}{\Delta t_2} \Rightarrow \Delta t_2 = \frac{\Delta R_2}{V_2} = \frac{50}{50}$

$= 1.0s$

t (s)	a (in/s2)	v (in/s)	ΔR (in)
0.00	0.0	0.0	0.00
0.00	100.0	0.0	0.00
0.13	100.0	12.5	0.78
0.25	100.0	25.0	3.13
0.38	100.0	37.5	7.03
0.50	100.0	50.0	12.50
0.50	0.0	50.0	12.50
1.50	0.0	50.0	87.50
1.50	-100.0	50.0	87.50
1.63	-100.0	37.5	92.97
1.75	-100.0	25.0	96.88
1.88	-100.0	12.5	99.22
2.00	-100.0	0.0	100.00
2.00	0.00	0.0	100.00



7-16

$\omega_2 = 300 rpm \left(\frac{\pi}{30}\right) = 31.42 \frac{rad}{sec}$, CCW

$a_A^n = (31.42)^2(8) = 7898\ in/s^2 \angle 50°$

$a_A^t = 0$

$a_B^n = (31.42)^2(18) = 17770\ in/s^2 \angle 70°$

$a_B^t = 0$



7-17

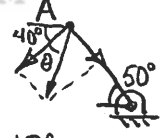
$\omega_2 = 200 rpm \left(\frac{\pi}{30}\right) = 20.94 \frac{rad}{s}$, CCW

$a_A^n = (20.94)^2(8) = 3507 \frac{in}{s^2} \angle 50°$

$a_A^t = (400)(8) = 3200\ in/s^2 \angle 40°$

$a_A = \sqrt{3507^2 + 3200^2}$

$= 4748\ in/s^2$



$\theta = \tan^{-1}\left(\frac{3507}{3200}\right) = 47.63°$

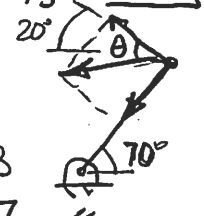
$a_A = 4748\ in/s^2 \angle 87.63°$

$a_B^n = (20.94)^2(18) = 7893\ in/s^2 \angle 70°$

$a_B^t = (400)(18) = 7200\ in/s^2 \angle 20°$

$a_B = \sqrt{7893^2 + 7200^2}$

$= 10684\ in/s^2$



$\theta = \tan^{-1}\left(\frac{7893}{7200}\right) = 47.63$

$a_B = 10684\ in/s^2 \angle 27.63°$

7-18

$\omega_2 = 300 rpm \left(\frac{\pi}{30}\right) = 31.42 \frac{rad}{sec}$, CCW

$a_A^n = (31.42)^2(8) = 7898\ in/s^2 \angle 50°$

$a_A^t = (800)(8) = 6400\ in/s^2 \angle 40°$

$a_A = \sqrt{7898^2 + 6400^2}$

$= 10166\ in/s^2$



$\theta = \tan^{-1}\left(\frac{7898}{6400}\right) = 50.98°$

$a_A = 10166\ in/s^2 \angle 10.98°$

$a_B^n = (31.42)^2(18) = 17770\ in/s^2 \angle 70°$

$a_B^t = (800)(18) = 14400\ in/s^2 \angle 20°$

$a_B = \sqrt{17770^2 + 14400^2}$

$= 22872\ in/s^2$



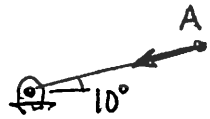
$\theta = \tan^{-1}\left(\frac{17770}{14400}\right) = 50.98°$

$a_B = 22872\ in/s^2 \angle 70.98°$

7-19

$\omega = 300 rpm \left(\frac{\pi}{30}\right) = 31.42 \frac{rad}{s}$, CW

$a_A^n = (31.42)^2(5) = 4935 \frac{in}{s^2} \angle 10°$



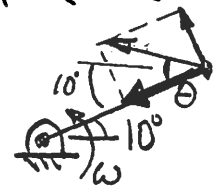
$a_A^t = 0$

7-20

$\omega = 300 \text{ rpm}\left(\frac{\pi}{30}\right) = 31.42 \frac{\text{rad}}{\text{s}}$, cw

$a_A^n = (31.42)^2(5) = 4935 \frac{\text{in}}{\text{s}^2}$ $10°$

$a_A^t = (300)(5) = 1500 \frac{\text{in}}{\text{s}^2}$ $80°$



$a_A = \sqrt{4935^2 + 1500^2} = 5158 \text{ in/s}^2$

$\theta = \tan^{-1}\left(\frac{1500}{4935}\right) = 16.9°$

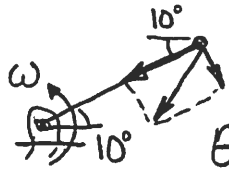
$a_A = 5158 \text{ in/s}^2$ $6.9°$

7-21

$\omega = 300 \text{ rpm}\left(\frac{\pi}{30}\right) = 31.42 \frac{\text{rad}}{\text{s}}$, cw

$a_A^n = (31.42)^2(5) = 4935 \text{ in/s}^2$ $10°$

$a_A^t = (300)(5) = 1500 \text{ in/s}^2$ $80°$



$a_A = \sqrt{4935^2 + 1500^2} = 5158 \text{ in/s}^2$

$\theta = \tan^{-1}(1500/4935) = 16.9°$

$a_A = 5158 \text{ in/s}^2$ $26.9°$

7-22

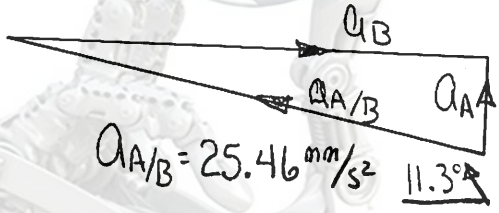


$V_{A/B} = V_A \to V_B$



$V_{A/B} = 12.20 \frac{\text{mm}}{\text{s}}$ $55.0°$

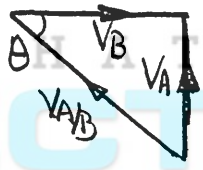
$a_{A/B} = a_A \to a_B$



$a_{A/B} = 25.46 \text{ mm/s}^2$ $11.3°$

7-23

$V_{A/B} = V_A \to V_B$

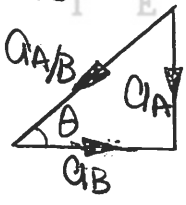


$V_{A/B} = \sqrt{(50)^2 + (42)^2} = 65.31 \text{ in/s}$

$\theta = \tan^{-1}\left(\frac{50}{42}\right) = 50°$

$V_{A/B} = 65.31 \text{ in/s}$ $50°$

$a_{A/B} = a_A \to a_B$

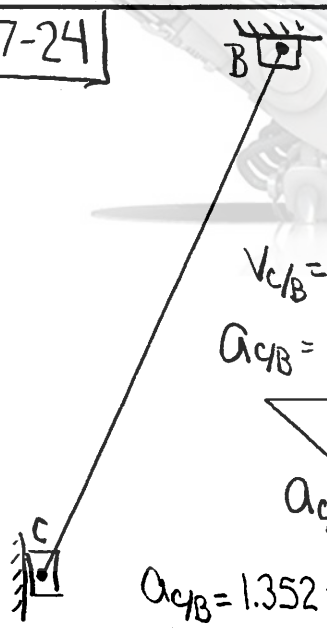


$a_{A/B} = \sqrt{(125)^2 + (48.6)^2} = 134.12 \text{ in/s}^2$

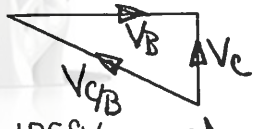
$\theta = \tan^{-1}(125/48.6) = 68.75°$

$a_{A/B} = 134.12 \text{ in/s}^2$ $68.75°$

7-24

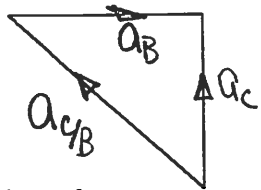


$V_{C/B} = V_C \to V_B$



$V_{C/B} = 1.105 \text{ ft/s}$ $25.17°$

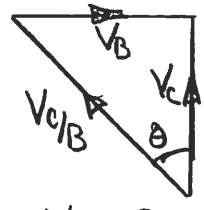
$a_{C/B} = a_C \to a_B$



$a_{C/B} = 1.352 \text{ ft/s}^2$ $42.30°$

7-25

$V_{C/B} = V_C \to V_B$

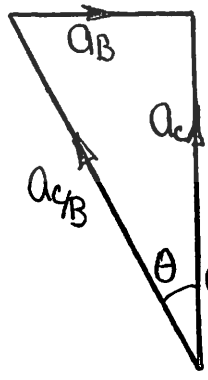


$V_{C/B} = \sqrt{2^2 + 2.85^2} = 3.482$

$\theta = \tan^{-1}\frac{2}{2.85} = 35.0°$

$V_{C/B} = 3.482 \text{ ft/s}$ $55°$

$a_{C/B} = a_C \to a_B$



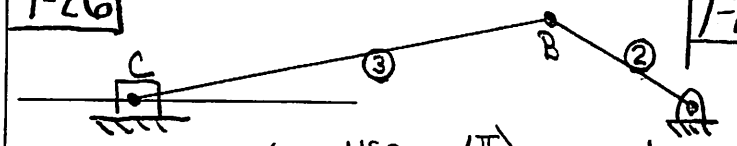
$a_{C/B} = \sqrt{1^2 + 8.51^2} = 8.569$

$\theta = \tan^{-1}\left(\frac{1}{8.51}\right) = 6.70°$

$a_{C/B} = 8.569 \text{ ft/s}^2$ $83.30°$

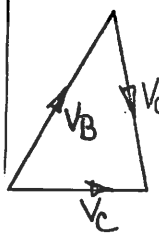
13-782 500 SHEETS, FILLER 5 SQUARE
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7-26



$\omega_2 = 1150 \text{ rpm}\left(\frac{\pi}{30}\right) = 120.43 \text{ rad/s, cw}$

$V_B = (120.43)(2") = 240.86 \text{ in/s} \nearrow 60°$

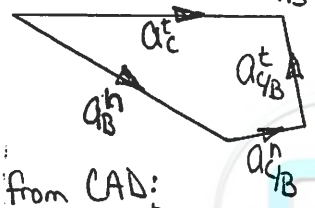


$V_C = V_B + V_{C/B}$

from CAD: $V_C = 162.43 \text{ in/s} \rightarrow$

$V_{C/B} = 212.13 \text{ in/s} \swarrow 78.46°$

$a_C^t = a_B^n + a_{C/B}^n + a_{C/B}^t$

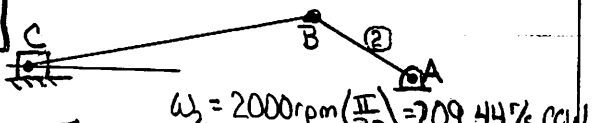


$a_B^n = (240.86)^2/2 = 29007 \text{ in/s}^2 \downarrow 30°$

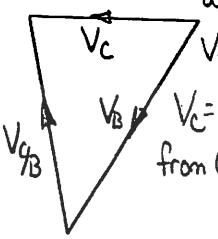
$a_{C/B}^n = (212.13)^2/5 = 9000 \text{ in/s}^2 \nearrow 11.54°$

from CAD: $a_C^t = 31341 \text{ in/s}^2 \rightarrow$ //

7-27



$\omega_2 = 2000 \text{ rpm}\left(\frac{\pi}{30}\right) = 209.44 \text{ rad/s, ccw}$



$V_B = (209.44)(2) = 418.88 \text{ in/s} \nearrow 60°$

$V_C = V_B + V_{C/B}$

from CAD $V_C = 283.5 \text{ in/s} \leftarrow$

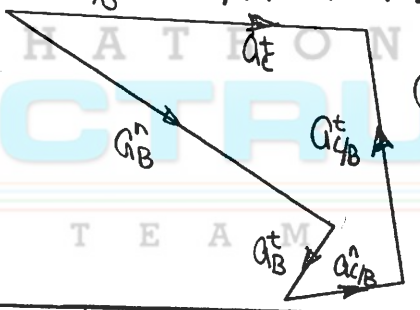
$V_{C/B} = 370.2 \text{ in/s} \nwarrow 78.46°$

$a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (418.88)^2/2 = 87730 \text{ in/s}^2 \downarrow 30°$

$a_B^t = (10000)(2) = 20000 \text{ in/s}^2 \swarrow 60°$

$a_{C/B}^n = (370.2)^2/(5) = 27409 \text{ in/s}^2 \nearrow 11.54°$



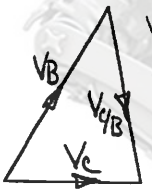
$a_C^t = 81461 \text{ in/s}^2 \rightarrow$ //

7-28



$\omega_2 = 1500 \text{ rpm}\left(\frac{\pi}{30}\right) = 157.08 \text{ rad/s, cw}$

$V_B = (157.08)(2) = 314.16 \text{ in/s} \nearrow 60°$

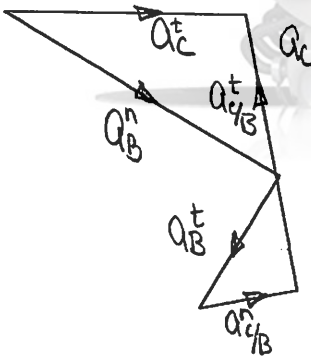


$V_C = V_B + V_{C/B}$

$V_C = 212.62 \text{ in/s} \rightarrow$

$V_{C/B} = 277.68 \text{ in/s} \swarrow 78.46$

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$



$a_B^n = (314.16)^2/2 = 49348 \text{ in/s}^2 \downarrow 30°$

$a_B^t = (12000)(2) = 24000 \text{ in/s}^2 \swarrow 60°$

$a_{C/B}^n = (277.68)^2/5 = 15421 \text{ in/s}^2 \nearrow 11.54°$

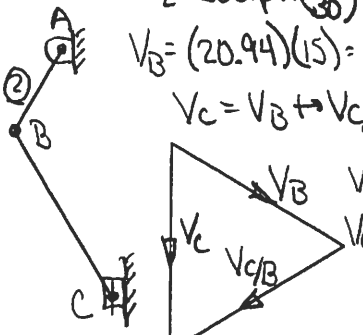
$a_C = 37194 \text{ in/s}^2 \rightarrow$ //

7-29

$\omega_2 = 200 \text{ rpm}\left(\frac{\pi}{30}\right) = 20.94 \text{ rad/s, ccw}$

$V_B = (20.94)(15) = 314.1 \text{ mm/s} \swarrow 30°$

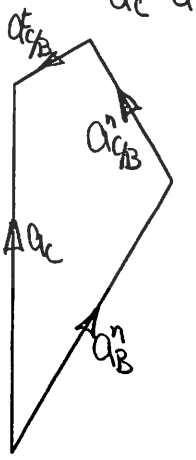
$V_C = V_B + V_{C/B}$



$V_C = 321.2 \text{ mm/s} \downarrow$

$V_{C/B} = 317.8 \text{ mm/s} \swarrow 31.11°$

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$



$a_B^n = \frac{(314.1)^2}{15} = 6577 \frac{\text{mm}}{\text{s}^2} \nearrow 60°$

$a_{C/B}^n = \frac{(317.8)^2}{30} = 3367 \frac{\text{mm}}{\text{s}^2} \nwarrow 58.89°$

$a_C = 7648 \frac{\text{mm}}{\text{s}^2} \uparrow$ //

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7-30

$\omega_2 = 300\text{rpm}\left(\frac{\pi}{30}\right)$
$= 31.42\ \text{rad/s}$, CW

$V_B = (31.42)(15)$
$= 471.24\ \text{mm/s}$ ↑ 30°

$V_C = V_B + V_{C/B}$

from CAD
$V_C = 481.89\ \text{mm/s}$ ↑
$V_{C/B} = 476.66\ \text{mm/s}$ ↗ 31.11°

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (471.24)^2/15 = 14804\ \text{mm/s}^2$ ↙ 60°
$a_B^t = 800(15) = 12000\ \text{mm/s}^2$ ↑ 30°
$a_{C/B}^n = (476.66)^2/30$
$= 7402\ \text{mm/s}^2$ ↖ 58.89°

from CAD
$a_C = 29271\ \text{mm/s}^2$ ↑

7-31

$\omega_2 = 200\text{rpm}\left(\frac{\pi}{30}\right)$
$= 20.97\ \text{rad/s}$, CW

$V_B = (20.97)(15)$
$= 314.16\ \text{mm/s}$ ↗ 60°

$V_C = V_B + V_{C/B}$

from CAD
$V_C = 118.24\ \text{mm/s}$ ↑
$V_{C/B} = 219.86\ \text{mm/s}$ ↙ 44.4°

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (314.16)^2/15 = 6580\ \text{mm/s}^2$ ↓ 30°
$a_B^t = 400(15) = 6000\ \text{mm/s}^2$ ↙ 60°
$a_{C/B}^n = (219.86)^2/30$
$= 1611\ \text{mm/s}^2$ ↖ 45.6°

from CAD
$a_C = 8874\ \text{mm/s}^2$ ↓

7-32

$\omega_2 = 80\text{rpm}\left(\frac{\pi}{30}\right)$
$= 8.378\ \text{rad/s}$, CW

$V_B = (8.378)(1.75)$
$= 14.66\ \text{in/s}$ ↘ 60°

$V_C = V_B + V_{C/B}$

from CAD
$V_C = 10.90\ \text{in/s}$ →
$V_{C/B} = 13.19\ \text{in/s}$ ↗ 74.29°

$a_C = a_B^n + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (14.66)^2/1.75$
$= 122.83\ \text{in/s}^2$ ↙ 30°
$a_{C/B}^n = (13.19)^2/6$
$= 29.00\ \text{in/s}^2$ ↘ 15.71°

from CAD
$a_C = 58.97\ \text{in/s}^2$ ←

7-33

$\omega_2 = 60\text{rpm}\left(\frac{\pi}{30}\right)$
$= 6.283\ \text{rad/s}$, CW

$V_B = (6.283)(1.75)$
$= 11.00\ \text{in/s}$ ↘ 60°

$V_C = V_B + V_{C/B}$

from CAD
$V_{C/B} = 9.89\ \text{in/s}$ ↗ 14.29°
$V_C = 8.18\ \text{in/s}$ →

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (11.00)^2/1.75$
$= 69.14\ \text{in/s}^2$ ↓ 30°
$a_B^t = (40)(1.75)$
$= 70.0\ \text{in/s}^2$ ↘ 60°
$a_{C/B}^n = (9.89)^2/6$
$= 16.30\ \text{in/s}^2$ ↘ 15.71°

from CAD $a_C = 18.83\ \text{in/s}^2$ →

MECHATRONICS SPECTRUM TEAM

7-34

$\omega_2 = 70 \text{ rpm} \left(\frac{\pi}{30}\right) = 7.33 \text{ rad/s, cw}$

$V_B = (7.33)(1.75) = 12.83 \text{ in/s} \angle 60°$

$V_C = V_B + V_{C/B}$

from CAD $V_{C/B} = 11.54 \text{ in/s} \angle 74.29$

$V_C = 9.54 \text{ in/s} \rightarrow$

$a_C = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (12.83)^2/1.75 = 94.06 \text{ in/s}^2 \angle 30°$

$a_B^t = (45)(1.75) = 78.75 \text{ in/s}^2 \angle 60°$

$a_{C/B}^n = (11.54)^2/6 = 22.20 \text{ in/s}^2 \angle 15.71°$

from CAD: $a_C = 103.73 \text{ in/s}^2 \leftarrow$

7-35

$\omega_2 = 90 \text{ rpm} \left(\frac{\pi}{30}\right) = 9.424 \text{ rad/s, cw}$

$V_B = (9.424)(4) = 37.70 \text{ in/s} \angle 60°$

$V_C = V_B + V_{C/B}$

from CAD:

$V_C = 75.40 \text{ in/s} \angle 18.53°$

$V_{C/B} = 53.35 \text{ in/s} \angle 9.37°$

$\omega_4 = 75.40/27 = 2.79 \text{ rad/s, cw}$

$a_C^n + a_C^t = a_B^n + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (37.7)^2/4 = 355.32 \text{ in/s}^2 \angle 30°$

$a_{C/B}^n = (53.35)^2/30 = 94.87 \text{ in/s}^2 \angle 80.63°$

$a_C^n = (75.4)^2/27 = 210.56 \text{ in/s}^2 \angle 71.47°$

from CAD $a_C^t = 72.61 \text{ in/s}^2 \angle 18.53°$

$\alpha_4 = (72.61)/27 = 2.69 \text{ rad/s}^2, \text{ ccw}$

7-36

$\omega_2 = 60 \text{ rpm} \left(\frac{\pi}{30}\right) = 6.28 \text{ rad/s, cw}$

$V_B = (6.28)(4) = 25.12 \text{ in/s} \angle 45°$

$V_C = V_B + V_{C/B}$

from CAD:

$V_{C/B} = 29.22 \text{ in/s} \angle 11.96°$

$V_C = 47.80 \text{ in/s} \angle 14.18°$

$\Rightarrow \omega_4 = 47.8/27 = 1.77 \text{ rad/s, cw}$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (25.12)^2/4 = 157.75 \text{ in/s}^2 \angle 45°$

$a_B^t = (30)(4) = 120 \text{ in/s}^2 \angle 45°$

$a_{C/B}^n = (29.22)^2/30 = 28.46 \text{ in/s}^2 \angle 78.04°$

$a_C^n = (47.80)^2/27 = 84.62 \text{ in/s}^2 \angle 75.82°$

from CAD

$a_C^t = 140.93 \text{ in/s}^2 \angle 14.18°$

$\alpha_4 = 140.93/27 = 5.22 \text{ rad/s}^2, \text{ CW}$

7-37

$\omega_2 = 40 \text{ rpm} \left(\frac{\pi}{30}\right) = 4.189 \text{ r/s, cw}$

$V_B = (4.189)(4) = 16.756 \text{ in/s} \angle 30°$

$V_C = V_B + V_{C/B}$

from CAD

$V_C = 19.436 \text{ in/s} \angle 4.77°$

$V_{C/B} = 34.529 \text{ in/s} \angle 11.06°$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (16.756)^2/4 = 70.19 \text{ in/s}^2 \angle 60°$

$a_B^t = (20)(4) = 80 \text{ in/s}^2 \angle 30°$

$a_{C/B}^n = (34.529)^2/30 = 39.74 \text{ in/s}^2 \angle 78.94°$

$a_C^n = (19.436)^2/27 = 13.99 \text{ in/s}^2 \angle 85.23°$

from CAD

$a_C^t = 244.03 \text{ in/s}^2 \angle 4.77°$

$\omega_2 = (19.436)/27 = .720 \text{ rad/s, ccw}$

$\alpha_2 = (244.03)/27 = 9.04 \text{ rad/s}^2, \text{ ccw}$

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7-38

$\omega_2 = 120\text{rpm}\left(\frac{\pi}{30}\right) = 12.566\ \text{rad/s}$, CCW

$V_B = (3")(12.566\text{r/s}) = 37.7"/s$ $\overline{40°}$

$V_C = V_B + V_{C/B}$

$V_C = 35.3"/s$ $\overline{44°}$

$V_{C/B} = 3.4"/s$ $\overline{4°}$

$\omega_4 = 35.3/6 = 5.9\ \text{rad/s}$ CCW

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (37.7)^2/3 = 473.7"/s^2$ $\overline{50°}$

$a_B^t = 0$

$a_{C/B}^n = (3.4)^2/10 = 1.2"/s^2$ $\overline{86°}$

$a_C^n = (35.3)^2/6 = 207.9"/s^2$ $\overline{46°}$

measuring:

$a_C^t = 271.9"/s^2$ $\overline{44°}$

$\alpha_4 = 271.9/6 = 45.2\ \text{rad/s}^2$, CCW

7-39

$\omega_2 = 150\text{rpm}\left(\frac{\pi}{30}\right) = 15.710\ \text{rad/s}$, CCW

$V_B = (3)(15.710) = 47.12"/s$

$V_C = V_B + V_{C/B}$

from CAD:

$V_{C/B} = 16.68"/s$ $\overline{6.51°}$

$V_C = 48.17"/s$ $\overline{69.87°}$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (47.12)^2/3 = 740.10"/s^2$ →

$a_B^t = (200)(3) = 600"/s^2$ ↓

$a_{C/B}^n = (16.68)^2/10 = 27.82"/s^2$ $\overline{83.49°}$

$a_C^n = (48.17)^2/6 = 386.72"/s^2$ $\overline{20.13°}$

from CAD:

$a_C^t = 404.72"/s^2$ $\overline{69.87°}$

$\omega_4 = (48.17)/6 = 8.028\ \text{rad/s}$, CCW

$\alpha_4 = (404.72)/6 = 15.934\ \text{rad/s}^2$, CCW

7-40

$\omega_2 = 100\text{rpm}\left(\frac{\pi}{30}\right) = 10.472\ \text{rad/s}$, CCW

$V_B = (10.472)(3) = 31.42"/s$ $\overline{60°}$

$V_C = V_B + V_{C/B}$

from CAD:

$V_{C/B} = 18.73"/s$ $\overline{10.74°}$

$V_C = 23.87"/s$ $\overline{83.54°}$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (31.42)^2/3 = 329.07"/s^2$ $\overline{30°}$

$a_B^t = (3)(100) = 300"/s^2$ $\overline{60°}$

$a_{C/B}^n = (18.73)^2/10 = 35.08"/s^2$ $\overline{79.26°}$

$a_C^n = (23.87)^2/6 = 94.96"/s^2$ $\overline{6.46°}$

from CAD: $a_C^t = 471.32"/s^2$ $\overline{83.54°}$

$\omega_4 = (23.87)/6 = 3.98\ \text{rad/s}$, CCW

$\alpha_4 = (471.32)/6 = 78.55\ \text{rad/s}^2$, CW

7-41

$\omega_2 = 20\text{rpm}\left(\frac{\pi}{30}\right) = 2.09\ \text{rad/s}$, CCW

$V_B = (2.09)(12) = 25.13"/s$ $\overline{15°}$

$V_C = V_B + V_{C/B}$

from CAD

$V_{C/B} = 3.94"/s$ $\overline{55.05°}$

$V_C = 24.07"/s$ $\overline{23.86°}$

$a_C^n + a_C^t = a_B^n + a_{C/B}^n + a_{C/B}^t$

$a_B^n = (25.13)^2/12 = 52.63"/s^2$ $\overline{75°}$

$a_{C/B}^n = (3.94)^2/30 = .52"/s^2$ $\overline{34.95°}$

$a_C^n = (24.07)^2/26 = 22.28$ $\overline{66.14°}$

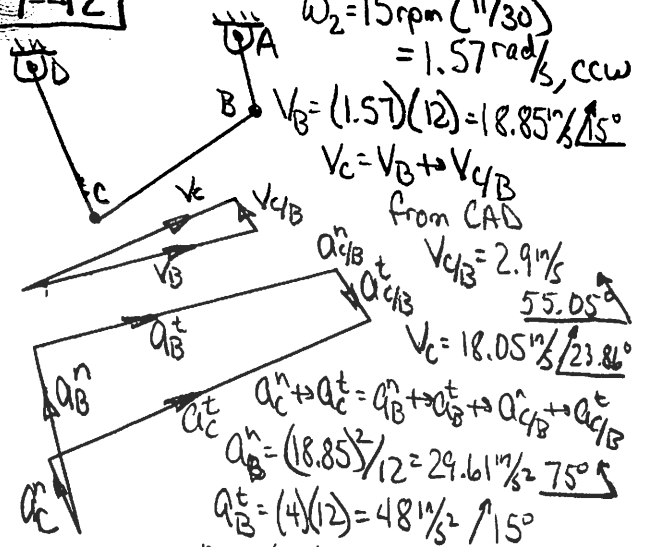
from CAD: $a_C^t = 14.46"/s^2$ $\overline{23.86°}$

$\omega_4 = (24.07)/26 = .928\ \text{rad/s}$, CCW

$\alpha_4 = (14.46)/26 = .556\ \text{rad/s}^2$, CCW

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7-42



$\omega_2 = 15 \text{ rpm}\left(\frac{\pi}{30}\right) = 1.57 \text{ rad/s, ccw}$

$V_B = (1.57)(12) = 18.85 \text{ in/s} \nearrow 15°$

$V_C = V_B + V_{C/B}$

from CAD $V_{C/B} = 2.9 \text{ in/s}$ $55.05°$

$V_C = 18.05 \text{ in/s} \nearrow 23.86°$

$a^n_C + a^t_C = a^n_B + a^t_B + a^n_{C/B} + a^t_{C/B}$

$a^n_B = (18.85)^2/12 = 29.61 \text{ in/s}^2 \; 75°$

$a^t_B = (4)(12) = 48 \text{ in/s}^2 \nearrow 15°$

$a^n_{C/B} = (2.90)^2/30 = .28 \text{ in/s}^2 \nearrow 34.95°$

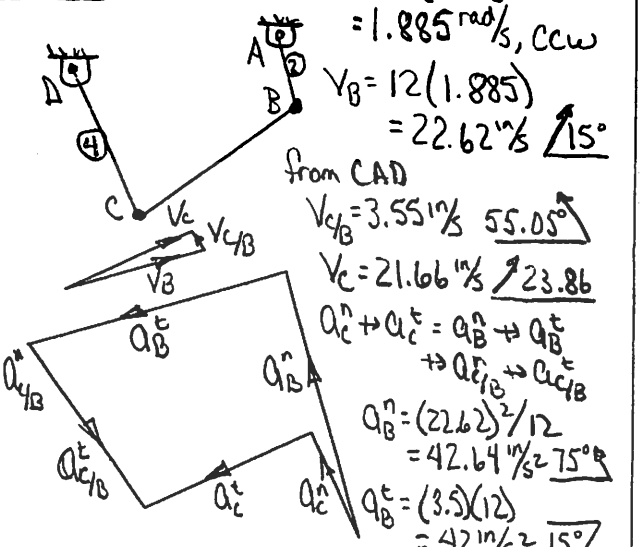
$a^n_C = (18.05)^2/26 = 12.53 \text{ in/s}^2 \; 66.14°$

from CAD: $a^t_C = 54.10 \text{ in/s}^2 \nearrow 23.86°$

$\omega_4 = (12.53)/26 = .482 \text{ rad/s, CW}$

$\alpha_4 = (54.10)/26 = 2.081 \text{ rad/s}^2, \text{ CW}$

7-43



$\omega_2 = 18 \text{ rpm}\left(\frac{\pi}{30}\right) = 1.885 \text{ rad/s, ccw}$

$V_B = 12(1.885) = 22.62 \text{ in/s} \nearrow 15°$

from CAD $V_{C/B} = 3.55 \text{ in/s} \; 55.05°$

$V_C = 21.66 \text{ in/s} \nearrow 23.86$

$a^n_C + a^t_C = a^n_B + a^t_B + a^n_{C/B} + a^t_{C/B}$

$a^n_B = (22.62)^2/12 = 42.64 \text{ in/s}^2 \; 75°$

$a^t_B = (3.5)(12) = 42 \text{ in/s}^2 \; 15°$

$a^n_{C/B} = (3.55)^2/30 = .42 \text{ in/s}^2 \; 34.95°$

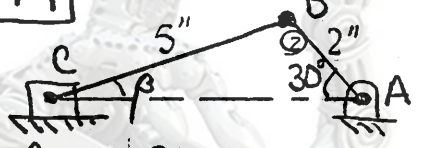
$a^n_C = (21.66)^2/26 = 18.04 \text{ in/s}^2 \; 66.14°$

from CAD: $a^t_C = 28.51 \text{ in/s}^2 \; 23.86°$

$\omega_4 = (21.66)/26 = .833 \text{ rad/s, CCW}$

$\alpha_4 = (28.51)/26 = 1.097 \text{ rad/s}^2, \text{ CW}$

7-44

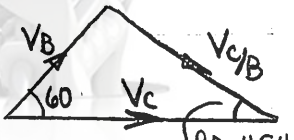


$\frac{\sin\beta}{2} = \frac{\sin 30}{5} \Rightarrow \beta = 11.54°$

$\omega_2 = 1800 \text{ rpm}\left(\frac{\pi}{30}\right) = 188.49 \text{ rad/s, CW}$

$V_B = (188.49)(2) = 376.99 \text{ in/s} \nearrow 60°$

$V_C = V_B + V_{C/B}$



$\frac{V_B}{\sin 78.46} = \frac{V_C}{\sin[180-(60+78.46)]}$

$V_C = 255.16 \text{ in/s} \rightarrow$

$\frac{V_B}{\sin 78.46} = \frac{V_{C/B}}{\sin 60}$

$V_{C/B} = 333.22 \text{ in/s} \searrow 78.46$

$a^n_B = \frac{(376.99)^2}{2} = 71061 \text{ in/s}^2 \searrow 30°$

$a^t_B = 12000(2) = 24000 \text{ in/s}^2 \nearrow 60°$

$a^n_{C/B} = \frac{(333.22)^2}{5} = 22207 \text{ in/s}^2 \nwarrow 11.54$

assume: $a^t_{C/B} \searrow 78.46$ $\quad a^t_C \rightarrow$

$a^t_C = a^n_B + a^t_B + a^n_{C/B} + a^t_{C/B}$

horiz:

$a^t_C = 71061\cos 30 + 24000\cos 60 + 22207\cos 11.54 + a^t_{C/B}\cos 78.46$

Vert:

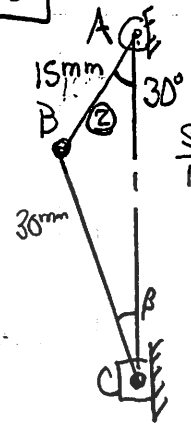
$0 = -71061\sin 30 + 24000\sin 60 + 22207\sin 11.54 - a^t_{C/B}\sin 78.46$

$a^t_{C/B} = -10516$

$= 10516 \text{ in/s}^2 \; 78.46$

$a_C = 93195 \text{ in/s}^2 \rightarrow$

7-45



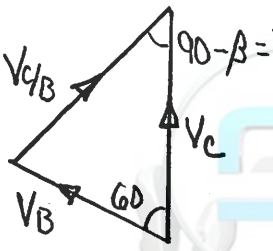
$\frac{\sin\beta}{15} = \frac{\sin 30}{30}$

$\beta = 14.48°$

$\omega_2 = 250 \text{ rpm}\,(\pi/30) = 26.18 \text{ rad/s, CW}$

$V_B = (26.18)(15) = 392.70 \text{ mm/s} \;\; 30°\nearrow$

$V_C = V_B + V_{C/B}$



$\frac{V_B}{\sin 60} = \frac{V_{C/B}}{\sin 75.52}$

$V_{C/B} = 439.05 \text{ mm/s} \;\; 14.48°\nearrow$

$\frac{V_B}{\sin 60} = \frac{V_C}{\sin[180-(60+75.52)]} \Rightarrow V_C = 317.72 \text{ mm/s} \uparrow$

$a_B^n = \frac{(392.7)^2}{15} = 10.281 \text{ m/s}^2 \;\; 60°\nearrow$

$a_B^t = (600)(15) = 9 \text{ m/s}^2 \;\; 30°\nwarrow$

$a_{C/B}^n = \frac{(439.05)^2}{30} = 6.425 \text{ m/s}^2 \;\; 75.52°\nearrow$

assume: $a_{C/B}^t \nearrow 14.48°$ $\quad a_C^t \uparrow$

$a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

horiz:

$0 = 10.281\cos 60 - 9\cos 30 - 6.425\cos 75.72 + a_{C/B}^t \cos 14.48°$

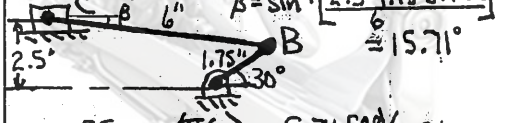
vert:

$a_C^t = 10.281\sin 60 + 9\sin 30 + 6.425\sin 75.52 + a_{C/B}^t \sin 14.48$

$a_{C/B}^t = 4.378 \text{ m/s}^2 \;\; 14.48°\nearrow$

$a_C^t = 20.719 \text{ m/s}^2 \uparrow$

7-46



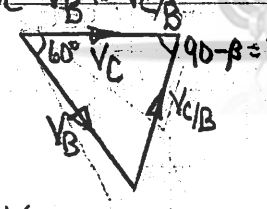
$6\sin\beta + 1.75\sin 30 = 2.5$

$\beta = \sin^{-1}\left[\frac{2.5 - 1.75\sin 30}{6}\right] \approx 15.71°$

$\omega_2 = 55 \text{ rpm}\,(\pi/30) = 5.76 \text{ rad/s, CW}$

$V_B = (5.76)(1.75) = 10.08 \text{ in/s} \;\; 60°\nearrow$

$V_C = V_B + V_{C/B}$



$\frac{V_B}{\sin 74.29} = \frac{V_{C/B}}{\sin 60}$

$V_{C/B} = 9.068 \text{ in/s} \;\; 74.29°\nearrow$

$\frac{V_B}{\sin 74.29} = \frac{V_C}{\sin[180-(60+74.29)]}$

$V_C = 7.495 \text{ in/s} \rightarrow$

$a_B^n = \frac{(10.08)^2}{1.75} = 58.061 \text{ in/s}^2 \;\; 30°\nwarrow$

$a_B^t = (35)(1.75) = 61.25 \text{ in/s}^2 \;\; 60°\nearrow$

$a_{C/B}^n = \frac{(9.068)^2}{6} = 13.705 \text{ in/s}^2 \;\; 15.71°\nwarrow$

assume $a_C^t \rightarrow$ $\quad a_{C/B}^t = \nearrow 74.29$

$a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

horiz:

$a_C^t = -58.061\cos 30 - 61.25\cos 60 - 13.705\cos 15.71 + a_{C/B}^t \cos 74.29$

vert:

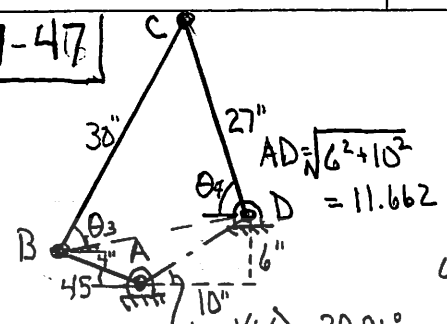
$0 = -58.061\sin 30 + 61.25\sin 60 + 13.705\sin 15.71 + a_{C/B}^t \sin 74.29$

$a_{C/B}^t = -28.800 = 28.800 \text{ in/s}^2 \;\; 74.29°$

$a_C^t = -130.700 = 130.700 \text{ in/s}^2 \leftarrow$

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7-47



$\omega_2 = 45 \text{ rpm}\left(\frac{\pi}{30}\right) = 4.71 \text{ rad/s, CW}$

$V_B = (4)(4.71) = 18.85 \text{ in/s} \angle 45°$

$\omega_4 = \frac{V_C}{27''} = 1.18 \frac{\text{rad}}{\text{s}}$, CW

$\angle BAD = 180 - (45 + 30.96) = 104.04°$

$BD = \sqrt{4^2 + 11.662^2 - 2(4)(11.662)\cos 104.04} = 13.215''$

$\angle ABD = \sin^{-1}\left(\frac{11.662}{13.215}\sin 104.04\right) = 58.88°$

$\angle ADB = 180 - (104.04 + 58.88) = 17.08°$

$\angle CBD = \cos^{-1}\left(\frac{30^2 + 13.215^2 - 27^2}{2(30)(13.215)}\right) = 64.16°$

$\angle BDC = \cos^{-1}\left(\frac{13.215^2 + 27^2 - 30^2}{2(13.215)(27)}\right) = 89.71°$

$\theta_3 = \angle ABD + \angle CBD - 45 = 78.04°$

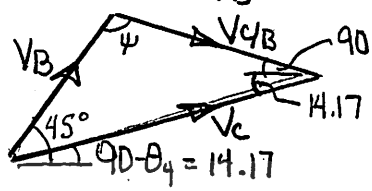
$\theta_4 = \angle ADB + \angle BDC - 30.96 = 75.83°$

$a_C^t = 120.23 \text{ in/s}^2 \angle 14.17$

$a_{C/B}^t = -4.55 = 4.55 \text{ in/s}^2 \quad 11.96$

$\alpha_4 = \frac{120.23}{27} = 4.453 \text{ rad/s}^2$, CW

$V_C = V_B + V_{C/B}$



$\frac{V_{C/B}}{\sin(45 - 14.17)} = \frac{V_B}{\sin(11.96 + 14.17)} \Rightarrow V_{C/B} = 21.94 \text{ in/s} \angle 11.96$

$\psi = 180 - [45 - 14.17] - [11.96 + 14.17] = 123.04°$

$\frac{V_C}{\sin(123.04)} = \frac{V_B}{\sin(11.96 + 14.17)} \Rightarrow V_C = 31.79 \text{ in/s} \angle 14.17$

$a_B^n = \frac{(18.85)^2}{4} = 88.83 \text{ in/s}^2 \angle 45$

$a_B^t = (25)(4) = 100 \text{ in/s}^2 \angle 45$

$a_{C/B}^n = \frac{(21.94)^2}{30} = 16.05 \text{ in/s}^2 \quad 78.04$

$a_C^n = \frac{(31.79)^2}{27} = 37.43 \text{ in/s}^2 \angle 75.83°$

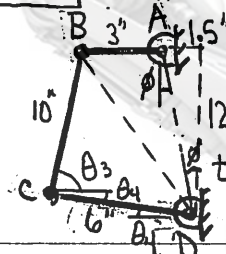
assume: $a_{C/B}^t \angle 11.96$, $a_C^t \angle 14.17$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

horiz: $37.43\cos 75.83 + a_C^t \cos 14.17 = 88.83\cos 45 + 100\cos 45 - 16.05\cos 78.04 + a_{C/B}^t \cos 11.96$

Vert: $-37.43\sin 75.83 + a_C^t \sin 14.17 = -88.83\sin 45 + 100\sin 45 - 16.05\sin 78.04 - a_{C/B}^t \sin 11.96$

7-48



$AD = \sqrt{1.5^2 + 12^2} = 12.093$

$\omega_2 = 130 \text{ rpm}\left(\frac{\pi}{30}\right) = 13.61 \text{ rad/s, CCW}$

$V_B = (3)(13.61) = 40.84 \text{ in/s} \downarrow$

$V_C = V_B + V_{C/B}$

$\psi = 180 - (\theta_3 + \theta_4) = 76.38°$

$\frac{V_{C/B}}{\sin\theta_4} = \frac{V_B}{\sin\psi} \Rightarrow V_{C/B} = 14.455 \text{ in/s} \quad 6.5°$

$\frac{V_C}{\sin\theta_3} = \frac{V_B}{\sin\psi} \Rightarrow V_C = 41.752 \text{ in/s} \quad 69.88$

$\angle BAD = 90 + 7.13 = 97.13°$

$BD = \sqrt{3^2 + 12.093^2 - 2(3)(12.093)\cos 97.13} = 12.816''$

$\angle ABD = \sin^{-1}\left(\frac{12.093}{12.816}\sin 97.13\right) = 69.44°$

$\angle ADB = 180 - (69.44 + 97.13) = 13.43°$

$\angle CBD = \cos^{-1}\left(\frac{10^2 + 12.816^2 - 6^2}{2(10)(12.816)}\right) = 27.07°$

$\angle BDC = \cos^{-1}\left(\frac{12.816^2 + 6^2 - 10^2}{2(12.816)(6)}\right) = 49.32°$

$\angle BCD = 180 - (27.07 + 49.32) = 103.62°$

$\theta_4 = 90 - (\angle BDC + \angle ADB + \phi) = 20.12°$

$\theta_3 = \angle BCD - \theta_4 = 83.50°$

$\omega_4 = \frac{V_C}{6''} = 6.96 \frac{\text{rad}}{\text{s}}$, CCW

$a_B^n = \frac{(40.84)^2}{3} = 555.97 \text{ in/s}^2 \rightarrow$

$a_B^t = (3)(180) = 540 \text{ in/s}^2 \uparrow$

$a_{C/B}^n = \frac{(14.455)^2}{10} = 20.89 \text{ in/s}^2 \angle 83.50°$

$a_C^n = \frac{(41.752)^2}{6} = 290.54 \text{ in/s}^2 \angle 20.12$

assume: $a_{C/B}^t \quad 6.5°$, $a_C^t \quad 69.88$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

horiz: $290.54\cos 20.12 - a_C^t \cos 69.88 = 555.97 + 20.89\cos 83.5 - a_{C/B}^t \cos 6.5$

Vert: $-290.54\sin 20.12 - a_C^t \sin 69.88 = 540 + 20.89\sin 83.50 + a_{C/B}^t \sin 6.5$

$a_{C/B}^t = -42.01 = 42.01 \text{ in/s}^2 \quad 6.5°$, $a_C^t = 708.70 \text{ in/s}^2 \quad 69.88$

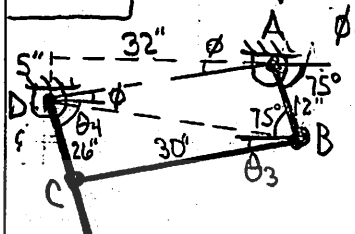
$\alpha = 708.7/6 = 118.12 \text{ rad/s}^2$, CCW

500 SHEETS, FILLER 5 SQUARE
50 SHEETS EYE-EASE® 5 SQUARE
100 SHEETS EYE-EASE® 5 SQUARE
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7-49

$AD = \sqrt{32^2+5^2} = 32.388''$

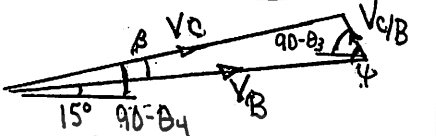
$\phi = \tan^{-1}\left(\frac{5}{32}\right) = 8.88°$



$\omega_2 = 12\text{ rpm}\left(\frac{\pi}{30}\right) = 1.257\text{ rad/s, ccw}$

$V_B = (1.257)(12) = 15.080\text{ in/s} \angle 15°$

$V_C = V_B + V_{C/B}$



$\omega_4 = \frac{V_C}{26} = .555\text{ rad/s, ccw}$ //

$\psi = 15 + 90 - \theta_3 = 70.04$ $\quad \beta = 90 - \theta_4 - 15 = 8.86°$

$\angle BAD = 180 - (75+\phi) = 96.12°$

$BD = \sqrt{32.388^2+12^2-2(32.388)(12)\cos 96.12} = 35.719''$

$\frac{V_{C/B}}{\sin\beta} = \frac{V_B}{\sin(180-\psi-\beta)} \Rightarrow V_{C/B} = 2.367 \angle 55.04$

$\frac{V_C}{\sin\psi} = \frac{V_B}{\sin(180-\psi-\beta)} \Rightarrow V_C = 14.44\text{ in/s} \angle 23.86$

$\angle ABD = \sin^{-1}\left(\frac{32.388}{35.719}\sin 96.12\right) = 64.37°$

$\angle ADB = 180° - (64.37+96.12) = 19.51°$

$\angle CBD = \cos^{-1}\left(\frac{30^2+35.719^2-26^2}{2(30)(35.719)}\right) = 45.59°$

$\angle BDC = \cos^{-1}\left(\frac{35.719^2+26^2-30^2}{2(35.719)(26)}\right) = 55.51°$

$\theta_3 = \angle CBD + \angle ABD - 75° = 34.96°$

$\theta_4 = \angle ADB + \angle BDC - \phi = 66.14°$

$a_B^n = \frac{(15.080)^2}{12} = 18.95\text{ in/s}^2 \angle 75°$

$a_B^t = (12)(3) = 36\text{ in/s}^2 \angle 15°$

$a_{C/B}^n = \frac{(2.367)^2}{30} = .187\text{ in/s}^2 \angle 34.96°$

$a_C^n = \frac{(14.44)^2}{26} = 8.02\text{ in/s}^2 \angle 66.14°$

assume $a_{C/B}^t \angle 55.04$

$a_C^t \angle 23.86$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

horiz: $8.02\cos 66.14 + a_C^t \cos 23.86 = -18.95\cos 75 + 36\cos 15 - .187\cos 34.96 - a_{C/B}^t \cos 55.04$

vert: $-8.02\sin 66.14 + a_C^t \sin 23.86 = 18.95\sin 75 + 36\sin 15 - .187\sin 34.96 + a_{C/B}^t \sin 55.04$

$a_{C/B}^t = -21.57 = 21.57\text{ in/s}^2 \angle 55.04$ ⇐

$a_C^t = 42.46\text{ in/s}^2 \angle 23.86$

$\alpha_4 = \frac{a_C^t}{26} = 1.633\text{ rad/s}^2\text{, ccw}$ //

7-50

$\omega_2 = 20\text{ rpm}\left(\frac{\pi}{30}\right) = 2.094\text{ rad/s, CW}$

$V_B = (2.094)(3.25) = 6.807\text{ in/s} \angle 30°$

$V_C = V_B + V_{C/B}$ $\quad V_C = 3.614\text{ in/s} \angle 70.57$

$V_X = V_B + V_{X/B}$ $\quad V_{C/B} = 4.693\text{ in/s} \leftarrow$

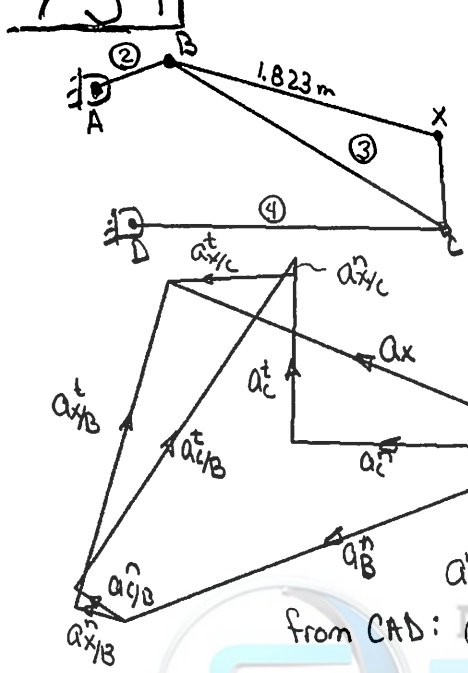
$V_X = V_C + V_{X/C}$ $\quad V_{X/B} = 5.628\text{ in/s} \angle 32.16$

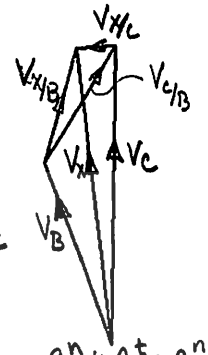
$V_{X/C} = 3.00\text{ in/s} \angle 88.63$

$V_X = 1.201\text{ in/s} \angle 19.81$

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7-54





$V_B = .5m\left[12rpm\left(\pi/30\right)\right] = .628 m/s$ ↑ 0°

$V_C = V_B + V_{C/B}$

$V_X = V_B + V_{X/B}$

$= V_C + V_{X/C}$

from CAD:

$V_C = .967 m/s$ ↑ 88.7°

$V_{C/B} = .445 m/s$ ↑ 55.9

$V_{X/B} = .386 m/s$ ↑ 73.6°

$V_{X/C} = .127 m/s$ ↓ 2.6°

$a^n_C + a^t_C = a^n_B + a^t_B + a^n_{C/B} + a^t_{C/B}$

$a_X = a^n_C + a^t_C + a^n_{X/C} + a^t_{X/C} = a^n_B + a^t_B + a^n_{X/B} + a^t_{X/B}$

$a^n_B = (.628)^2/.5 = .789 m/s^2$ ↓ 20 $\quad a^t_B = 0$

$a^n_{C/B} = (.445)^2/2.1 = .094 m/s^2$ ↖ 34.1 $\quad a^n_C = (.967)^2/2.0 = .468 m/s^2$ ↓ 1.8°

$a^n_{X/C} = (.127)^2/.6 = .027 m/s^2$ ↓ 87.4 $\quad a^n_{X/B} = .082 m/s^2$ ↖ 16.4°

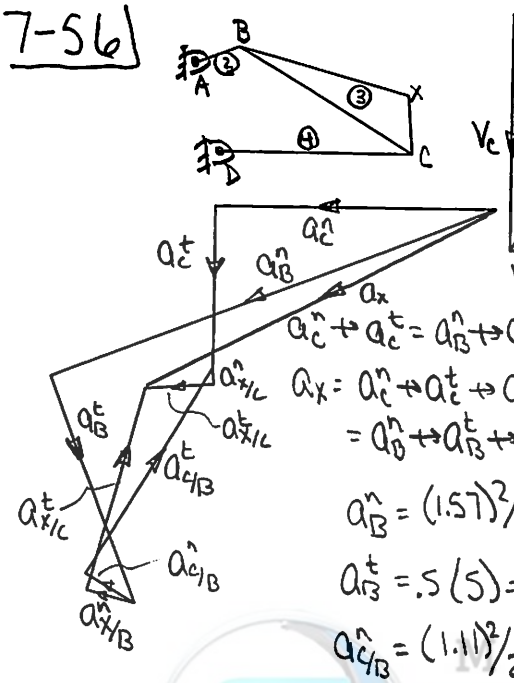
from CAD: $a^t_{C/B} = .639 m/s^2$ ↑ 55.9, $a^t_C = .293 m/s^2$ ↑ 88.7, $a^t_{X/B} = .545$ ↑ 73.6°

$a^t_{X/C} = .204 m/s^2$ ↓ 2.6, $a_X = .721 m/s^2$ ↑ 22.5°

$\alpha_4 = .293/2.0 = .147 rad/s^2$, CCW

SPECTRUM TEAM

7-56



$\omega_2 = 30(\pi/30) = 3.14 \text{ rad/s, cw}$

$v_B \quad v_B = (.5)(3.14) = 1.57 \text{ m/s} \searrow 70$

$v_C = v_B + v_{C/B}$ } from CAD

$v_X = v_B + v_{X/B}$
$= v_C + v_{X/C}$ }

$v_C = 2.42 \text{ m/s} \quad 89 \searrow$

$v_{C/B} = 1.11 \text{ m/s} \quad 58 \swarrow$

$v_{X/B} = .96 \text{ m/s} \quad 74 \swarrow$

$v_{X/C} = .32 \text{ m/s} \quad 3 \swarrow$

$v_X = 2.42 \text{ m/s} \quad 84 \swarrow$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_X = a_C^n + a_C^t + a_{X/C}^n + a_{X/C}^t$
$= a_B^n + a_B^t + a_{X/B}^n + a_{X/B}^t$

$a_B^n = (1.57)^2/.5 = 4.93 \text{ m/s}^2 \quad 20 \swarrow$

$a_B^t = .5(5) = 2.50 \text{ m/s}^2 \quad 70 \searrow$

$a_{C/B}^n = (1.11)^2/2.1 = .59 \text{ m/s}^2 \quad 32 \nwarrow$

$a_C^n = (2.418)^2/2 = 2.92 \text{ m/s}^2 \quad 1° \uparrow$

$a_{X/B}^n = (.965)^2/1.823 = .51 \text{ m/s}^2 \quad 16° \nwarrow$

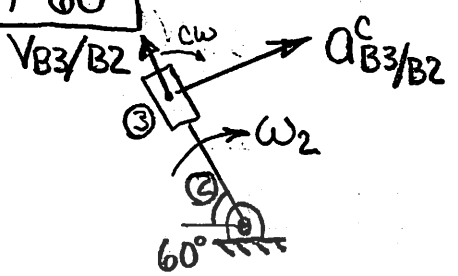
$a_{X/C}^n = (.318)^2/.6 = .17 \text{ m/s}^2 \quad 87° \swarrow$

measuring: $a_X = 4.08 \text{ m/s}^2 \quad 26° \swarrow$ // $a_C^t = 1.66 \text{ m/s}^2 \quad 89° \swarrow$ $\alpha_4 = 1.66/2 = .83 \text{ rad/s, cw}$

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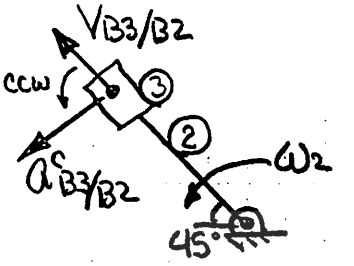
SPECTRUM TEAM

7-60



$$a^c_{B3/B2} = 2\, V_{B3/B2}\, \omega_2$$
$$= 2(15\ mm/s)(30\ rad/s)$$
$$= 900\ mm/s^2 \angle 30°$$

7-61



$$a^c_{B3/B2} = 2V_{B3/B2}\, \omega_2$$
$$= 2(15\ mm/s)(30\ rad/sec)$$
$$= 900 mm/s^2 \angle 45°$$

7-62



$$a^c_{B3/B2} = 2\, V_{B3/B2}\, \omega_2$$
$$= 2(15\ mm/s)(30\ rad/s)$$
$$= 900\ mm/s^2 \angle 60°$$

7-63

MECHATRONICS SPECTRUM TEAM

Answers to the Chapter 7 Case Study Questions:

Case 7-1.

1. As crankshaft I rotates 30 degrees from the position shown, slide J will move upward.
2. As crankshaft I rotates a few more than 30 degrees, the crank will lose contact with tab K, and J will fall.
3. Component C serves to catch tab G and prevents slide J from falling to a lowest position.
4. After approximately 270 degrees of crankshaft rotation, crank D will contact lever C and release slide J. Slide J will encounter a free-fall.
5. Pin B serves as a stop, determining a lowest position for slide J.
6. The mechanism raises slide J in a controlled manner. Slide J will then fall slightly and remain an upward position for a set length of time. Eventually slide J will be released and free-fall to a lower position and remain there for a short time.

MECHATRONICS SPECTRUM TEAM

Case 7-2.

1. As the rotating machine table turns counterclockwise, component G pushes roller F causing a clockwise rotation of link E.
2. Spring K provides resistance for clockwise rotation of link E, maintaining contact between F and G.
3. As the rotating machine table turns counterclockwise and link E rotates clockwise, Link D is pulled towards the right.
4. Spring L creates a force that tends to reduce the angle between link D and E.
5. The hood on D and groove on B is called a ratchet and pawl arrangement. As the ratchet wheel turns, the pawl falls into the groove, allowing rotation is a single direction (clockwise in this case).
6. Item H is a ball detent mechanism. It is used to hold wheel B in a fixed position, until sufficient force is applied to overcome the spring loaded ball.
7. As the rotary machine table rotates, barrel B advances clockwise one position, releasing a rivet into an assembly machine.

Case 7-3.

1. As link J rotates 90 degrees clockwise from the position shown, bellcrank H will rotate counterclockwise.
2. As link J rotates 90 degrees clockwise from the position shown, pusher E and plate C move toward the left.
3. As link J rotates 90 degrees clockwise from the position shown, pin S will also move toward the left.
4. As link J rotates 90 degrees clockwise from the position shown, guide pin R will move along the groove, left and upward.
5. As link J rotates 90 degrees clockwise from the position shown, bellcrank P moves toward the left and rotates counterclockwise.

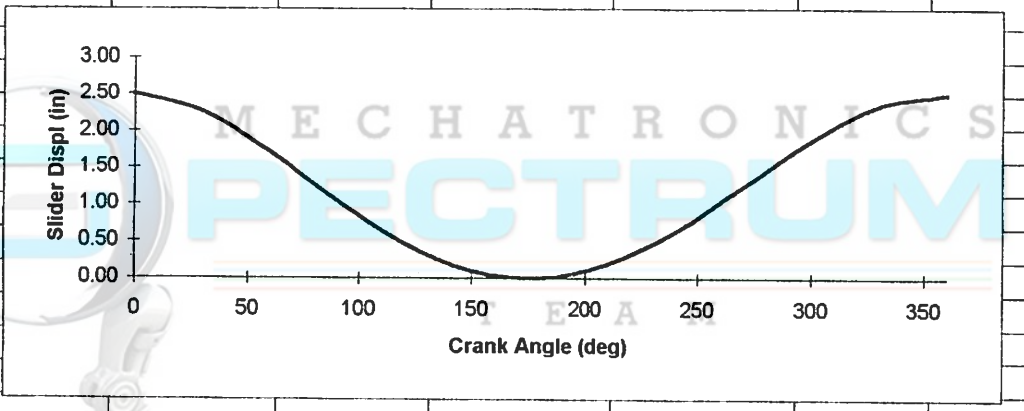
6. As link J rotates 90 degrees clockwise from the position shown, slide L and plate M will stay nearly stationary as the motion due to the translation of S is counterbalanced by the rotation of P. Further rotation of the link J will move the slide L toward the left, as the profile of the groove changes and imparts a clockwise rotation of bellcrank P.
7. A connecting link N is required as link P rotates and drives slide L, which is constrained to horizontal sliding.
8. As link J rotates 90 degrees clockwise from the position shown, slide M and C will be nearly parallel.
9. The continual motion of the device will move plates M and C together on the left side of channel B, and move plate C alone on the right side of the channel while plte M remains nearly stationary.

MECHATRONICS
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Chapter 8

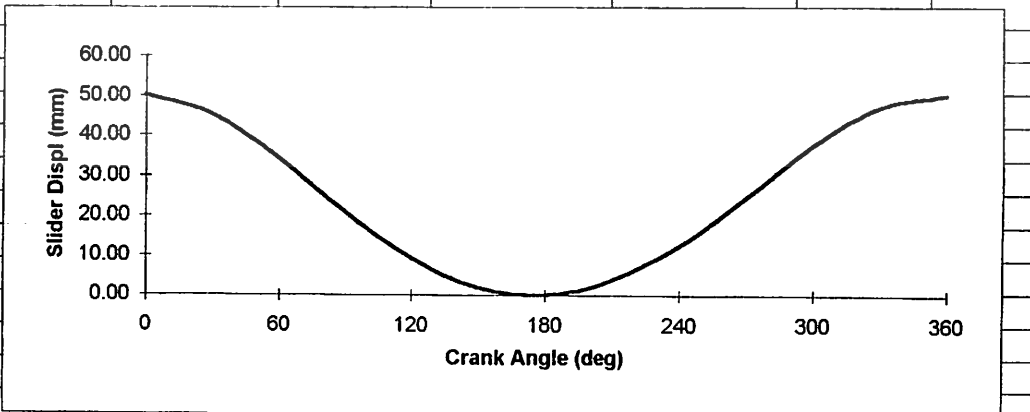
8-1

Crank Ang (deg)	Theta3 (deg)	Gamma (deg)	Slider Dist (in)	Slider Displ (in)	Offset	0.50	in
0	4.10	175.90	8.2321	2.5000	Link2	1.25	in
30	9.25	140.75	7.9915	2.2594	Link3	7.00	in
60	13.07	106.93	7.4438	1.7117			
90	14.48	75.52	6.7777	1.0456			
120	13.07	46.93	6.1938	0.4617			
150	9.25	20.75	5.8265	0.0944			
180	4.10	-4.10	5.7321	0.0000			
210	-1.02	-28.98	5.9164	0.1843			
240	-4.77	-55.23	6.3507	0.6186			
270	-6.15	-83.85	6.9597	1.2276			
300	-4.77	-115.23	7.6007	1.8686			
330	-1.02	-148.98	8.0814	2.3493			
360	4.10	-184.10	8.2321	2.5000			



8-2

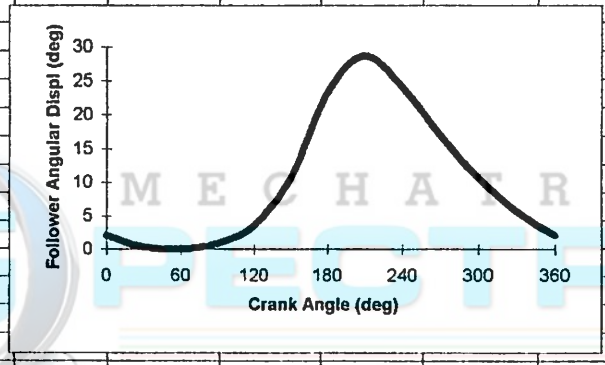
Crank Ang (deg)	Theta3 (deg)	Gamma (deg)	Slider Dist (mm)	Slider Displ (mm)	Offset	10.000	mm
0	4.10	175.90	164.64	50.00	Link2	25.000	mm
30	9.25	140.75	159.83	45.19	Link3	140.000	mm
60	13.07	106.93	148.88	34.24			
90	14.48	75.52	135.55	20.91			
120	13.07	46.93	123.88	9.24			
150	9.25	20.75	116.53	1.89			
180	4.10	-4.10	114.64	0.00			
210	-1.02	-28.98	118.33	3.69			
240	-4.77	-55.23	127.01	12.37			
270	-6.15	-83.85	139.19	24.55			
300	-4.77	-115.23	152.01	37.37			
330	-1.02	-148.98	161.63	46.99			
360	4.10	-184.10	164.64	50.00			



MECHATRONICS SPECTRUM

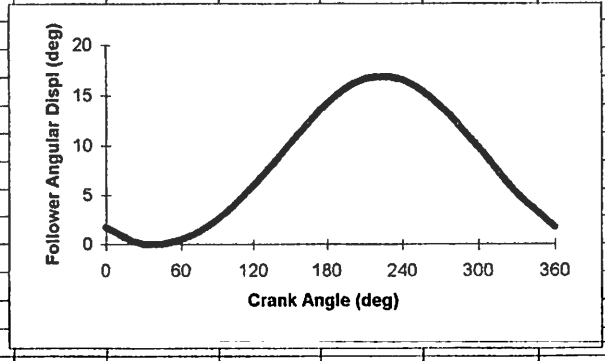
8-3

θ 2	θ 3	γ	θ 4	Δθ 4	L 1	750.0	mm
(deg)	(deg)	(deg)	(deg)	(deg)	L 2	50.0	mm
0	-176	46	5.4	2.1	L 3	750.0	mm
30	-173	53	3.7	0.4	L 4	75.0	mm
60	-171	69	3.3	0.0			
90	-170	88	4.1	0.8			
120	-172	108	6.8	3.5			
150	-174	123	14.0	10.7			
180	-176	130	26.4	23.1			
210	-177	123	32.0	28.7			
240	-178	108	27.4	24.1			
270	-178	88	20.4	17.1			
300	-178	69	14.0	10.7			
330	-177	53	8.9	5.6			
360	-176	46	5.4	2.1			



8-4

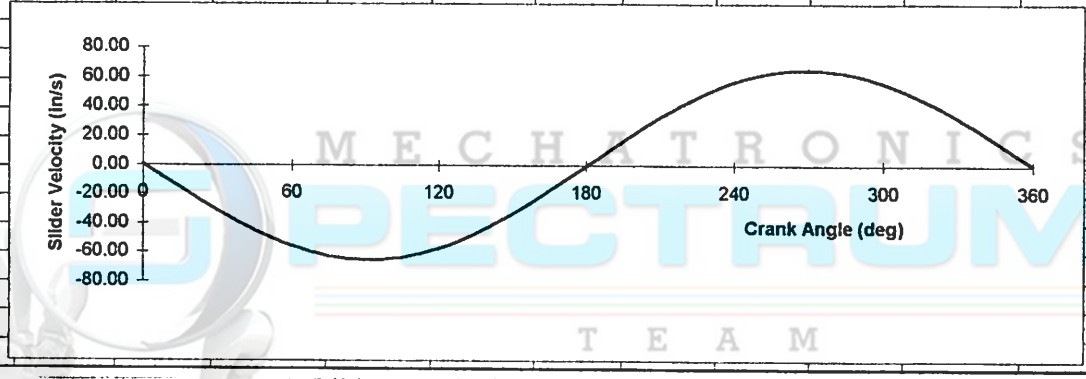
θ 2	θ 3	γ	θ 4	Δθ 4	L 1	14.0	mm
(deg)	(deg)	(deg)	(deg)	(deg)	L 2	1.0	mm
0	-169	36	12.4	1.7	L 3	16.0	mm
30	-167	39	10.7	0.0	L 4	4.0	mm
60	-164	46	11.1	0.4			
90	-163	54	13.2	2.5			
120	-163	62	16.6	5.9			
150	-164	67	20.9	10.2			
180	-166	68	24.9	14.2			
210	-168	67	27.3	16.6			
240	-169	62	27.2	16.5			
270	-171	54	24.7	14.0			
300	-171	46	20.6	9.9			
330	-171	39	16.0	5.3			
360	-169	36	12.4	1.7			



MECHATRONICS SPECTRUM TEAM

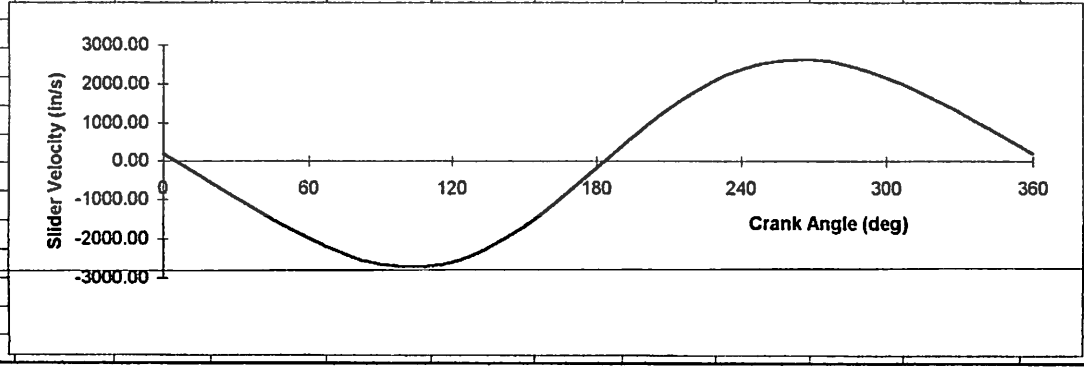
8-5

θ 2	ω 2	θ 3	γ	L 4	S 4	ω 3	V 4	α 3	a 4	L 1	1.250	in
(deg)	(rad/s)	(deg)	(deg)	(mm)	(mm)	(rad/s)	(mm/s)	(rad/s2)	(mm/s2)	L 2	3.250	in
0	20.00	0.43	179.57	170.75	156.54	-0.39	0.5	-0.4	-1325	L 3	167.500	in
30	20.00	0.98	149.02	170.29	156.08	-0.34	-31.5	3.9	-1156	ω 2(0)	20.00	rad/s
60	20.52	1.39	118.61	169.08	154.87	-0.19	-55.5	6.7	-684	α 2	0.00	rad/s2
90	21.02	1.54	88.46	167.44	153.23	0.00	-65.0	7.8	-35			
120	21.51	1.39	58.61	165.83	151.62	0.19	-57.1	6.7	616			
150	21.99	0.98	29.02	164.66	150.45	0.34	-33.5	3.9	1096			
180	22.47	0.43	-0.43	164.25	150.04	0.39	-0.5	0.0	1275			
210	22.93	-0.13	-29.87	164.68	150.47	0.34	32.6	-3.9	1105			
240	23.38	-0.54	-59.46	165.87	151.66	0.19	56.6	-6.7	633			
270	23.82	-0.68	-89.32	167.49	153.28	0.00	65.0	-7.8	-16			
300	24.26	-0.54	-119.46	169.12	154.91	-0.19	56.0	-6.7	-667			
330	24.69	-0.13	-149.87	170.31	156.10	-0.34	32.4	-3.9	-1146			
360	25.11	0.43	-180.43	170.75	156.54	-0.39	0.5	0.0	-1325			



8-6

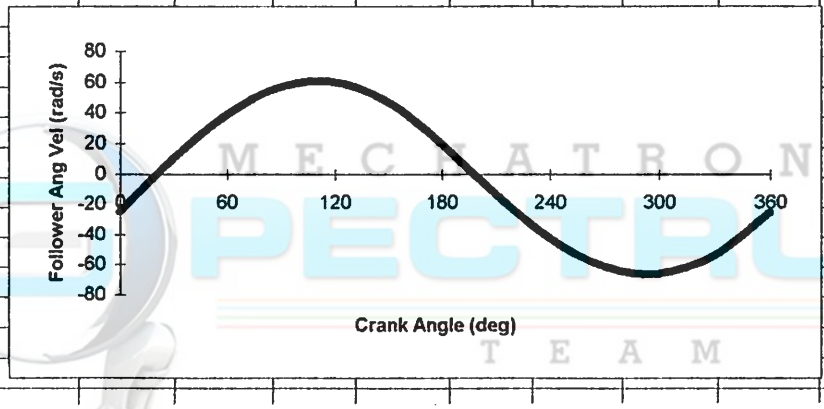
θ 2	ω 2	θ 3	γ	L 4	S 4	ω 3	V 4	α 3	a 4	L 1	30.000	mm
(deg)	(rad/s)	(deg)	(deg)	(mm)	(mm)	(rad/s)	(mm/s)	(rad/s2)	(mm/s2)	L 2	75.000	mm
0	35.00	4.10	175.90	493.93	479.72	-6.27	188.0	-3.5	-108220	L 3	420.000	mm
30	36.47	9.25	140.75	479.49	465.28	-5.48	-942.3	100.0	-102536	ω 2(0)	35.00	rad/s
60	36.96	13.07	106.93	446.63	432.42	-3.21	-1968.7	187.7	-74466	α 2	100.00	rad/s2
90	37.46	14.48	75.52	406.66	392.45	0.00	-2625.0	225.9	-31222			
120	37.94	13.07	46.93	371.63	357.42	3.21	-2577.9	206.0	15669			
150	38.42	9.25	20.75	349.59	335.38	5.48	-1682.7	131.4	54481			
180	38.90	4.10	-4.10	343.93	329.72	6.27	-188.0	20.7	74805			
210	39.37	-1.02	-28.98	354.98	340.77	5.41	1353.1	-94.4	70301			
240	39.83	-4.77	-55.23	381.04	366.83	3.14	2382.9	-182.0	41957			
270	40.29	-6.15	-83.85	417.58	403.37	0.00	2625.0	-220.0	-2401			
300	40.74	-4.77	-115.23	456.04	441.83	-3.14	2163.7	-199.9	-50544			
330	41.19	-1.02	-148.98	484.88	470.67	-5.41	1271.9	-125.4	-89063			
360	41.63	4.10	-184.10	493.93	479.72	-6.27	188.0	-15.1	-107871			



MECHATRONICS SPECTRUM TEAM

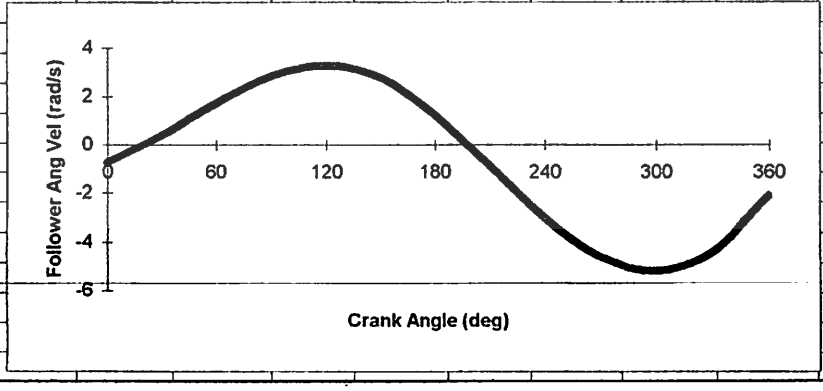
8-7

θ 2 (deg)	ω2 (rad/s)	θ 3 (deg)	γ (deg)	θ 4 (deg)	Δθ 4 (deg)	ω3 (rad/s)	ω4 (rad/s)	L 1	9.0	in
								L 2	1.0	in
0	200	-162	46	25.1	14.4	11.7	-25.0	L 3	10.0	in
30	200	-158	49	21.7	11.0	-3.8	9.9	L 4	3.5	in
60	200	-154	56	21.7	11.0	-15.0	38.9	ω 2(0)	200	rad/s
90	200	-153	64	24.8	14.1	-20.1	56.6	α 2	0	rad/s2
120	200	-154	72	30.6	19.9	-21.0	59.8			
150	200	-157	78	38.1	27.4	-19.0	46.8			
180	200	-160	80	45.3	34.6	-14.4	20.0			
210	200	-163	78	49.7	39.0	-6.9	-12.8			
240	200	-165	72	49.9	39.2	3.7	-42.2			
270	200	-166	64	45.9	35.2	15.4	-61.5			
300	200	-166	56	39.1	28.4	23.9	-66.3			
330	200	-165	49	31.4	20.7	23.2	-53.9			
360	200	-162	46	25.1	14.4	11.7	-25.0			



8-8

θ 2 (deg)	ω2 (rad/s)	θ 3 (deg)	γ (deg)	θ 4 (deg)	Δθ 4 (deg)	ω3 (rad/s)	ω4 (rad/s)	L 1	360.0	mm
								L 2	40.0	mm
0	6.0	-162	46	25.1	14.4	0.35	-0.75	L 3	400.0	mm
30	7.5	-158	49	21.7	11.0	-0.14	0.37	L 4	140.0	mm
60	8.8	-154	56	21.7	11.0	-0.66	1.72	ω 2(0)	6	rad/s
90	9.9	-153	64	24.8	14.1	-1.00	2.81	α 2	20	rad/s2
120	10.9	-154	72	30.6	19.9	-1.15	3.27			
150	11.9	-157	78	38.1	27.4	-1.13	2.77			
180	12.7	-160	80	45.3	34.6	-0.92	1.27			
210	13.5	-163	78	49.7	39.0	-0.47	-0.86			
240	14.3	-165	72	49.9	39.2	0.26	-3.01			
270	15.0	-166	64	45.9	35.2	1.16	-4.60			
300	15.7	-166	56	39.1	28.4	1.87	-5.20			
330	16.3	-165	49	31.4	20.7	1.90	-4.40			
360	17.0	-162	46	25.1	14.4	0.99	-2.12			



MECHATRONICS SPECTRUM TEAM

8-9.

SLIDER-CRANK KINEMATICS (length, deg, length/sec, rad/sec)

*** Crank ***			*** Coupler ***			*** Slider ***			Transmission
Angle	Speed	Accel.	Angle	Speed	Accel	Length	Speed	Accel	Angle
0	4.00	0.0	3.28	-0.57	-0.55	59.9	1.7	-136	176.7
30	4.00	0.0	7.39	-0.50	1.18	58.6	-11.6	-125	142.6
60	4.00	0.0	10.42	-0.29	2.03	55.4	-23.2	-84	109.6
90	4.00	0.0	11.54	0.00	2.33	51.4	-30.0	-24	78.5
120	4.00	0.0	10.42	0.29	2.03	47.9	-28.7	36	49.6
150	4.00	0.0	7.39	0.50	1.18	45.6	-18.4	83	22.6
180	4.00	0.0	3.28	0.57	0.02	44.9	-1.7	103	-3.3
210	4.00	0.0	-0.82	0.49	-1.15	46.0	15.4	90	-29.2
240	4.00	0.0	-3.82	0.29	-1.99	48.6	27.0	49	-56.2
270	4.00	0.0	-4.92	0.00	-2.29	52.3	30.0	-10	-85.1
300	4.00	0.0	-3.82	-0.29	-1.99	56.1	25.0	-71	-116.2
330	4.00	0.0	-0.82	-0.49	-1.15	59.0	14.6	-118	-149.2
360	4.00	0.0	3.28	-0.57	0.02	59.9	1.7	-137	-183.3
Maximum		0.0	11.5	0.6	2.3	59.9	30.0	102.8	176.7
Minimum		0.0	-4.9	-0.6	-2.3	44.9	-30.0	-135.5	-149.2

8-10.

SLIDER-CRANK KINEMATICS (length, deg, length/sec, rad/sec)

*** Crank ***			*** Coupler ***			*** Slider ***			Transmission
Angle	Speed	Accel.	Angle	Speed	Accel	Length	Speed	Accel	Angle
0	10.00	10.0	4.37	-1.80	-1.55	617.5	71.8	-11026	175.6
30	10.51	10.0	9.54	-1.65	8.92	599.1	-350.2	-11653	140.5
60	11.00	10.0	13.37	-1.01	18.60	557.8	-772.4	-9280	106.6
90	11.46	10.0	14.79	0.00	24.34	507.6	-1077.6	-4201	75.2
120	11.91	10.0	13.37	1.10	23.82	463.8	-1102.8	2349	46.6
150	12.34	10.0	9.54	1.94	16.04	436.3	-749.0	8588	20.5
180	12.76	10.0	4.37	2.29	2.20	429.5	-91.7	12470	-4.4
210	13.16	10.0	-0.76	2.04	-14.02	443.5	633.0	12292	-29.2
240	13.56	10.0	-4.52	1.22	-27.80	476.4	1154.0	7525	-55.5
270	13.94	10.0	-5.90	0.00	-34.97	522.2	1310.1	-948	-84.1
300	14.31	10.0	-4.52	-1.28	-32.87	570.4	1111.6	-11033	-115.5
330	14.67	10.0	-0.76	-2.27	-20.89	606.4	673.5	-19911	-149.2
360	15.02	10.0	4.37	-2.70	-1.24	617.5	107.9	-24972	-184.4
Maximum		10.0	14.8	2.3	24.3	617.5	1310.1	12469.8	175.6
Minimum		10.0	-5.9	-2.3	-35.0	429.5	-1102.8	-19910.8	-149.2

MECHANISMS SPECTRUM TEAM

8-11.

FOUR BAR KINEMATICS (deg, rad/sec)

*** Crank ***			*** Coupler ***		*** Follower ***		Transmission
Angle	Speed	Accel.	Angle	Speed	Angle	Speed	Angle
0	150.0	0.0	18.5	8.8	25.1	-18.8	46.4
30	150.0	0.0	22.5	-2.9	21.7	7.4	49.1
60	150.0	0.0	25.6	-11.2	21.7	29.2	55.9
90	150.0	0.0	26.7	-15.1	24.8	42.4	64.4
120	150.0	0.0	25.7	-15.7	30.6	44.9	72.3
150	150.0	0.0	23.2	-14.2	38.1	35.1	77.9
180	150.0	0.0	20.2	-10.8	45.3	15.0	79.9
210	150.0	0.0	17.4	-5.2	49.7	-9.6	77.9
240	150.0	0.0	15.3	2.8	49.9	-31.7	72.3
270	150.0	0.0	14.1	11.6	45.9	-46.1	64.4
300	150.0	0.0	14.0	17.9	39.1	-49.8	55.9
330	150.0	0.0	15.4	17.4	31.4	-40.4	49.1
360	150.0	0.0	18.5	8.8	25.1	-18.8	46.4
Maximum		0.0	26.7	17.9	49.9	44.9	79.9
Minimum		0.0	14.0	-15.7	21.7	-49.8	46.4

8-12.

FOUR BAR KINEMATICS (deg, rad/sec)

*** Crank ***			*** Coupler ***		*** Follower ***		Transmission
Angle	Speed	Accel.	Angle	Speed	Angle	Speed	Angle
0	360.0	20.0	16.4	77.06	18.0	-154.29	21.8
30	360.0	20.0	32.3	-59.41	11.1	-16.04	30.2
60	360.1	20.0	42.8	-93.24	13.0	82.34	46.6
90	360.1	20.0	43.9	-97.02	20.4	163.10	63.5
120	360.1	20.0	39.0	-94.38	34.0	204.42	78.1
150	360.1	20.0	31.6	-92.05	53.5	178.34	88.3
180	360.2	20.0	24.2	-88.46	72.6	83.12	92.0
210	360.2	20.0	18.1	-74.86	83.9	-41.95	88.3
240	360.2	20.0	13.5	-39.69	84.8	-150.07	78.1
270	360.3	20.0	10.5	24.32	76.4	-222.67	63.5
300	360.3	20.0	8.8	110.38	60.1	-260.26	46.6
330	360.3	20.0	9.4	170.63	38.0	-255.34	30.2
360	360.3	20.0	16.4	77.14	18.0	-154.44	21.8
Maximum		20.0	43.9	170.6	84.8	204.4	92.0
Minimum		20.0	8.8	-97.0	11.1	-260.3	21.8

MECHATRONICS SPECTRUM TEAM

Answers to the Chapter 8 Case Study Questions:

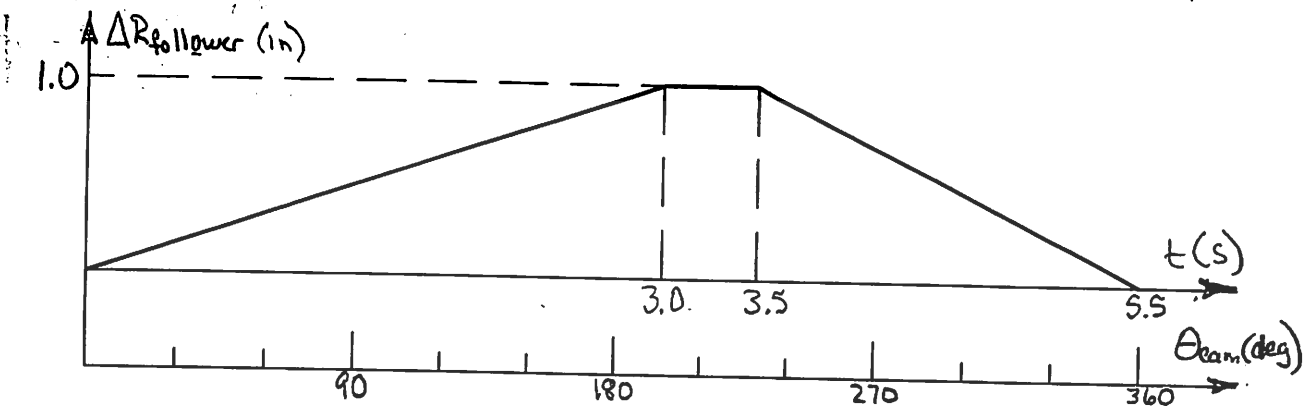
Case 8-1.

1. As slide bar E is pulled to the left, link D rotates clockwise.
2. As slide bar E is pulled to the left, slide block I moves upward.
3. As pulley J is rotated, being eccentric from the center of rotates crank pin C will move along a circular path.
4. As pulley J rotates, slide bar E will rotate about it's about axis of shaft A.
5. Moving slide bar E to the left will increase the eccentricity of crank pin C. Crank pin C will move along a circular path of a larger radius.
6. If sleeve F is keyed to the housing, it remains stationary encasing shaft A.
7. Item G is an enveloping gear that mates with a worm gear.
8. As item G rotates, the threads thrust sleeve F either left or right.
9. As item G rotates, slide bar E is moved either left or right, depending on the direction of rotation of G.
10. This mechanism can be used to adjust the crank length, which the shaft is spinning.

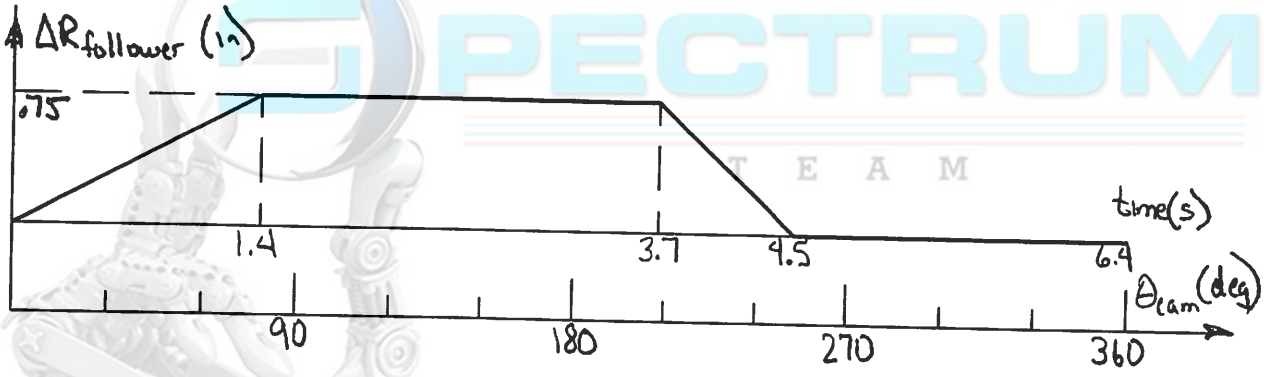


MECHATRONICS
SPECTRUM
TEAM

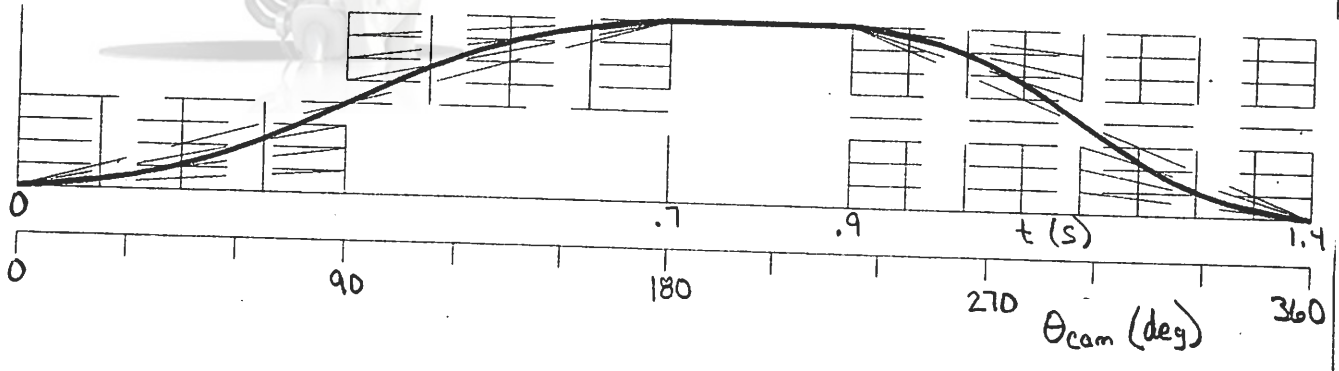
9-1 $\omega_{cam} = \frac{1\,rev}{\Sigma T_i} = \frac{1\,rev}{5.5\,sec} = .182\,\frac{rev}{s} = 10.91\,rpm$



9-2 $\omega_{cam} = \frac{1\,rev}{\Sigma T_i} = \frac{1\,rev}{6.4\,s} = .156\,rev/s = 9.38\,rpm$

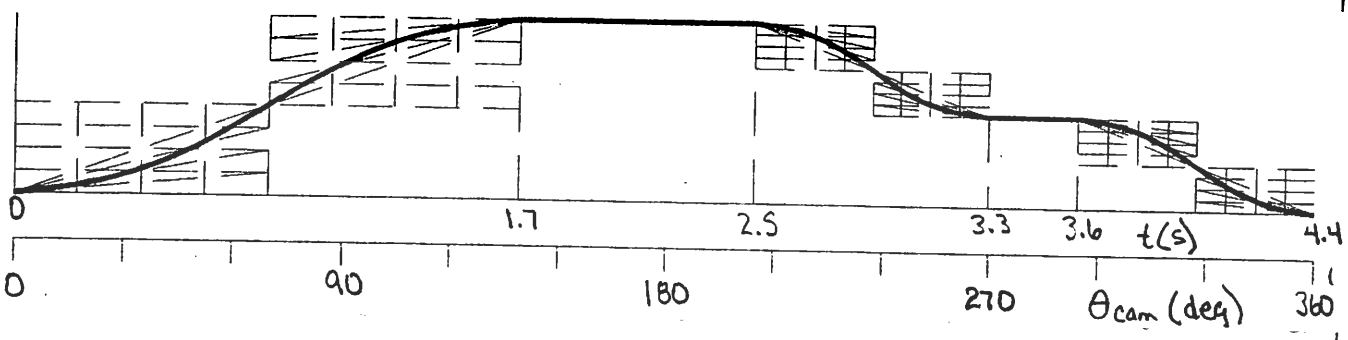


9-3 $\omega_{cam} = \frac{1\,rev}{\Sigma T_i} = \frac{1\,rev}{1.4\,sec} = 0.714\,\frac{rev}{sec} = 42.8\,rpm$



MECHATRONICS SPECTRUM TEAM

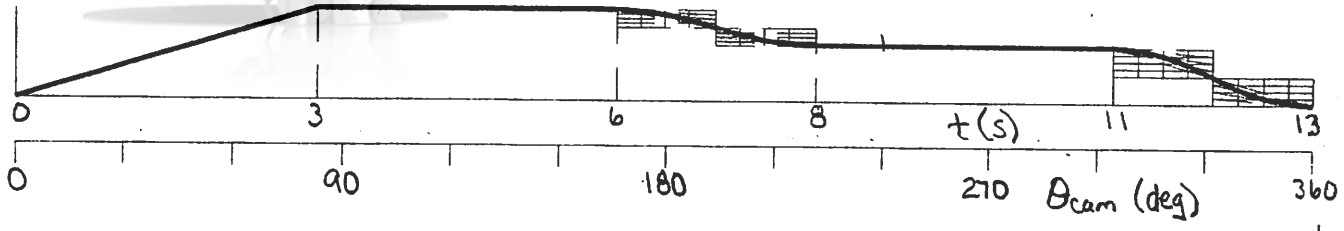
9-4 $\omega_{cam} = \frac{1\,rev}{\Sigma T_i} = \frac{1\,rev}{4.4s} = .227\,rev/s = 13.6\,rpm$



9-5

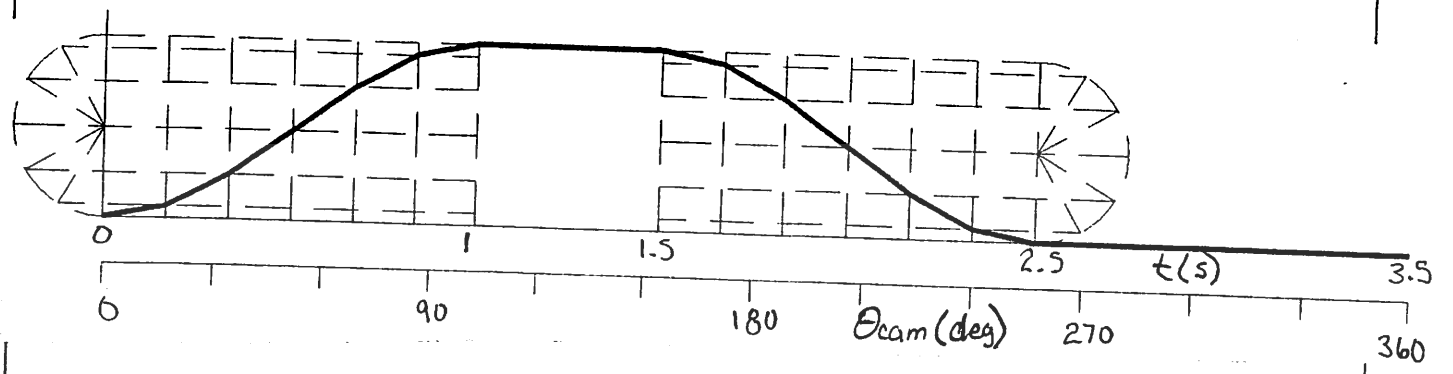
MECHATRONICS SPECTRUM TEAM

9-6 $\omega_{cam} = \frac{1\,rev}{13\,sec} = .077\,rev/sec = 4.6\,rpm$

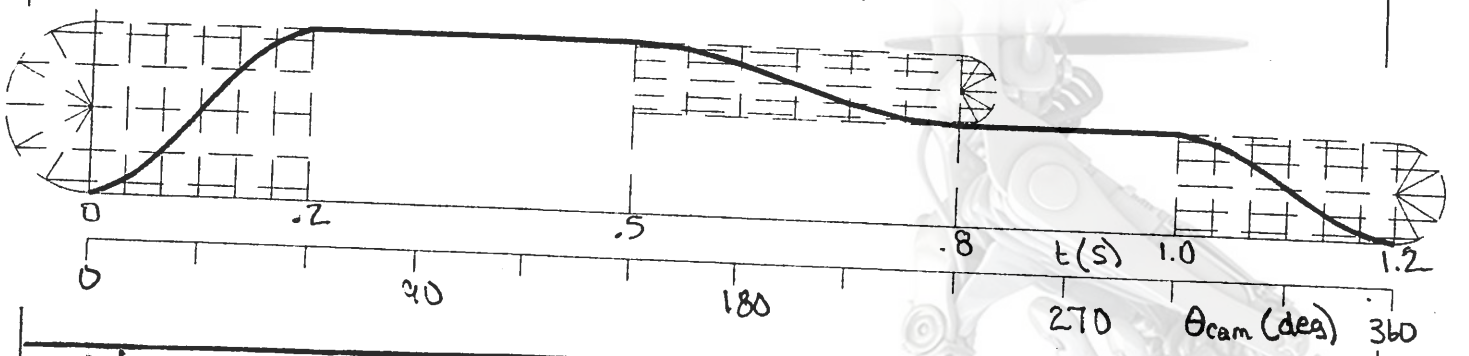


National Brand
13-782 500 SHEETS, FILLER 5 SQUARE
42-381 50 SHEETS EYE-EASE® 5 SQUARE
42-382 100 SHEETS EYE-EASE® 5 SQUARE
42-389 200 SHEETS EYE-EASE® 5 SQUARE
42-392 100 RECYCLED WHITE 5 SQUARE
42-399 200 RECYCLED WHITE 5 SQUARE
Made in U.S.A.

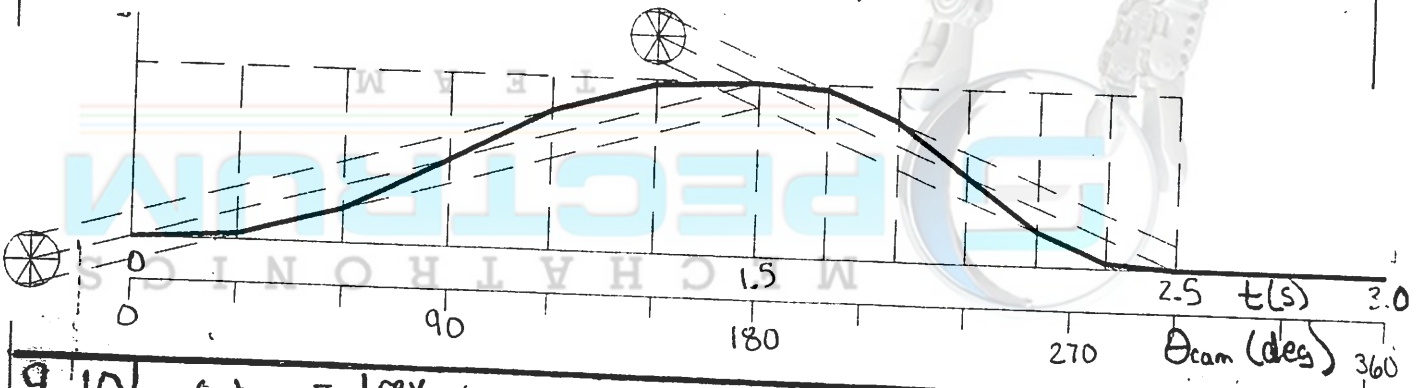
9-7 $\omega_{cam} = \frac{1\ rev}{3.5\ sec} = 17.1\ rpm$



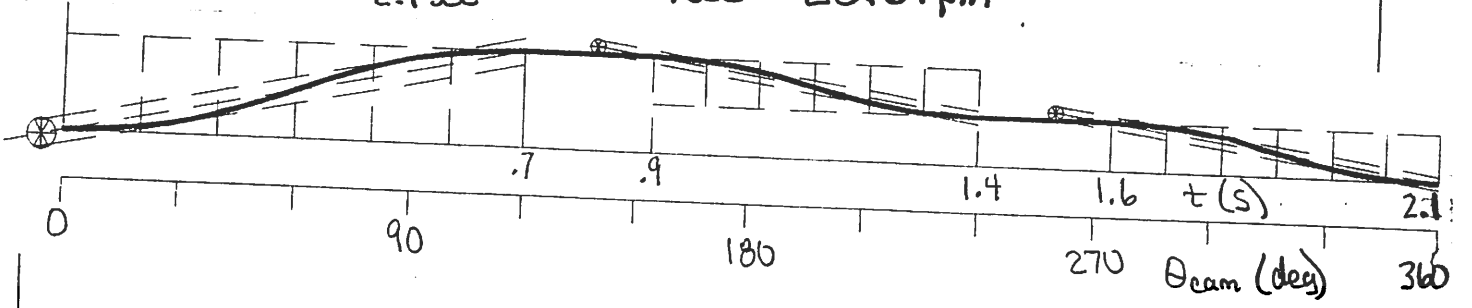
9-8 $\omega_{cam} = \frac{1\ rev}{1.2\ sec} = .833\ rev/sec = 50\ rpm$



9-9 $\omega_{cam} = \frac{1\ rev}{3\ sec} = .333\ rev/s = 20\ rpm$



9-10 $\omega_{cam} = \frac{1\ rev}{2.1\ sec} = .476\ rev/sec = 28.6\ rpm$

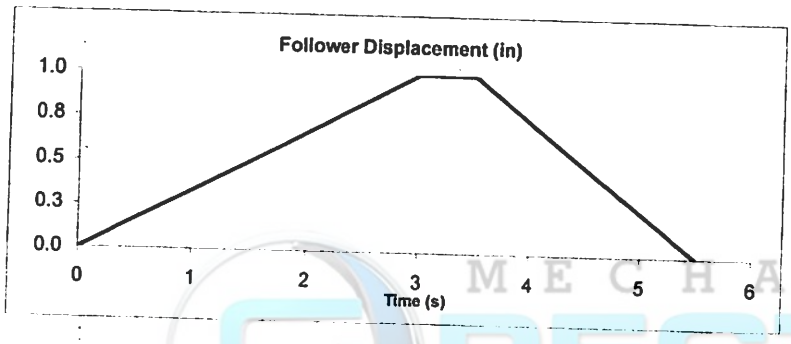


9-14

Since fall occurs in less time, max vel will occur then

$V_{max} = H/T = 1/2 = .5\ in/s$

$a_{max} = \infty$

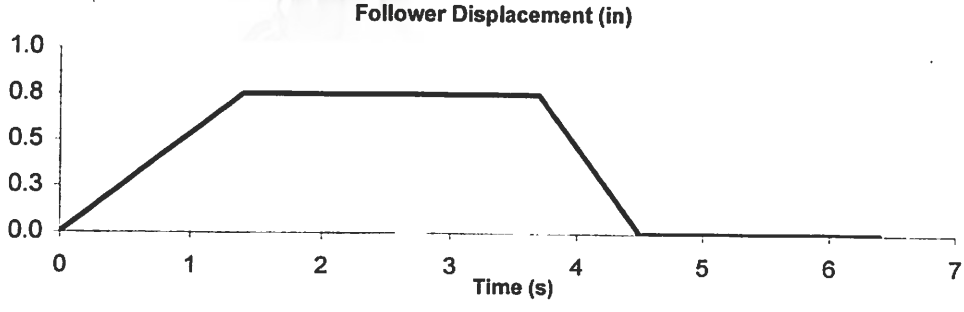


time (s)	ΔR (in)
0	0
0.0	0.000
0.5	0.167
1.0	0.333
1.5	0.500
2.0	0.667
2.5	0.833
3.0	1.000
3.0	1.000
3.0	1.000
3.5	1.000
3.5	1.000
3.5	1.000
4.0	0.750
4.5	0.500
5.0	0.250
5.5	0.000
5.5	0.000

9-15

Since fall occurs in less time, max vel will occur then

$V_{max} = H/T = .75/.8 = .938\ m/s$



time (s)	ΔR (in)
0	0
0.0	0.000
0.4	0.188
0.7	0.375
1.1	0.563
1.4	0.750
1.4	0.750
1.4	0.750
3.7	0.750
3.7	0.750
3.7	0.750
3.9	0.563
4.1	0.375
4.3	0.188
4.5	0.000
4.5	0.000
4.5	0.000
6.4	0.000

13-782 42-381 42-382 42-389 42-390 42-399 500 SHEETS, FILLER 5 SQUARE 50 SHEETS EYE-EASE® 5 SQUARE 100 SHEETS EYE-EASE® 5 SQUARE 200 SHEETS EYE-EASE® 5 SQUARE 100 RECYCLED WHITE 5 SQUARE 200 RECYCLED WHITE 5 SQUARE Made in U.S.A. National® Brand

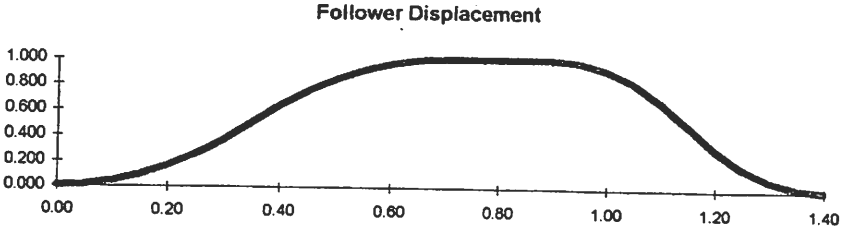
MECHATRONICS SPECTRUM TEAM

9-16

Since fall occurs in less time, max vel & accel will occur then

$V_{max} = 2H/T = 2(1in)/.5sec = 4 in/s$

$a_{max} = 4H/T^2 = 4(1in)/(.5sec)^2 = 16 in/s^2$



time (sec)	s (mm)
0.00	0.000
0.05	0.010
0.10	0.041
0.15	0.092
0.20	0.163
0.25	0.255
0.30	0.367
0.35	0.500
0.35	0.500
0.40	0.633
0.45	0.745
0.50	0.837
0.55	0.908
0.60	0.959
0.65	0.990
0.70	1.000
0.70	1.000
0.90	1.000
0.90	1.000
0.95	0.980
1.00	0.920
1.05	0.820
1.10	0.680
1.15	0.500
1.15	0.500
1.20	0.320
1.25	0.180
1.30	0.080
1.35	0.020
1.40	0.000
1.40	0.000

9-17

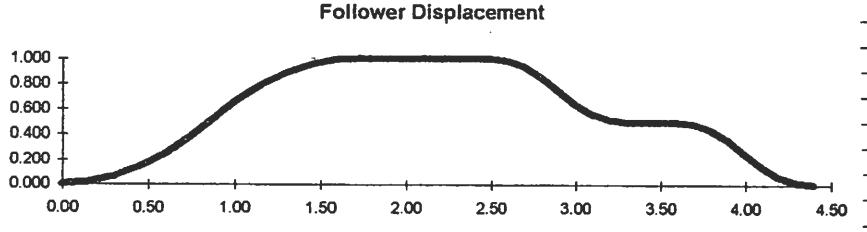
Both fall sequences are identical & exhibit max vel & accel

$V_{max} = 2H/T = 2(.5in)/(.8sec)$

$= 1.25 in/s$

$a_{max} = 4H/T^2 = 4(.5in)/(.8sec)^2$

$= 3.13 in/s^2$



time (sec)	s (mm)
0.00	0.000
0.15	0.016
0.30	0.062
0.45	0.140
0.60	0.249
0.75	0.389
0.85	0.500
0.85	0.500
1.00	0.661
1.15	0.791
1.30	0.889
1.45	0.957
1.60	0.993
1.70	1.000
1.70	1.000
2.50	1.000
2.50	1.000
2.60	0.984
2.70	0.938
2.80	0.859
2.90	0.750
2.90	0.750
3.00	0.641
3.10	0.563
3.20	0.516
3.30	0.500
3.30	0.500
3.60	0.500
3.60	0.500
3.70	0.484
3.80	0.438
3.90	0.359
4.00	0.250
4.00	0.250
4.10	0.141
4.20	0.063
4.30	0.016
4.40	0.000
4.40	0.000

MECHATRONICS SPECTRUM TEAM

500 SHEETS, FILLER 5 SQUARE
50 SHEETS EYE-EASE 5 SQUARE
100 SHEETS EYE-EASE 5 SQUARE
200 SHEETS EYE-EASE 5 SQUARE
100 RECYCLED WHITE 5 SQUARE
200 RECYCLED WHITE 5 SQUARE
13-782
42-381
42-382
42-385
42-392
42-399
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9-18

13-782 500 SHEETS, FILLER 5 SQUARE
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42-392 100 RECYCLED WHITE 5 SQUARE
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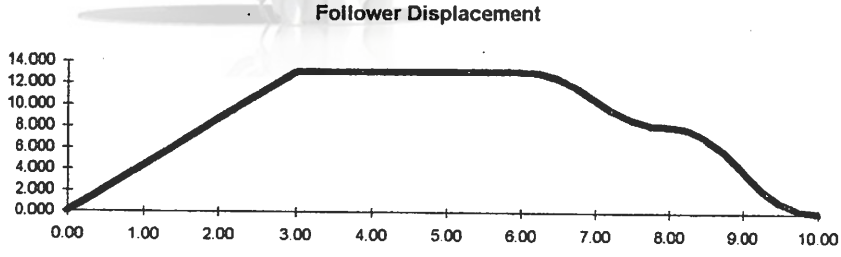
MECHATRONICS SPECTRUM TEAM

9-19

max vel & accel occur
In the final fall sequence

$V_{max} = 2H/T = 2(8mm)/2sec = 8mm/s$

$a_{max} = 4H/T^2 = 4(8mm)/(2sec)^2 = 8mm/s^2$



time (sec)	s (mm)
0.00	0.000
0.00	0.000
0.50	2.167
1.00	4.333
1.50	6.500
2.00	8.667
2.50	10.833
3.00	13.000
3.00	13.000
3.00	13.000
6.00	13.000
6.00	13.000
6.25	12.844
6.50	12.375
6.75	11.594
7.00	10.500
7.00	10.500
7.25	9.406
7.50	8.625
7.75	8.156
8.00	8.000
8.00	8.000
8.25	7.750
8.50	7.000
8.75	5.750
9.00	4.000
9.00	4.000
9.25	2.250
9.50	1.000
9.75	0.250
10.00	0.000
10.00	0.000

9-20

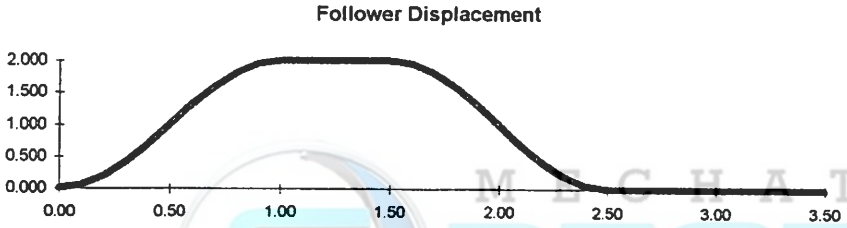
Rise and fall exhibit identical kinematics

$V_{max} = \pi H / 2T = \pi(2\text{ in}) / 2(1\text{ sec})$

$= 3.14\text{ in/s}$

$a_{max} = \pi^2 H / 2T^2 = \pi(2\text{ in}) / 2(1\text{ sec})^2$

$= 9.87\text{ in/s}^2$



time (sec)	s (mm)
0.00	0.000
0.10	0.049
0.20	0.191
0.30	0.412
0.40	0.691
0.50	1.000
0.60	1.309
0.70	1.588
0.80	1.809
0.90	1.951
1.00	2.000
1.00	2.000
1.50	2.000
1.50	2.000
1.60	1.951
1.70	1.809
1.80	1.588
1.90	1.309
2.00	1.000
2.10	0.691
2.20	0.412
2.30	0.191
2.40	0.049
2.50	0.000
2.50	0.000
3.50	0.000

9-21

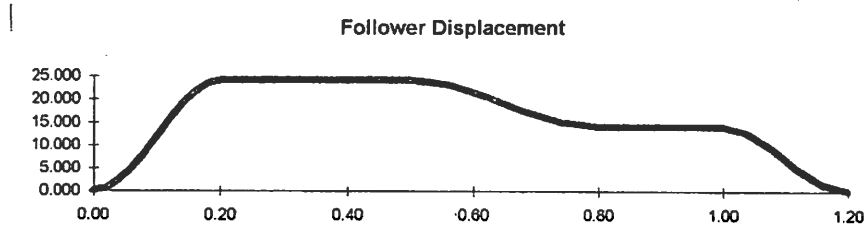
Max velocity occurs in first sequence

$V_{max} = \pi H / 2T = \pi(24\text{mm}) / 2(.2\text{sec})$

$= 188.5\text{ mm/s}$

$a_{max} = \pi^2 H / 2T^2 = \pi^2(24\text{mm}) / 2(.2\text{sec})^2$

$= 2961\text{ mm/s}^2$



time (sec)	s (mm)
0.00	0.000
0.02	0.587
0.04	2.292
0.06	4.947
0.08	8.292
0.10	12.000
0.12	15.708
0.14	19.053
0.16	21.708
0.18	23.413
0.20	24.000
0.20	24.000
0.50	24.000
0.50	24.000
0.56	23.045
0.62	20.545
0.68	17.455
0.74	14.955
0.80	14.000
0.80	14.000
1.00	14.000
1.00	14.000
1.04	12.663
1.08	9.163
1.12	4.837
1.16	1.337
1.20	0.000
1.20	0.000

MECHATRONICS SPECTRUM TEAM

9-22

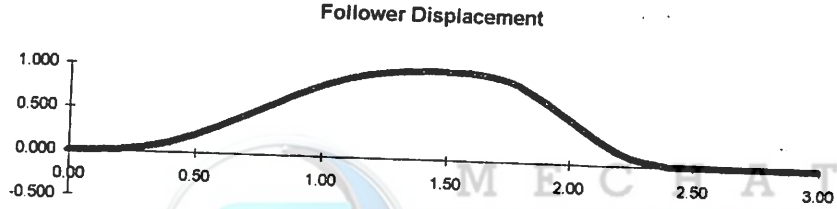
Max velocity is during fall sequence

$V_{max} = 2H/T = 2(1\,in)/1\,sec$

$= 2\ in/s$

$a_{max} = 2\pi H/T^2 = 2\pi(1\,in)/(1\,sec)^2$

$= 6.28\ in/s^2$



time (sec)	s (mm)
0.00	0.000
0.10	0.002
0.20	0.015
0.30	0.049
0.40	0.108
0.50	0.196
0.60	0.306
0.70	0.434
0.80	0.566
0.90	0.694
1.00	0.804
1.10	0.892
1.20	0.951
1.30	0.985
1.40	0.998
1.50	1.000
1.60	0.994
1.70	0.951
1.75	0.909
1.80	0.851
1.90	0.694
2.00	0.500
2.10	0.306
2.20	0.149
2.25	0.091
2.30	0.049
2.40	0.006
2.50	0.000
3.00	0.000

9-23

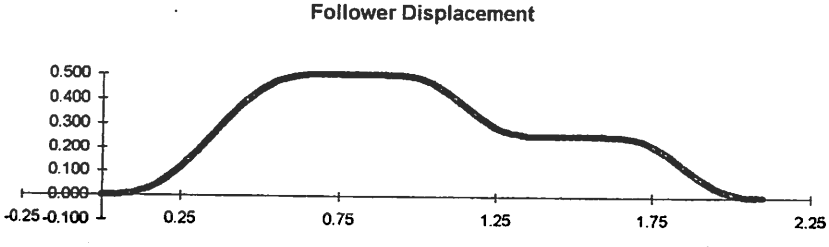
Max velocity is during rise sequence

$V_{max} = 2H/T = 2(.5\,in)/.7\,sec$

$= 1.43\ in/s$

$a_{max} = 2\pi H/T^2 = 2\pi(.5\,in)/(.7\,sec)^2$

$= 6.41\ in/s^2$

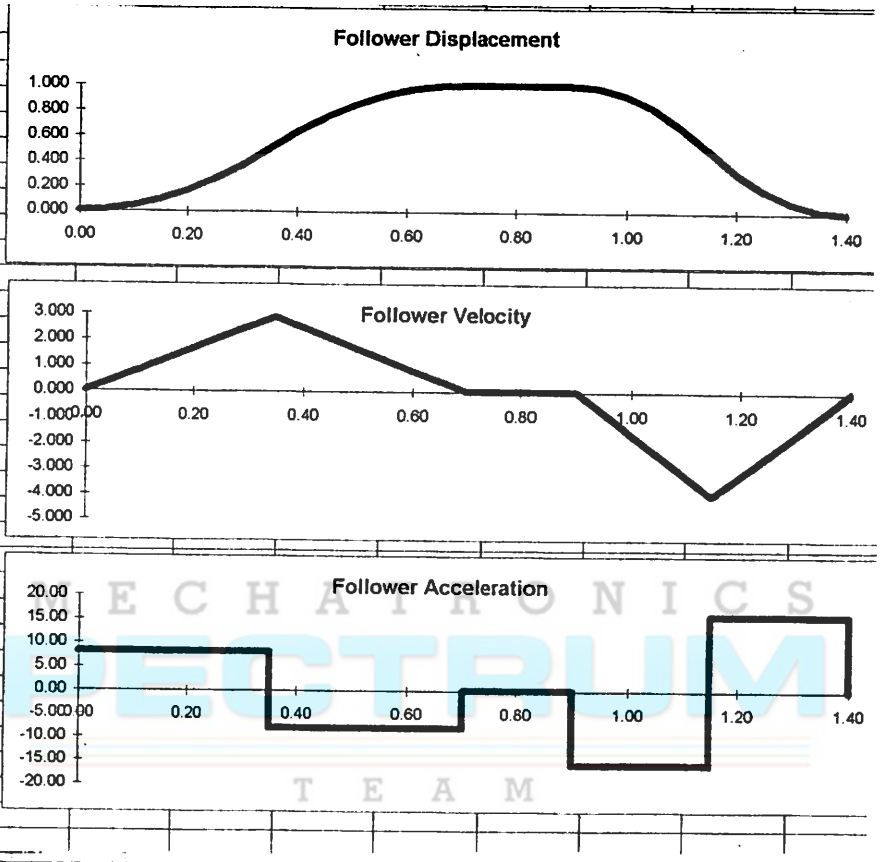


time (sec)	s (mm)
0.00	0.000
0.05	0.001
0.10	0.009
0.15	0.030
0.18	0.045
0.20	0.065
0.25	0.116
0.30	0.180
0.35	0.250
0.40	0.320
0.45	0.384
0.50	0.435
0.53	0.455
0.55	0.470
0.60	0.491
0.65	0.499
0.70	0.500
0.90	0.500
0.95	0.498
1.00	0.488
1.03	0.477
1.05	0.463
1.10	0.423
1.15	0.375
1.20	0.327
1.25	0.287
1.28	0.273
1.30	0.262
1.35	0.252
1.40	0.250
1.60	0.250
1.65	0.248
1.70	0.238
1.73	0.227
1.75	0.213
1.80	0.173
1.85	0.125
1.90	0.077
1.95	0.037
1.98	0.023
2.00	0.012
2.05	0.002
2.10	0.000

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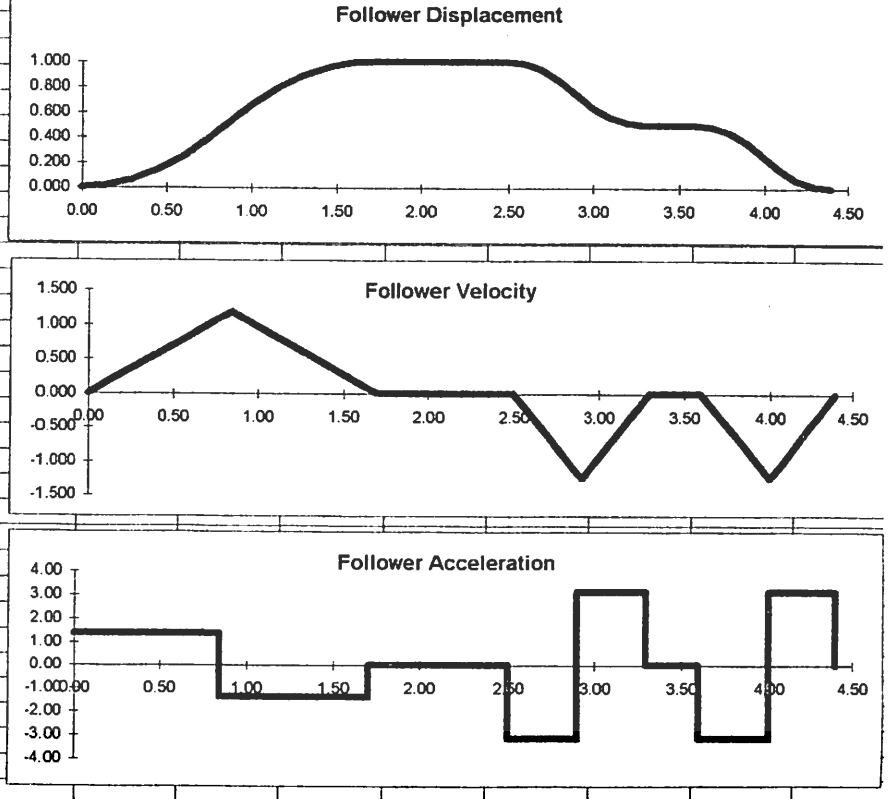
9-29

time (sec)	s (mm)	v (mm/s)	a (mm/s^2)
0.00	0.000	0.000	8.16
0.05	0.010	0.408	8.16
0.10	0.041	0.816	8.16
0.15	0.092	1.224	8.16
0.20	0.163	1.633	8.16
0.25	0.255	2.041	8.16
0.30	0.367	2.449	8.16
0.35	0.500	2.857	8.16
0.35	0.500	2.857	-8.16
0.40	0.633	2.449	-8.16
0.45	0.745	2.041	-8.16
0.50	0.837	1.633	-8.16
0.55	0.908	1.224	-8.16
0.60	0.959	0.816	-8.16
0.65	0.990	0.408	-8.16
0.70	1.000	0.000	-8.16
0.70	1.000	0.000	0.00
0.90	1.000	0.000	0.00
0.90	1.000	0.000	-16.00
0.95	0.980	-0.800	-16.00
1.00	0.920	-1.600	-16.00
1.05	0.820	-2.400	-16.00
1.10	0.680	-3.200	-16.00
1.15	0.500	-4.000	-16.00
1.15	0.500	-4.000	16.00
1.20	0.320	-3.200	16.00
1.25	0.180	-2.400	16.00
1.30	0.080	-1.600	16.00
1.35	0.020	-0.800	16.00
1.40	0.000	0.000	16.00
1.40	0.000	0.000	0.00



9-30

time (sec)	s (mm)	v (mm/s)	a (mm/s^2)
0.00	0.000	0.000	1.38
0.15	0.016	0.208	1.38
0.30	0.062	0.415	1.38
0.45	0.140	0.623	1.38
0.60	0.249	0.830	1.38
0.75	0.389	1.038	1.38
0.85	0.500	1.176	1.38
0.85	0.500	1.176	-1.38
1.00	0.661	0.969	-1.38
1.15	0.791	0.761	-1.38
1.30	0.889	0.554	-1.38
1.45	0.957	0.346	-1.38
1.60	0.993	0.138	-1.38
1.70	1.000	0.000	-1.38
1.70	1.000	0.000	0.00
2.50	1.000	0.000	0.00
2.50	1.000	0.000	-3.13
2.60	0.984	-0.313	-3.13
2.70	0.938	-0.625	-3.13
2.80	0.859	-0.938	-3.13
2.90	0.750	-1.250	-3.13
2.90	0.750	-1.250	3.13
3.00	0.641	-0.938	3.13
3.10	0.563	-0.625	3.13
3.20	0.516	-0.312	3.13
3.30	0.500	0.000	3.13
3.30	0.500	0.000	0.00
3.60	0.500	0.000	0.00
3.60	0.500	0.000	-3.13
3.70	0.484	-0.313	-3.13
3.80	0.438	-0.625	-3.13
3.90	0.359	-0.938	-3.13
4.00	0.250	-1.250	-3.13
4.00	0.250	-1.250	3.13
4.10	0.141	-0.938	3.13
4.20	0.063	-0.625	3.13
4.30	0.016	-0.313	3.13
4.40	0.000	0.000	3.13
4.40	0.000	0.000	0.00



MECHATRONICS SPECTRUM TEAM

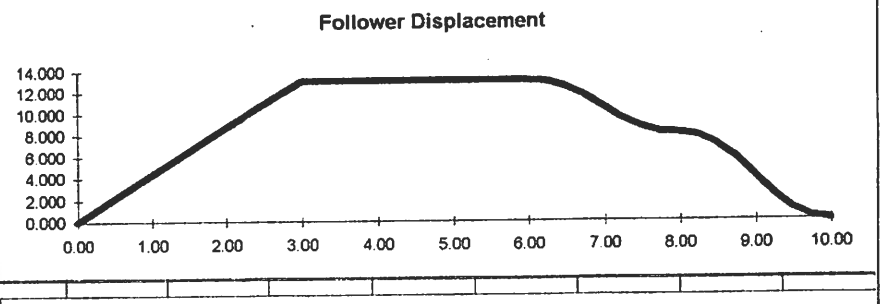
9-31

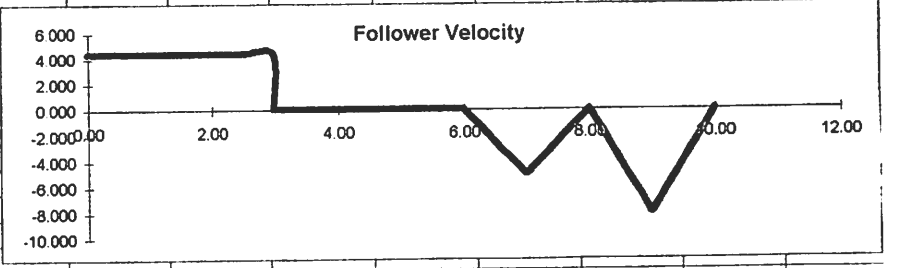
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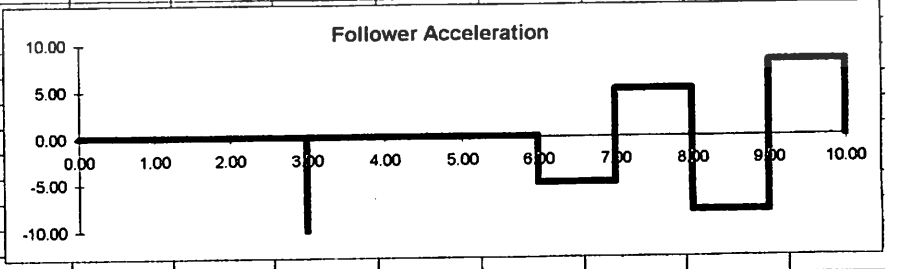
MECHATRONICS
SPECTRUM
TEAM

9-32

time	s	v	a
(sec)	(mm)	(mm/s)	(mm/s^2)
0.00	0.000	0.000	10.00
0.00	0.000	4.333	0.00
0.50	2.167	4.333	0.00
1.00	4.333	4.333	0.00
1.50	6.500	4.333	0.00
2.00	8.667	4.333	0.00
2.50	10.833	4.333	0.00
3.00	13.000	4.333	0.00
3.00	13.000	0.000	-10.00
3.00	13.000	0.000	0.00
6.00	13.000	0.000	0.00
6.00	13.000	0.000	-5.00
6.25	12.844	-1.250	-5.00
6.50	12.375	-2.500	-5.00
6.75	11.594	-3.750	-5.00
7.00	10.500	-5.000	-5.00
7.00	10.500	-5.000	5.00
7.25	9.406	-3.750	5.00
7.50	8.625	-2.500	5.00
7.75	8.156	-1.250	5.00
8.00	8.000	0.000	5.00
8.00	8.000	0.000	-8.00
8.25	7.750	-2.000	-8.00
8.50	7.000	-4.000	-8.00
8.75	5.750	-6.000	-8.00
9.00	4.000	-8.000	-8.00
9.00	4.000	-8.000	8.00
9.25	2.250	-6.000	8.00
9.50	1.000	-4.000	8.00
9.75	0.250	-2.000	8.00
10.00	0.000	0.000	8.00
10.00	0.000	0.000	0.00

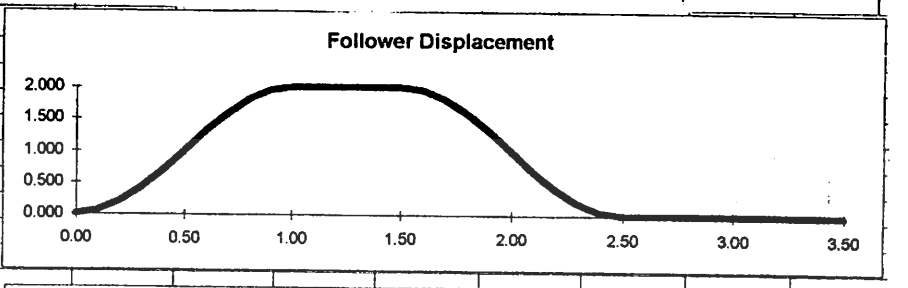


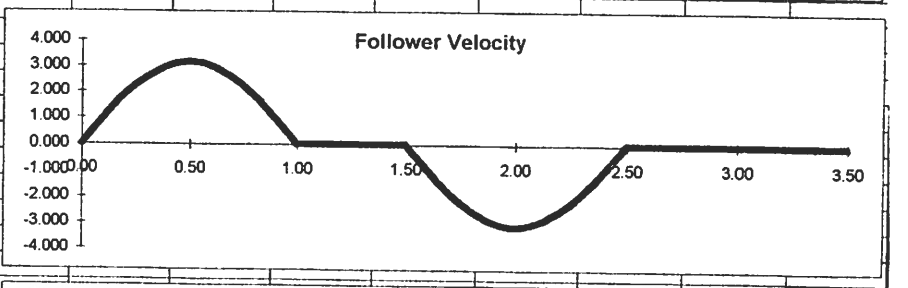


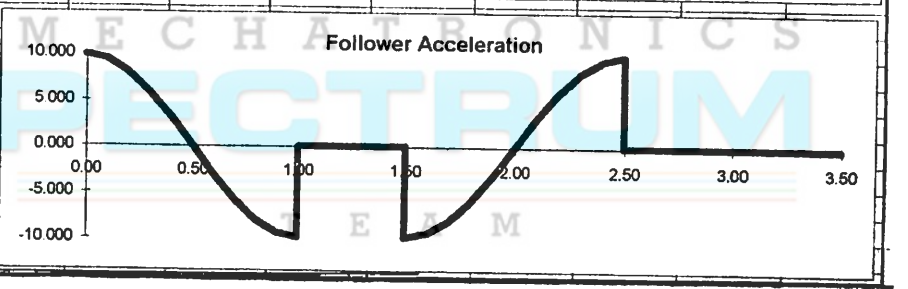


9-33

time (sec)	s (mm)	v (mm/s)	a (mm/s²)
0.00	0.000	0.000	9.870
0.10	0.049	0.971	9.387
0.20	0.191	1.847	7.985
0.30	0.412	2.542	5.801
0.40	0.691	2.988	3.050
0.50	1.000	3.142	0.000
0.60	1.309	2.988	-3.050
0.70	1.588	2.542	-5.801
0.80	1.809	1.847	-7.985
0.90	1.951	0.971	-9.387
1.00	2.000	0.000	-9.870
1.00	2.000	0.000	0.000
1.50	2.000	0.000	0.000
1.50	2.000	0.000	-9.870
1.60	1.951	-0.971	-9.387
1.70	1.809	-1.847	-7.985
1.80	1.588	-2.542	-5.801
1.90	1.309	-2.988	-3.050
2.00	1.000	-3.142	0.000
2.10	0.691	-2.988	3.050
2.20	0.412	-2.542	5.801
2.30	0.191	-1.847	7.985
2.40	0.049	-0.971	9.387
2.50	0.000	0.000	9.870
2.50	0.000	0.000	0.000
3.50	0.000	0.000	0.000

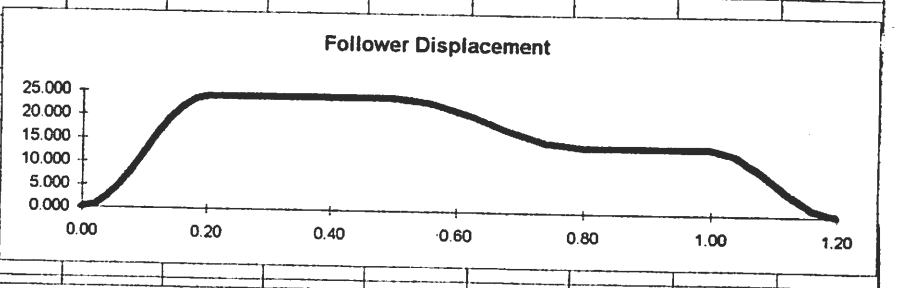


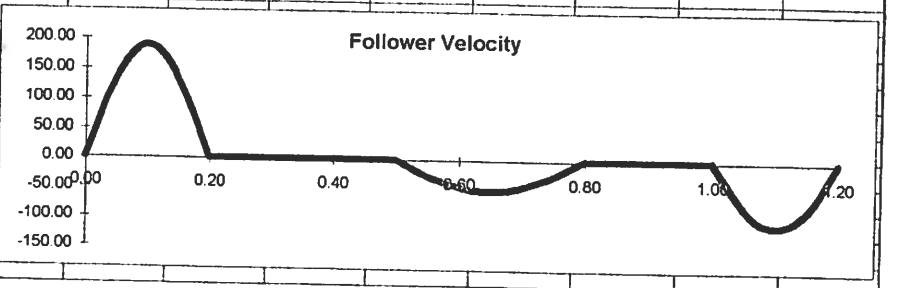


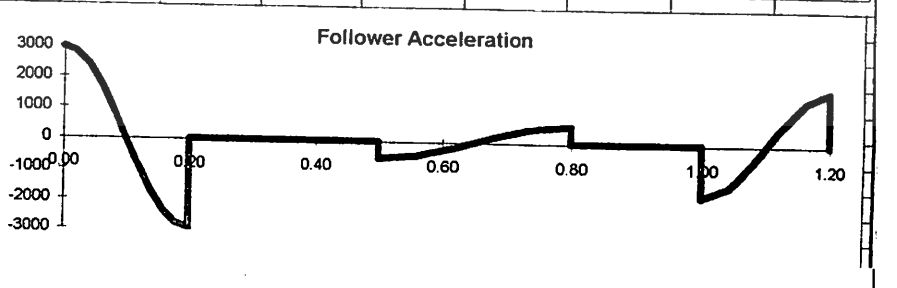


9-34

time (sec)	s (mm)	v (mm/s)	a (mm/s²)
0.00	0.000	0.00	2961
0.02	0.587	58.25	2816
0.04	2.292	110.79	2395
0.06	4.947	152.50	1740
0.08	8.292	179.27	915
0.10	12.000	188.50	0
0.12	15.708	179.27	-915
0.14	19.053	152.50	-1740
0.16	21.708	110.79	-2395
0.18	23.413	58.25	-2816
0.20	24.000	0.00	-2961
0.20	24.000	0.00	0
0.50	24.000	0.00	0
0.50	24.000	0.00	-548
0.56	23.045	-30.78	-444
0.62	20.545	-49.80	-169
0.68	17.455	-49.80	169
0.74	14.955	-30.78	444
0.80	14.000	0.00	548
0.80	14.000	0.00	0
1.00	14.000	0.00	0
1.00	14.000	0.00	-1727
1.04	12.663	-64.63	-1397
1.08	9.163	-104.57	-534
1.12	4.837	-104.57	534
1.16	1.337	-64.63	1397
1.20	0.000	0.00	1727
1.20	0.000	0.00	0



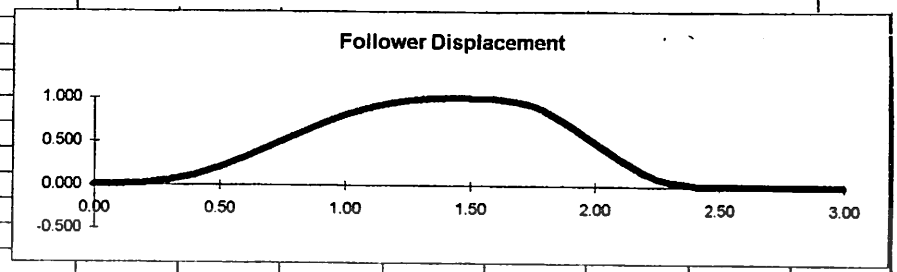


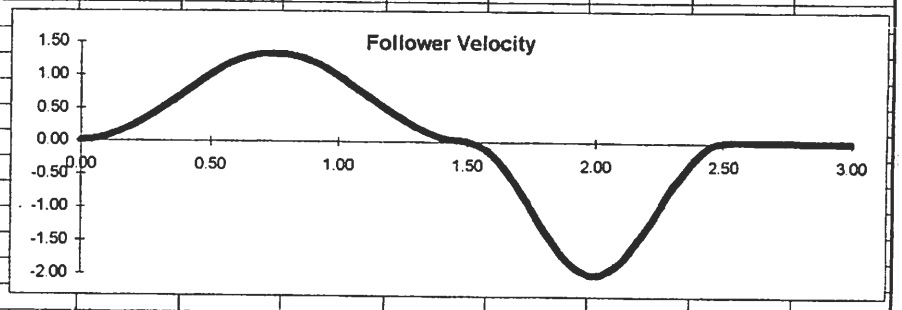


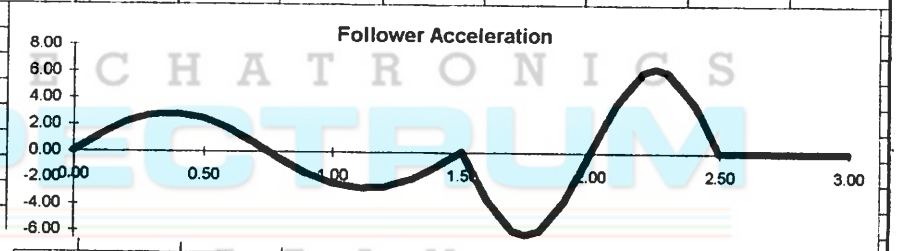
MECHATRONICS SPECTRUM TEAM

9-35

time (sec)	s (mm)	v (mm/s)	a (mm/s²)
0.00	0.000	0.00	0.00
0.10	0.002	0.06	1.14
0.20	0.015	0.22	2.08
0.30	0.049	0.46	2.66
0.40	0.108	0.74	2.78
0.50	0.196	1.00	2.42
0.60	0.306	1.21	1.64
0.70	0.434	1.32	0.58
0.80	0.566	1.32	-0.58
0.90	0.694	1.21	-1.64
1.00	0.804	1.00	-2.42
1.10	0.892	0.74	-2.78
1.20	0.951	0.46	-2.66
1.30	0.985	0.22	-2.08
1.40	0.998	0.06	-1.14
1.50	1.000	0.00	0.00
1.60	0.994	-0.19	-3.69
1.70	0.951	-0.69	-5.98
1.75	0.909	-1.00	-6.28
1.80	0.851	-1.31	-5.98
1.90	0.694	-1.81	-3.69
2.00	0.500	-2.00	0.00
2.10	0.306	-1.81	3.69
2.20	0.149	-1.31	5.98
2.25	0.091	-1.00	6.28
2.30	0.049	-0.69	5.98
2.40	0.006	-0.19	3.69
2.50	0.000	0.00	0.00
3.00	0.000	0.00	0.00

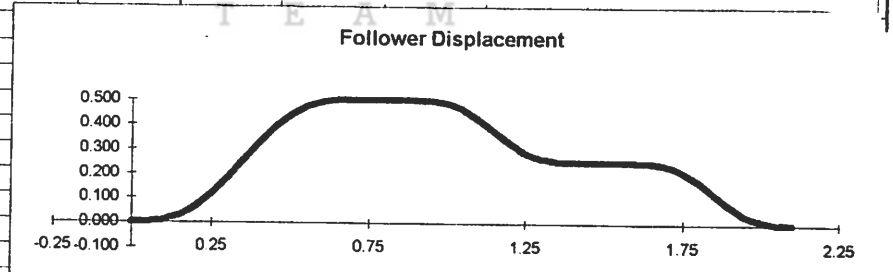


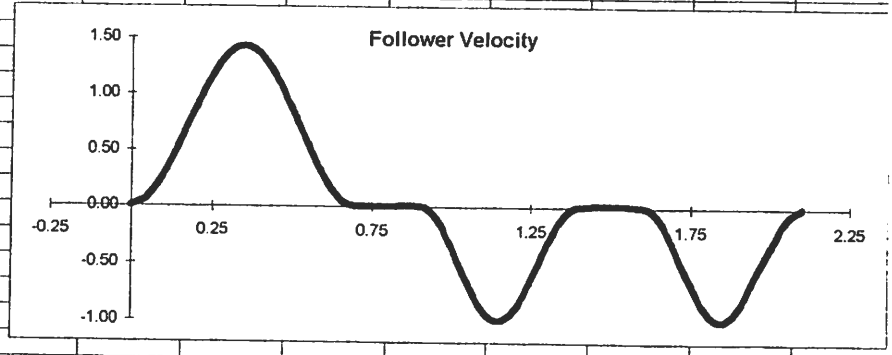


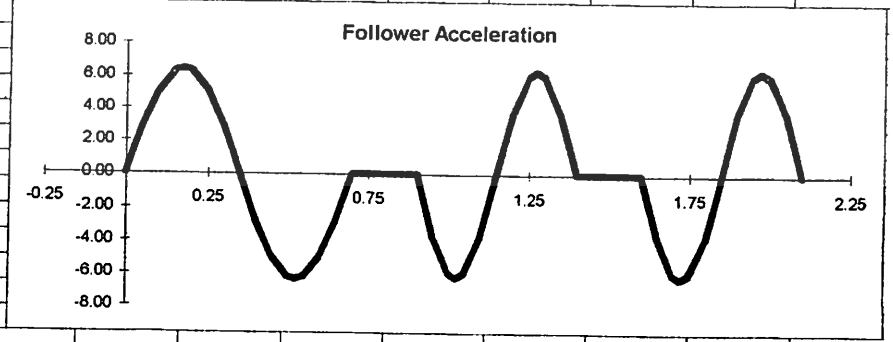


9-36

time (sec)	s (mm)	v (mm/s)	a (mm/s²)
0.00	0.000	0.00	0.00
0.05	0.001	0.07	2.78
0.10	0.009	0.27	5.01
0.15	0.030	0.56	6.25
0.18	0.045	0.71	6.41
0.20	0.065	0.87	6.25
0.25	0.116	1.16	5.01
0.30	0.180	1.36	2.78
0.35	0.250	1.43	0.00
0.40	0.320	1.36	-2.78
0.45	0.384	1.16	-5.01
0.50	0.435	0.87	-6.25
0.53	0.455	0.71	-6.41
0.55	0.470	0.56	-6.25
0.60	0.491	0.27	-5.01
0.65	0.499	0.07	-2.78
0.70	0.500	0.00	0.00
0.90	0.500	0.00	0.00
0.95	0.498	-0.10	-3.69
1.00	0.488	-0.35	-5.98
1.03	0.477	-0.50	-6.28
1.05	0.463	-0.65	-5.98
1.10	0.423	-0.90	-3.69
1.15	0.375	-1.00	0.00
1.20	0.327	-0.90	3.69
1.25	0.287	-0.65	5.98
1.28	0.273	-0.50	6.28
1.30	0.262	-0.35	5.98
1.35	0.252	-0.10	3.69
1.40	0.250	0.00	0.00
1.60	0.250	0.00	0.00
1.65	0.248	-0.10	-3.69
1.70	0.238	-0.35	-5.98
1.73	0.227	-0.50	-6.28
1.75	0.213	-0.65	-5.98
1.80	0.173	-0.90	-3.69
1.85	0.125	-1.00	0.00
1.90	0.077	-0.90	3.69
1.95	0.037	-0.65	5.98
1.98	0.023	-0.50	6.28
2.00	0.012	-0.35	5.98
2.05	0.002	-0.10	3.69
2.10	0.000	0.00	0.00

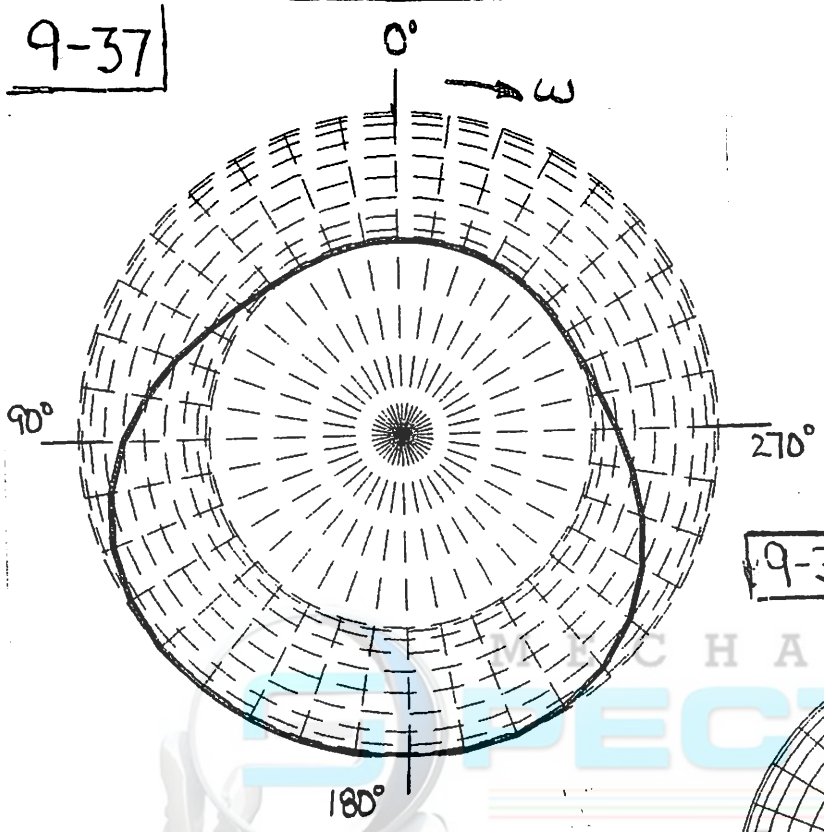




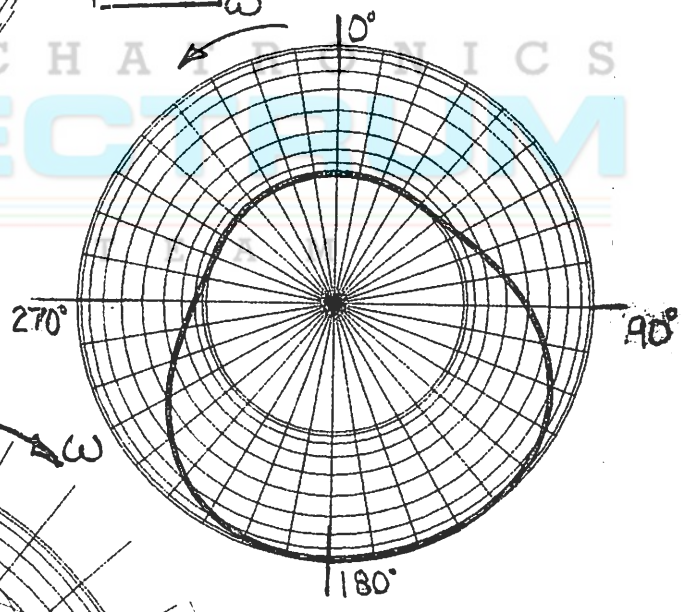


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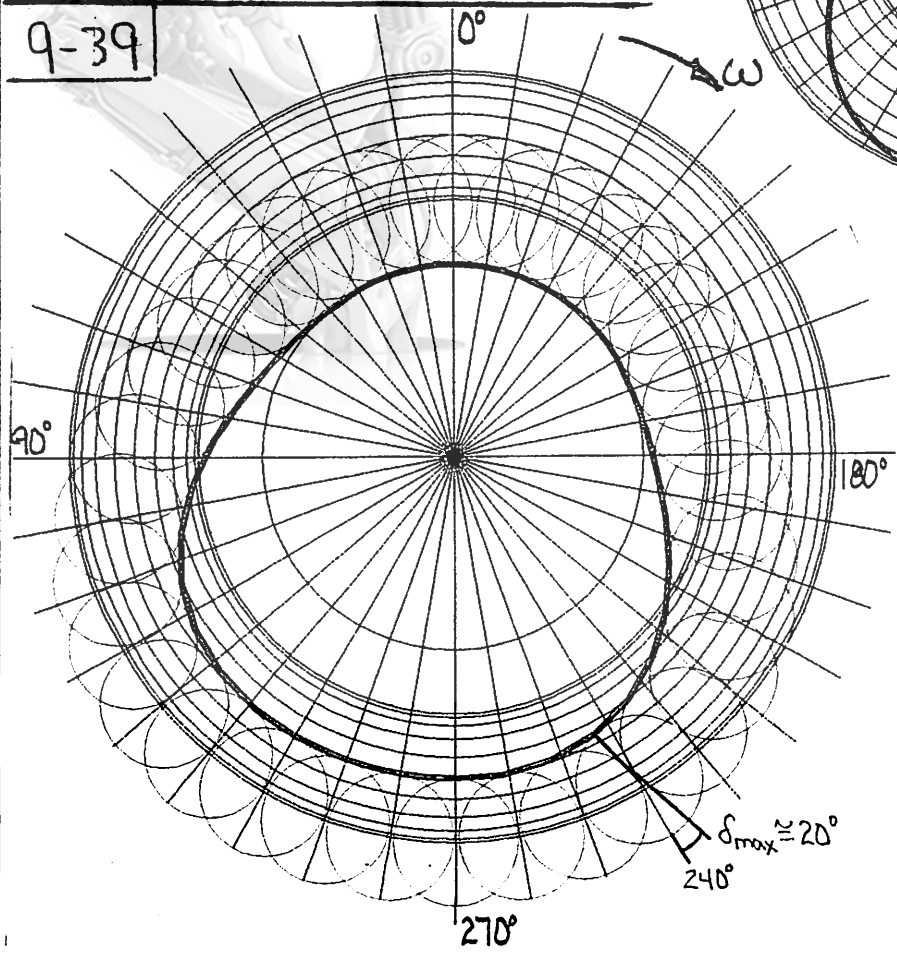
9-37



9-38

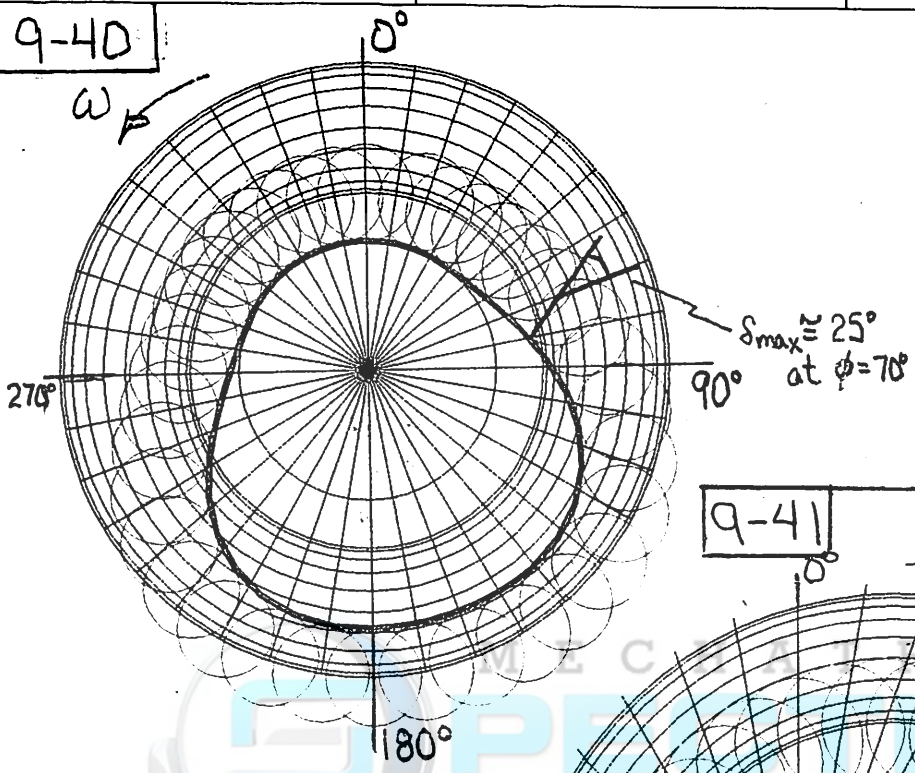


9-39

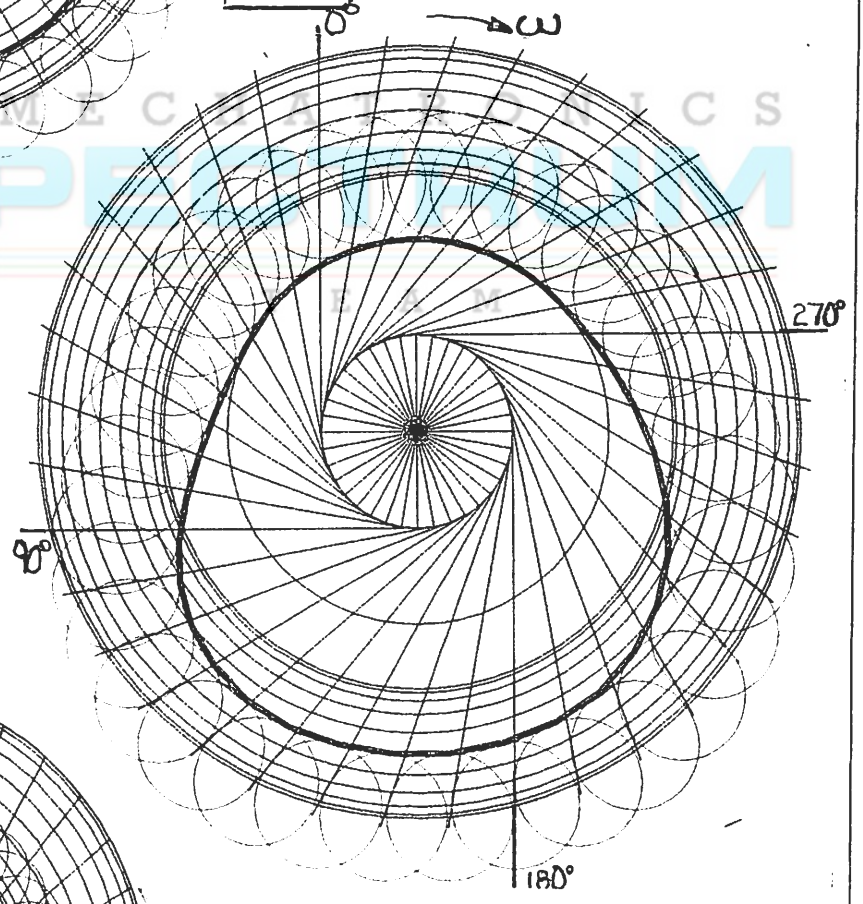


13-782 500 SHEETS, FILLER 5 SQUARE
42-391 50 SHEETS EYE-EASE® 5 SQUARE
42-392 100 SHEETS EYE-EASE® 5 SQUARE
42-389 200 SHEETS EYE-EASE® 5 SQUARE
42-382 200 RECYCLED WHITE 5 SQUARE
42-387 500 RECYCLED WHITE 5 SQUARE
42-389 200 RECYCLED WHITE 5 SQUARE
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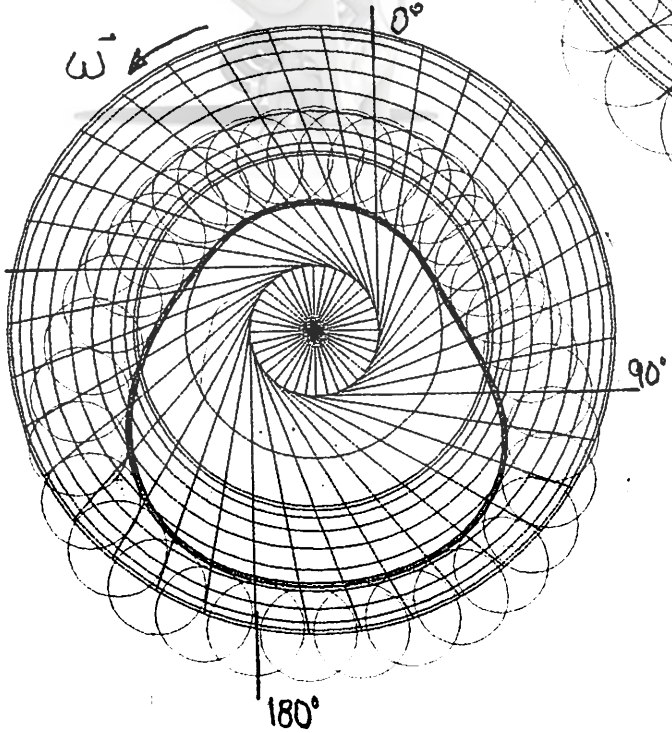
9-40



9-41



9-42



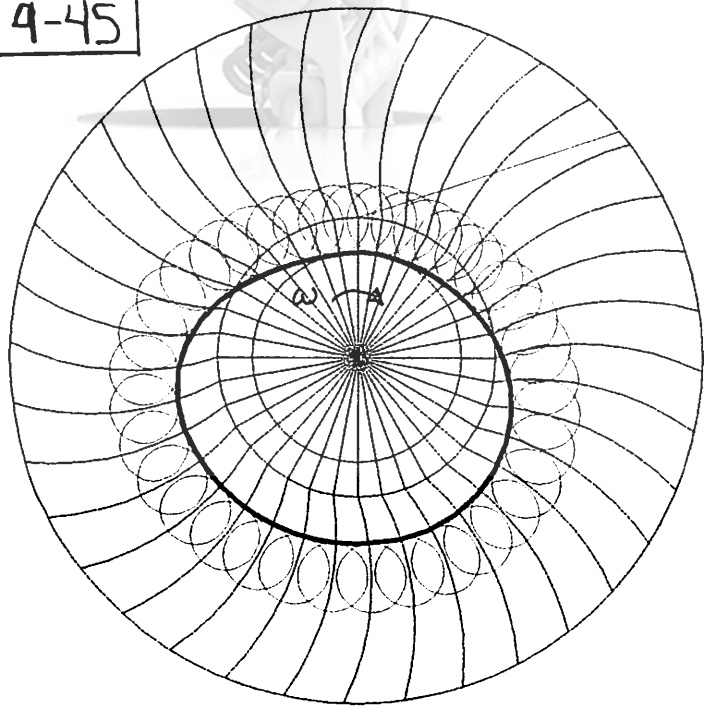
13-782 500 SHEETS, FILLER 5 SQUARE
42-381 50 SHEETS EYE-EASE® 5 SQUARE
42-382 100 SHEETS EYE-EASE® 5 SQUARE
42-389 200 SHEETS EYE-EASE® 5 SQUARE
42-392 100 RECYCLED WHITE 5 SQUARE
42-399 200 RECYCLED WHITE 5 SQUARE
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9-43

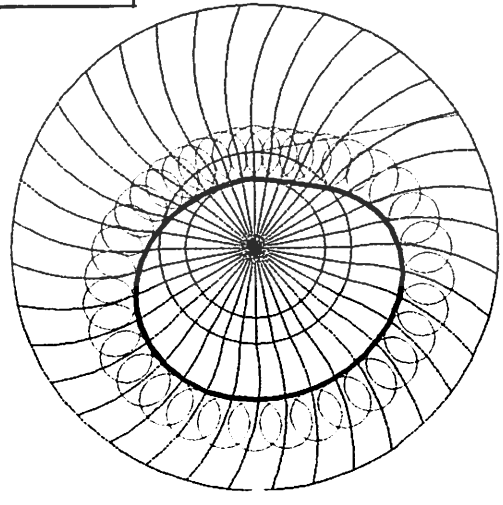
13-782 500 SHEETS, FILLER 5 SQUARE
42-381 50 SHEETS EYE-EASE® 5 SQUARE
42-382 100 SHEETS EYE-EASE® 5 SQUARE
42-388 200 SHEETS EYE-EASE® 5 SQUARE
42-398 100 RECYCLED WHITE 5 SQUARE
42-399 200 RECYCLED WHITE 5 SQUARE
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9-45



9-46

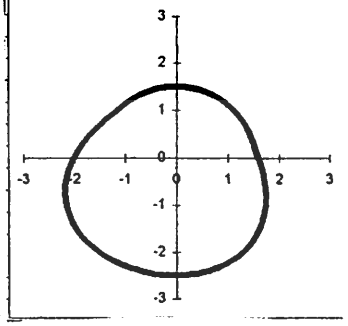


9-47

ϕ	s	Rx	Ry
0	0.000	1.500	0.000
10	0.000	1.477	-0.260
20	0.000	1.410	-0.513
30	0.000	1.299	-0.750
40	0.004	1.152	-0.967
50	0.029	0.983	-1.171
60	0.091	0.796	-1.378
70	0.196	0.580	-1.594
80	0.337	0.319	-1.809
90	0.500	0.000	-2.000
100	0.663	-0.376	-2.130
110	0.804	-0.788	-2.165
120	0.909	-1.205	-2.086
130	0.971	-1.588	-1.893
140	0.996	-1.912	-1.604
150	1.000	-2.165	-1.250
160	1.000	-2.349	-0.855
170	1.000	-2.462	-0.434
180	1.000	-2.500	0.000
190	0.996	-2.458	0.433
200	0.971	-2.322	0.845
210	0.909	-2.086	1.205
220	0.804	-1.765	1.481
230	0.663	-1.390	1.657
240	0.500	-1.000	1.732
250	0.337	-0.628	1.726
260	0.196	-0.295	1.670
270	0.091	0.000	1.591
280	0.029	0.266	1.506
290	0.004	0.514	1.413
300	0.000	0.750	1.299
310	0.000	0.964	1.149
320	0.000	1.149	0.964
330	0.000	1.299	0.750
340	0.000	1.410	0.513
350	0.000	1.477	0.260
360	0.000	1.500	0.000

9-48

Rb = 1.5

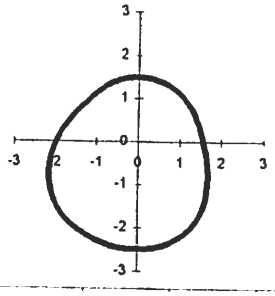


9-49

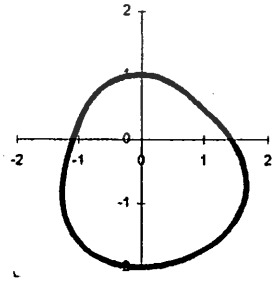
ϕ	α	s	Rx	Ry
0	0.0	0.000	0.000	1.500
10	0.0	0.000	-0.260	1.477
20	0.0	0.000	-0.513	1.410
30	0.7	0.000	-0.755	1.296
40	4.1	0.004	-0.995	1.130
50	9.9	0.029	-1.232	0.922
60	16.1	0.091	-1.464	0.686
70	20.2	0.196	-1.682	0.428
80	21.8	0.337	-1.876	0.142
90	20.5	0.500	-2.032	-0.175
100	16.9	0.663	-2.126	-0.522
110	12.1	0.804	-2.140	-0.890
120	7.0	0.909	-2.059	-1.259
130	2.8	0.971	-1.878	-1.607
140	0.4	0.996	-1.601	-1.915
150	0.0	1.000	-1.250	-2.165
160	0.0	1.000	-0.855	-2.349
170	0.0	1.000	-0.434	-2.462
180	-0.4	1.000	-0.004	-2.500
190	-2.7	0.996	0.410	-2.463
200	-6.8	0.971	0.791	-2.346
210	-11.7	0.909	1.122	-2.146
220	-16.1	0.804	1.388	-1.869
230	-19.3	0.663	1.572	-1.535
240	-20.5	0.500	1.672	-1.167
250	-19.1	0.337	1.696	-0.791
260	-15.3	0.196	1.665	-0.428
270	-9.6	0.091	1.598	-0.084
280	-4.0	0.029	1.513	0.231
290	-0.7	0.004	1.415	0.509
300	0.0	0.000	1.299	0.750
310	0.0	0.000	1.149	0.964
320	0.0	0.000	0.964	1.149
330	0.0	0.000	0.750	1.299
340	0.0	0.000	0.513	1.410
350	0.0	0.000	0.260	1.477
360	0.0	0.000	0.000	1.500

Smax

Rb = 1.5
Rf = 0.5



Rb = 1.0
Rf = 0.375



9-50

ϕ	α	s	Rx	Ry
0	0.0	0.000	0.000	1.000
10	0.0	0.000	0.174	0.985
20	0.0	0.000	0.342	0.940
30	1.0	0.000	0.505	0.863
40	5.9	0.004	0.676	0.746
50	14.2	0.029	0.856	0.598
60	22.3	0.091	1.040	0.436
70	27.2	0.196	1.222	0.262
80	28.6	0.337	1.393	0.063
90	26.5	0.500	1.539	-0.167
100	21.6	0.663	1.640	-0.429
110	15.4	0.804	1.674	-0.715
120	8.8	0.909	1.628	-1.007
130	3.5	0.971	1.496	-1.285
140	0.6	0.996	1.280	-1.531
150	0.0	1.000	1.000	-1.732
160	0.0	1.000	0.684	-1.879
170	0.0	1.000	0.347	-1.970
180	-0.6	1.000	0.004	-2.000
190	-3.5	0.996	-0.324	-1.970
200	-8.6	0.971	-0.623	-1.875
210	-14.8	0.909	-0.878	-1.712
220	-20.3	0.804	-1.075	-1.484
230	-24.6	0.663	-1.200	-1.211
240	-26.5	0.500	-1.250	-0.914
250	-25.3	0.337	-1.235	-0.620
260	-21.0	0.196	-1.179	-0.344
270	-13.6	0.091	-1.102	-0.088
280	-5.8	0.029	-1.022	0.142
290	-1.0	0.004	-0.946	0.338
300	0.0	0.000	-0.866	0.500
310	0.0	0.000	-0.766	0.643
320	0.0	0.000	-0.643	0.766
330	0.0	0.000	-0.500	0.866
340	0.0	0.000	-0.342	0.940
350	0.0	0.000	-0.174	0.985
360	0.0	0.000	0.000	1.000

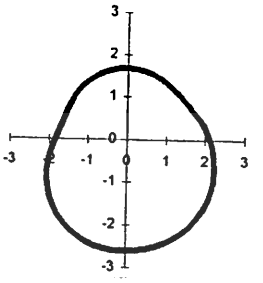
Smax

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9-51

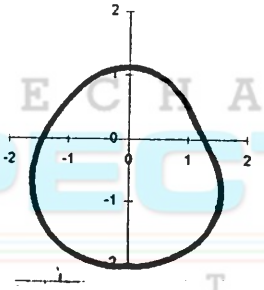
e =	0.75
Rb =	1.5
Rf =	0.5

ϕ	α	s	Rx	Ry
0	0.0	0.000	-0.750	1.500
10	0.0	0.000	-0.999	1.347
20	0.0	0.000	-1.218	1.153
30	-0.4	0.000	-1.396	0.926
40	-2.6	0.004	-1.524	0.685
50	-6.4	0.029	-1.620	0.453
60	-11.1	0.091	-1.713	0.234
70	-15.5	0.196	-1.822	0.007
80	-18.7	0.337	-1.937	-0.257
90	-19.7	0.500	-2.029	-0.582
100	-18.0	0.663	-2.051	-0.966
110	-14.1	0.804	-1.964	-1.384
120	-8.6	0.909	-1.754	-1.792
130	-3.6	0.971	-1.431	-2.140
140	-0.6	0.996	-1.034	-2.391
150	0.0	1.000	-0.600	-2.540
160	0.0	1.000	-0.150	-2.606
170	0.0	1.000	0.304	-2.592
180	0.6	1.000	0.755	-2.500
190	3.6	0.996	1.203	-2.323
200	8.8	0.971	1.623	-2.045
210	14.4	0.909	1.970	-1.663
220	18.6	0.804	2.195	-1.200
230	20.6	0.663	2.276	-0.702
240	19.7	0.500	2.217	-0.219
250	16.3	0.337	2.050	0.202
260	11.6	0.196	1.828	0.542
270	6.6	0.091	1.594	0.808
280	2.6	0.029	1.372	1.026
290	0.4	0.004	1.156	1.223
300	0.0	0.000	0.924	1.400
310	0.0	0.000	0.667	1.539
320	0.0	0.000	0.390	1.631
330	0.0	0.000	0.100	1.674
340	0.0	0.000	-0.192	1.666
350	0.0	0.000	-0.478	1.607
360	0.0	0.000	-0.750	1.500



9-52

e =	0.5
Rb =	1.0
Rf =	0.375



ϕ	α	s	Rx	Ry
0	0.0	0.000	0.500	1.000
10	0.0	0.000	0.666	0.898
20	0.0	0.000	0.812	0.769
30	0,2	0.000	0.934	0.615
40	1.2	0.004	1.034	0.443
50	2.7	0.029	1.121	0.265
60	3.8	0.091	1.208	0.091
70	3.6	0.196	1.304	-0.083
80	2.0	0.337	1.406	-0.273
90	-0.1	0.500	1.500	-0.499
100	-1.7	0.663	1.553	-0.770
110	-2.2	0.804	1.529	-1.073
120	-1.7	0.909	1.409	-1.378
130	-0.8	0.971	1.192	-1.646
140	-0.1	0.996	0.901	-1.850
150	0.0	1.000	0.567	-1.982
160	0.0	1.000	0.214	-2.050
170	0.0	1.000	-0.145	-2.056
180	0.1	1.000	-0.501	-2.000
190	0.8	0.996	-0.844	-1.878
200	1.9	0.971	-1.156	-1.677
210	2.8	0.909	-1.404	-1.394
220	3.0	0.804	-1.558	-1.048
230	2.0	0.663	-1.604	-0.676
240	0.1	0.500	-1.549	-0.316
250	-1.8	0.337	-1.424	0.002
260	-2.7	0.196	-1.262	0.267
270	-2.3	0.091	-1.091	0.485
280	-1.1	0.029	-0.928	0.664
290	-0.2	0.004	-0.773	0.812
300	0.0	0.000	-0.616	0.933
310	0.0	0.000	-0.445	1.026
320	0.0	0.000	-0.260	1.087
330	0.0	0.000	-0.067	1.116
340	0.0	0.000	0.128	1.111
350	0.0	0.000	0.319	1.072
360	0.0	0.000	0.500	1.000

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9-61

$\beta_0 = \frac{360}{4} = 90°$

$\gamma_0 = \frac{180-90}{2} = 45°$

$a = (3\text{in})\sin 45 = 2.12\text{ in}$

$\psi = 180 - \frac{90}{2} + 25 = 160°$

$r = \sqrt{2.12^2 + 3^2 - (2)(2.12)(3)\cos(180-160)}$

$= 1.242\text{ in}$

$\beta = \sin^{-1}\left[\frac{2.12}{1.242}\sin 20\right] = 35.7°$

$\omega_{in} = 60\text{rpm}\left(\frac{\pi}{30}\right) = -6.28\text{ rad/s}$

$\omega_{out} = \frac{2.12}{1.242}(-6.28)\cos(35.7-160) = -6.04\frac{\text{r}}{\text{s}}$ CCW

$\alpha_{out} = -\left(\frac{2.12}{1.242}\right)(-6.28)^2\sin(35.7-160) + \left(\frac{2.12}{1.242}\right)^2(-6.28)^2\sin(71.4-320)$

$= 163.1\text{ rad/sec}^2$ CCW

9-62

$\beta_0 = \frac{360}{5} = 70°$

$\gamma_0 = \frac{180-70}{2} = 54°$

$a = 60\sin 54 = 35.2\text{mm}$

$\psi = 180 - 70/2 + 20 = 164°$

$r = \sqrt{(35.2)^2 + (60)^2 - 2(35.2)(60)\cos(180-164)}$

$= 27.85\text{ mm}$

$\beta = \sin^{-1}\left[\frac{35.2}{27.85}\sin 16\right] = 20.4°$

$\omega_{in} = 70\text{rpm}\left(\frac{\pi}{30}\right) = -7.33\text{ rad/s}$

$\omega_{out} = \frac{35.2}{27.85}(-7.33)\cos(20.4-164) = -7.46\frac{\text{r}}{\text{s}^2}$ CCW

$\alpha_{out} = -\left(\frac{35.2}{27.85}\right)(-7.33)^2\sin(20.4-164) + \left(\frac{35.2}{27.85}\right)^2(-7.33)^2\sin(40.8-328)$

$= 122.7\text{ rad/sec}^2$ CCW

9-63

$\beta_0 = \frac{360}{6} = 60$

$\gamma_0 = \frac{180-60}{2} = 60$

$a = (4\text{in})\sin 60 = 2.0\text{ in}$

$\psi = 180 - \frac{60}{2} + 25 = 175°$

$r = \sqrt{(2.0)^2 + (4.0)^2 - 2(2)(4)\cos(180-175)}$

$= 2.02\text{ in}$

$\beta = \sin^{-1}\left[\frac{2.0}{4.0}\sin 5\right] = 4.96°$

$\omega_{in} = 90\text{rpm}\left(\frac{\pi}{30}\right) = 9.42\text{ rad/s}$

$\omega_{out} = \frac{2}{4}(9.42)\cos(4.96-175) = 9.21\text{ rad/s}$

$\alpha_{out} = -\left(\frac{2}{4}\right)(9.42)^2\sin(4.96-175) + \left(\frac{2}{4}\right)^2(9.42)^2\sin(9.92-350)$

$= 45.06\text{ rad/sec}^2$

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Answers to the Chapter 9 Case Study Questions:

Case 9-1.

1. As shaft G turns clockwise, item E moves upward.
2. Connection between item E and F is a rack and pinion gear joint.
3. Counterweight K balances the attack of papers sitting on item J.
4. As paper is used, the weight of the stack decreased. The radius of item H changes to reduce the moment, and hence the force countering the weight of the papers.
5. A pin is used to join shaft G with cam H, constraining both to rotate together.
6. For smaller stacks of paper, springs are used as counterbalances. They share the common feature of reducing the supporting force at lower deflections.

Case 9-2.

1. As cam D rotates clockwise, the immediate motion of link B will be clockwise rotation, followed by a dwell.
2. Cam D is a plate cam.
3. Item C is a roller mounted on a pivoted follower.
4. Item F is a spring.
5. Spring F that attempts to maintain alignment of links A and B.
6. Item E is a screw.
7. Screw E serves as a stop to limit the motion of item A.
8. Item B is a pivoted follower. It moves together with item A, until stop E is reached. When A is in contact with the stop, the pivot point for link B changes and follower C traces a shortened radius.
9. Lengthening screw E will cause the pivot point to move from the bottom of link A to the top of link A for a greater portion of the cycle.
10. Shortening screw E will cause the pivot point to move from the bottom of link A to the top of link A for a reduced portion of the cycle.

Case 9-3.

1. As rod C starts to slide downward, cam E rotates counterclockwise.
2. As rod C starts to slide downward and cam E rotates counterclockwise, plunger H is forced downward.
3. The strip of sheet metal clamped at W is contacted by plunger H and formed into an inverted channel shape (⊔) .
4. As rod C continues to slide downward, follower proceeds P past the lobe of cam E, and plunger H raises.
5. As rod C continues to slide downward, the angled surfaces D contact slides I and force them inward.
6. As slides I move inward, they contact the sides of the steel strip (already formed as an inverted channel) and bend them over (⊔).
7. As rod C starts to move upward, the slides return to the position shown, and follower P once again rides over the lobe of cam E. The plunger H again strikes the top of the steel, forming an open box profile over (⊏⊐).

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8. The mechanism converts a strip of steel into a box profile with a downward and upward stroke of rod C.
9. The springs contacting slides I are return springs. They return the slides to the position shown when not being contacted by the angled surfaces D.
10. Spring under item K give the base some compliance. As the strip steel W is first contacted, a single material thickness is trapped between H and K. On the second strike stroke, two material thicknesses are trapped between H and K.
11. The linear reciprocation of C can be actuated by a slider-crank mechanism, another cam system, a hydraulic cylinder, along with a variety of other linkages and mechanisms.

Case 9-4.

1. As gear K rotates clockwise, link F (which is at its limiting rightward posstion) rotates clockwise.
2. Links J and F are configured as shaper-crank mechanism and exhibit quick-return motion.
3. As gear K rotates clockwise, the motion of slide D is slowly to the right and a quick return to the left.
4. As gear K rotates clockwise, gear N rotates counterclockwise.
5. As gear K rotates clockwise, link Q does not move for the next 180 degrees of rotation, then rotates counterclockwise.
6. Component P is a plate cam.
7. Item V is constrained to vertical translation.
8. Link Q and link V are coupled with a connecting link.
9. As gear K rotates clockwise the workpiece X is slid to the right, then lifted upward. Timing is controlled since all motion is achieved with a single actuator.
10. Pick and place mechanisms are used throughout a manufacturing facility.

MECHATRONICS SPECTRUM TEAM

Chapter 10

10-1

$d = \frac{N}{P_d} = \frac{18}{12} = 1.5$ in

$d_b = d\cos\phi = 1.5\cos(20) = 1.409$ in

$a = \frac{1}{P_d} = .083 \Rightarrow d_a = d + 2a = 1.667$ in

$b = \frac{1.25}{P_d} = .104 \Rightarrow d_b = d - 2b = 1.292$ in

$p = \pi d/N = \pi \frac{1.5}{18} = .262$ in

10-2

$d = \frac{N}{P_d} = \frac{48}{8} = 6$ in

$d_b = d\cos\phi = 6\cos 20 = 5.638$ in

$a = \frac{1}{P_d} = .125 \Rightarrow d_a = d + 2a = 6.25$ in

$b = \frac{1.25}{P_d} = .156 \Rightarrow d_b = d - 2b = 5.688$ in

$p = \pi d/N = \pi\ 6/48 = 0.393$ in

10-3

$d = N/P_d = 40/16 = 2.50$ in

$d_b = d\cos\phi = 2.5\cos 14.5 = 2.420$ in

$a = \frac{1}{P_d} = .063 \Rightarrow d_a = d + 2a = 2.625$ in

$b = \frac{1.25}{P_d} = .078 \Rightarrow d_b = d - 2b = 2.344$ in

$p = \pi d/N = \pi(2.5)/40 = 0.196$ in

10-4

$d = m \cdot N = (4)(21) = 84$ mm

10-5

$d_1 = \frac{18}{12} = 1.5$ $\quad d_2 = \frac{42}{12} = 3.5$ in

$C = \frac{1}{2}(1.5 + 3.5) = 2.5$ in

$P_b = \pi(1.5)\cos 20/18 = .246$

$L = .400$ $\quad$ Contact Ratio $= \frac{.400}{.246} = 1.63$

$N_1 = 18$ can mate with any gear (no interfer)

$\Delta B = \frac{.1}{P_d} - \frac{.4}{P_d} = -.025$ in

$\Delta C \cong \Delta B/(2\tan\phi) = -.034$ in

$C_{adj} = C + \Delta C = 2.466$ in

10-6

$d_1 = \frac{12}{4} = 3$ in $\quad d_2 = \frac{28}{4} = 7$ in

$C = \frac{1}{2}(3 + 7) = 5$ in

$P_b = \pi(3)\cos 20/12 = .738$

$L = 1.128 \Rightarrow$ Contact Ratio $= \frac{1.128}{.738} = 1.53$

$N_1 = 12 \Rightarrow$ interference problems with any gear

$\Delta B = \frac{.1}{P_d} - \frac{.4}{P_d} = -.075$ in

$\Delta C \cong \Delta B/(2\tan\phi) = -.103$ in

$C_{adj} = C + \Delta C = 4.897$ in

10-7

$d_1 = \frac{18}{48} = .375$ in $\quad d_2 = \frac{48}{48} = 1$ in

$C = \frac{1}{2}(.375 + 1) = .688$ in

$P_b = \pi(.375)\cos 25/18 = .059$ in

$L = .086 \Rightarrow$ Contact Ratio $= \frac{.086}{.059} = 1.46$

$N_1 = 18$ can mate w/any gear (no interference)

$\Delta B = \frac{.1}{P_d} - \frac{.4}{P_d} = -.006$

$\Delta C \cong \Delta B/(2\tan\phi) = -.007$

$C_{adj} = C + \Delta C = .681$ in

10-8

$d_1 = \frac{16}{16} = 1$ in $\quad d_2 = \frac{72}{16} = 4.5$ in

$C = \frac{1}{2}(1 + 4.5) = 2.75$ in

$P_b = \pi(1)\cos 14.5/16 = .177$ in

$L = .261$ in $\Rightarrow$ contact ratio $= \frac{.261}{.177} = 1.47$

$N_1 = 16 \Rightarrow$ interference problems with any gear

$\Delta B = \frac{.1}{P_d} - \frac{.4}{P_d} = -.018$ in

$\Delta C \cong \Delta B/2\tan\phi = -.020$ in

$C_{adj} = C + \Delta C = 2.729$ in

10-9

$d_1 = \frac{18}{12} = 1.5$ in $\quad d_2 = \frac{48}{12} = 4$ in

$C_1 = \frac{1}{2}(4 - 1.5) = 1.25$ in

10-10

$d_1 = \frac{15}{20} = .75$ $\quad d_2 = \frac{60}{20} = 3$ in

$C = \frac{1}{2}(3 - .75) = 1.125$ in

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10-11 $C = \frac{1}{2}\left(\frac{N_1}{P_d} + \frac{N_2}{P_d}\right)$

$4.125 = \frac{1}{2}\left(\frac{18}{P_d} + \frac{48}{P_d}\right) \Rightarrow P_d = 8$

$d_1 = \frac{18}{8} = 2.25$ in $\quad d_2 = \frac{48}{8} = 6$ in

10-12 $C = \frac{1}{2}\left(\frac{N_1}{P_d} + \frac{N_2}{P_d}\right)$

$3.25 = \frac{1}{2}\left(\frac{20}{P_d} + \frac{45}{P_d}\right) \Rightarrow P_d = 10$

$d_1 = \frac{20}{10} = 2$ in $\quad d_2 = \frac{45}{10} = 4.5$ in

10-13 $d_1 = \frac{18}{8} = 2.25$ in $\quad d_2 = \frac{64}{8} = 8$ in

$C = \frac{1}{2}(8 - 2.25) = 2.875$ in

10-14 $d_1 = \frac{24}{12} = 2$ in $\quad d_2 = \frac{108}{12} = 9$ in

$C = \frac{1}{2}(9 - 2) = 3.5$ in

10-15 $\omega_2 = \frac{N_1}{N_2}\omega_1 = \frac{18}{64}(1150)$

$= 323.4$ rpm ccw

$= 33.87$ rad/sec

$r_2 = \frac{1}{2}\left(\frac{64}{8}\right) = 4$ in

$V_t = r_2\omega_2 = (4\text{ in})\left(33.87 \frac{\text{rad}}{\text{sec}}\right) = 135.5 \frac{\text{in}}{\text{sec}}$

10-16 $\omega_2 = \frac{150}{60}(1725) = 4312.5$ rpm ccw

$= 45.16$ rad/sec

$r_2 = \frac{1}{2}\left(\frac{60}{20}\right) = 1.5$ in

$V_t = r_2\omega_2 = (1.5\text{ in})\left(45.16 \frac{\text{rad}}{\text{sec}}\right) = 67.7 \frac{\text{in}}{\text{sec}}$

10-17 $\omega_2 = \frac{24}{42}(850) = 425$ rpm ccw

$= 44.51$ rad/sec

$r_2 = \frac{1}{2}\left(\frac{42}{6}\right) = 3.5$ in

$V_t = (3.5\text{ in})\left(44.51 \frac{\text{rad}}{\text{s}}\right) = 155.8 \frac{\text{in}}{\text{sec}}$

10-18 $\omega_2 = \frac{24}{144}(1725) = 287.5$ rpm ccw

$= 30.11 \frac{\text{rad}}{\text{sec}}$

$r_2 = \frac{1}{2}\left(\frac{144}{24}\right) = 3$ in

$V_t = (3\text{ in})\left(30.11 \frac{\text{rad}}{\text{s}}\right) = 90.3 \frac{\text{in}}{\text{sec}}$

10-19 $VR = 5 = \frac{d_2}{d_1} \Rightarrow d_2 = 5d_1$

$C = 12 = \frac{1}{2}(d_1 + 5d_1) \Rightarrow d_1 = 4.0$ in $\quad d_2 = 20.0$ in

$N = P_d d \Rightarrow N_1 = 40$ teeth

$N_2 = 200$ teeth

10-20 $VR = 4 \Rightarrow d_2 = 4.0\, d_1$

$C = 3.75 = \frac{1}{2}(d_1 + 4d_1) \Rightarrow d_1 = 1.5$ in $\quad d_2 = 6.0$ in

$N = P_d d \Rightarrow N_1 = 24$ teeth

$N_2 = 96$ teeth

10-21 $VR = 8 \Rightarrow d_2 = 8d_1$

$C = 2.25 = \frac{1}{2}(d_1 + 8d_1) \Rightarrow d_1 = .5$ in $\quad d_2 = 4$ in

$N = P_d d \Rightarrow N_1 = 16$ teeth

$N_2 = 128$ teeth

10-22 $VR = 4 \Rightarrow d_2 = 4d_1$

$C = 5 = \frac{1}{2}(d_1 + 4d_1) \Rightarrow d_1 = 2$ in

Could use: $d_2 = 8$ in

$P_d = 8$

$N_1 = 16$ & $N_2 = 64$

Would avoid interference, but power capacity must be examined

10-23 $VR = 6 \Rightarrow d_2 = 6d_1$

$C = 3.5 = \frac{1}{2}(d_1 + 6d_1) \Rightarrow d_1 = 1$ in $\quad d_2 = 6$ in

Could use:

$P_d = 16$

$N_1 = 16$ $\quad N_2 = 96$

Would avoid interference but power capacity must be examined

10-24 $VR = 3 \Rightarrow d_2 = 3d_1$

$C = 10 = \frac{1}{2}(d_1 + 3d_1) \Rightarrow d_1 = 5$ in $\quad d_2 = 15$ in

Could use:

$P_d = 4$

$N_1 = 20$ $\quad N_2 = 60$

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10-25 From table 9.5

5 hp & $n_1 = 1800$ rpm

$\Rightarrow P_d = 12$ //

10-26 From table 9.5

2.5 hp & $n_1 = 1500$ rpm

$\Rightarrow P_d = 12$ //

10-27 From table 9.5

8 hp & $n_1 = 1500$ rpm $\Rightarrow P_d = 8$ //

$VR = \frac{n_1}{n_2} = \frac{1500}{200} = 7.5$

if $N_1 = 16$ $N_2 = 120$ (problems with interference)

$N_1 = 18$ // $N_2 = 135$ //

$d_1 = 2.25$ in $d_2 = 16.875$ in

10-28 From table 9.5

10 hp & $n_1 = 800$ rpm $\Rightarrow P_d = 6$ //

$VR = \frac{800}{180} = 4.44$

if $N_1 = 18$ // $N_2 = 80$ //

$\Rightarrow n_2 = \frac{18}{80}(800) = 180$ rpm

$d_1 = 3.0$ in // $d_2 = 13.33$ in //

10-29 From table 9.5

1 hp & 1725 rpm $\Rightarrow P_d = 20$

$VR = \frac{1725}{560} = 3.08$

if $N_1 = 16$ // $N_2 = 48$ //

$\Rightarrow n_2 = \frac{16}{48}(1725) = 575$ rpm (5 rpm low)

$d_1 = 0.8$ in // $d_2 = 2.4$ in //

10-30 From table 9.5

10 hp & 1175 rpm $\Rightarrow P_d = 8$

$VR = \frac{1175}{230} = 5.11$

if $N_1 = 16$ $N_2 = 82$

$\Rightarrow n_2 = \frac{16}{82}(1175) = 229$ rpm

$d_1 = 2$ in // $d_2 = 10.25$ in //

10-31 From table 9.5

10 hp & 1175 rpm $\Rightarrow P_d = 8$

$VR = \frac{1175}{170} = 6.9 \approx 7 \Rightarrow d_2 = 7d_1$

$C = 8\text{ in} = \frac{1}{2}(d_1 + d_2) = \frac{1}{2}(d_1 + 7d_1)$

$\Rightarrow d_1 = 2$ in $d_2 = 14$ in

$N_1 = 16$ $N_2 = 112$

(interference problems)

Could try stronger steel & use $P_d = 10$ //

$d_1 = 2$ in $d_2 = 14$ in

$N_1 = 20$ // $N_2 = 140$ //

$n_2 = 1175/7 = 168$ rpm

10-32 From table 9.5

3 hp & 1750 rpm $\Rightarrow P_d = 12$

$VR = \frac{1750}{290} = 6.03 \approx 6 \Rightarrow d_2 = 6d_1$

$C = 14 = \frac{1}{2}(d_1 + d_2) = \frac{1}{2}(d_1 + 6d_1)$

$\Rightarrow d_1 = 4$ in $d_2 = 24$ in

$N_1 = 48$ // $N_2 = 288$ //

$n_2 = 1750/6 = 292$ rpm

10-33 From table 9.5

20 hp & 825 rpm $\Rightarrow P_d = 5$

$VR = \frac{825}{205} = 4.02 \approx 4 \Rightarrow d_2 = 4d_1$

$C = 20 = \frac{1}{2}(d_1 + 4d_1)$

$\Rightarrow d_1 = 8$ in $d_2 = 32$ in

$N_1 = 40$ // $N_2 = 160$ //

$n_2 = 825/4 = 206$ rpm

10-34 $d = \frac{18}{24} = .75$ in

$\Delta s = (d)(\Delta\theta)/2$

$\Delta\theta = 2\Delta s/d = 2(5\text{ in})/.75\text{ in}$

$= 13.3$ radians $\left(\frac{1\text{ rev}}{2\pi\text{ rad}}\right)$

$= 2.12$ revolutions //

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10-35 $d = \frac{20}{16} = 1.25$ in

$\Delta\theta = 2\Delta s/d = 2(3)/1.25 = 4.8$ rad

$\Delta\theta = 4.8 \text{ rad}\left(\frac{1 \text{ rev}}{2\pi \text{ rad}}\right) = .76$ revolutions //

10-36 $d = \frac{18}{8} = 2.25$ in

$\Delta s = \Delta\theta\, d/2 = (3 \text{ rev})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)\left(\frac{2.25 \text{ in}}{2}\right)$

$= 42.4$ in. //

10-37 $d = \frac{24}{12} = 2$ in

$\Delta s = (5 \text{ rev})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)\left(\frac{2 \text{ in}}{2}\right) = 31.4$ in //

10-38 $d = \frac{18}{12} = 1.5$ in

$V = \omega\frac{d}{2} \Rightarrow \omega = 2V/d = \frac{2(50 \frac{\text{in}}{\text{min}})}{1.5 \text{ in}}$

$\omega = 66.7 \frac{\text{rad}}{\text{min}}\left(\frac{1 \text{ rev}}{2\pi \text{ rad}}\right) = 10.6$ rpm //

10-39 $d = \frac{20}{10} = 2$ in

$\omega = 80 \text{ rpm}\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) = 502 \frac{\text{rad}}{\text{min}}$

$V = \omega\frac{d}{2} = 502 \frac{\text{in}}{\text{min}}$ //

10-40 $d_1 = \frac{N_1}{P_d} = \frac{16}{8} = 2$ in //

$d_2 = \frac{N_2}{P_d} = \frac{32}{8} = 4$ in //

$P_d^n = P_d/\cos\psi = 8/\cos 45 = 11.3$ //

$p^n = p\cos\psi = \frac{\pi d}{N}\cos\psi$

$p_1^n = \frac{\pi(2)}{16}\cos 45 = .277 \text{ in} = p_2^n$

no interference // (must be larger than 7 teeth)

10-41 $d_1 = \frac{16}{12} = 1.33$ in //

$d_2 = \frac{48}{12} = 4$ in //

$P_d^n = 12/\cos 30 = 13.8$ //

$p^n = \frac{\pi}{P_d}\cos\psi = \frac{\pi}{12}\cos 30 = .226$ in //

interference problems // (should be larger than 21 teeth)

10-42 $VR = \frac{40}{20} = 2.0$

$C = (20+40)/2(8) = 3.75$ in

Helical Gears

$C = (N_2+N_1)/2P_d\cos\psi$

$= (N_2+N_1)/2(8)\cos\psi$

$3.75 = 3.0N_1/16(\cos\psi)$

$\cos\psi = N_1/20$

can use:

$N_1 = 19$ // $N_2 = 38$ // $P_d^n = 8$ $\psi = 18.2°$ //

$P_d = P_d^n\cos\psi = 7.6$ //

10-43 $VR = \frac{54}{18} = 3.0$

$C = (18+54)/2(12) = 3$ in

Helical gears

$C = 4N_1/2(12)\cos\psi = 3$ in

$\cos\psi = N_1/18$

Can use

$N_1 = 16$ // $N_2 = 48$ // $P_d^n = 12$

$\psi = 27.3°$ $P_d = 12\cos 27.3$

$= 10.67$ //

10-44 $VR = \frac{75}{20} = 3.75$ //

$\tan\gamma_p = \sin 90/(\cos 90 + 3.75)$

$\gamma_p = 14.93°$ //

$\tan\gamma_G = \sin 90/(\cos 90 + \frac{1}{3.75})$

$\gamma_G = 75.07°$ //

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10-45 $VR = \frac{75}{20} = 3.75$

$\tan \gamma_p = \sin 60 / (\cos 60 + 3.75)$

$\gamma_p = 11.5°$

$\tan \gamma_G = \sin 60 / (\cos 60 + \frac{1}{3.75})$

$\gamma_G = 48.5°$

10-46 $VR = \frac{90}{18} = 5.0$

$\tan \gamma_p = \sin 75 / (\cos 75 + 5)$

$\gamma_p = 10.4°$

$\tan \gamma_G = \sin 75 / (\cos 75 + \frac{1}{5})$

$\gamma_G = 64.6°$

10-47 $VR = \frac{3600}{60} = 60$

Single thread:

$N_G = 60$ teeth. $p_w = \frac{\pi}{16} = .1963$ in

$\lambda = 5° \Rightarrow d_w = \frac{\pi \tan 5}{.1963} = 1.40$ in

$d_G = \frac{60}{16} = 3.75$ in $C = \frac{1.4 + 3.75}{2} = 2.575$ in

10-48 $VR = \frac{1800}{18} = 100$

Double thread

$N_G = \frac{100}{2} = 50$ teeth $p_w = \frac{\pi}{12} = .2618$

$\lambda = 5° \Rightarrow d_w = \frac{\pi \tan 5}{2(.2618)} = .525$ in

$d_G = \frac{50}{12} = 4.167$ in $C = \frac{4.167 + .525}{2} = 2.35$ in

10-49 $VR = \frac{3600}{40} = 90$

double thread

$N_G = 45$ teeth $p_w = \frac{\pi}{20} = .1571$ in

$\lambda = 5° \Rightarrow d_w = \frac{\pi \tan 5}{2(.1571)} = .875$ in

$d_G = \frac{45}{20} = 2.25$ in $C = \frac{2.25 + .875}{2} = 1.563$ in

10-50 $\omega_5 = \left(\frac{18}{72}\right)\left(\frac{16}{48}\right)(1200) = 100$ rpm cw

$C = \frac{18 + 72}{2(10)} + \frac{(16 + 48)}{2(8)} = 8.5$ in

10-51 $\omega_5 = \left(\frac{20}{6(10)}\right)\left(\frac{2(8)}{48}\right)(1800) = 200$ rpm ccw

$C = \frac{20 + 6(10)}{2(10)} + \frac{2(8) + 48}{2(8)} = 8$ in

10-52

$\omega_8 = \left(\frac{15}{90}\right)\left(\frac{15}{75}\right)\left(\frac{75}{75}\right)\left(\frac{15}{60}\right)(3600)$

$= 30$ rpm cw

$C = \frac{15 + 90}{2(16)} + \frac{15 + 75}{2(12)} + \frac{75}{(12)} + \frac{15 + 60}{2(8)} = 17.97$ in

10-53

$\omega_8 = \left(\frac{16}{8(16)}\right)\left(\frac{1.5}{50/10}\right)\left(\frac{50/10}{5.5}\right)\left(\frac{1.5(8)}{56}\right)(1200 \text{ ccw})$

$= 8.22$ rpm ccw

$C = \frac{16 + 8(16)}{2(16)} + \frac{1.5 + 5.5}{2} + \frac{50}{10} + \frac{1.5(8) + 56}{2(8)}$

$= 17.25$ in

10-54

$S_4 = \left(\frac{20/16}{8}\right)\left(\frac{1.25}{2}\right)(2\pi \text{ rad})$

$= 0.98$ in

$C = \frac{20/16 + 8}{2} = 4.625$ in

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10-55

$$\omega_1 = \left(60\frac{in}{min}\right)\left(\frac{2}{2.5\,in}\right)\left(\frac{5.5(20)}{18}\right)$$
$$= 244\ rpm$$

10-56

$$\Delta S = \frac{2.5}{(75/10)}\left(\frac{24}{2P_d}\right)(1\ rev) = .5\ in$$
$$P_d = 8$$

10-57

$$\omega_4 = \frac{(16/16)}{8}\left(\frac{20}{50}\right)(1800)$$
$$= 90\ rpm$$

10-58

$$\omega_1 = \frac{4}{(17/20)}\left(\frac{36}{18}\right)(380\ rpm)$$
$$= 3576\ rpm$$

10-59

$$\omega_5 = \left(\frac{41}{45}\right)\left(\frac{1.6}{80}\right)(1800)$$
$$= 9\ rpm$$

$$C = \frac{(18/16)+6}{2} + \frac{6+(80/16)}{2} = 9.06\ in$$

10-60

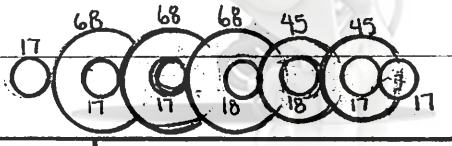
$$\omega_1 = \left(\frac{54}{18}\right)\left(\frac{60}{2}\right)(28\ rpm)$$
$$= 2520\ rpm$$

$$C = \frac{(18/12)+6}{2} + \frac{6+(54/12)}{2} = 9\ in$$

10-61

$$VR_{max} = \frac{75}{17} = 4.4$$

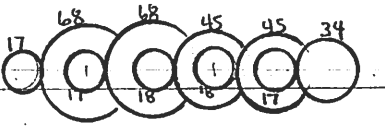
$$400 = (4)(4)(4)(2.5)(2.5)(1)$$
$$= \left(\frac{68}{17}\right)\left(\frac{68}{17}\right)\left(\frac{68}{17}\right)\left(\frac{45}{18}\right)\left(\frac{45}{18}\right)\left(\frac{17}{17}\right)$$



10-62

$$VR_{max} = \frac{75}{17} = 4.4$$

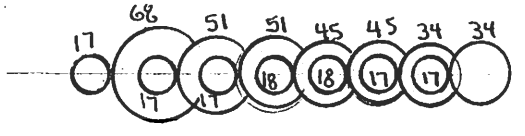
$$-200 = (4)(4)(2.5)(2.5)(2)(\ )$$
$$= \left(\frac{68}{17}\right)\left(\frac{68}{17}\right)\left(\frac{45}{18}\right)\left(\frac{45}{18}\right)\left(\frac{34}{17}\right)$$



10-63

$$VR_{max} = \frac{75}{17} = 4.4$$

$$-900 = (4)(3)(3)(2.5)(2.5)(2)(2)$$
$$= \left(\frac{68}{17}\right)\left(\frac{51}{17}\right)\left(\frac{51}{17}\right)\left(\frac{45}{18}\right)\left(\frac{45}{18}\right)\left(\frac{34}{17}\right)\left(\frac{34}{17}\right)$$

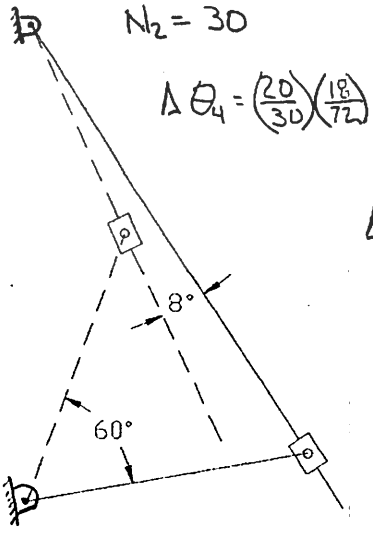


10-64

$N_1 = d_1 P_d = 20$
$N_2 = 30$
$N_3 = 18$
$N_4 = 18(4) = 72$

$$\Delta\theta_4 = \left(\frac{20}{30}\right)\left(\frac{18}{72}\right)(1\ rev) = 0.167\ rev$$
$$= 60^\circ\ CW$$

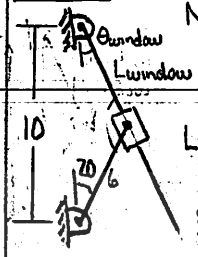
$\Delta\theta_{window} = 8^\circ$ CCW



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10-65 $N_1 = d_1 P_d = 20$ $N_3 = 18$

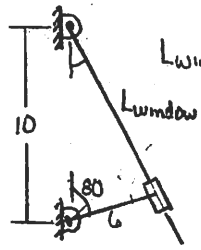
$N_2 = 30$ $N_4 = 18(4) = 72$



$\Delta\theta_4 = \left(\frac{20}{30}\right)\left(\frac{18}{72}\right)(1\text{ rev}) = .167\text{ rev} = 60^\circ$ cw

$L_{window} = \sqrt{10^2 + 6^2 - 2(10)(6)\cos 20} = 4.82$ in

$\frac{\sin\theta_{window}}{6} = \frac{\sin 20}{4.82} \Rightarrow \theta_{window} = 25.1^\circ$



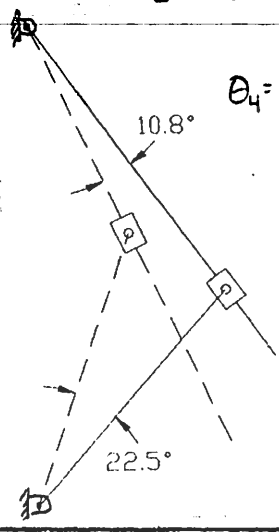
$L_{window} = \sqrt{10^2 + 6^2 - 2(10)(6)\cos 80} = 10.73$ in

$\frac{\sin\theta_{window}}{6} = \frac{\sin 80}{10.73} \Rightarrow \theta_{window} = 33.4^\circ$

$\Delta\theta_{window} = 25.1 - 33.4^\circ = -8.22^\circ = 8.22^\circ$ ccw

10-66 $N_1 = .75(32) = 24$ $N_3 = 16$

$N_2 = 48$ $N_4 = 4(32) = 128$

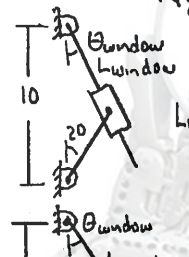


$\theta_4 = \left(\frac{24}{48}\right)\left(\frac{16}{128}\right)(1\text{ rev}) = .0625\text{ rev} = 22.5^\circ$ cw

$\Delta\theta_{window} = 10.8^\circ$ ccw

10-67 $N_1 = .75(32) = 24$ $N_3 = 16$

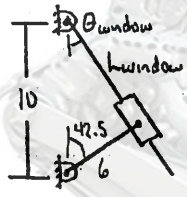
$N_2 = 48$ $N_4 = 4(32) = 128$



$\Delta\theta_4 = \left(\frac{24}{48}\right)\left(\frac{16}{128}\right)(1\text{ rev}) = .0625\text{ rev} = 22.5^\circ$ cw

$L_{window} = \sqrt{10^2 + 6^2 - 2(10)(6)\cos 20} = 4.82$ in

$\frac{\sin\theta_{window}}{6} = \frac{\sin 20}{4.82} \Rightarrow \theta_{window} = 25.1^\circ$



$L_{window} = \sqrt{10^2 + 6^2 - 2(10)(6)\cos 42.5} = 6.89$ in

$\frac{\sin\theta_{window}}{6} = \frac{\sin 42.5}{6.89} \Rightarrow \theta_{window} = 36.03^\circ$

$\Delta\theta_{window} = 25.1 - 36.03 = -10.84^\circ = 10.84^\circ$ ccw

10-68 $N_1 = 16$ $N_3 = 2(16) = 32$ $N_4 = 5(16) = 80$

	1	2	3	4
Step 1	+1	0	$-16/32$	$-16/80$
Step 2	−1	−1	−1	−1
	0	−1	−1.5	−1.2

$\omega_1 = 0$

$\omega_2 = \left(\frac{-1}{-1}\right)1800\text{ rpm} = 1800\text{ rpm}$ cw

$\omega_3 = \left(\frac{-1.5}{-1}\right)(1800) = 2700\text{ rpm}$ cw

$\omega_4 = \left(\frac{-1.2}{-1}\right)(1800) = 2160\text{ rpm}$ cw

10-69 $N_1 = 18$ $N_3 = 2.5(12) = 30$ $N_4 = 6.5(12) = 78$

	1	2	3	4
Step 1	$-78/18$	0	$78/30$	+1
Step 2	−1	−1	−1	−1
	−5.33	−1	1.6	0

$\omega_1 = \frac{-5.33}{-1}(800) = 4264\text{ rpm}$ ccw

$\omega_2 = \frac{-1}{-1}(800) = 800\text{ rpm}$ ccw

$\omega_3 = \frac{1.6}{-1}(800) = 1280\text{ rpm}$ cw

$\omega_4 = 0$

10-70 $N_1 = 1.25(16) = 20$ $N_3 = 42$ $N_4 = 21$

$N_5 = 32$ $N_6 = 144$

	1	2	3	4	5	6
Step 1	+1	0	$-20/42$	$+20/21$	$+20/21$	$\frac{32}{144}\left(\frac{20}{21}\right)$
Step 2	−1	−1	−1	−1	−1	−1
	0	−1	−1.476	−.048	−.048	.212

$\omega_1 = 0$

$\omega_2 = \left(\frac{-1}{-1}\right)(680) = 680\text{ rpm}$ cw

$\omega_3 = \left(\frac{-1.476}{-1}\right)(680) = 1004\text{ rpm}$ cw

$\omega_4 = \left(\frac{-.048}{-1}\right)(680) = 33\text{ rpm}$ cw

$\omega_5 = \omega_4 = 33\text{ rpm}$ cw

$\omega_6 = \left(\frac{.212}{-1}\right)(680) = 144\text{ rpm}$ ccw

10-71 $N_1 = 1(20) = 20$ $N_3 = 45$ $N_4 = 20$
$N_5 = 30$ $N_6 = 150$

	1	2	3	4	5	6
Step 1	$+\frac{20}{20}\left(\frac{150}{30}\right)$	0	$-\frac{45}{20}\left(\frac{150}{30}\right)$	$\frac{150}{30}$	$\frac{150}{30}$	+1
Step 2	-1	-1	-1	-1	-1	-1
	4	-1	-12.25	4	4	0

$\omega_1 = \frac{4}{-1}(1125) = 4500$ rpm cw//

$\omega_2 = \left(\frac{-1}{-1}\right)(1125) = 1125$ rpm ccw//

$\omega_3 = \frac{-12.25}{-1}(1125) = 13{,}781$ rpm ccw//

$\omega_4 = \frac{4}{-1}(1125) = 4500$ rpm cw//

$\omega_5 = \omega_4 = 4500$ rpm cw//

$\omega_6 = 0$//

10-72 $N_1 = 24$ $N_2 = 48$ $N_3 = 25(10) = 25$
$N_4 = 35$

	1	2	Carrier	3	4
Step 1	$-\frac{48}{24}$	+1	0	$\frac{-48}{24}$	$+\frac{25}{35}\left(\frac{48}{24}\right)$
Step 2	-1	-1	-1	-1	-1
	-3	0	-1	-3	+.42

$\omega_1 = \frac{-3}{-1}(900) = 2700$ rpm cw//

$\omega_2 = 0$//

$\omega_{carrier} = \frac{-1}{-1}(900) = 900$ rpm cw//

$\omega_3 = \frac{-3}{-1}(900) = 2700$ rpm cw//

$\omega_4 = \frac{.42}{-1}(900) = 378$ rpm ccw//

10-73 $N_1 = 25$ $N_2 = 4(10) = 40$
$N_3 = 2.5(8) = 20$ $N_4 = 32$

	1	2	Carrier	3	4
Step 1	$-\frac{40}{25}$	+1	0	$-\frac{40}{25}$	$+\frac{20}{32}\left(\frac{40}{25}\right)$
Step 2	-1	-1	-1	-1	-1
	-2.6	0	-1	-2.6	0

Output ~~shaft~~ will not rotate!

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Answers to the Chapter 10 Case Study Questions:

Case 10-1.

1. As segment gear A rotates counterclockwise, rack C is pushed to the left.
2. As segment gear A rotates counterclockwise pushing rack C to the left, gear B rotates counterclockwise.
3. Once tooth E disengages with the rack, the leading tooth on segment gear A will engage gear B, driving gear D clockwise.
4. For continuous counterclockwise motion of segment gear A, segment gear B reciprocates counter and clockwise.
5. When engaged with gear A, rack C is pushed left. When disengaged from A, but engaged with B, rack C is pushed back to the right.
6. This mechanism creates reciprocating linear (rack C) and rotation (gear B) with a uniform continuously rotary input.
7. The constant engaging and disengaging can cause shock and vibration problems, if the mechanism is driven at high speed or with substantial loads.

Case 10-2.

1. As link A moves to the left, link B rotates counterclockwise.
2. As link A moves to the left, gear C rotates counterclockwise.
3. As link A moves to the left, gear D rotates clockwise.
4. As link A moves to the left, link E rotates clockwise.
5. As link A moves to the left, link F rotates clockwise. Note that the configuration of links E and F form a shaper-crank mechanism and exhibit quick-return motion. It the position shown, the mechanism is within the slower portion of its stroke.
6. As link A moves to the left, link G moves to the left. However, due to the quick-return nature of the arrangement, link G moves slower than link A.
7. As link A reciprocates back and forth, link G also reciprocates but with a smaller stroke.
8. If link E were rotated 90 degrees, then link G would reciprocate with a larger stroke than the stroke of link A.

Case 10-3.

1. As gear A rotates clockwise, gear B rotates counterclockwise.
2. As gear A rotates clockwise and gear B rotates counterclockwise, gear C rotates clockwise.
3. As gear A rotates clockwise and gear B rotates counterclockwise, gear D rotates clockwise.
4. After the handle is rotated upward, gear B disengages with gear A but gear C engages with gear A.
5. As gear A continues to rotate clockwise, gear C rotates counterclockwise, gear B rotates clockwise and gear D rotates counterclockwise.
6. The mechanism produces forward and reverse rotation of an output gear (D) from a constant direction of an input gear (A).

7. Engaging and disengaging gears while transferring power causes abrupt shock loading on the gear teeth.

Case 10-4.

1. As gear B rotates clockwise, gear C rotates counterclockwise.
2. As gear B rotates clockwise, gear D rotates clockwise.
3. Link J, gear C and link E (along with the frame) create a four-bar mechanism. The rotation of gear C actuates the mechanism and causes link J to oscillate, pivoting at A.
4. The center of gear D will trace an arc, centered at A.
5. Piston G is constrained to horizontal sliding.
6. The mechanism creates a large stroke for piston G.

Case 10-5.

1. As shaft G rotates as shown, gear H rotates counterclockwise, when viewing the mechanism from a right side view.
2. Gears F, J and H are bevel gears.
3. Item A is a nut that is pushed leftward with the counterclockwise (as viewed from a right side view) rotation of screw C.
4. As item A contacts color L, the screw is pushed to the right. Item K disengages with gear H, and contacts gear J. This action results in the reversal of screw C and nut A.
5. Item O is a spring-loaded ball detent, which holds screw C from axial translation, until enough force is created to overcome the spring force and move the screw.
6. Setscrews are located on collars L and Q to alter the stroke of nut A.
7. The mechanism creates reciprocating, constant velocity, motion of nut A between collars L and Q. This type of mechanism is common in winding machinery (such as cable on a spool).

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11-1 $\omega_2 = \frac{D_1}{D_2}\omega_1$

$= \frac{3.5}{8}(1725) = 755 \text{ rpm, cw}$

$\omega_1 = 1725(\pi/30) = 180.64 \text{ rad/s}$

$V_b = (180.64)(3.5/2) = 316 \text{ in/s } (60/12)$

$= 1581 \text{ ft/min}$

11-2 $\omega_2 = \frac{D_1}{D_2}\omega_1$

$= \frac{5}{12}(1150) = 479 \text{ rpm, cw}$

$\omega_1 = 1150(\pi/30) = 120.42 \text{ rad/s}$

$V_b = (120.42)(5/2) = 301 \text{ in/s } (60/12)$

$= 1505 \text{ ft/min}$

11-3 $\omega_1 = \frac{D_2}{D_1}\omega_2$

$= \frac{10}{4}(1200) = 3000 \text{ rpm, cw}$

$\omega_2 = 1200(\pi/30) = 125.66 \text{ rad/s}$

$V_b = (125.66)(10/2) = 628 \text{ in/s } (60/12)$

$= 3142 \text{ ft/min}$

11-4 $\omega_2 = \frac{D_1}{D_2}\omega_1$

$= \frac{2.5}{9}(1125) = 312.5 \text{ rpm, ccw}$

$\omega_1 = 1125(\pi/30) = 117.81 \text{ rad/s}$

$V_b = (117.81)(2.5/2) = 147 \text{ in/s } (60/12)$

$= 736. \text{ ft/min}$

11-5 $\omega_2 = \frac{D_1}{D_2}\omega_1$

$= \frac{6.5}{10.5}(1750) = 1083 \text{ rpm, ccw}$

$\omega_1 = 1750(\pi/30) = 183.26 \text{ rad/s}$

$V_b = (183.26)(6.5/2) = 595.6 \text{ in/s } (60/12)$

$= 2978 \text{ ft/min}$

11-6 $\omega_1 = \frac{D_2}{D_1}\omega_2$

$= \frac{9}{6}(1800) = 2700 \text{ rpm, ccw}$

$\omega_2 = 1800(\pi/30) = 188.50 \text{ rad/s}$

$V_b = (188.50)(9/2) = 848.2 \text{ in/s } (60/12)$

$= 4241 \text{ ft/min}$

11-7 $8 < C_{ideal} < 3(3.5+8) = 34.5$

$C_{act} = 23$ (within range)

$L = 2(23) + \frac{\pi}{2}(8+3.5) + \frac{(8-3.5)^2}{4(23)}$

$= 64.28 \text{ in}$

$\theta_1 = 180 - 2\sin^{-1}\left[\frac{8-3.5}{2(23)}\right] = 168.8°$

11-8 $12 < C_{ideal} < 3(5+12) = 51$

$B = 4(72) - 2\pi(5+12) = 181.19$

$C = \frac{181.19 + \sqrt{(181.19)^2 - 32(12-5)^2}}{16}$

$= 22.375 \text{ in}$ (within range)

$\theta_1 = 180 - 2\sin^{-1}\left[\frac{12-5}{2(22.375)}\right] = 162.0°$

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11-9 $12 < C_{ideal} < 3(8+12) = 60$

$B = 4(88) - 2\pi(8+12) = 226.33$

$$C = \frac{226.33 + \sqrt{(226.33)^2 - 32(12-8)^2}}{16}$$

$= 28.221$ in // (within range)

$\theta_1 = 180 - 2\sin^{-1}\left[\frac{12-8}{2(28.221)}\right] = 171.9°$ //

11-10 $24 < C_{ideal} < 3(8+24) = 96$

$B = 4(104) - 2\pi(8+24) = 214.94$

$$C = \frac{214.94 + \sqrt{(214.94)^2 - 32(24-8)^2}}{16}$$

$= 25.618$ in // (within range)

$\theta_1 = 180 - 2\sin^{-1}\left[\frac{24-8}{2(25.618)}\right] = 143.6°$ //

11-11 10hp & 3550rpm ⇒ 3V belt

one possible design:

use $D_1 = 2.5$ in //

$2.5(3550/900) = 9.86$, use $D_2 = 10.6$ in //

$10.6 < C_{ideal} < 3(2.5+10.6) = 39.3$

try $C \cong 25$ in

$L = 2(25) + \frac{\pi}{2}(2.5+10.6) + \frac{(10.6-2.5)^2}{4(25)}$

$= 71.233$ in

Use standard 71 in //

$B = 4(71) - 2\pi(2.5+10.6) = 201.69$

$$C = \frac{201.69 + \sqrt{201.69^2 - 32(10.6-2.5)^2}}{16}$$

$= 24.882$ in //

$\omega_1 = 3550(\pi/30) = 371.76$

$V_b = (371.76)(2.5/2) = 464.7$ in/s $(60/12)$

$= 2323$ ft/min //

11-12 25hp & 950rpm ⇒ 5V

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11-13 | 5hp & 1150rpm ⇒ 3V belt

11-14 | 10hp & 2000rpm ⇒ 3V belt

11-15 | 70hp & 3000rpm ⇒ 5V belt

11-16 | 2hp & 200rpm ⇒ 3V belt

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11-17 $\omega_1 = \frac{N_2}{N_1}\omega_2$

$= \frac{84}{19}(180) = 795.8 \text{ rpm, cw}$

$\omega_2 = 180(\pi/30) = 18.84 \text{ rad/s}$

$D_2 = .75/\sin(180/84) = 20.058 \text{ in}$

$V_c = (18.84)(20.058/2) = 188.9 \text{ in/s}\,(60/12)$

$= 944.7 \text{ ft/min}$

use bath lubrication

11-18 $\omega_2 = \frac{N_1}{N_2}\omega_1$

$= \frac{19}{50}(220) = 83.6 \text{ rpm, cw}$

$\omega_1 = 220(\pi/30) = 23.04 \text{ rad/s}$

$D_1 = .5/\sin(180/19) = 3.037 \text{ in}$

$V_c = (23.04)(3.037/2) = 34.99 \text{ in/s}\,(60/12)$

$= 175.0 \text{ ft/min}$

use manual lubrication

11-19 $\omega_2 = \frac{N_2}{N_1}\omega_1$

$= \frac{27}{68}(180) = 71.47 \text{ rpm, cw}$

$\omega_1 = 180(\pi/30) = 18.84 \text{ rad/s}$

$D_1 = 1/\sin(180/27) = 8.614 \text{ in}$

$V_c = (18.84)(8.614/2) = 81.14 \text{ in/s}\,(60/12)$

$= 405.7 \text{ ft/min}$

use manual lubrication

11-20 $\omega_1 = \frac{N_2}{N_1}\omega_2$

$= \frac{76}{25}(250) = 760 \text{ rpm, cw}$

$\omega_2 = 250(\pi/30) = 26.18 \text{ rad/s}$

$D_2 = 1.25/\sin(180/76) = 30.25 \text{ in}$

$V_c = (26.18)(30.25/2) = 395.9 \text{ in/s}\,(60/12)$

$= 1979 \text{ ft/min}$

use oil stream lubrication

11-21 $D_1 = \frac{.75}{\sin(180/17)} = 4.082 \text{ in}$

$D_2 = \frac{.75}{\sin(180/56)} = 13.376 \text{ in}$

$C = \frac{.75}{4}\left[120 - \frac{(17+56)}{2} + \sqrt{\left\{120 - \frac{(17+56)}{2}\right\}^2 - \frac{8(56-17)^2}{4\pi^2}}\right]$

$= 30.963 \text{ in}$

$\theta_1 = 180 - 2\sin^{-1}\left(\frac{.75(56-17)}{2(30.963)}\right) = 123.6°$

$\theta_2 = 180 + 2\sin^{-1}\left(\frac{.75(56-17)}{2(30.963)}\right) = 236.4°$

11-22 $D_1 = \frac{1}{\sin(180/19)} = 6.076 \text{ in}$

$D_2 = \frac{1}{\sin(180/64)} = 20.380 \text{ in}$

$C = \frac{1}{4}\left[140 - \frac{(19+64)}{2} + \sqrt{\left\{140 - \frac{(19+64)}{2}\right\}^2 - \frac{8(64-19)^2}{4\pi^2}}\right]$

$= 48.724 \text{ in}$

$\theta_1 = 180 - 2\sin^{-1}\left[\frac{1(64-19)}{2(48.724)}\right] = 125.0°$

$\theta_2 = 180 + 2\sin^{-1}\left[\frac{1(64-19)}{2(48.724)}\right] = 235.0°$

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11-23

$$D_1 = \frac{.5}{\sin(180/21)} = 3.355 \text{ in}$$

$$D_2 = \frac{.5}{\sin(180/84)} = 13.372 \text{ in}$$

$$C = \frac{.5}{4}\left[200 - \frac{(21+84)}{2} + \sqrt{\left\{200 - \frac{(21+84)}{2}\right\}^2 - \frac{8(84-21)^2}{4\pi^2}}\right]$$

$$= 36.531 \text{ in}$$

$$\theta_1 = 180 - 2\sin^{-1}\left(\frac{.5(84-21)}{2(36.531)}\right) = 128.9°$$

$$\theta_2 = 180 + 2\sin^{-1}\left[\frac{.5(84-21)}{2(36.531)}\right] = 231.1°$$

11-24

$$D_1 = \frac{1.5}{\sin(180/25)} = 11.968 \text{ in}$$

$$D_2 = \frac{1.5}{\sin(180/72)} = 34.388 \text{ in}$$

$$C = \frac{1.5}{4}\left[120 - \frac{(25+72)}{2} + \sqrt{\left\{120 - \frac{(25+72)}{2}\right\}^2 - \frac{8(72-25)^2}{4\pi^2}}\right]$$

$$= 52.424 \text{ in}$$

$$\theta_1 = 180 - 2\sin^{-1}\left(\frac{1.5(72-25)}{2(52.424)}\right) = 95.5°$$

$$\theta_2 = 180 + 2\sin^{-1}\left[\frac{1.5(72-25)}{2(52.424)}\right] = 264.5°$$

11-25 40 hp & 350 rpm

⇒ use No. 100 Chain

11-26 30 hp & 550 rpm

⇒ use No. 80 Chain

One possible design:

$N_1 = 11$ teeth

$N_2 = \frac{550}{100}(11) = 60.5 \cong 60$ teeth

$\omega_2 = \frac{11}{60}(550) = 100.83$ rpm

$D_1 = \frac{1}{\sin(180/11)} = 3.55$ in

$D_2 = \frac{1}{\sin(180/60)} = 19.11$ in

$V_c = \frac{\pi(3.55)(550)}{12} = 511$ ft/min

$30p < a < 50p$ ⇒ try $C = 40p$

$$L = 2(40) + \frac{1}{2}(11+60) - \frac{(60-11)^2}{4\pi(40)} = 117.02$$

use $L = 117$ links

$$C_{actual} = \frac{1}{4}\left[117 - \frac{71}{2} + \sqrt{\left[117 - \frac{71}{2}\right]^2 - \frac{8(49)^2}{4\pi^2}}\right]$$

$$= 39.93 \text{ in}$$

11-27 | 130hp & 600rpm ⇒ No 140 chain

11-28 | 50hp & 600rpm ⇒ No 100 chain

11-29 | 8 hp & 300rpm ⇒ No. 60 chain

11-30 | 10hp & 425rpm ⇒ No 60 chain

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Answers to the Chapter 11 Case Study Questions:

Case 11-1.

1. As sprocket A drives clockwise, sprocket B is driven clockwise.
2. As sprocket A drives clockwise, the instantaneous motion of pin C (which is attached to the chain) is moving left and upward toward sprocket A.
3. As sprocket A drives clockwise, the instantaneous motion of yoke D (which is constrained to horizontal sliding) is toward the left at constant velocity.
4. In roughly 1/2 revolution, the portion of the chain that contains pin C will engage with sprocket A.
5. As the portion of the chain that contains pin C engages on sprocket A, rod E will decelerate
6. As the portion of the chain that contains pin C moves onto the top of sprocket A, rod E will change directions then accelerate toward the right. Upon further rotation and disengagement of the portion of chain that contains pin C, rod E will move toward the right with constant velocity.
7. The mechanism creates reciprocating, constant velocity, motion of rod E.

Case 11-2.

1. As handle H rotates, shaft I rotates with the handle.
2. As handle H rotates, the right half sheave rotates threading itself onto (or off) the left half.
3. As handle H rotates, two half-sheaves are brought either together or apart.
4. Item J is a key and transfers rotational power from the sheave to shaft A.
5. Item F guides shaft I.
6. Item G is a spring, which keeps engagement of shaft I with half-sheave C.
7. This devise is a variable radius sheave, commonly used in CVTs (continuously variable transmissions).

Case 11-3.

1. As tab B is forced upward into lever C, item D is forced downward.
2. As tab B is forced upward into lever C, item D will move downward into a slot integral with sheave G. The resulting action is that sheave G will no longer be able to rotate.
3. As tab B is forced upward and sheave G is no longer able to rotate, the paint shaking machine will stop operation.
4. The purpose of this device is stop operation of the machine when the door is raised.
5. Sheave E has many notches for more immediate response to opening the door, stopping the shaking machine more quickly.
6. Spring H is a return spring to keep D in an upward position until forced downward.
7. This mechanism is a safety device.

12-1 $\frac{1}{4}$-20 $\Rightarrow p = \frac{1}{20} = .05$ in

$\lambda = \tan^{-1}\frac{N_t p}{\pi D} = \tan^{-1}\frac{(1)(.05)}{\pi(.2175)}$

$= 4.18°$

$\tan\lambda = .07 < .2$

threads are self locking

12-2 $\frac{1}{4}$-28 $\Rightarrow p = \frac{1}{28} = .0357$

$\lambda = \tan^{-1}\left[\frac{(1)(.0357)}{\pi(.2268)}\right] = 2.87°$

$\tan\lambda = .05 < .2$

threads are self locking

12-3 M16×2.0 $\Rightarrow p = 2$ mm

$\lambda = \tan^{-1}\left[\frac{(1)(2)}{\pi(14.68)}\right] = 2.48°$

$\tan\lambda = .043 < .2$

threads are self-locking

12-4 1-$\frac{1}{8}$-5 $\Rightarrow p = \frac{1}{5} = .20$ in

$\lambda = \tan^{-1}\left[\frac{(1)(.2)}{\pi(.9967)}\right] = 3.65°$

$\tan\lambda = .064 < .2$

threads are self-locking

12-5 $\frac{1}{4}$-20 $\Rightarrow p = \frac{1}{20} = .05$ in

$\Delta R = N_t p \Delta\theta = (1)(.05\,\text{in/rev})(3\,\text{rev})$

$= .15$ in //

12-6 ACME 2-4 $\Rightarrow p = \frac{1}{4} = .25$ in

$\Delta R = N_t p \Delta\theta = (1)(.25\,\text{in/rev})(10\,\text{rev})$

$= 2.5$ in //

12-7 ACME 1-5 $\Rightarrow p = \frac{1}{5} = .20$ in

$\Delta R = N_t p \Delta\theta = (1)(.2\,\text{in/rev})(10\,\text{rev})$

$= 2.5$ in ↓

12-8 $\frac{1}{2}$-13 $\Rightarrow p = \frac{1}{13} = .077$ in

$\Delta\theta_{nut} = \frac{N_{in.}}{N_{out}}\Delta\theta_{in} = \left(\frac{1}{5}\right)(10\,\text{rev})$

$= 2$ rev

$\Delta R = N_t p \Delta\theta = (1)(.077\,\text{in/rev})(2\,\text{rev})$

$= .154$ in

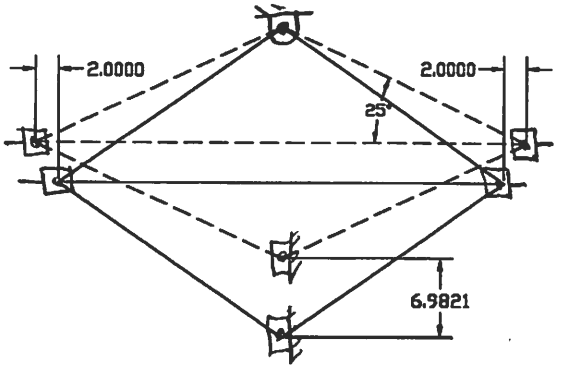
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12-9 ACME ½-10 ⇒ $p = \frac{1}{10} = .1$ in

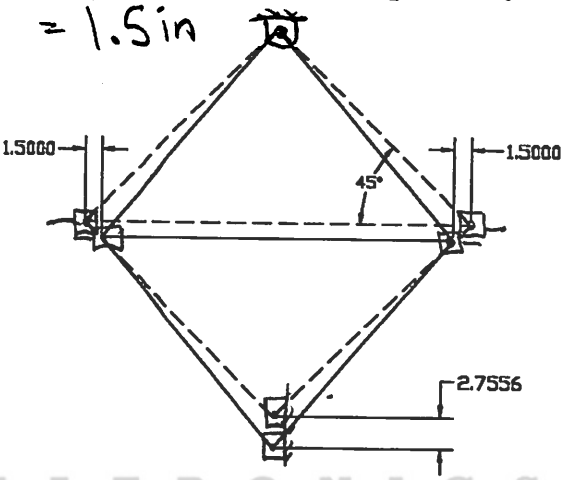
$\Delta R_{nut} = N_t p \Delta\theta = (1)(.1\text{ in/rev})(20\text{ rev})$
$= 2$ in



$\Delta R_{plate} = 6.982$ in ↓

12-10 ACME ½-10 ⇒ $p = \frac{1}{10} = .1$ in

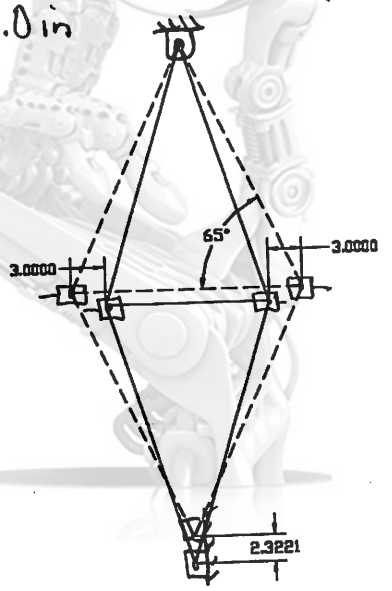
$\Delta R = N_t p \Delta\theta = (1)(.1\text{ in/rev})(15\text{ rev})$
$= 1.5$ in



$\Delta R_{plate} = 2.756$ in ↓

12-11 ACME ½-10 ⇒ $p = \frac{1}{10} = .1$ in

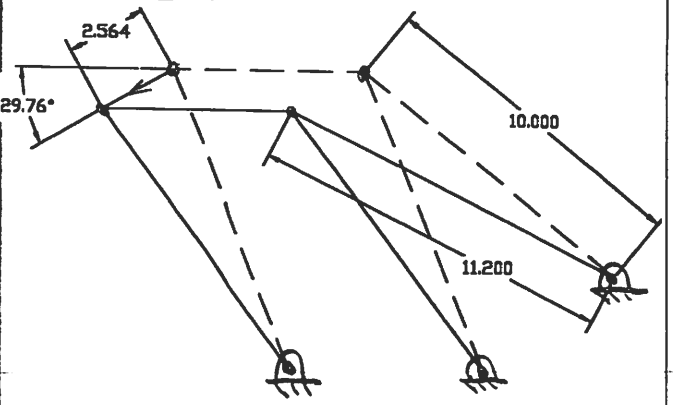
$\Delta R_{nut} = N_t p \Delta\theta = (1)(.1\text{ in/rev})(30\text{ rev})$
$= 3.0$ in



$\Delta R_{plate} = 2.322$ in ↓

12-12 ¾-10 ⇒ $p = \frac{1}{10} = .1$ in

$\Delta R = N_t p \Delta\theta = (1)(.1\text{ in})(12\text{ in})$
$= 1.2$ in

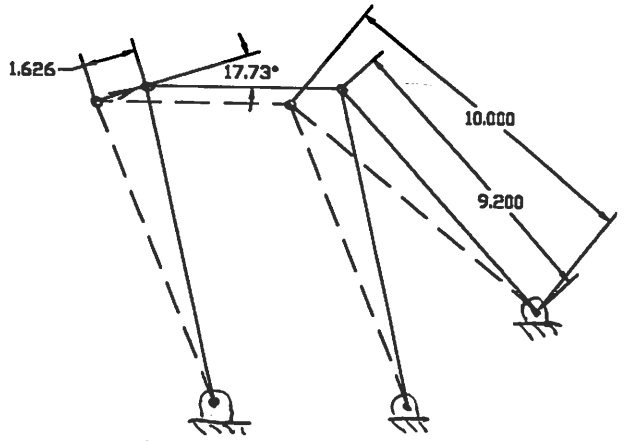


$\Delta R_{platform} = 2.564$ in 29.76° ↙

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12-13 $3/4\text{-}10 \Rightarrow p = \frac{1}{10} = 0.1\text{ in}$

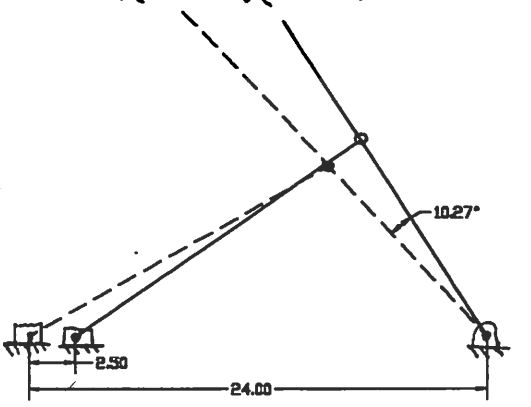
$\Delta R = (1)(.1\text{ in/rev})(8\text{ rev}) = .8\text{ in}$



$\Delta R_{platform} = 1.626\text{ in} \nearrow 17.73°$

12-14 $3/4\text{-}10 \Rightarrow p = \frac{1}{10} = 0.1\text{ in}$

$\Delta R_{nut} = (1)(.1\text{ in/rev})(25\text{ rev}) = 2.5\text{ in}$

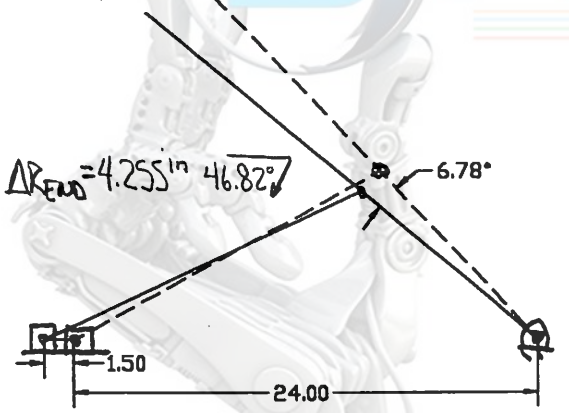


$\Delta\Theta_{platform} = 10.27°$, CW

$\Delta R_{end} = 6.445\text{ in} \nearrow 28.3°$

12-15 $3/4\text{-}10 \Rightarrow p = \frac{1}{10} = 0.1\text{ in}$

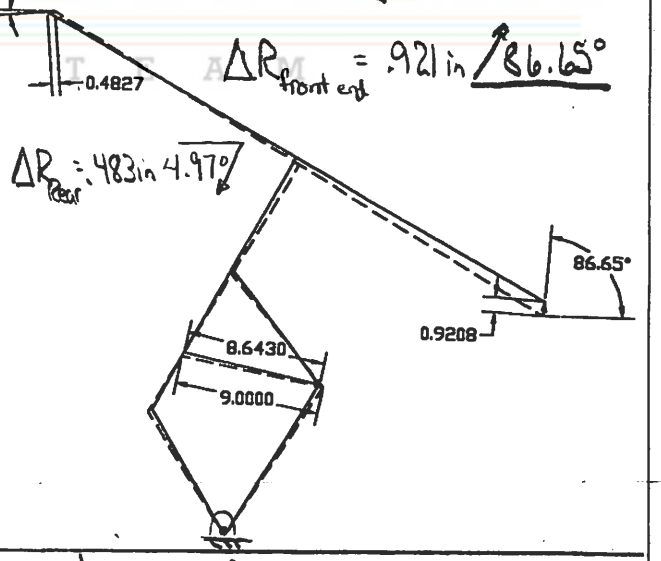
$\Delta R_{nut} = (1)(.1\text{ in/rev})(15\text{ rev}) = 1.5\text{ in}$



$\Delta\Theta_{platform} = 6.78°$, CCW

12-16 $7/8\text{-}14 \Rightarrow p = \frac{1}{14} = .0714\text{ in}$

$\Delta R = (1)(.0714\text{ in/rev})(5\text{ rev}) = .357\text{ in}$



12-17 $7/8\text{-}14 \Rightarrow p = \frac{1}{14} = .0714\text{ in}$

$\Delta R = (1)(.0714\text{ in/rev})(30\text{ rev}) = 2.142\text{ in}$

12-18 ACME $2\text{-}4 \Rightarrow p = \frac{1}{4} = .25\text{ in}$

$V_{nut} = N_t\, p\, \omega = (1)(.25\text{ in/rev})(40\text{ rev/min})$

$= 10\text{ in/min}\left(\frac{1\text{ min}}{60\text{ sec}}\right)$

$= .167\text{ in/s} \downarrow$

12-19 ACME 1-5 ⇒ $p = \frac{1}{5} = .20$ in

$V_{nut} = N_t\, p\, \omega$

$= (1)(.20^{in}/_{rev})(1200^{rev}/_{min})$

$= 240^{in}/_{min}\left(\frac{1\,min}{60\,sec}\right)$

$= 4^{in}/_{sec}$

$V_{door\ bottom} = 4^{in}/_{s}\ \uparrow$

12-20 $\omega_{in} = 600$ rpm

$\omega_{nut} = \left(\frac{1}{5}\right)(600) = 120$ rpm

½-13 UNC ⇒ $p = \frac{1}{13} = .0769$ in

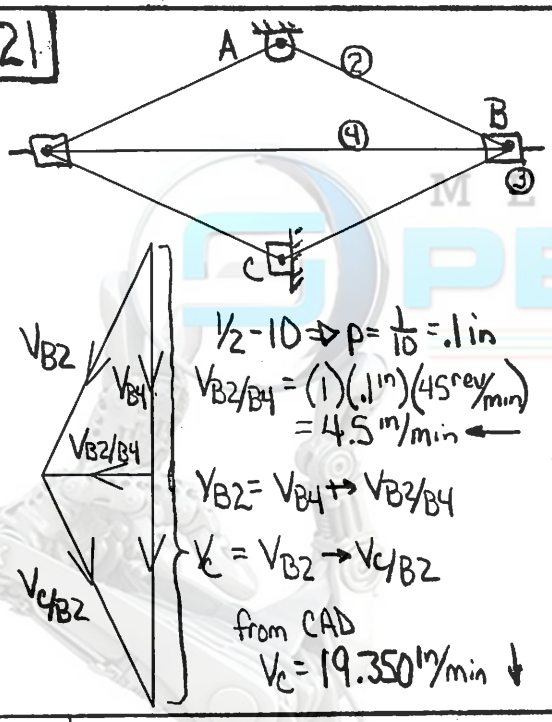
$V_{table} = N_t\, p\, \omega$

$= (1)(.0769^{in}/_{rev})(120^{rev}/_{min})$

$= 9.23^{in}/_{min}\left(\frac{1\,min}{60\,sec}\right)$

$= .154^{in}/_{sec}\ \uparrow$

12-21



½-10 ⇒ $p = \frac{1}{10} = .1$ in

$V_{B2/B4} = (1)(.1^{in})(45^{rev}/_{min})$

$= 4.5^{in}/_{min}\ \leftarrow$

$V_{B2} = V_{B4} \leftrightarrow V_{B2/B4}$

$V_C = V_{B2} \rightarrow V_{C/B2}$

from CAD

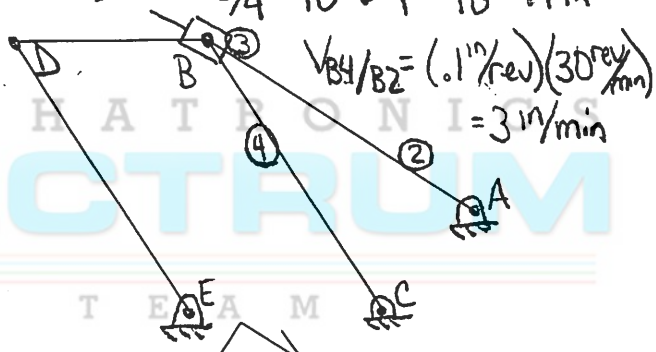
$V_C = 19.350^{in}/_{min}\ \downarrow$

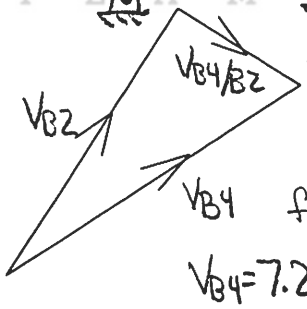
12-22

3/4-10 ⇒ $p = \frac{1}{10} = .1$ in

$V_{B4/B2} = (.1^{in}/_{rev})(30^{rev}/_{min})$

$= 3^{in}/_{min}$





$V_{B4} = V_{B2} \leftrightarrow V_{B4/B2}$

V_{B4} from CAD:

$V_{B4} = 7.236^{in}/_{min}\ \angle 32.75°$

12-23

12-24

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12-25 ACME 2-4 $\Rightarrow p = \frac{1}{4} = .25$ in

$\omega_1 = 40 \text{rpm}\left(\frac{1 \text{min}}{60 \text{sec}}\right) = .667 \text{ rev/sec}$

$\omega_2 = 0$

$\alpha = (0 - .667)/1.7 = -.392 \text{ rev/sec}^2$

$a_{nut} = N_t\, p\, \alpha$

$= (1)(.25 \text{ in/rev})(.392 \text{ rev/sec}^2)$

$= .098 \text{ in/s}^2 \uparrow$ (opposite vel)

12-26 ACME 1-5 $\Rightarrow p = \frac{1}{5} = .20$ in

$\omega_1 = 0$

$\omega_2 = 1200 \text{rpm}\left(\frac{1 \text{min}}{60 \text{sec}}\right) = 20 \text{ rev/sec}$

$\alpha = (20 - 0)/.7 = 28.57 \text{ rev/sec}^2$

$a_{nut} = N_t\, p\, \alpha$

$= (1)(.20 \text{ in/rev})(28.57 \text{ rev/sec}^2)$

$= 5.714 \text{ in/s}^2 \uparrow$ (same dir as vel)

12-27 $(\omega_{nut})_1 = 0$

$(\omega_{in})_2 = 1200 \text{rpm}$

$(\omega_{nut})_2 = \left(\frac{1}{5}\right)(1200) = 240 \text{rpm}\left(\frac{1 \text{min}}{60 \text{sec}}\right)$

$= 4 \text{ rev/sec}$

$\Delta\omega^2 = 2\alpha\Delta\theta \Rightarrow \alpha = \frac{\Delta\omega^2}{2\Delta\theta} = \frac{(4 \text{ rev/s})^2}{2(20 \text{rev})}$

$= .40 \text{ rev/sec}^2$

$a_{nut} = N_t\, p\, \alpha \qquad p = \frac{1}{13} = .0769$ in

$= (1)(.0769 \text{ in/rev})(.40 \text{ rev/sec}^2)$

$= .0307 \text{ in/sec}^2 \uparrow$ (same dir as vel)

12-28

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12-29 ACME $\frac{1}{2}\times 10$ $p = .10\text{ in} = L$ $D = .4306$

$$T = \frac{500(.4306)}{2}\left[\frac{.1 + \pi(.2)(.4306)}{\pi(.4306) - (.2)(.1)}\right]$$

$$= 29.9 \text{ in lbs}$$

12-30

$$e = \frac{.1}{\pi(.4306)}\left[\frac{\pi(.4306)(.968) - (.2)(.1)}{\pi(.2)(.4306) + (.1)(.968)}\right]$$

$$= .259 = 26\%$$

12-31 $\lambda = \tan^{-1}\frac{2(3\text{mm})}{\pi(25\text{mm})} = 7.25$ $D \approx 25$

$\tan\lambda = .127$ self-locking for general purpose

$$T = \frac{(4000\text{N})(25\text{mm})}{2}\left[\frac{10 + \pi(.15)(25)}{\pi(25) - .15(10)}\right] = 14136 \text{ N mm}$$

$$T = \frac{(4000\text{N})(25\text{mm})}{2}\left[\frac{\pi(.15)(25) - 10}{\pi(25) + .15(10)}\right] = 1113 \text{ N mm}$$

$$e = \frac{10}{\pi 25}\left[\frac{\pi(25)\cos 14.5 - .15(10)}{\pi .15(25) + 10\cos 14.5}\right] = .442$$

$$= 44.2\%$$

12-32 $\tan\lambda = \frac{2(.2)}{\pi(.88)} = .144$ $D = .8726$ close! to self locking for general purpose

$\lambda = 8.19°$

$$T = \frac{2000(.88)}{2}\left[\frac{.4 + \pi(.15)(.88)}{\pi(.88) - .15(.4)}\right] = 248 \text{ in lbs}$$

$$T = \frac{2000(1)}{2}\left[\frac{\pi(.15)(1) - .4}{\pi(1) + .15(.4)}\right] = 17.8 \text{ in lbs}$$

$$e = \frac{.4}{\pi(.88)}\left[\frac{\pi(.88)\cos 14.5 - .15(.4)}{\pi(.15)(.88) + .4\cos 14.5}\right] = .472$$

$$= 47.2\%$$

12-33 $D = .4306\text{in}$ $p = .077\text{ in}$

$$T = \frac{75(.4306)}{2}\left[\frac{.077 + \pi(.15)(.4306)}{\pi(.4306) - .15(.077)}\right] = 3.37 \text{ in lbs}$$

12-34

$$e = \frac{.077}{\pi(.4306)}\left[\frac{\pi(.4306)\cos 30 - .15(.077)}{\pi(.15)(.4306) + .077\cos 30}\right] = .245$$

$$= 24.5\%$$

12-35

$$S_{nut} = (L_B - L_C)\,\theta_{screw}$$

$$\frac{S_{nut}}{\theta_{screw}} = .5\frac{\text{mm}}{\text{rev}} = L_B - L_C$$

Thread B use M 10 x 1.50

Thread C use M 6 x 1.00

12-36

$$S_{nut} = (L_B - L_C)\,\theta_{screw}$$

$$\frac{S_{nut}}{\theta_{screw}} = .25\frac{\text{mm}}{\text{rev}} = L_B - L_C$$

Thread B use M 10 x 1.50

Thread C use M 8 x 1.25

12-37

$$S_{nut} = (L_B - L_C)\,\theta_{screw}$$

$$\frac{S_{nut}}{\theta_{screw}} = .05\frac{\text{in}}{\text{rev}} = L_B - L_C$$

Thread B use $\frac{3}{4}$-10 $p = .10$

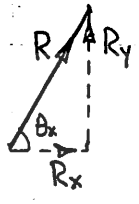
Thread C use $\frac{1}{4}$-20 $p = .05$

Chapter 13

13-1

horiz: $160 - 140\cos 25 = 33.1^N = R_x$

vert: $100 + 140\sin 25 = 159.2^N = R_y$



$R = \sqrt{R_x^2 + R_y^2} = 162.6^N$

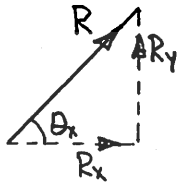
$\theta_x = \tan^{-1}(R_y/R_x) = 78.2°$

$R = 162.6^N \angle 78.2°$

13-2

horiz: $160 - 140\cos 65 = 100.8^N = R_x$

vert: $100 + 140\sin 65 = 226.9^N = R_y$



$R = \sqrt{R_x^2 + R_y^2} = 248.3^N$

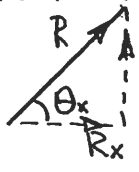
$\theta_x = \tan^{-1}(R_y/R_x) = 66.0°$

$R = 248.3^N \angle 66°$

13-3

horiz: $160 - 140\cos 105 = 196.2^N = R_x$

vert: $100 + 140\sin 105 = 235.2^N = R_y$

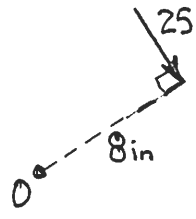


$R = \sqrt{R_x^2 + R_y^2} = 306.3^N$

$\theta_x = \tan^{-1}(R_y/R_x) = 50.1°$

$R = 306.3^N \angle 50.1°$

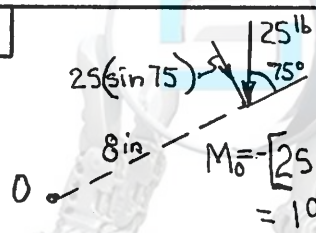
13-4



$M_O = -25^{lb}(8\,in)$

$= 200$ in lbs CW

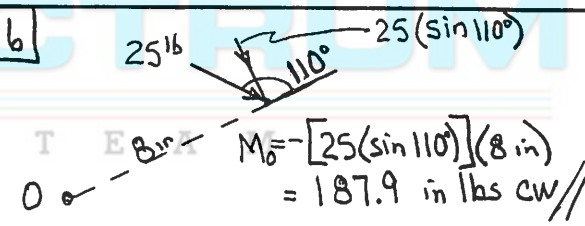
13-5



$M_O = -[25(\sin 75)](8\,in)$

$= 193.2$ in lbs CW

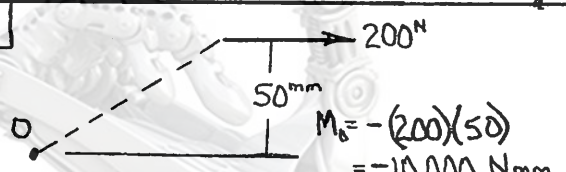
13-6



$M_O = -[25(\sin 110°)](8\,in)$

$= 187.9$ in lbs CW

13-7

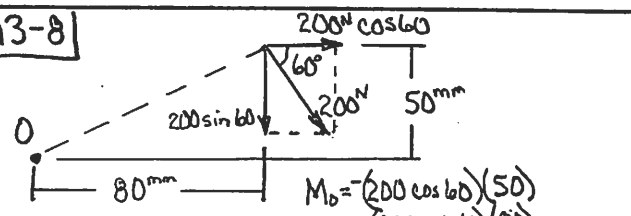


$M_O = -(200)(50)$

$= -10{,}000$ Nmm

$= 10$ Nm CW

13-8

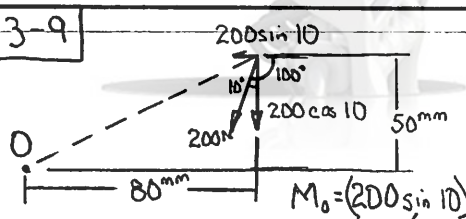


$M_O = -(200\cos 60)(50) - (200\sin 60)(80)$

$= -18{,}856$ N mm

$= 18.8$ Nm CW

13-9

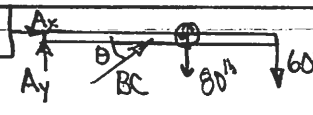


$M_O = (200\sin 10)(50) - (200\cos 10)(80)$

$= -14020$ N mm

$= 14.0$ Nm CW

13-10



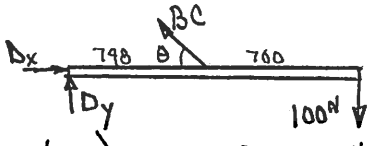
$\theta = \tan^{-1}\left(\frac{2.5}{1.5}\right) = 59°$

$+\circlearrowleft \sum M_A = 0 \quad (BD\sin 59°)1.5 - 600(1.5) - 80(3.75) = 0$

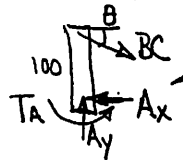
$BD = 3733$ lb Compression

13-11

$\theta = \tan^{-1}\left(\frac{500}{748}\right) = 33.7°$



$+\circlearrowleft M_D = 0$: $BC\sin\theta(748) - 100(1448) = 0 \Rightarrow BC = 349^N$

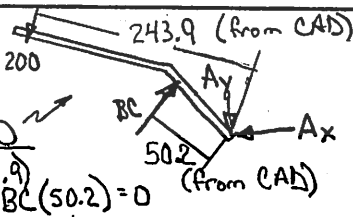


$+\circlearrowleft \sum M_A = 0$: $-BC\cos\theta(100) + T_A = 0$

$T_A = +29026$ N mm

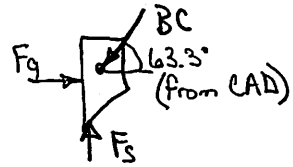
$= 29.0$ Nm CCW

13-12



$+\circlearrowleft \sum M_A = 0$: $200(243.9) - BC(50.2) = 0$

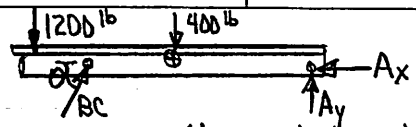
$BC = 971.7$ N



$+\uparrow \sum F_y = 0$: $F_s - BC\sin 63.3 = 0$

$F_s = 868^N \uparrow$

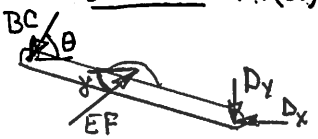
13-13



$\theta = \tan^{-1}\left(\frac{18}{10}\right) = 60.9°$

$\circlearrowleft \Sigma M_A = 0$: $+1200(80) + 400(44) - (BC\sin\theta)(72) = 0$

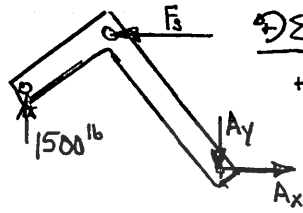
$BC = 1805.7$ lb



$\gamma = \tan^{-1}\left(\frac{18}{24}\right) = 36.9°$

$\circlearrowleft \Sigma M_D = 0$: $+(BC\cos\theta)(24) + (BC\sin\theta)(58) - (EF\cos\gamma)(18) - (EF\sin\gamma)(24) = 0 \Rightarrow EF = 3908.7$ lb

13-14



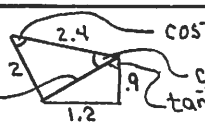
$\circlearrowleft \Sigma M_A = 0$

$+F_s(5) - 1500(4) = 0$

$F_s = 1200$ lbs (compression)

13-15

geom.

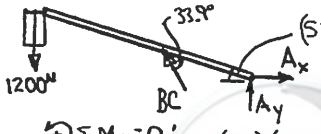


$\sqrt{.9^2 + 1.2^2} = 1.34$

$\cos^{-1}\left(\frac{2^2 + 2.4^2 - 1.34^2}{2(2)(2.4)}\right) = 33.9°$

$\cos^{-1}\left(\frac{2.4^2 + 1.34^2 - 2^2}{2(2.4)(1.34)}\right) = 56.4°$

$\tan^{-1}\left(\frac{1.2}{.9}\right) = 53.1°$

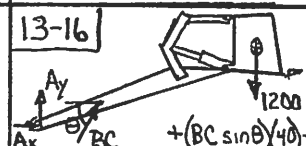


$(53.1 + 56.4) - 90 = 19.5°$

$\circlearrowleft \Sigma M_A = 0$: $+(1200)(5\cos 19.5) - (BC\sin 33.9)(2.4) = 0$

$BC = 4225$ N (compression)

13-16

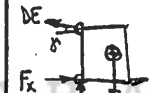


$\theta = \tan^{-1}\left(\frac{50}{40}\right) = 51.3°$

$\circlearrowleft \Sigma M_A = 0$:

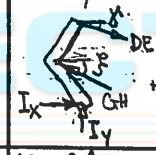
$+(BC\sin\theta)(40) - (BC\cos\theta)(20) - (1200)(112) = 0$

$BC = 7182$ lbs (Compression)



$\circlearrowleft \Sigma M_E = 0$: $(DE\cos\gamma)(20) - 1200(12) = 0 \Rightarrow DE = 745$ lbs

$\gamma = \tan^{-1}\left(\frac{8}{30}\right) = 14.9°$



$\circlearrowleft \Sigma M_I = 0$: $+(GH\cos\beta)(16) + (GH\sin\beta)(10) - (DE\cos\gamma)(28) = 0$

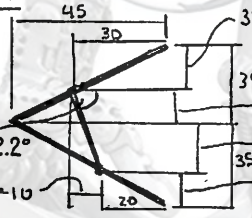
$\beta = \tan^{-1}\left(\frac{16}{30}\right) = 28.1°$

$GH = 2137$ lb (Compression)

13-17

geom

from similar triangles



$35\left(\frac{30}{45}\right) = 23.3$

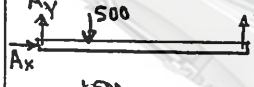
$35 - 23.3 = 11.7$

$35 - 15.6 = 19.4$ } 31.1

$\tan^{-1}\left(\frac{31.4}{10}\right) = 72.2°$

$30 - 20 = 10$

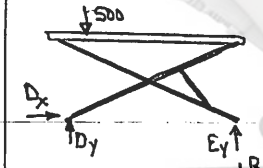
$35\left(\frac{20}{45}\right) = 15.6$



$\circlearrowleft \Sigma M_A = 0$: $-500(30) + B_y(90) = 0$ $\quad B_y = 166.6$

$\uparrow \Sigma F_y = 0$ $\quad A_y + B_y - 500 = 0$ $\quad A_y = 333.4$

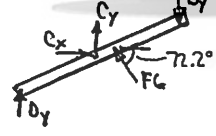
$\Sigma F_x = 0$ $\quad A_x = 0$



$D_y = 333.4$

$E_y = 166.6$

$D_x = 0$

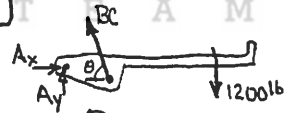


$\circlearrowleft \Sigma M_C = 0$

$-B_y(45) - D_y(45) + (FG\cos 72.2)(11.7) + (FG\sin 72.2)(15) = 0$

$FG = 1260$ lbs (compression)

13-18

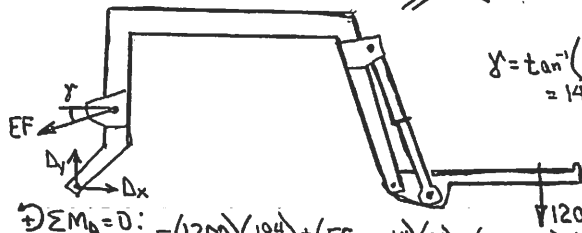


$\theta = \tan^{-1}\left(\frac{68}{32}\right) = 64.8°$

$\circlearrowleft \Sigma M_A = 0$:

$-1200(64) + (BC\sin 64.8)(18) - (BC\cos 64.8)(8) = 0$

$BC = 5962$ lbs (tension)

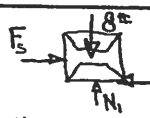


$\gamma = \tan^{-1}\left(\frac{18}{72}\right) = 14.0°$

$\circlearrowleft \Sigma M_D = 0$: $-(1200)(184) + (EF\cos 14)(23) - (EF\sin 14)(10) = 0$

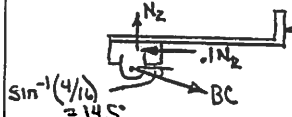
$EF = 11097$ lbs (tension)

13-19



$N_1 = 8$ oz

$F_s = .25N_1 = 2$ oz



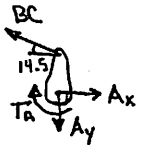
$\uparrow \Sigma F_y = 0$ $\quad N_2 - BC\sin 14.5 = 0$

$\rightarrow \Sigma F_x = 0$ $\quad -F_s - .1N_2 + BC\cos 14.5 = 0$

solving:

$BC = 2.12$ lbs

$N_2 = .53$ lbs



$\circlearrowleft \Sigma M_A = 0$:

$-T_A + (BC\cos 14.5)4 = 0$

$T_A = 8.21$ in lbs CW

500 SHEETS, FILLER 5 SQUARE
50 SHEETS EYE-EASE® 5 SQUARE
100 SHEETS EYE-EASE® 5 SQUARE
200 SHEETS EYE-EASE® 5 SQUARE
42-381 100 RECYCLED WHITE 5 SQUARE
42-382 200 RECYCLED WHITE 5 SQUARE
42-399
13-782
Made in U.S.A.
National® Brand

Chapter 14

14-1

$W = mg = (2.3\,kg)(9.81\,m/s^2) = 22.56\,N$

14-2

$m = \frac{W}{g} = \frac{4.5\,lb}{32.2\,ft/s^2} = 0.14\,slugs$

14-3

$I = mk^2 = \frac{W}{g}k^2$

$= \frac{(4.5\,lb)}{(32.2\,ft/s^2)}\left(\frac{5}{12}\,ft\right)^2 = .024\,lb\,ft\,s^2$

$= 0.29\,lb\,in\,s^2$

14-4

$I = mk^2$

$= (6\,kg)(150\,mm)^2$

$= 135000\,kg\,mm^2 = 0.135\,kg\,m^2$

14-5

$I_x = \frac{1}{2}mr^2 = \frac{W}{2g}r^2$

$= \frac{(5\,lb)}{2(386.4\,in/s^2)}\left(\frac{.625\,in}{2}\right)^2$

$= 0.00063\,lb\,in\,s^2$

$k = \sqrt{\frac{I}{m}} = \sqrt{\frac{.00063}{(5/386.4)}} = 0.221\,in$

14-6

$I_x = \frac{1}{2}mr^2$

$= \frac{1}{2}\left[100\,kg\left(\frac{50\,mm}{2}\right)^2\right]$

$= 31{,}250{,}000\,kg\,mm^2 = 31.25\,kg\,m^2$

$k = \sqrt{\frac{31.25\,kg\,m^2}{100\,kg}} = .56\,m = 56\,mm$

14-7

$I_x = \frac{1}{2}mr^2 = \frac{W}{2g}r^2$

$= \frac{48\,lb}{2(32.2\,ft/s^2)}\left(\frac{2\,ft}{2}\right)^2 = 0.745\,lb\,ft\,s^2$

14-8

$I_z = \frac{1}{12}\left[m(3r^2 + l^2)\right]$

$= \frac{1}{12}\left\{\left(\frac{48\,lb}{32.2\,ft/s^2}\right)\left[3\left(\frac{2\,ft}{2}\right)^2 + (3\,ft)^2\right]\right\}$

$= 1.49\,lb\,ft\,s^2$

14-9

$I_z = \frac{1}{12}\left[m(3r^2 + l^2)\right]$

$= \frac{1}{12}\left\{\left(\frac{2\,lb}{386.4\,in/s^2}\right)\left[3\left(\frac{1.25\,in}{2}\right)^2 + (14\,in)^2\right]\right.$

$= .085\,lb\,in\,s^2$

$I_{z'} = I_z + md^2 = .085 + \left(\frac{2}{386.4}\right)(3)^2$

$= 0.132\,lb\,in\,s^2$

14-10

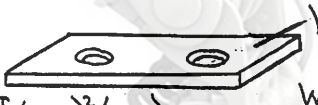
$I_z = \frac{1}{12}\left[ml^2\right]$

$= \frac{1}{12}\left[6\,kg\,(0.4\,m)^2\right] = .08\,kg\,m^2$

$I_{z'} = I_z + md^2 = .08 + 6(.12)^2$

$= 0.166\,kg\,m^2$

14-11



$V_{block} = (3.25)(.75)(.125)$

$= .305\,in^3$

$W_{block} = (.183)(.305)$

$= .056\,lb$

$V_{hole} = \frac{\pi}{4}(.375)^2(.125)$

$= .014\,in^3$

$W_{hole} = (.183)(.014) = .0025\,lb$

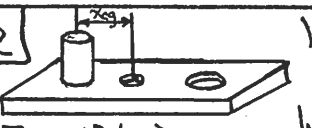
$I_{block} = \frac{1}{12}\left[\frac{.056}{386.4}(.75^2 + 3.25^2)\right] = .000134\,lb\,in\,s^2$

$I_{hole} = \frac{1}{2}\left[\frac{.0025}{386.4}(.1875)^2\right] = 1.14\times10^{-7}\,lb\,in\,s^2$

$I = .000134 - 2\left[1.14\times10^{-7} + \frac{.0025}{386.4}(.875)^2\right]$

$= .000133\,lb\,in\,s^2$

14-12



$V_{block} = (3.25)(.75)(.125)$

$= .305\,in^3$

$W_{block} = (.183)(.305)$

$= .056\,lb$

$V_{stud} = \frac{\pi}{4}(.375)^2(.75)$

$= .0828\,in^3$

$W_{stud} = (.183)(.0828)$

$= .015\,lb$

$V_{hole} = \frac{\pi}{4}(.375)^2(.125)$

$= .014\,in^3$

$W_{hole} = (.183)(.014) = .0025\,lb$

$\bar{x}_{cg} = \frac{(.015)(0) + (.056)(.875) - (.0025)(1.75)}{.015 + .056 - .0025}$

$= .65\,in$

$I_{block} = \frac{1}{12}\left[\frac{.056}{386.4}(.75^2 + 3.25^2)\right] = .000134\,lb\,in\,s^2$

$I_{hole} = \frac{1}{2}\left[\frac{.0025}{386.4}(.1875)^2\right] = 1.14\times10^{-7}\,lb\,in\,s^2$

$I_{stud} = \frac{1}{2}\left[\frac{.015}{386.4}(.1875)^2\right] = 6.82\times10^{-7}\,lb\,in\,s^2$

$I_{total} = .000134 + \frac{.056}{386.4}(.875 - .65)^2$

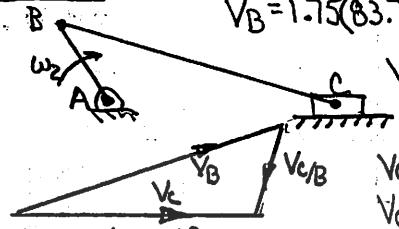
$+ 6.82\times10^{-7} + .015/386.4\,(.65)^2$

$-\left[1.14\times10^{-7} + .0025/386.4\,(1.75 - .65)^2\right] = .000135\,lb\,in\,s^2$

14-13

$\omega_2 = 800 \text{ rpm}\left(\frac{\pi}{30}\right) = 83.77 \text{ rad/s}$

$V_B = 1.75(83.77) = 146.6 \text{ in/s} \angle 25°$



$V_C = V_B + V_{C/B}$

from CAD

$V_{C/B} = 63.4 \text{ in/s} \quad 76.9°$

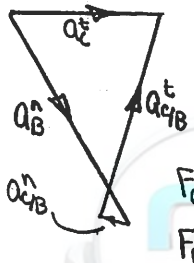
$V_C = 118.0 \text{ in/s} \rightarrow$

$a_B^n = (146.6)^2/1.75 = 12280 \text{ in/s}^2 \quad 65°$

$a_{C/B}^n = (63.4)^2/7 = 574.2 \text{ in/s}^2 \quad 13.1°$

$a_C^n = 0$

$a_C^t = a_B^n + a_{C/B}^n + a_{C/B}^t$



from CAD

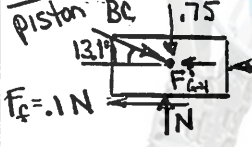
$a_C^t = 7187.3 \text{ in/s}^2 \rightarrow$

$a_{C/B}^t = 11292.6 \text{ in/s}^2 \quad 76.9°$

$F_{G4}^i = \frac{.75}{386.4}(7187.3) = 13.91 \text{ lb} \leftarrow$

$F_p = 70\left[\frac{\pi}{4}(1.5)^2\right] = 123.7 \text{ lb} \leftarrow$

FBD piston



$\Sigma F_x = 0$:

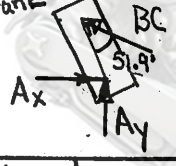
$BC\cos 13.1 - .1N - F_p - F_{G4}^i = 0$

$\Sigma F_y = 0$:

$-BC\sin 13.1 - .75 + N = 0$

$\Rightarrow BC = 144.7 \text{ lb}$

FBD crank



$\circlearrowleft \Sigma M_A = 0$

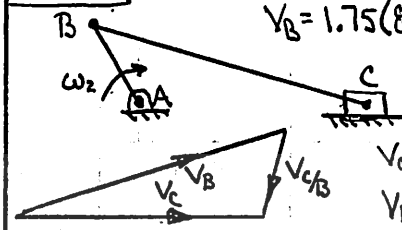
$-T_A + (BC\sin 59.1)(1.75) = 0$

$T_A = 217.3 \text{ in lbs CW}$

14-14

$\omega_2 = 800 \text{ rpm}\left(\frac{\pi}{30}\right) = 83.77 \text{ rad/s}$

$V_B = 1.75(83.77) = 146.6 \text{ in/s} \angle 25°$



$V_C = V_B + V_{C/B}$

from CAD

$V_{C/B} = 63.4 \text{ in/s} \quad 76.9°$

$V_B = 118.0 \text{ in/s} \rightarrow$

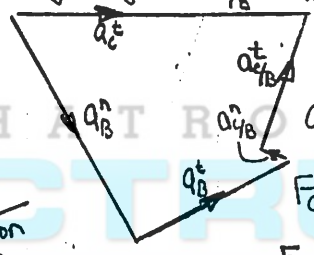
$a_B^n = (146.6)^2/1.75 = 12280 \text{ in/s}^2 \quad 65°$

$a_{C/B}^n = (63.4)^2/7 = 574.2 \text{ in/s}^2 \quad 13.1°$

$a_C^n = 0$

$a_B^t = (1.75)(5000) = 8750 \text{ in/s}^2 \quad 25°$

$a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$



from CAD

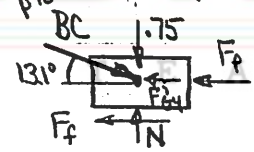
$a_C^t = 14257 \text{ in/s}^2 \rightarrow$

$a_{C/B}^t = 7496 \text{ in/s}^2 \quad 76.9°$

$F_{G4}^i = \frac{.75}{386.4}(14257) = 26.7 \text{ lb} \leftarrow$

$F_p = 70\left[\frac{\pi}{4}(1.5)^2\right] = 123.7 \text{ lb} \leftarrow$

FBD piston



$\Sigma F_x = 0$:

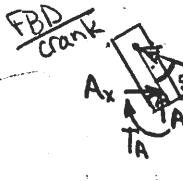
$BC\cos 13.1° - F_p - F_{G4}^i - .1N = 0$

$\Sigma F_y = 0$:

$N - BC\sin 13.1 - .75 = 0$

$\Rightarrow BC = 148.3 \text{ lb}$

FBD crank



$\circlearrowleft \Sigma M_A = 0$

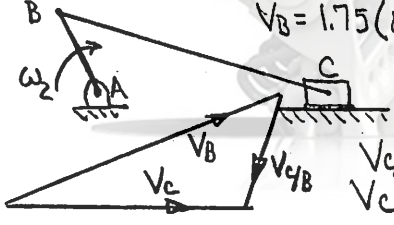
$-T_A + (BC\sin 59.1)(1.75) = 0$

$T_A = 222.7 \text{ in lb CW}$

14-15

$\omega_2 = 800 \text{ rpm}\left(\frac{\pi}{30}\right) = 83.77 \text{ rad/s}$

$V_B = 1.75(83.77) = 146.6 \text{ in/s} \angle 25°$



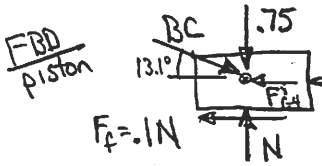
$V_C = V_B + V_{C/B}$

from CAD

$V_{C/B} = 63.4 \text{ in/s} \quad 76.9°$

$V_C = 118.0 \text{ in/s} \rightarrow$

FBD piston



$F_p = 70\left[\frac{\pi}{4}(1.5)^2\right] = 123.7 \text{ lb} \leftarrow$

$\Sigma F_x = 0$:

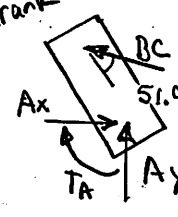
$BC\cos 13.1 - F_{G4}^i - F_p - .1N = 0$

$\Sigma F_y = 0$:

$-BC\sin 13.1 - .75 + N = 0$

$\Rightarrow BC = 130.4 \text{ lb}$

FBD crank



$\circlearrowleft \Sigma M_A = 0$

$-T_A + (BC\sin 59.1)(1.75) = 0$

$T_A = 195.7 \text{ in lb CW}$

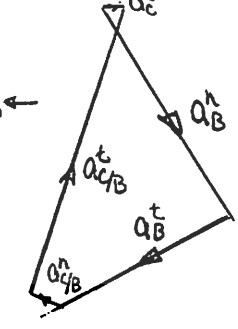
$a_B^n = (146.6)^2/1.75 = 12280 \text{ in/s}^2 \quad 65°$

$a_{C/B}^n = (63.4)^2/7 = 574.2 \text{ in/s}^2 \quad 13.1°$

$a_C^n = 0$

$a_B^t = (1.75)(5000) = 8750 \text{ in/s}^2 \quad 25°$

$a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$



from CAD

$a_C^t = 118 \text{ in/s}^2 \rightarrow$

$a_{C/B}^n = 15089 \text{ in/s}^2 \quad 76.9°$

$F_{G4}^i = \frac{.75}{386.4}(118) = .23 \text{ lb} \leftarrow$

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100 RECYCLED WHITE 5 SQUARE
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14-16

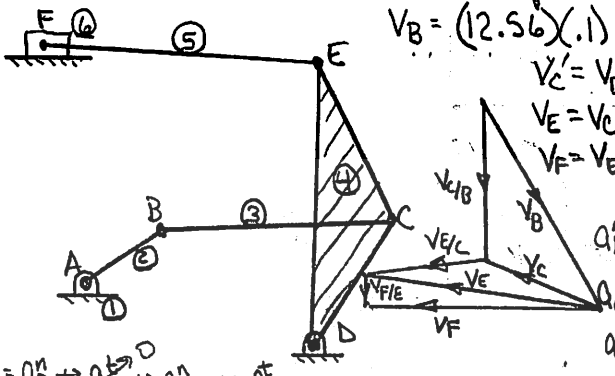
$\omega = 120 \text{ rpm} = 12.56 \text{ r/s}$ CCW

$V_B = (12.56)(.1) = 1.256 \text{ m/s}$ ↗ 60°

$V_C = V_B + V_{C/B}$

$V_E = V_C + V_{E/C}$

$V_F = V_E + V_{F/C}$



from CAD

$V_C = .619 \text{ m/s}$ ↘ 22.7°

$V_{C/B} = .851 \text{ m/s}$ ↘ 86.2

$V_E = 1.238 \text{ m/s}$ ↗ 7.6°

$V_{E/C} = .660 \text{ m/s}$ ↘ 6.5°

$V_F = 1.228 \text{ m/s}$ ←

$V_{F/E} = .164 \text{ m/s}$ ↘ 89.5°

$a_B^n = \frac{(1.256)^2}{.1} = 15.77 \text{ m/s}^2$ ↘ 30°

$a_C^n = \frac{(.619)^2}{.3} = 1.28 \text{ m/s}^2$ ↘ 67.3°

$a_{C/B}^n = \frac{(.851)^2}{.4} = 1.81 \text{ m/s}^2$ ↘ 3.8°

$a_E^n = \frac{(1.238)^2}{.6} = 2.55 \text{ m/s}^2$ ↘ 82.4°

$a_{E/C}^n = \frac{(.660)^2}{.32} = 1.361 \text{ m/s}^2$ ↘ 83.5

$a_{F/E}^n = \frac{(.164)^2}{.65} = .04 \text{ m/s}^2$ ↘ 5°

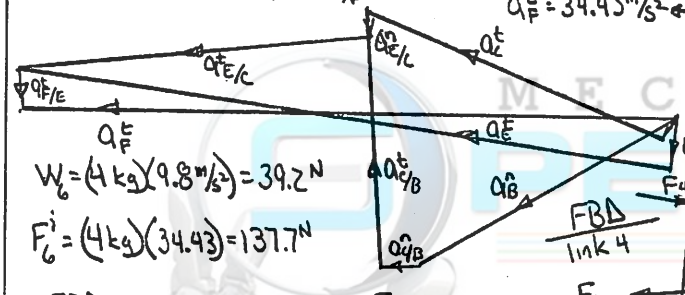
$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_E^n + a_E^t = a_C^n + a_C^t + a_{E/C}^n + a_{E/C}^t$

$a_F^t = a_E^n + a_E^t + a_{E/F}^n + a_{E/F}^t$

from CAD

$a_F^t = 34.43 \text{ m/s}^2$ ←



$W_6 = (4 \text{ kg})(9.8 \text{ m/s}^2) = 39.2 \text{ N}$

$F_6^i = (4 \text{ kg})(34.43) = 137.7 \text{ N}$

FBD link 6

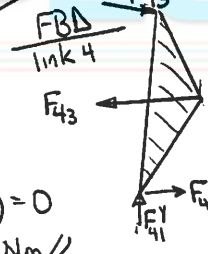


$\Sigma F_y = 0$: $-39.2 + F_{61} + F_{65} \sin .5° = 0$

$\Sigma F_x = 0$: $137.7 + .15(F_{61}) - F_{65} \cos .5° = 0$

$F_{65} = 143.6 \text{ N}$

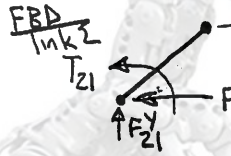
FBD link 4



$\Sigma M_D = 0$ $F_{43}(.2684) - F_{45}(.5954) = 0$

$F_{43} = 318.6 \text{ N}$ from CAD

FBD link 2



$\Sigma M_A = 0$

$T_{21} - F_{23}(.0441) = 0$

$T_{21} = 14.04 \text{ Nm}$ //

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_E^n + a_E^t = a_C^n + a_C^t + a_{E/C}^n + a_{E/C}^t$

$a_F^t = a_E^n + a_E^t + a_{F/E}^n + a_{F/E}^t$

14-17

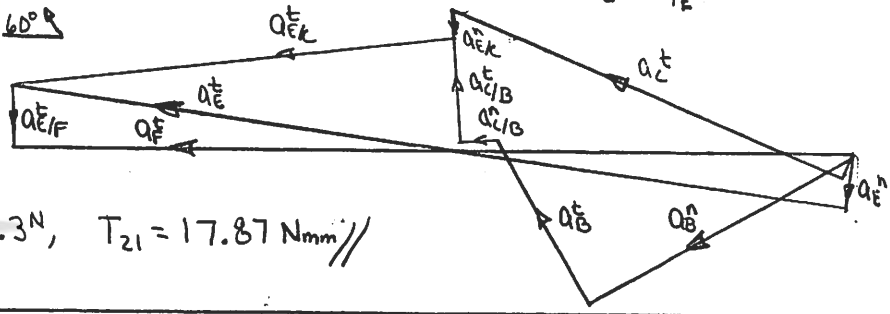
Using analysis from 14-16, with:

$\alpha_2 = 100 \text{ r/s}^2$ CCW

$a_B^t = (100)(.1) = 10 \text{ m/s}^2$ ↗ 60°

$a_F^t = 44.21 \text{ m/s}^2$ ←

$F_6^i = (4 \text{ kg})(44.21 \text{ m/s}^2) = 176.9 \text{ N}$



Using FBD's from 14-16:

$F_{65} = 182.7 \text{ N}$, $F_{43} = 405.3 \text{ N}$, $T_{21} = 17.87 \text{ Nmm}$ //

14-18

Using analysis from 14-16, with:

$\alpha_2 = 100 \text{ r/s}^2$ CW

$a_B^t = (100)(.1) = 10 \text{ m/s}^2$ ↗ 60°

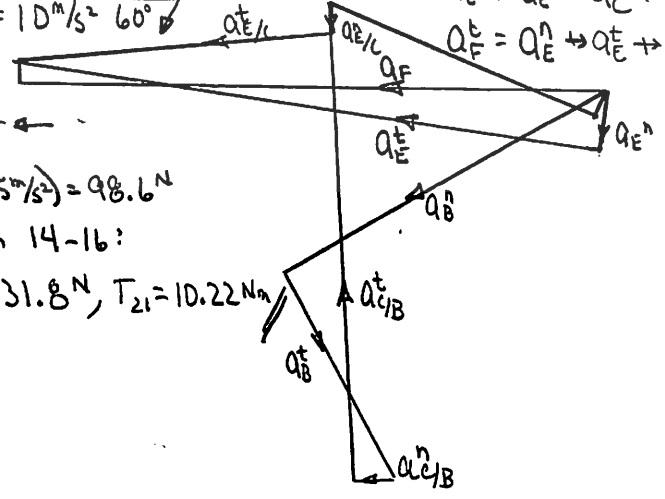
$a_F^t = 24.65 \text{ m/s}^2$ ←

$F_6^i = (4 \text{ kg})(24.65 \text{ m/s}^2) = 98.6 \text{ N}$

$a_C^n + a_C^t = a_B^n + a_B^t + a_{C/B}^n + a_{C/B}^t$

$a_E^n + a_E^t = a_C^n + a_C^t + a_{E/C}^n + a_{E/C}^t$

$a_F^t = a_E^n + a_E^t + a_{F/E}^n + a_{F/E}^t$



Using FBD's from 14-16:

$F_{65} = 109.5 \text{ N}$, $F_{43} = 231.8 \text{ N}$, $T_{21} = 10.22 \text{ Nm}$ //

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